

# Annual Report

2018-19



**ICAR-National Institute for Plant Biotechnology**

Lal Bahadur Shastri Centre, Pusa Campus

New Delhi - 110012

([www.nrcpb.res.in](http://www.nrcpb.res.in))



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## Preface



It is a matter of great pride that ICAR-NRCPB has been elevated to the level of National Institute for Plant Biotechnology (ICAR-NIPB). The Institute has come a long way since its inception in 1985 as Biotechnology Centre under ICAR-IARI. It was upgraded to an independent National Research Centre on Plant Biotechnology in 1993 and now as ICAR-NIPB in 2019. This brings us much greater national responsibility to conduct basic and applied research, and human resource development in Molecular Biology and Plant Biotechnology.

Our Institute is doing basic and strategic research on different aspects of crop improvement in five mandate crops, viz. rice, wheat, pigeonpea, chickpea, and mustard. Some of the major achievements of 2018-19 are summarized below:

For QTL mapping of salinity stress tolerance in wild rice, a linkage map covering 12 linkage groups with 1137 markers spanning 1309.3 cM map distance was constructed using 75 BC1 derived from IR64 NKWR173 (*O. rufipogon*) cross. Based on sequence analysis, 31 potential candidate genes underlying the major heat tolerance QTL regions on chromosomes 5 and 9 of Nagina 22 and IR 64 were identified for heat stress tolerance in rice. One polymorphic SSR marker each that can be used for introgression of heat QTLs (*qSTIY5.1*, *qSTIPSS9.1*) in nine popular rice varieties was identified. Two polymorphic CAPS markers were developed of which one was validated in the mapping population for yield and spikelet fertility under heat stress. Genetic diversity and population structure of 94 deepwater rice varieties were studied using 50K rice SNP gene chip. Forty one additive QTLs, two large-effect QTLs and twenty six minor effect QTLs, were identified for seedling stage salinity tolerance in aromatic short grain rice of Assam. To differentiate 22 *Oryza* species is chloroplast genome based Indel markers were developed. A novel bacterial isolate, *Pseudomonas*

*aeruginosa*, was identified as an effective biocontrol agent against *R. solani* and five other plant pathogens. Standard Operating Protocols (SOP) for genetic fidelity testing were developed using IRAP and ISSR markers for tissue culture raised banana, sugarcane, potato, date palm and bamboo plants. In an ongoing effort, total 791 accessions of wild rice have been collected from different parts of India, and a database web portal has been developed. Another comprehensive database, RiceMetaSysB, has been developed for identification of blast and bacterial leaf blight responsive genes in rice using publicly available microarray datasets.

Four entries of wheat, viz. HD 3289, HD 3294, HD 3295, HD 3296, were included in AICW&BIP trials (NIVT). Wheat variety, HD 3237 (Timely sown Restricted Irrigated condition (NWPZ)), was released and notified by CVRC in 2018. Three QTLs for leaf spot disease resistance in wheat were transferred to HD 2733 and PBW 343 through marker-assisted back crossing. The BC 3F1 plants showed up to 94.29% genetic similarity with the recipient parent. The derived lines exhibited significantly higher level of resistance against spot blotch under field condition. Transcript expression profiling of *GPAT* gene in two wheat cultivars, viz. HD 2967 and Raj 3765 was carried out at reproductive stage. A full coding region of *GPAT* was cloned from variety Raj 3765. A protocol for regeneration and transformation of bread wheat cv. HD 2967 was standardized by over expressing a *RBP* gene. A drought inducible promoter *PM19* was cloned and characterized from wheat genotype Raj 3765. Expression studies of one high affinity nitrate transporter gene (*TaNRT2.1*) and its probable component partner proteins (*TaNAR2.1* and *TaNAR2.2*) were conducted in wheat.

Small RNA sequencing of chickpea varieties ICC 4958 and ICC 1882 contrasting for drought tolerance was carried out at flowering stage under control

and drought stress conditions. Total 658 conserved miRNAs and 186 novel miRNAs were identified. From 335 differentially expressed miRNAs, eleven conserved and six novel miRNAs were validated in root tissue of the two genotypes.

A set of important flowering related genes, namely Ft and CCT were identified and characterized in pigeonpea. Different pigeonpea varieties were evaluated for total protein content and cooking qualities revealing good variation. To understand the molecular mechanism of pod borer resistance, transcriptome and proteome variations in pod borer-challenged leaves of the tolerant wild relative and sensitive cultivated pigeonpea genotypes were analyzed. A recombinant binary vector was constructed and thirty two transgenic tobacco lines were developed for combinatorial silencing of *H. armigera* acetylcholinesterase gene, *Ace1*, and 20 hydroxy ecdysone receptor gene, *EcR*. Eleven transgenic tobacco lines demonstrated moderate to high level of resistance against *H. armigera* in detached leaf insect bioassay.

Two *B. juncea*, accessions, namely IC 265495 and IC 313380, were registered with NBPGR (registration no. INGR 18032 and INGR 18033, respectively) for white rust resistance. ICAR-NIPB is one of the collaborators in development of these germplasms. Evaluations of the synthetic and introgression lines were done for enlarging genepool of *Brassica juncea*. Diversity analysis of advanced resynthesized and introgression lines showed that the new lines are more diverse than the existing genotypes. Resynthesized *B. juncea* lines were also evaluated for yield and yield-related traits on these locations for stability analysis. The advance introgression lines for *Alternaria* resistance were screened in hotspot locations and under controlled condition at ICAR-NRCPB, Delhi. The potential of legume lectins from chickpea, pigeonpea, cowpea,

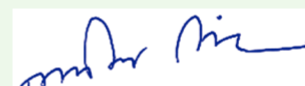
moth bean and lentil was explored vis-à-vis snowdrop lectin by expressing the lectin proteins *in vitro* and performing diet and bioassays for aphid and *Helicoverpa*, respectively. The LL and CHPL proteins were found to be more effective against aphids. The pupal mortality of *Helicoverpa* was maximum in LL and PPL. CRISPR/Cas9 vectors were developed to knockout *CENH3* gene in *B. juncea* and *B. oleracea*. A GFP-tailswap vector was also developed to rescue these *CENH3* knockout lines.

With regards to HRD activities, five NRCPB staffs received training from different national institutes of repute in 2018-19. Total thirty eight Ph. D. and 14 M. Sc. students are currently registered in the discipline of Molecular Biology and Biotechnology with IARI PG School. Four Ph. D. and seven M. Sc. students were awarded with doctoral and master's degree, respectively. More than 50 students from different universities and institutes across the country were trained in various aspects of biotechnology at the Institute through their thesis research project.

I would like to record my sincere thanks to all the scientists, administrative and technical staff, students, research fellows and contractual supporting staff for their contribution in research and other institutional activities of the Institute. My special thanks are due to Dr. Debasis Pattanayak, Dr. Monika Dalal, Dr. Rohini Sreevathsa, Dr. Subodh K. Sinha, Dr. Naveen C. Gupta, Dr. Deepak S. Bist, Dr. Sandhya Sanand and Dr. Rampal S. Niranjana for their help in compiling and editing of the annual report.

I am grateful to Dr. Trilochan Mohapatra, Secretary, DARE and Director General, ICAR; Dr. Anand K. Singh, Deputy Director General (Crop Sciences) and Dr. Devendra K. Yadav, Assistant Director General (Seeds), ICAR, for their constant support and guidance in overall functioning of the Institute.

Date: June 2019  
New Delhi



(N. K. Singh)

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# EXECUTIVE SUMMARY

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## Executive Summary

The committed and sustained efforts of the ICAR-National Institute for Plant Biotechnology (ICAR-NIPB) in conducting basic, applied, strategic and anticipatory research for crop improvement have culminated into high standard research discoveries, which are aptly reflected in quality publications and patent applications during the year 2018-19. Salient achievements of the Institute under five research programmes: (i) Search and Deployment of Genes for Stress Tolerance and Grain Quality in Rice, (ii) Adaptation of Wheat to Climate Change Induced Stresses, (iii) Improvement of Stress Tolerance in Chickpea, (iv) Stress Tolerance and Quality Improvement in Pigeonpea, and (v) Biotechnological Approaches for Brassica Improvement; and contributions of the Institute in Human Resource Development (HRD) in the year are given below:

Total 791 accessions of wild rice had been collected from different states of India, and a database web portal had been developed (nksingh.nationalprof.in:8080/iwrdb). All the information regarding collection such as village, block, district, state, longitude and latitude from where the accessions were collected and characterization of particular accessions is stored in the database. Another comprehensive database, RiceMetaSysB, was developed for identification of blast and bacterial leaf blight responsive genes in rice using publicly microarray datasets.

A total of 75 BILs was derived from IR64 X NKS173 (*O. rufipogon*) cross for the construction of high density linkage map and QTL analysis for salinity stress tolerance. A map of 1137 markers covering 12 linkage groups was constructed. The intraspecific map generated here spanned 1309.3 cM distance of the rice genome. Maximum number of markers were assigned to linkage groups 1 and 2 *i.e.* 175 and 152 respectively, while minimum number of markers were mapped to linkage group 9 and 10.

Genome wide association analysis revealed that 10 novel SNPs, distributed into six different chromosomes, were significantly associated with

submergence tolerance. Analysis of 119 genes in the major heat QTL regions, identified on chromosomes 5 and 9 using whole genome sequencing of Nagina 22 and IR 64, revealed that 34 genes harboured SNPs while 31 of them had one or more non-synonymous changes suggesting them as the potential candidates for heat stress tolerance in rice.

Five WRKY genes, viz. *WRKY7*, *WRKY58*, *WRKY62*, *WRKY64*, *WRKY76*, were identified to be highly expressing only after infection by virulent pathogen strain of *Maganporthe oryzae* suggesting their probable role in providing resistance against panicle blast disease in rice.

At least one polymorphic SSR that can be used for introgression of each heat QTL (qSTIY5.1, qSTIPSS9.1) in nine popular rice varieties had been identified. Further, two polymorphic CAPS markers were developed for the same, of which CAPS\_HT\_5.1 was validated in the mapping population for yield and spikelet fertility under heat stress tolerance.

Genetic diversity and population structure of total 94 deepwater rice landraces were studied using 50K rice SNP gene chip. Forty one additive QTLs, two large-effect QTLs and twenty six minor effect QTLs, were identified for seedling stage salinity tolerance in aromatic short grain rice of Assam. Two mapping population segregating for sheath blight resistance phenotype were developed. Fifteen Indel markers able to differentiate 22 *Oryza* species based on PCR amplicon size difference were identified.

A novel bacterial isolate, *Pseudomonas aeruginosa*, was identified as an effective biocontrol agent against *R. solani* and five other plant pathogens.

Standard Operating Protocols (SOP) for genetic fidelity testing were developed using IRAP and ISSR markers systems for detecting somaclonal variations amongst tissue culture raised banana, sugarcane, potato, date palm and bamboo plants.

Four entries of wheat, viz. HD 3289, HD3294, HD3295, HD3296, were included in AICW&BIP Trials (NIVT) during 2017-18. Wheat variety, HD 3237

(Timely Sown Restricted Irrigated Condition (NWPZ), released and notified by CVRC 2018.

Three wheat QTLs, Qsb.bhu-2A-Xgwm425 from SW89-5422, Qsb.bhu-2A and Qsb.bhu-5B from Chirya3, for leaf spot disease resistance in wheat were transferred to HD2733 and PBW343. Foreground and whole genome background selection were done in BC3F1 generation. The progenies showed up to 94.29% genetic similarity with the recipient parent. The derived lines exhibited significantly higher level of resistance against spot blotch under field condition.

*GPAT*, a multi gene family, encodes enzymes involved in several lipid biosynthetic pathways that play important biological roles in plant growth and development. Transcript expression profiling of *GPAT* was carried out in two wheat cultivars, viz. HD 2967, a popular high yielding variety, and Raj 3765, a heat tolerant cultivar at five generative stages starting from flag leaf induction. The results showed positive modulation of *GPAT* gene under high temperature in wheat implicating its functional role in pollen development, gametogenesis, and early grain development vide induction of phospholipid and TAG synthesis. A full coding region of *GPAT* was cloned from variety Raj3765 and structurally confirmed by various *in silico* analyses.

A protocol for regeneration and transformation of bread wheat cvHD2967 was standardized by over expressing a *RBP* gene. The putative transformants are being molecularly characterized in the subsequent generation. Continuing with the hormonal and heat priming work initiated a few years ago, the primed plants in advanced generation was evaluated in field conditions. Studies have been initiated to evaluate acquired and transgenerational thermotolerance in wheat.

EST-SSR markers were developed for pearl millet research from the available heat stress responsive transcriptome pearl millet data (NCBI SRA Accession No.: SRP151237). Fifty SSR motifs and 9 STS markers were validated in 24 diverse genotypes that were varied in their high temperature stress tolerance ability. Four EST-SSRs markers (15.38%) and one STS marker (16.66%) displayed polymorphism respectively. The PIC (Polymorphic Information Content) of newly developed EST-SSRs was in the range of 0.207 to 0.662 with an average value of 0.435. Based on the five polymorphic markers, the 24 pearl

millet genotypes could be clustered into 5 main and 11 sub clusters. Annotation of the five transcripts for which polymorphism was detected, were found to code for genes related to stress response and signaling pathway. The EST-SSRs and STS markers thus analyzed would augment the existing SSR marker resource and find their applications in diversity assessment and marker-aided breeding programs for genetic improvement of pearl millet.

A drought inducible promoter *PM19* was cloned and characterized from wheat genotype Raj3765. qRT-PCR analysis revealed that *PM19* is significantly induced in root as compared to shoot under drought stress.

In the quest of identifying key genes of nitrate-uptake and its translocation and best/contrasting wheat genotypes for this trait, expression studies of one of the important high affinity nitrate transporter gene (TaNRT2.1) and genes of its probable component partner proteins (TaNAR2.1 and TaNAR2.2) were conducted. Expression of both the NAR2s was observed under N-starved conditions. Expression of three key genes, presumably involved in nitrate-uptake (HATS and LATS categories) and root-shoot translocation, was dependent on external N-concentration and varied in different genotypes.

Four small RNA libraries from mature chickpea plants of drought tolerant var., ICC4958, and drought-susceptible var., ICC1882, at flowering stage, under control and drought stress conditions, were constructed and sequenced. Sequence analyses revealed presence of 658 miRNAs belonging to 37 conserved families and 186 novel miRNAs. Analysis revealed 335 differentially expressed miRNAs under drought stress in both tolerant and susceptible genotypes. Seventeen miRNAs, including eleven conserved and six novel miRNAs were validated in root tissue of both genotypes.

Transcriptome analysis was performed in chickpea genotypes contrasting with respect to seed protein content. In all, four cDNA libraries were generated using genotypes contrasting for seed protein content. Overall, 98.06% of the high-quality reads were mapped to the chickpea genome. About 400 significantly differentially expressed genes were found in genotypes with high and low seed protein content. The small RNA libraries were also made for identification of miRNAs and target genes in genotypes with high- and low-seed protein content.

A comprehensive survey of the pigeonpea genome was carried out to identify and characterize the flowering related genes. A set of important genes such as Ft and CCT were identified, characterized and their role was shown in flowering pathways that are either photoperiod dependent and independent.

Different pigeonpea varieties were evaluated for total protein content and cooking qualities. Some preliminary information suggests that protein content ranges from 10 to 26 %. Similarly cooking quality parameters including swelling capacity and swelling index etc are important measures for further evaluation of this trait.

Molecular mechanism of pod borer resistance in the pigeonpea wild relative, *Cajanus platycarpus* was studied. Transcriptomic and proteomic variations in pod borer-challenged leaves of the wild relative and the cultivated pigeonpea were analyzed. Differentially expressed genes were cloned and are now validated for pod borer resistance.

Combinatorial host-delivered-RNAi was employed to test pesticidal efficacy against *Helicoverpa armigera*. A recombinant binary vector cassette, pBI121::IR-EcR-Ace1-Com, was constructed for combinatorial silencing of *H. armigera* acetylcholinesterase gene, *Ace1*, and 20 hydroxy ecdysone receptor gene, *EcR*. Thirty two transgenic tobacco lines, obtained after screening of putative transformants by PCR and RT-PCR using gene specific and *nptII* specific primers, were tested for *H. armigera* resistance by detached leaf insect bioassay. Eleven transgenic tobacco lines demonstrated moderate to high level resistance. Growth retardation of larvae fed on these 11 transgenic tobacco leaves was observed.

Two germplasm of *B. juncea*, IC265495 and IC313380, were registered with NBPGR (registration no. INGR18032 and INGR18033, respectively) for white rust resistance. ICAR-NIPB is one of the collaborators in development of these germplasms.

Evaluations of the synthetic and introgression lines were done for enlarging genepool of *Brassica juncea*. The advanced resynthesized (RBJ1 to RBJ 92) and introgression lines (ERJ1 to ERJ156) were analyzed for genetic diversity along with the parents, and results showed that the new lines are more diverse than the existing genotypes. Resynthesized *B. juncea* lines were also evaluated for yield and yield-related traits on these locations for stability analysis.

The advance introgression lines for the *Alternaria* resistance were screened under natural condition at GBPUAT, Pantangar & ICAR-IARI, RS, Pusa, Samastipur, Bihar and under controlled condition at ICAR-NRCPB, Delhi.

The potential of legume lectins from chickpea, pigeonpea, cowpea, mothbean and lentil was explored vis-à-vis snowdrop lectin. These legume lectin proteins were functionally validated by expressing the lectin proteins *in vitro* and their inclusion in the diet for aphid and *Helicoverpa* bioassay. The LL and CHPL proteins were found to be more effective amongst the five legume lectins with the aphid mortality of 59.25 to 81.47% against aphids. The LC<sub>50</sub> dose on fourth day was as low as 0.004 ug/g of diet for LL protein compared to snowdrop lectin which was 7.102 ug/g of diet. This was also seen in the transmission electron microscopy of lectin treated foregut of aphid which showed the damage level. Similarly, the LC<sub>50</sub> dose for *Helicoverpa* larvae on day 8 was found to be 0.580 ug/g of diet. The pupal mortality was maximum in LL (20%) and PPL and LL having maximum of malformed adult emergence (20%).

The expression of *WRKY33* transcript was found to be higher in leaf tissues compared to other tissues in both *Brassica juncea* and *Camelina sativa*. The transcript expression level of *CsWRKY33* gene was found to be higher in different tissues as compared to *BjWRKY33* transcripts (more than 1.8 folds in leaf tissues).

CRISPR/Cas9 vectors were developed to knockout *CENH3* gene in *B. juncea* and *B. oleracea*. Further a GFP-tailswap vector was also developed to rescue these *CENH3* knockout lines. Genome size of some wild mustard species was estimated through flow cytometry.

With regards to HRD activities, five NRCPB staff (three scientists and two administrative staff) received training from different national institutes of repute in 2018-19 under HRD programme. Thirty eight Ph. D. and 14 M. Sc. students were registered in the discipline of Molecular Biology and Biotechnology in 2018-19. Four Ph. D. and seven M. Sc. students were awarded with doctoral and master's degree, respectively. More than fifty students from different universities and institutes across the country were trained in various aspects of biotechnology at the Centre.



# **About the Institute**

**NIPB and the Mandate**

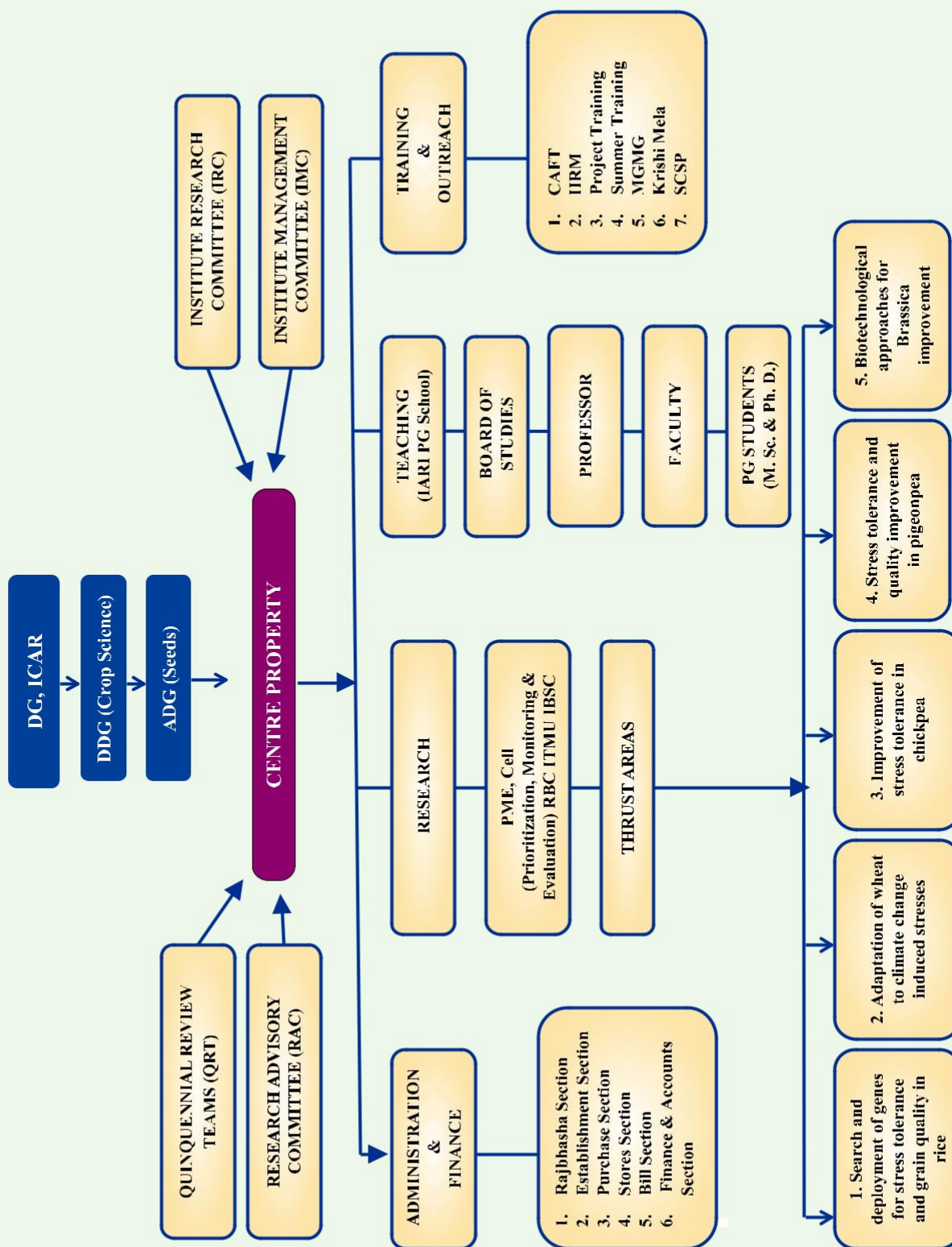
**Staff Position**

**Financial Statement**

**Resource Generation**

**Personnel**

# ORGANOGRAM OF ICAR-NIPB



## ICAR-NIPB

**N**ational Institute for Plant Biotechnology (NIPB)

is the premiere research institution of the Indian Council of Agricultural research (ICAR), engaged in molecular biology and biotechnology research. The Biotechnology Centre, established in 1985, as part of the Indian Agricultural Research Institute (IARI), was upgraded to a National Research Centre on Plant Biotechnology in the year 1993, with a vision to impart the biotechnology advantage to the National Agricultural Research System (NARS). The name of the Institute has been Changed to National Institute for Plant Biotechnology (NIPB) in 2019. NIPB has acquired, in the past, an excellent infrastructure in terms of equipment and other physical facilities and also a high degree of scientific competence. Development of transgenic crops for biotic and abiotic stress management, exploitation of heterosis through marker and genomic approaches, marker assisted selection and molecular breeding of major crops for productivity and quality enhancement, search for novel genes and promoters for efficient native and transgene expression are the major activities taken up by the centre. There is now considerable emphasis on structural and functional genomics of crop species such as rice, wheat, chickpea, pigeonpea, and mustard in the centre. In addition to research, the centre is contributing significantly to

competent human resource development by way of offering regular M.Sc. and Ph.D. programmes by partnering with PG School, IARI.

In order to develop strong inter-and intra-institutional linkages for promoting and strengthening plant biotechnology research in the ICAR system, the Centre has been identified as the lead centre. It has established strong linkages with various research institutes in the country including ICAR, CSIR, State Agricultural Universities and CGIAR institutes like ICRISAT. The center is now placing considerable emphasis on the development of products, processes, patents and research publications in journals with high impact factor. It also encourages and practices Public-Private Partnership (PPP) mode for commercializing the products of genetic engineering.

### Mandate

- Basic plant molecular biology research for understanding molecular basis of biological processes
- Coordination and capacity building for devising tools and techniques of biotechnology and genetic engineering for crop improvement

### Staff Strength of the Centre:

| Staff                    | Sanctioned | Filled | Vacant |
|--------------------------|------------|--------|--------|
| Scientific               | 33+1       | 31     | 2+1    |
| Technical                | 14         | 10     | 04     |
| Administrative           | 18         | 15     | 03     |
| Skilled Supporting Staff | 01         | -      | 01     |
| Total Strength           | 67         | 56     | 11     |

## Financial Statement 2018-19

(Rs. In lacs)

| Institute Grant                     |                |                |
|-------------------------------------|----------------|----------------|
|                                     | Allocation     | Utilization    |
| Capital                             | 143.00         | 142.91         |
| <b>Revenue</b>                      |                |                |
| Establishment                       | 996.91         | 993.62         |
| Pension & other retirement benefits | 215.00         | 214.99         |
| Travelling Allowances               | 5.60           | 5.58           |
| Research and Operational Expenses   | 144.70         | 144.64         |
| Administrative Expenses             | 348.85         | 348.64         |
| Miscellaneous Expenses              | 1854.91        | 1851.23        |
| <b>Total</b>                        | <b>1854.91</b> | <b>1851.23</b> |

## Network projects on transgenic crops, National Institute for Plant Biotechnology (NIPB)

|                                   | Allocation    | Utilization   |
|-----------------------------------|---------------|---------------|
| Capital                           | 31.00         | 30.91         |
| Travelling Allowances             | 9.91          | 9.91          |
| Research and Operational Expenses | 306.67        | 306.67        |
| <b>Total</b>                      | <b>347.58</b> | <b>347.49</b> |

## Resource Generation

|                                                               |                |
|---------------------------------------------------------------|----------------|
| Sales of Farm Produce                                         | 0.052          |
| License Fee                                                   | -              |
| Leave Salary and Pension Contribution                         | 8.050          |
| Interest Earned on Short Term Deposits                        | 46.660         |
| Income Generated from Internal Resource Generation (Trg. etc) | 20.170         |
| Miscellaneous Receipts                                        | 36.450         |
| <b>Total</b>                                                  | <b>111.382</b> |

## Fund Received through Externally Funded Projects

|                            |                |
|----------------------------|----------------|
| Externally Funded Projects | 1118.05        |
| Consultancy Projects       | 0.00           |
| <b>Total</b>               | <b>1118.05</b> |

# Personnel

## Scientific Staff

## Area of Interest

Prof. N. K. Singh  
nksingh@nrpcb.org  
nksingh4@gmail.com

Different aspects of structural, functional and comparative genomics with special emphasis on rice, wheat, pigeonpea, tomato and *Rhizobium*; mapping genes for yield and quality traits in rice, wheat and pigeonpea; high resolution mapping of loci for salinity tolerance and grain characteristics in rice

Dr. Sarvjeet Kaur  
dr\_sarvajeetkaur@yahoo.com

Isolation and characterization of quorum quenching genes from *Bacillus thuringiensis* and other bacteria for plant disease control; isolation of *Cry1*- and *Cry 2*-type genes from native *B. thuringiensis* isolates from diverse habitats in India for development of transgenic crops

Dr. Anita Grover  
anitagrover@hotmail.com

Plant-fungus interaction using *Brassica juncea*- *Alternaria brassicae* system, isolation of defence genes and development of transgenics of *B. juncea* for increased resistance to *Alternaria* blight

Dr. Rekha Kansal  
rekhakansal@hotmail.com

Isolation of genes encoding lectins, protease inhibitors and amylase inhibitors from legumes, like chickpea, pigeonpea, peas and mothbean, for imparting improved tolerance to sucking insect pests (*Lipaphis erysimi*)

| Scientific Staff                                        | Area of Interest                                                                                                                                                                                                                                                                                   |                                                                                       |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Dr. Sanjay Singh<br>sanjay_singh777@yahoo.com           | Marker assisted introgression of gene(s)/QTLs for resistance to Bipolaris leaf spot ( <i>Bipolaris sorokiniana</i> ) in wheat                                                                                                                                                                      |    |
| Dr. J. C. Padaria<br>jasdeep_kaur64@yahoo.co.in         | Cloning and characterization of abiotic stress responsive genes from different tolerant plant systems as Pennisetum, Zizyphus and Prosopis for development of transgenic wheat adaptable to climate change induced abiotic stresses                                                                |    |
| Dr. Pranab Kumar Mandal<br>pranabkumarmandal@gmail.com  | Molecular and biochemical basis of nitrogen use efficiency in wheat                                                                                                                                                                                                                                |   |
| Dr. Debasis Pattanayak<br>debasispattanayak@yahoo.co.in | Pod borer resistance management employing host-delivered RNAi, development of RNA-based gene silencing tools for crop improvement and functional genomics, and deciphering RNAi pathways as development and stress regulators                                                                      |  |
| Dr. R. C. Bhattacharya<br>ramcharan99@yahoo.com         | Identification and mobilization of genes conferring resistance to insect herbivores and fungal pathogens into the elite crop cultivars through development of transgenics, identification of defence related signal peptides and molecular characterization of systemic wound signalling in plants |  |



| Scientific Staff                            | Area of Interest                                                                                                                                                                                                                                         |  |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Dr. Pradeep Kumar Jain<br>jainpmb@gmail.com | Isolation of plant genes and promoters for important traits like drought, temperature, blight and wilt in chickpea employing functional genomics                                                                                                         |  |
| Dr. S. Barthakur<br>sbthakur@yahoo.com      | Isolation and functional characterization of genes and promoters for genetic engineering for abiotic stress tolerance in plants                                                                                                                          |  |
| Dr. K. S. Gaikwad<br>kish2012@yahoo.com     | Development of genomic resources including deep transcriptome analysis in crops; molecular analysis on abiotic stress, particularly salinity and thermo-tolerance, in rice and pigeonpea                                                                 |  |
| Dr. Kanika Kumar<br>kanika@rediffmail.com   | Abiotic stress management in plants through the microbes associated with them by exploiting PGPRs, role of microbial genes in reducing the concentration of stress ethylene in plants, structural and functional genomics of <i>Mesorhizobium ciceri</i> |  |
| Dr. Monika Dalal<br>monika@nrcpb.org        | Functional genomics for abiotic stress tolerance especially drought and salinity stress in wheat                                                                                                                                                         |  |

| Scientific Staff                             | Area of Interest                                                                                                                          |                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Dr. Tapan K. Mondal<br>mondaltk@yahoo.com    | Understanding molecular mechanisms of salinity stress tolerance in rice and its wild relatives                                            |    |
| Dr. Vandana Rai<br>vandnrai2006@gmail.com    | Discovering new genes and alleles for drought and salinity tolerance in rice and wheat using transcriptomics, proteomics and metabolomics |    |
| Dr. Prasanta K. Dash<br>pdas@nrcpb.org       | Isolation of insecticidal genes and promoters from Indian legumes, manipulating pathways for yield enhancements                           |   |
| Dr. Rhitu Rai<br>rhitunrcpb@yahoo.com        | Molecular characterization of plant pathogenesis and plant disease resistance mechanisms                                                  |  |
| Dr. Subodh Kumar Sinha<br>subsinha@gmail.com | Molecular analysis of nitrogen use efficiency in wheat and microRNA-mediated regulation of nitrogen metabolism                            |  |



| Scientific Staff                                    | Area of Interest                                                                                                                                           |                                                                                       |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Dr. Rohini Sreevathsa<br>rohinisreevathsa@gmail.com | Meristem-based <i>in planta</i> transformation for biotic stress tolerance using different insecticidal proteins (ICPs) from <i>Bacillus thuringiensis</i> |                                                                                       |
| Dr. Ashish Kumar<br>ashish.pathology@gmail.com      | Development of white rust resistance in mustard, pathotyping of <i>Albugo candida</i> isolates and introgression of gene(s) for white rust resistance      |    |
| Dr. Navin Chandra Gupta<br>guptanc@nrcpb.org        | Molecular and bioinformatics approaches to comprehend the role of novel genes and promoters, development of transgenic mustard with high oil content       |   |
| Dr. S.V. Amitha CR Mithra<br>amitha@nrcpb.org       | QTL mapping for grain traits and abiotic stress tolerance in rice using SSR and SNP markers                                                                |  |
| Dr. Amolkumar U. Solanke<br>amolsgene@nrcpb.org     | Genomics and transgenicsto understand biotic and abiotic stress tolerance, molecular biology of fruit development                                          |  |
|                                                     |                                                                                                                                                            |  |

| Scientific Staff                                  | Area of Interest                                                                                                                 |                                                                                       |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Mr. Ramawatar<br>ram_nrcpb@nrcpb.org              | Functional genomics to understand the role of non-coding RNA in plant pathogen interaction and disease development               |    |
| Dr. Dinabandhu Behera<br>aumreetam@rediffmail.com | Structural, functional and comparative genomics of crop plants                                                                   |    |
| Mr. Mahesh Rao<br>amritkushinagar@gmail           | Improvement of <i>Brassica</i> spp. using tissue culture, molecular marker, wide hybridization and gene introgression approaches |   |
| Ms. Nimmy M.S.<br>nimmybiotech@gmail.com          | Isolation and characterization of abiotic stress responsive gene(s) and promoter(s) from chickpea                                |  |
| Dr. Deepak Singh Bisht<br>deebisht@gmail.com      | Molecular identification of resistance and susceptibility genes in rice- <i>Rhizoctonia solani</i> system                        |  |

**Scientific Staff****Area of Interest**

Dr. Anshul Watts  
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Development and maintenance of genetic stocks for  
heterosis breeding in *Brassica juncea*

Dr. Sandhya  
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Functional genomics for milling, cooking and protein  
quality in pigeonpea



Dr. S.R. Bhat  
ICAR, Emeritus Scientist

Dr. C. Kole  
Raja Raman Fellow

| ICAR Emeritus Scientist    |                         |
|----------------------------|-------------------------|
| Dr. S. R. Bhat             |                         |
| Technical Staff            | Administrative Staff    |
| Sh. Ram Niwas Gupta        | Sh. Rajesh Kumar Sharma |
| Sh. R. K. Narula           | Sh. Suresh Kumar Sharma |
| Dr. Krishan Pal            | Sh. Krishan Dutt        |
| Smt. Sandhya Rawat         | Sh. A.K Jain            |
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| Dr. Pankaj Kumar           | Sh. Vipin Kumar         |
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| Sh. Deepak kumar Rathor    | Sh. Mohit Sikka         |
| Ms. Reeta                  | Ms. Nidhi Nailwal       |
| Ms. Megha                  | Smt. Rekha Chauhan      |
| Sh. Anshul kumar Verma     | Sh. Rajesh Kumar Pal    |
|                            | Sh. Kunal Mann          |
|                            | Ms. Priyanka Nayar      |
|                            | Sh. Mitravesh Chaudhary |

# Research Achievements

1. Search and Deployment of Genes for Stress Tolerance and Grain Quality in Rice
2. Adaptation of Wheat to Climate Change Induced Stresses
3. Improvement of Stress Tolerance in Chickpea
4. Stress Tolerance and Quality Improvement in Pigeonpea
5. Biotechnological Approaches for Brassica Improvement



# 1. Search and Deployment of Genes for Tolerance in

## Allele mining for agronomically important genes in wild rice germplasm and stress tolerant land races of rice growing in the hot spots

(Balwant Singh, Kabita Tripathi, Vandna Rai., Nagendra Kumar Singh)

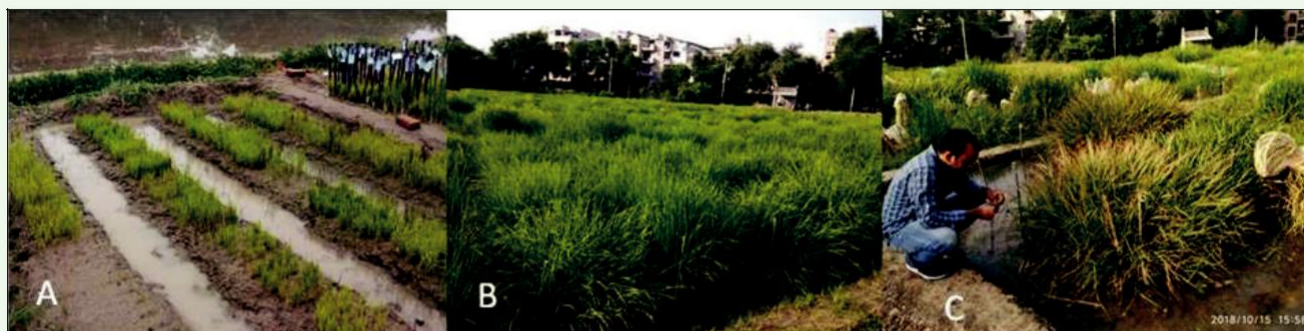
Wild progenitors of rice are important source of tolerance/resistance genes governing for both biotic and abiotic stress phenotype. They are well adapted to extreme habitats including flooding, drought, saline and acidic soil conditions and hence must have genes conferring tolerance to these extreme environmental conditions. Based on the soil types, rainfall patterns, temperature and water availability India has been divided into 15 agro-climatic zones and wild rice grows naturally in all the agro-climatic zones. All the zones are important areas of rice diversity where large numbers of wild rice grow in their natural habitats, around farmers' field and also within field. The wild rice genotypes have survived for thousands of years in the nature and therefore must also possess large repertoire of genes for resistance to various diseases and insect pests.

Total 157 accessions were collected from 19 districts of six states namely Assam, Chhattisgarh, Goa, Karnataka, Odisha and Uttar Pradesh (UP) (Fig. 1.1 and Fig. 1.2). Out of 157 accessions, 25 was collected from 3 districts of Assam, one from Chhattisgarh, four from NorthGoa district of Goa, 46 from five districts of Karnataka, 40 from four districts of Odisha and 41 from eight districts of Uttar Pradesh. Four

accessions from Lakhimpur, 20 from Dhamaji and one Eastsiang district of Assam; seven from Unnao, five from Raebareli, three from Barabanki, six from Gonda, five from Shravasti, eight from Bahraich, six from Lakhimpur Khiri, one Sitapur district of UP; 18 accessions were collected from Koraput, 14 from Nabarangpur, 7 from Sambalpur and 1 from Bargarh district of Odisha. One accession was collected from Mahasamund district of Chattishgarh. Out of 46 accessions collected from karnatak, 34 were collected from Uttar kannad, 2 accessions from Karwar, 7 from Udapi and 2 from Manglore district. Total 4 accessions were collected from North goa district of Goa.

Total 250 of the earlier collected wild rice accessions were multiplied at Indian Agriculture Research Institute, New Delhi during *Kharif* 2018 along with three populations developed for anaerobic germination and salt tolerance. In additions, 564 accessions were planted at Banras Hindu University seed multiplication Accessions were planted in a plot size of 60 x 60 cm area each with plot-to-plot spacing of 50 cm. Till now, total 791 accessions of wild rice have been collected from different state of India, and a database web portal has been developed (nksingh.nationalprof.in:8080/iwrdb). All the information regarding collection such as village, block, district, state, longitude and latitude from where the accessions were collected and characterization of particular accessions is stored in database

Fig. 1.1: Collection site visited during 2018 (A. Uttar Pradesh, B. Odisha, C. Karnataka and D. Assam).

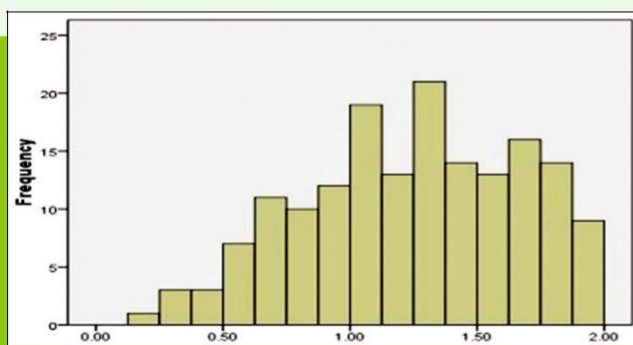


**Fig. 1.2:** Wild rice seedlings in raised bed nursery (A); and Field multiplication of accessions (B & C) in 60x60cm plot area each at Indian Agriculture Research Institute, New Delhi.

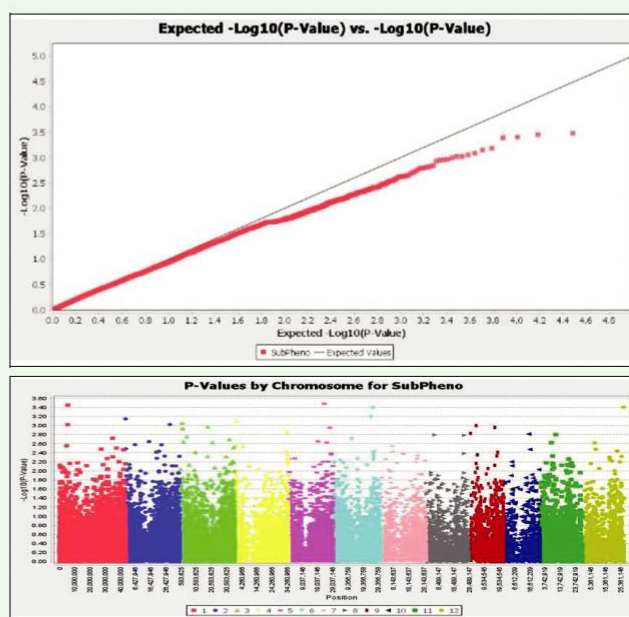
## Genome wide association analysis for submergence tolerance

(Balwant Singh, Kabita Tripathi, Vandna Rai and Nagendra Kumar Singh)

Out of 791 ORSC wild rice accessions collected from different agro-climatic region of the country, 288 accessions were phenotyped for submergence tolerance. Seeds of individual accessions were directly seeded into submergence phenotyping tank and grown for 30 days. Thirty days old seedlings were submerged for 18 days and data was recorded for survival after 21 days of de-submergence. Total 18 accessions were found tolerant, 35 moderate tolerant and 246 were susceptible. Data was normalised for distribution in individual class (Fig. 1.3). In order to perform GWAS SNP genotyping was conducted for individual accessions using OsSNPnks chip. Genetic structure was evaluated using structure software. Compressed mixed liner model (MLM) GWAS analysis Tassel5 software. A filter was set for SNP having MAF more than 5%. Total 29000 SNP was finally selected for GWAS analysis. GWAS analysis showed total 10 SNPs distributed into six different chromosomes were significantly associated with submergence tolerance (Fig. 1.4). None of the SNP was closed to known Sub1 QTL on chromosome 9 suggesting novel source of flooding tolerance.



**Fig. 1.3:** Histogram of frequency distribution of phenotypic data for submergence tolerance.



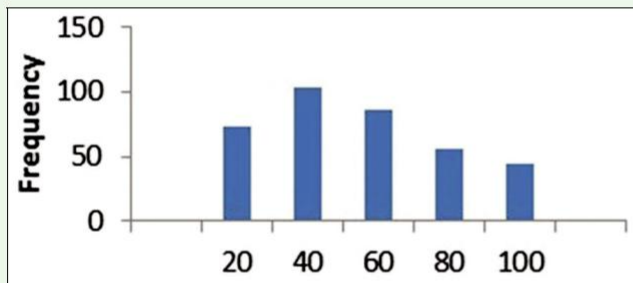
**Fig. 1.4:** GWAS for submergence tolerance; A) QQ plot, B) Manhattan Plot.

## Genome wide association analysis for anaerobic germination

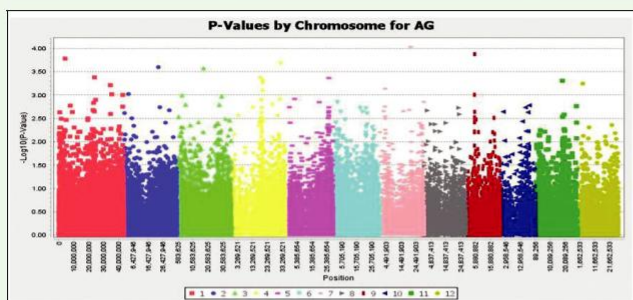
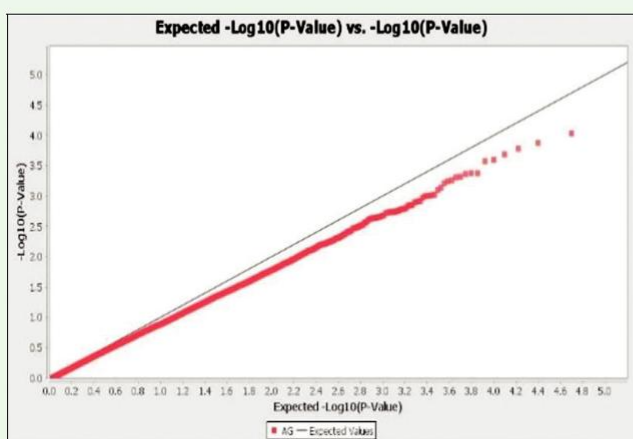
(Balwant Singh, Vandna Rai and Nagendra Kumar Singh)

Out of 791 ORSC wild rice accessions collected from different agro-climatic region of the country, 442 Indian ORSC wild rice accessions were phenotyped for anaerobic germination tolerance. For phenotyping seeds were directly sown in magenta boxes having 1 cm of soil with 10 seeds per accessions per box. Experiment was performed in two replications. Boxes were filled with water for further 10cm and scoring was conducted after 21 days. Total 42 accessions were found to be tolerant, 103 were moderate and 297 were susceptible. Data was normalised for distribution in individual class (Fig. 1.5). In order to perform GWAS, SNP genotyping was conducted for individual accessions using OsSNPnks chip. Genetic structure was evaluated using structure software. Compressed mixed liner

model (MLM) GWAS analysis Tassel5 software. A 5% filter was set for SNP having MAF more than 5%. Total 29000 SNP was finally selected for GWAS analysis. Total 16 SNPs distributed into five different chromosomes were significantly associated with AG tolerance (Fig. 1.6).



**Fig. 1.5:** Histogram of frequency distribution of phenotypic data for anaerobic germination tolerance.



**Fig. 1.6:** GWAS for anaerobic germination tolerance; A) QQ plot, B) Manhattan Plot.

## Development of species – specific marker for the wild rice species

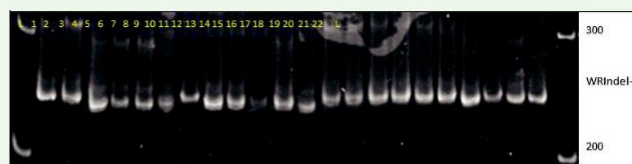
Kabita Tripathi, (Balwant Singh, Vandna Rai and Nagendra Kumar Singh)

For finding species-specific Indel marker, the whole chloroplast genome sequences of 22 *Oryza* species were downloaded from the NCBI GenBank. Out of the 22 species, there are eight species genome AAtype, three species each of genome CC and CCDD types, two species each of genome GG and HHJJ

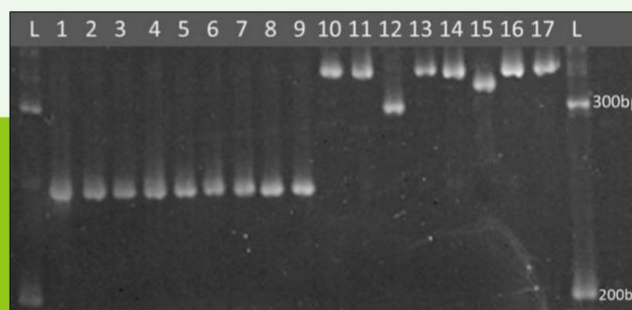
types, and one species each of genome BB, EE, FF and BBCC, types. Chloroplast genome alignment was completed using Mauve 2.3.1 and Mafft 7 software tools. After analysis, total 8829 indels were identified among 58 accessions. InDels having more than 4 bp lengths were considered for marker design and total 15 such InDel markers were designed with flanking PCR primers (Table 1.1). A single PCR amplicon of expected size was amplified for each marker. Out of the 15 InDels, four were present in genic region and 11 were in the intergenic region (Table 1.1). Among the four genic markers, WRindel-1 and WRindel-3 were located in *rps16* gene, WRindel-7 in *trnK-UUU* gene and WRindel-10 in *infA* gene. From in silico analysis 15 InDel markers were able to differentiate each of the 22 *Oryza* species based on PCR amplicon size differences (Table 1.2). The 15 InDel markers were validated for amplification of expected size PCR products in 96 accessions of 18 different *Oryza* species, including 33 cultivars and 63 wild rice accessions (Fig. 1.7A-C).



**Fig. 1.7A:** Indel marker Wrindel-10 shows *O.alta* specific amplicon and *O. latifolia* specific amplicon difference from other species. L, 100bp DNA Ladder; 1-3, *O. alta*; 4-8, *O. latifolia*; 9-17, *O. Sativa*.



**Fig. 1.7B:** Indel marker WRindel-2 showed difference in amplicon size between *O. sativa indica* and *O. sativa japonica*. L, 100bp DNA Ladder; 1-12, 20-22 *O. Sativa indica*; 13-19, *O. sativa japonica*.



**Fig. 1.7C:** Indel marker WRindel-8 in 17 different *Oryza* species. L, 100bp DNA Ladder. AA genome: 1, *O. Glaberrima*; 2, *O. barthii*; 3, *O. nivara*; 4, *O. rufipogon*; 5, *O. longistaminata*; 6, *O. meridionalis*; 7, *O. glumipetula*; 8, *O. Sativa indica*; 9, *O. sativa japonica*. CC genome: 10, *O. Officinalis*; 11, *O. eichengiri*. BB genome: 12, *O. Punctata*. BBCC genome: 13, *O. Minuta*. EE genome: 14, *O. austrelansis*. FF genome: 15, *O. brachyantha*. CCDD genome: 16, *O. latifolia*; 17, *O. alta*.

**Table 1.1: Details of InDel markers with primer sequence, chloroplast genomic positions, lengths of InDels and their genic or intergenic location.**

| Marker name | Primer Sequence           | Product size* | Indel position* | Indel size | In gene/ between genes            |
|-------------|---------------------------|---------------|-----------------|------------|-----------------------------------|
| NKS_KT1     | TCGTGTTTACTTGTTCGGGAAT    | 150           | 5004            | 7          | In rps16                          |
|             | GGAAATGCTTTATCGAAGGAAA    |               |                 |            |                                   |
| NKS_KT2     | ACTATCTTAGCCCCCTTGCAC     | 246           | 12602           | 4          | Between itrnG-UCC and trnfM-CAU   |
|             | TATAAGGCCAAATTGAGCAGGT    |               |                 |            |                                   |
| NKS_KT3     | TCTCTAAACTCTTTCTTGCCTCA   | 100           | 4967            | 6          | In rps16                          |
|             | CAAGTAAACACGATTTTCAACTGTC |               |                 |            |                                   |
| NKS_KT4     | CAGAACAGGGTGTTCATTTT      | 150           | 3397            | 5          | Between matK and rps16            |
|             | CAATTGGCCTGCAAAAATAAAG    |               |                 |            |                                   |
| NKS_KT5     | AAATTGGCTATCGTCTTTTCAA    | 245           | 108578          | 34         | Between psaC and ndhE             |
|             | AGGAAATCAACTCGTATCAATCA   |               |                 |            |                                   |
| NKS_KT6     | ATCGAGATCGTTTTCGCTTAAC    | 207           | 6367            | 17         | Between rps16 and trnQ-UUG        |
|             | AATCCAGTGTCAAGTCCAGTCA    |               |                 |            |                                   |
| NKS_KT7     | CATTTTCAGAATGGAAACTGTGA   | 185           | 1652            | 26         | Between trnK and UUU              |
|             | TTGATGTTTCGCCAAAACAAC     |               |                 |            |                                   |
| NKS_KT8     | TGTATGCAGCCTAATGAGGAG     | 249           | 50100           | 86         | Between ndhC and trnV-UAC         |
|             | GCCTGCGCCATAAAATAGAG      |               |                 |            |                                   |
| NKS_KT9     | ACCAATCAGAATCCCTTGAGAA    | 181           | 14466           | 8          | Between trnfM-CAU and trnT-GGU    |
|             | TGTCTGTTTTGGTGAGAGGTTG    |               |                 |            |                                   |
| NKS_KT10    | AAACCCTCGAAAGTAATTTTGC    | 238           | 76482           | 31         | In infA                           |
|             | CAGAATTGGGGGAGAAGTCTTA    |               |                 |            |                                   |
| NKS_KT11    | GATTTTCTCTTCAGCAGGCATT    | 173           | 8195            | 5          | Between trnS-GCU and psbD         |
|             | CATTCAAGGAGTCAGGTCCACT    |               |                 |            |                                   |
| NKS_KT12    | ATTTCGGCTCCTTTATGGGTATT   | 219           | 13101           | 5          | Between trnfM-CAU and trnT-GGU    |
|             | ACCATCAGCATATTGTTTCAGC    |               |                 |            |                                   |
| NKS_KT13    | ATCCGGTTGGAATCATAAAGTG    | 273           | 16837           | 24         | Between psbM and petN             |
|             | TAAGCATGAAAGATTCGCTGGT    |               |                 |            |                                   |
| NKS_KT14    | GCTATATATCCTCGCGGAAGTC    | 238           | 66790           | 5          | Between rpl20 and rps12           |
|             | CATAGAAACGATGGAACCCACT    |               |                 |            |                                   |
| NKS_KT15    | AACGTTAGCCGCAGCCTAT       | 235           | 31303           | 19         | intergenic region (atpI and atpH) |
|             | GCCCCTCCCTAATGCCTA        |               |                 |            |                                   |

\*Size and position of the InDel based on reference chloroplast genome of rice cultivar 'Hassawi' (JN861109.1)

**Table 1.2: Genome-specific and species-specific InDel markers validated to distinguish 22 *Oryza* species showing PCR amplicon size in base pairs.**

| Genome type | Species                   | WRindel-1 | WRindel-2 | WRindel-3 | WRindel-4 | WRindel-5 | WRindel-6 | WRindel-7 | WRindel-8 | WRindel-9 | WRindel-10 | WRindel-11 | WRindel-12 | WRindel-13 | WRindel-14 | WRindel-15 |
|-------------|---------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|------------|------------|------------|------------|------------|------------|
| AA          | <i>O. sativa</i> Indica   | 150       | 246       | 100       | 150       | 245       | 207       | 185       | 249       | 181       | 238        | 173        | 219        | 273        | 238        | 235        |
| AA          | <i>O. sativa</i> Japonica | 143       | 250       | 100       | 150       | 245       | 207       | 185       | 249       | 181       | 241        | 176        | 219        | 273        | 238        | 235        |
| AA          | <i>O. nivara</i>          | 143       | 246       | 106       | 150       | 245       | 207       | 185       | 249       | 181       | 261        | 176        | 219        | 298        | 257        | 250        |
| AA          | <i>O. rufipogon</i>       | 143       | 246       | 100       | 150       | 245       | 207       | 185       | 249       | 181       | 240        | 175        | 219        | 273        | 238        | 235        |
| AA          | <i>O. glaberrima</i>      | 143       | 246       | 100       | 150       | 279       | 207       | 185       | 249       | 181       | 240        | 176        | 219        | 273        | 238        | 235        |
| AA          | <i>O. barthii</i>         | 143       | 246       | 100       | 155       | 279       | 207       | 185       | 249       | 181       | 240        | 176        | 219        | 273        | 238        | 235        |
| AA          | <i>O. glumipatula</i>     | 143       | 246       | 100       | 150       | 245       | 207       | 185       | 249       | 193       | 240        | 176        | 219        | 272        | 257        | 231        |
| AA          | <i>O. meridionalis</i>    | 143       | 246       | 100       | 150       | 245       | 207       | 185       | 249       | 170       | 240        | 180        | 219        | 272        | 257        | 235        |
| AA          | <i>O. longistaminata</i>  | 143       | 246       | 100       | 150       | 245       | 207       | 185       | 249       | 170       | 240        | 176        | 219        | 273        | 257        | 235        |
| BB          | <i>O. punctata</i>        | 143       | 247       | 100       | 150       | 245       | 213       | 185       | 309       | 168       | 240        | 177        | 224        | 273        | 257        | 250        |
| BBCC        | <i>O. minuta</i>          | 143       | 247       | 100       | 150       | 245       | 0         | 185       | 333       | 169       | 240        | 177        | 224        | 274        | 257        | 250        |
| CC*         | <i>O. rhizomatis</i>      | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 311       | 170       | 217        | 233        | 219        | 298        | 257        | 231        |
| CC          | <i>O. officinalis</i>     | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 326       | 170       | 217        | 233        | 219        | 298        | 257        | 240        |
| CCO.        | <i>eichengeri</i>         | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 326       | 170       | 217        | 233        | 219        | 298        | 257        | 250        |
| CCDD        | <i>O. latifolia</i>       | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 334       | 170       | 269        | 232        | 219        | 274        | 257        | 250        |
| CCDD        | <i>O. alta</i>            | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 334       | 170       | 238        | 233        | 219        | 298        | 257        | 250        |
| CCDD*       | <i>O. grandiglumis</i>    | 143       | 247       | 100       | 150       | 242       | 213       | 185       | 334       | 170       | 238        | 234        | 219        | 274        | 257        | 250        |
| EE          | <i>O. austrelansis</i>    | 144       | 247       | 100       | 150       | 243       | 207       | 185       | 327       | 186       | 240        | 192        | 219        | 272        | 257        | 231        |
| FF          | <i>O. brachyantha</i>     | 136       | 190       | 87        | 154       | 261       | 175       | 180       | 301       | 0         | 192        | 176        | 219        | 299        | 258        | 0          |
| GG*         | <i>O. granulata</i>       | 144       | 247       | 100       | 147       | 262       | 229       | 185       | 329       | 172       | 191        | 177        | 219        | 293        | 263        | 0          |
| GG*         | <i>O. meyeriana</i>       | 144       | 247       | 100       | 147       | 262       | 229       | 185       | 328       | 172       | 191        | 177        | 219        | 293        | 258        | 0          |
| HHJJ*       | <i>O. longiglumis</i>     | 143       | 190       | 100       | 147       | 270       | 212       | 211       | 325       | 172       | 191        | 178        | 219        | 298        | 258        | 0          |
| HHJJ*       | <i>O. ridleyi</i>         | 144       | 184       | 100       | 147       | 270       | 212       | 211       | 330       | 172       | 191        | 178        | 219        | 298        | 254        | 0          |

\*= Not studied

## Mapping of QTLs for salt tolerance in wild rice introgression lines

(Sarita Kumari, Dhriti Satya, Balwant Singh, Krishnamurthy SL, Vandna Rai and Nagendra Kumar Singh)

Seventy five BC1F2 lines developed from IR64×NKS173 were screened for their reproductive stage salt-tolerance at CSSRI, Karnal. The total four salt tolerance parameters were analyzed under salt stress stress which includes yield and yield related

traits like days to 50% flowering, plant height, panicle length and grain yield. The data indicate significant variations for all four salt tolerance parameters under salt stress in the 75 BC1F1 derived from IR64 X NKS173 (*O. rufipogon*) parent population. Thus, renders a fitting population for QTL analysis. Moreover, the frequency distribution for phenotypic values remained continuous with normal or skew-normal distributions (Fig. 1.8).

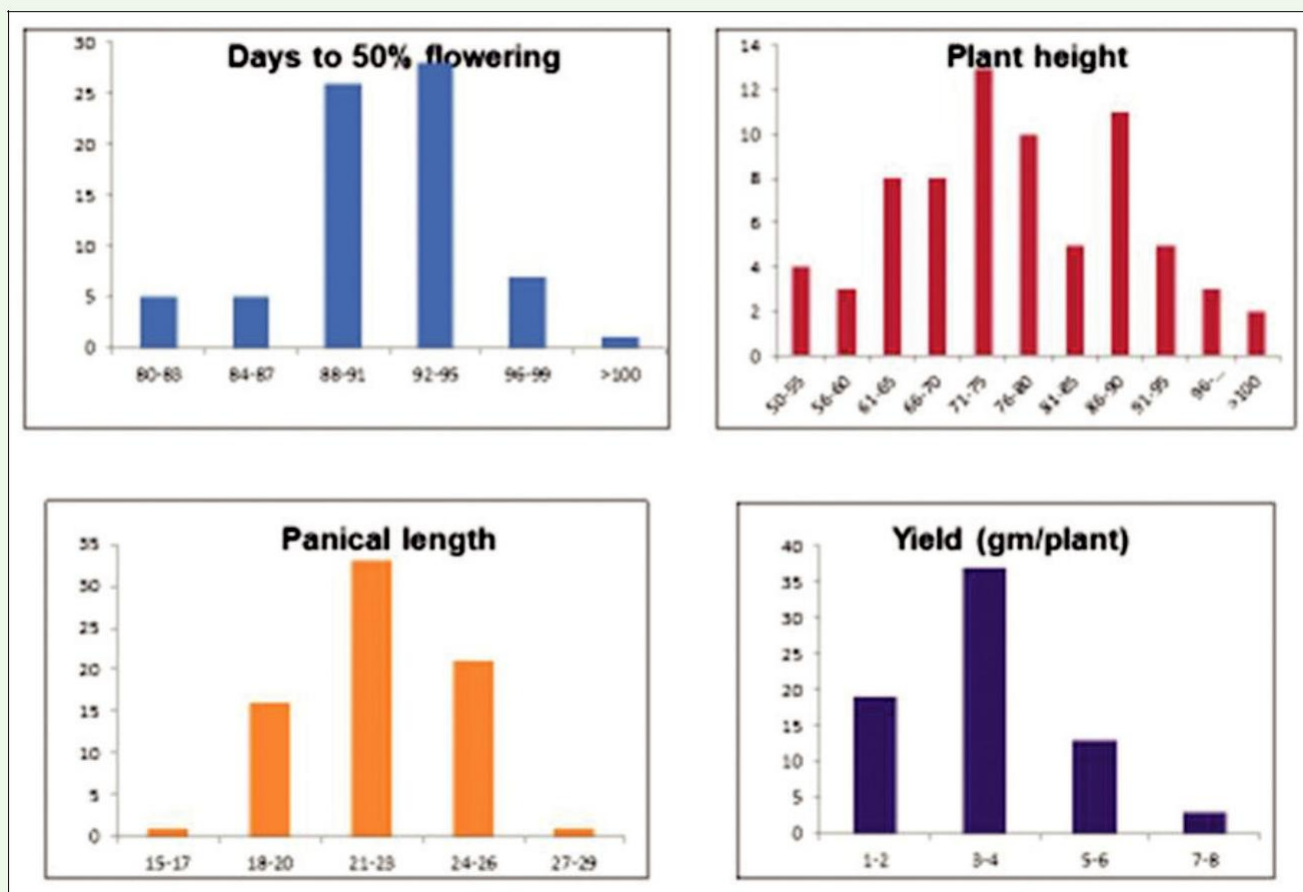


Fig. 1.8: Frequency distribution for four different agronomic traits.

## A high-density genetic linkage map for salinity stress tolerance

(Sarita Kumari, Dhriti Satya, Sangeeta Singh, Nagendra Vandna Rai and Kumar Singh)

A total of 75 BILs was derived from IR64 X NKS173 (*O. rufipogon*) cross for the construction of data for salinity stress tolerance and the analysis of a genetic linkage map. The BC1F1 population (IR64/NKSWR173) of rice of 74 plants was used for the construction of high density linkage map and QTL analysis. The genotyping data was extracted for all the polymorphic (9119) SNP loci. Many of SNP markers belong to same gene so the removal of duplicate markers gave 4034 SNP. These 4034 marker were tested for segregation distortion leading to a final number of 2703 non distorted markers. The construction of linkage map was done with the help of Joinmap 4.0 software. Linkage groups were identified by grouping of markers at a minimum LOD threshold of 3 to a maximum of 10 with a step of 0.5. The groups were converted to maps at LOD 3 using the regression algorithm with recombination frequency of 0.25 and performing a ripple after

adding 2 loci. Kosambi's mapping function was used to calculate map distance in centiMorgan (cM). A map of 1137 markers covering 12 linkage groups was constructed. The intraspecific map generated here spanned 1309.3 cM distance of the rice genome. The constructed linkage groups differ from each other with respect to total number of markers mapped, total cM distance and average marker interval. Maximum number of markers were assigned to linkage groups 1 and 2 i.e. 175 and 152 respectively, while minimum number of markers were mapped to linkage group 9 and 10. QTL analysis for traits was carried out using WinQTL Cartographer v2.5<sub>3</sub>. In order to find out the association of each marker to a trait, composite interval mapping (CIM) using the ZmapQTL standard model 6 with a window size of 10 cM and 2cM walk speed was used. For estimating a genome-wide LOD threshold score for QTL ( $p = 0.05$ ), a 1000-permutation test was performed. The additive effect and the percentage of phenotypic variation explained by each putative QTL were estimated by the Composite interval mapping method. All the results were tabulated in excel sheet (Table 1.3 and Table 1.4; Fig. 1.9).



Table 1.3: Quantitative trait loci (QTLs) associated with the traits: PL (Panicle length), PH (Plant height), DF (Days to fifty percent Flowering), yield (gm/plant; YP) and mapped on the linkage map of rice (IR64 X NKS173). A negative value indicates that given trait is derived from IR64 and a positive number indicates that the trait is derived from NKS173.

| S. No. | Chr No. | QTL Name | QTL Position | Marker            | CI (cM)       | Marker Interval                        | LOD Score | Additive effect | R2(%) |
|--------|---------|----------|--------------|-------------------|---------------|----------------------------------------|-----------|-----------------|-------|
| 1      | 2       | qDF2.1   | 181.51       | CSCWR-Os02g57690  | 180.4 - 183.1 | SCR200-Os02g57650 to SCR100-Os02g57830 | 3.682953  | 31.1158         | 14    |
| 2      | 4       | qDF4.1   | 55.58        | CSCWR-Os04g55690  | 53.9 - 58.2   | SCR100-Os04g55360 to CSCWR-Os04g53700  | 3.37025   | -27.2486        | 13    |
| 3      | 8       | qPH8.1   | 102.86       | CSCWR-Os08g39380  | 100.6 - 110.8 | CSCWR-Os08g39250 to SCR100-Os08g40630  | 3.39848   | 30.1595         | 15    |
| 4      | 11      | qPL11.1  | 44.61        | SCR200-Os11g32720 | 40.4 - 47.4   | SCR200-Os11g34150 to SCR200-Os11g32360 | 3.923996  | 33.0281         | 16    |
| 5      | 11      | qYP11.1  | 44.61        | SCR200-Os11g32720 | 40.4 - 47.4   | SCR200-Os11g34150 to SCR200-Os11g32360 | 4.738328  | -39.9775        | 22    |

Table 1.4: Distribution of 1137 SNP markers on the twelve linkage groups of rice BC1F1 population (IR64/NKSWR173).

| Linkage group | No of markers | Position (cM) |
|---------------|---------------|---------------|
| LG1           | 175           | 141.9         |
| LG2           | 152           | 184.4         |
| LG3           | 142           | 101.2         |
| LG4           | 111           | 71.3          |
| LG5           | 108           | 127.9         |
| LG6           | 110           | 85.2          |
| LG7           | 80            | 67.3          |
| LG8           | 102           | 158           |
| LG9           | 13            | 59.0          |
| LG10          | 26            | 125           |
| LG11          | 71            | 108.3         |
| LG12          | 47            | 79.8          |
| Total         | 1137          | 1309.3        |

## Genetic improvement of deep-water rice of Assam

(Megah Rohila, Abhijit Majumdar, Nagendra Kumar Singh and Tapan Kumar Mondal)

Assam, a state of North Eastern India, has varied Agro-climatic zones including substantial submergence area which are mostly inundated by flood water where the farmers grow typical deepwater rice known as “Bao dhan”. In the present study, genetic diversity and population structure of total 94 deepwater rice landraces were analysed using 50K rice SNP gene chip. The genetic relatedness

was evaluated by means of phylogenetic tree which grouped all 94 landraces in 8 distinct clusters. Principal coordinate analysis showed deepwater rice was distributed according to population structure. The analysis of molecular variance showed maximum diversity of 64%, distributed among the individuals. Based on population structure analysis, these genotypes were stratified into four sub-populations,

likewise the haplotype based phylogenetic analysis revealed that these genotypes were clustered into five distinct groups. To the best of our knowledge, the present study is the first comprehensive report on molecular characterization of this distinct deepwater rice of Assam, India using the most informative and conclusive 50K rice SNP gene chip. The evaluation of genetic diversity and haplotype based phylogenetic analysis together with deepwater rice provides some useful guides for assisting plant breeders in selecting suitable genetically diverse parents for deepwater stress tolerance rice breeding program.

## Comprehensive evaluation of candidate reference genes for qRT-PCR studies of gene expression in *Oryza coarctata*

(Soni Chorasias, Alok Kumar Pandai, Hameet Kaur and Tapan Kumar Mondal)

*Oryza coarctata* is the only wild halophytic species among the 24 species of the genus *Oryza*. It can tolerate the level of salinity up to a maximum limit of 40 ds/m. It has unique morphological, anatomical as well as physiological adaptations, which help it to survive under high saline-submerged conditions. Thus, it is considered to be a reservoir of salinity

tolerant genes. The selection of suitable reference genes for a species and its validation with target stress-responsive genes through quantitative real-time (qRT)-PCR is pre-requisite for studying the gene expression of that species under a particular condition. Therefore, the present study, deal with the expression of eight candidate genes namely, ubiquitin 2 (*UBQ2*), histone3 (*H3*), beta tubulin ( $\beta$ -*Tub*), alpha tubulin ( $\alpha$ -*Tub*), glyceraldehyde-3-phosphate dehydrogenase (*GADPH*), eukaryotic initiation factor -4-alpha (*Eif-4- $\alpha$* ), cyclophilin (*CYP*) and actin (*ACT*) under different abiotic stresses such as heat, cold, drought, salinity, ABA (Absciscic acid) and salinity with submergence (dual stress). The expression of these candidate genes were analyzed through 3 different statistical algorithms, namely, Bestkeeper, Normfinder, geNorm and Refinder along with commonly used basic statistical method. The analysis revealed that actin and eukaryotic initiation factors -4-alpha are both the most stable as reference

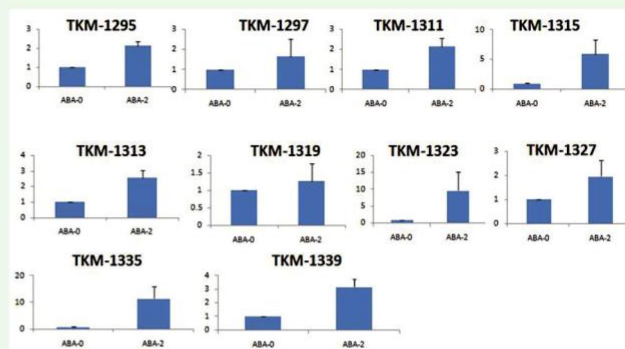
genes, whereas glyceraldehyde-3-phosphate dehydrogenase and ubiquitin 2 are both the least stable all across the samples. Therefore, this information best of our knowledge is first report and will be useful for studying the abiotic stress related gene expression in this species.

### Identification and characterization of novel stress responsive genes from *Oryza coarctata* Roxb.

(Soni Chorasias, Jyoti, Harmeet Kaur, Illa Mukul Tiwari and Tapan Kumar Mondal)

*O. coarctata* Roxb. is a halophytic wild rice, popularly known as Asian wild rice, grows naturally in the coastal region of South-East Asian countries. This plant experiences saline submergence condition for most of its life. Thus, it is expected that this plant might have some useful salinity tolerant alleles. Thus from the genome sequence data of *O. coarctata* novel and unique genes were identified. In total, we had identified and selected 24 unique genes. Out of 24 genes, only 15 genes were amplified with gene specific primers and sequenced for confirmation.

The differential expression by Q-RT PCR of these genes were analyzed under different stress conditions and found that 10 genes were differentially expressed under salinity stress. Experiments are underway to further characterize the same (Fig. 1.10).



**Fig. 1.10:** The relative expression levels of ten selected uncharacterized novel transcript of *O. coarctata* under 2h ABA treated. Y axis represents fold change.

### Identification and mapping of quantitative trait loci (QTL) for seedling stage salinity tolerance of aromatic short grain rice (*Oryza sativa* L. var. Kolajoha-1) of Assam, India

(Abhshek Mazumder, Megha Rohila, Deepak Singh Bisht, Nagendra Kumar Singh and Tapan Kumar Mondal)

Salinity is one of the most prominent abiotic stress which limits rice production worldwide. Many investigations have been carried out to discover quantitative trait loci (QTL) for traits associated with salinity tolerance in rice throughout the world. However, utilization of those reported QTLs in rice breeding programs is limited due to large confidence interval for the position of QTL. In order to develop a high-density rice genetic linkage map based on single-nucleotide polymorphisms identified throughout the genome for discovery of QTLs and candidate genes for salinity tolerance at seedling stage, We have evaluated a population of 68 recombinant inbred lines (RILs) developed from a cross between Ranjit and Kolajoha-1 variety of rice for ten different traits related to salinity tolerance. A total of 3649 SNP markers generated by genotyping-by-sequencing (GBS) method have been mapped to 1259 recombination points. The genetic map has a total map length of 1201 cM with an average resolution of 0.95 cM between markers. For ten traits, a total of forty one additive QTLs have been identified of which two are large-effect QTLs and twenty six are minor effect QTLs. Epistatic QTL mapping has identified several pairs of QTLs that contribute significantly in the variation of traits among the RILs.

## Marker development for major heat stress tolerance QTLs and a novel, consistent and minor QTL for drought stress tolerance and their validation in rice

(Sourav Kumar, Chandra Prakash, Amolkumar U Solanke, Nagendra Kumar Singh, SV Amitha Mithra)

Incidence of high temperature at reproductive stage especially during anthesis, fertilization and initial grain filling can prove highly detrimental to rice productivity. Introgression of known, major quantitative trait loci (QTLs) for heat stress tolerance, through marker assisted selection (MAS), can facilitate development of heat tolerant varieties. Two major heat stress tolerance QTLs (qSTIY5.1, qSTIPSS9.1), identified in our laboratory and reported last year, in the background of a heat tolerant upland *aus* variety, Nagina 22, are yet to be used in crop improvement. Similarly, a novel and minor but consistent QTL for drought tolerance (qTS4.1) has also been identified in rice using the same mapping population in our laboratory. As these QTLs were detected using SNP based genetic map, there is a need to develop robust PCR based codominant markers such as simple sequence repeats (SSRs) and cleaved amplified polymorphic sequences (CAPS) to enable their use in rice breeding for heat tolerance through marker assisted breeding.

Fort this, SSRs in nearly 1 Mb region flanking (500 Kb on each side) each of the two QTLs was mined and a total of 18 di and trinucleotide markers were identified. Of these, nine SSRs could be amplified and validated in a set of 11 widely cultivated upland and lowland rice varieties (Vandana, Rasi, Naveen, IR 64, SambhaMahsuri, Jyoti, Swarna Sub 1, Swarna, Pusa 1609, Pusa Basmati 1) including Nagina 22, the donor of the selected QTLs. Sufficient SSR polymorphism to allow introgression of the heat tolerant QTLs could be identified across all varieties except for Pusa Basmati 1 in case of qSTIPSS9. (Table 1.5). Further, a total of 393 SNPs and 23 InDels were identified between the parents used for detection of these QTLs, Nagina 22 and IR 64, in the QTL regions. STS markers developed based on the InDels could not be validated. Three CAPS markers were designed using the InDels and one of them (CAPS\_HT\_5.1), could be validated across all the 11 varieties. This CAPS marker was further used to genotype the mapping population derived from N22 x IR64. Use of this marker in the mapping population validated the association between this marker and heat stress tolerance. Two tailed t-test using the appropriate phenotype and the marker data in the mapping population revealed significant associations between CAPS\_HT\_5.1 with stress tolerant and susceptibility index under heat stress.

**Table 1.5:** Polymorphism survey of SSRs developed for the introgression of heat and drought tolerance QTLs in rice.

| QTL        | SSR         | Rice Varieties |     |         |     |         |       |      |      |    |        |
|------------|-------------|----------------|-----|---------|-----|---------|-------|------|------|----|--------|
|            |             | IR64           | PB1 | PB 1609 | MTU | Vandana | Lalat | SSB1 | Rasi | SM | Swarna |
| qSTIY5.1   | 5.1U1       | N              | N   | P       | N   | N       | N     | N    | N    | N  | N      |
|            | 5.1D1       | P              | P   | P       | P   | P       | P     | P    | P    | P  | P      |
|            | 5.1D2       | N              | N   | N       | N   | N       | N     | N    | N    | N  | N      |
|            | CAPS_HT_5.1 | P              | P   | P       | P   | P       | P     | P    | P    | P  | P      |
| qSTIPSS9.1 | 9.1U1       | N              | N   | N       | N   | P       | N     | N    | P    | N  | N      |
|            | 9.1U2       | N              | N   | N       | N   | P       | N     | N    | P    | N  | N      |
|            | 9.1D1       | P              | N   | P       | P   | P       | P     | P    | P    | P  | P      |
|            | CAPS_HT_9.1 | P              | P   | P       | P   | P       | P     | P    | P    | P  | P      |
| qTS4.1     | 4U1         | P              | P   | N       | N   | P       | N     | N    | N    | N  | N      |
|            | 4D1         | N              | N   | N       | N   | N       | N     | N    | N    | N  | N      |
|            | 4D2         | N              | N   | N       | N   | P       | N     | N    | P    | N  | N      |
|            | CAPS_DT_4.1 | P              | P   | P       | P   | N       | P     | P    | P    | P  | P      |

## Whole genome sequencing of Nagina 22 and IR64 and its use in bioinformatics analysis of heat tolerance QTL regions

(Sureshkumar V, Bablee Kumari Singh, Amolkumar U Solanke, Nagendra Kumar Singh, SV Amitha Mithra)

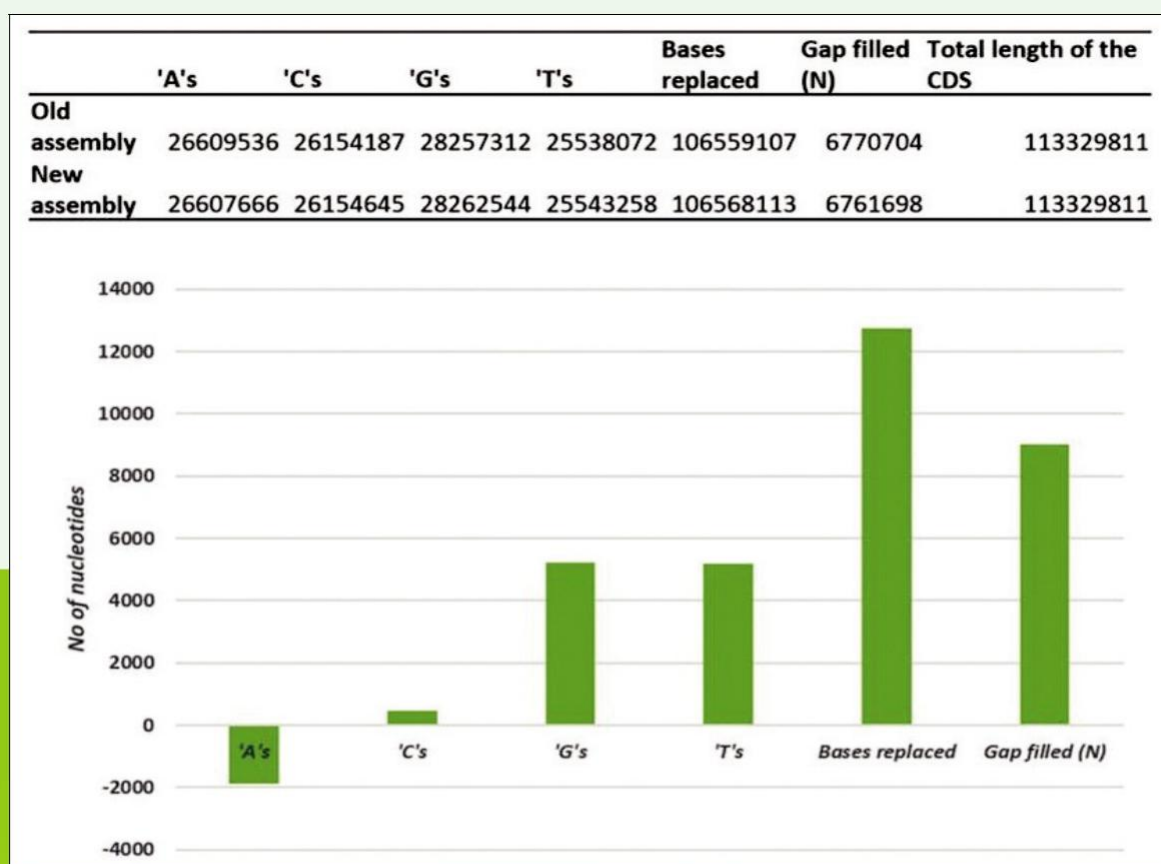
In the previous year, we reported the whole genome sequencing of Nagina 22 and IR 64. Meanwhile we generated more data from Nagina 22, nearly 69,765,000 reads amounting to 16.3x coverage with the total. After trimming of adaptors and removal of low-quality reads, a total of 64 million reads were available for improving the Nagina22 assembly and coverage. As this data is not from

the mutants but N22 itself, the reads were directly mapped on the previously assembled N22 reference genome. This exercise could improve the coverage by 0.01%. Since the contribution of the mutant reads was much higher than N22 in the earlier version, the nucleotide composition in the final assembly also changed for 12,746 nucleotides. The number of genes with complete coverage (no 'N's) has also increased to additional 60 genes (Fig. 1.11). Using the WGS data of N22 and IR64, the major heat QTLs identified on chromosomes 5 and 9 were analyzed for the SNPs in the genes and the subsequent amino acid changes (Table 1.6).

**Table 1.6:** Nucleotide and amino acid variations between Nagina 22 and IR64 in the heat tolerance QTL regions identified on chromosomes 5 and 9.

| Heat tolerance QTL | Number of genes in the QTL interval | Number of genes with SNP changes | Number of SNPs | Number of genes with amino acid changes | Number of amino acid changes | Number of genes with > 1 amino acid changes | Number of genes with > 2 amino acid changes |
|--------------------|-------------------------------------|----------------------------------|----------------|-----------------------------------------|------------------------------|---------------------------------------------|---------------------------------------------|
| qSTIY5.1           | 54                                  | 17                               | 91             | 16                                      | 35                           | 7                                           | 5                                           |
| qSTIPSS9.1         | 65                                  | 17                               | 70             | 15                                      | 22                           | 6                                           | 1                                           |
| Total              | 119                                 | 34                               | 161            | 31*                                     | 57                           | 13                                          | 6                                           |

\*The 31 genes are being analyzed for expression under heat stress.



**Fig. 1.11:** Improvement of the CDS assembly of the Nagina 22 sequences over that of Sevanthi et al., (2018). The improvement was made possible by generating whole genome sequencing data of the WT at 16.3x depth using Illumina platform.

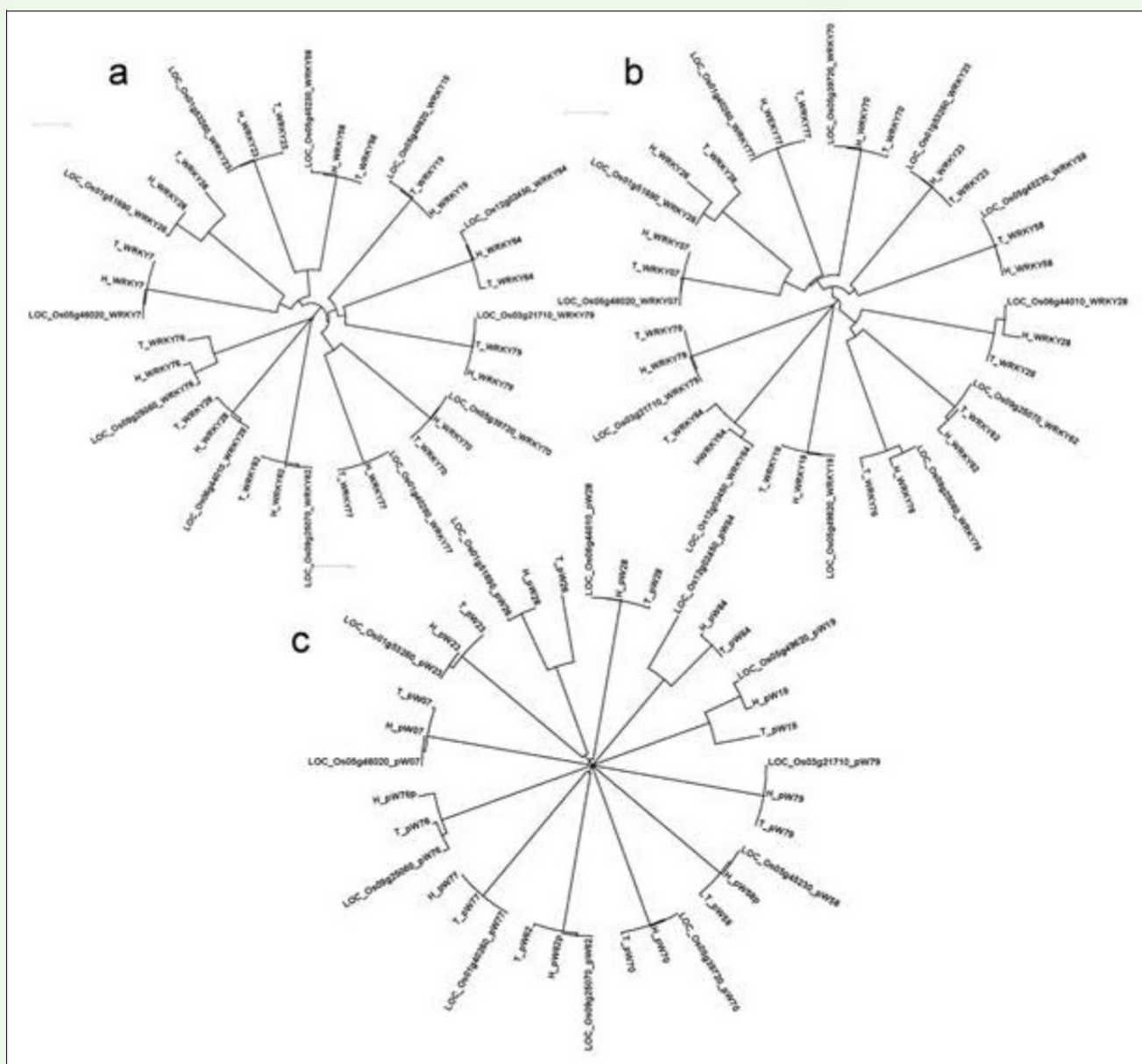
## Characterization of *Magnaporthe*-responsive *WRKY* genes for panicle blast resistance in rice

(Bipratip Dutta, Vishesh Kumar, Prakash G, SV Amitha Mithra and Amolkumar U. Solanke)

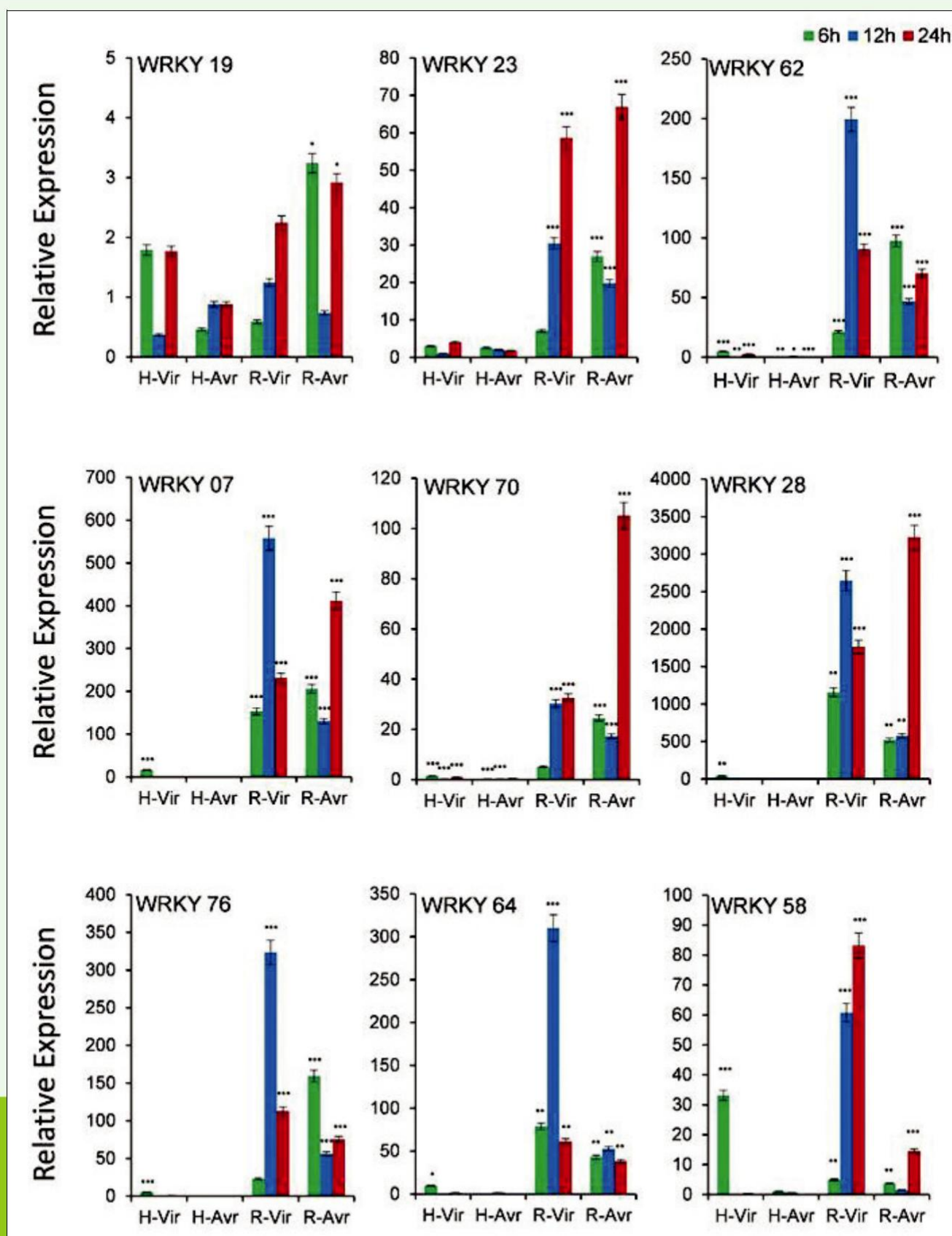
Every year large amount of rice produce is ruined with several biotic and abiotic stresses. Among biotic stresses, fungal pathogen *Magnaporthe oryzae* is the leading stressor causing rice blast disease. This fungus is highly adaptive and can affect the plants at all developmental stages, but as panicle is the economic part of rice, panicle blast causes huge yield loss sometimes up to 100%. The best way to tackle this disease is use of blast resistant cultivars. These cultivars through R genes, signal transduction molecules and appropriate transcription factors activate pathways which provide resistance. Among all transcription factors which are involved in plant defense pathways especially during blast resistance, WRKY family proteins are one of the major players. Thus, R genes with broad spectrum resistance and alleles of WRKY genes supporting for resistance can help to develop blast resistant cultivars, which can last for longer period in the field. Therefore, the research was undertaken to identify panicle blast responsive WRKY genes, their promoters and structural variations between blast resistant and susceptible genotypes, and to study their expression pattern in contrasting rice genotypes upon infection by the pathogen.

In this study, twelve panicle blast responsive WRKY genes were identified from publicly available and our own datasets, and studied structural variations at both nucleotide and amino acid level along with their 1 kb promoter regions. We observed

considerable variations in the genomic region as well as in the promoter regions. Variations are in the form of SNPs and InDels, long regions of insertions or deletions have also been observed in several WRKY genes. It observed that most of the variations were present in Tetep than HP2216 when compared to Nipponbare (Fig. 1.12). While studying multiple sequence alignment at protein level, conservation of WRKY domain was observed in all the WRKY genes. Large number of variations at amino acid level between Tetep and HP2216 were observed in case of WRKY26, WRKY28, WRKY62, WRKY64 and WRKY76. The variations identified in the promoter regions of WRKY genes may be contributing to the differential expression pattern of these genes in the two studied genotypes. In fact, promoter sequences of three genes, WRKY19, WRKY26 and WRKY64 are found to be completely different in Tetep after 500 bp upstream of ATG. The expression analysis of the WRKY genes by quantitative real time PCR showed that WRKY19 was expressed in both the genotypes against both virulent and avirulent pathogen strains of *Magnaporthe oryzae* (Fig. 1.13). WRKY23, WRKY28, WRKY70 were upregulated only in RIL4 against both the strains, but the expression was more against avirulent strain. WRKY7, WRKY58, WRKY62, WRKY64, WRKY76 were highly expressed in RIL4 especially against the virulent pathogen strain. These five genes might be the probable candidates for providing resistance against the virulent pathogen strain. Overall, in the present study, we identified 12 panicle blast responsive genes along with structural variations, and their expression pattern in panicle blast resistant and susceptible genotypes with both virulent and avirulent strains of *M. oryzae*.



**Fig. 1.12:** Phylogenetic analysis of 12 differentially expressed WRKY genes in blast experiments using a) complete genes, b) proteins and c) 1 kb promoter regions in three genotypes Tetep, HP2216 and Nipponbare using ClustalX2.1 tool.



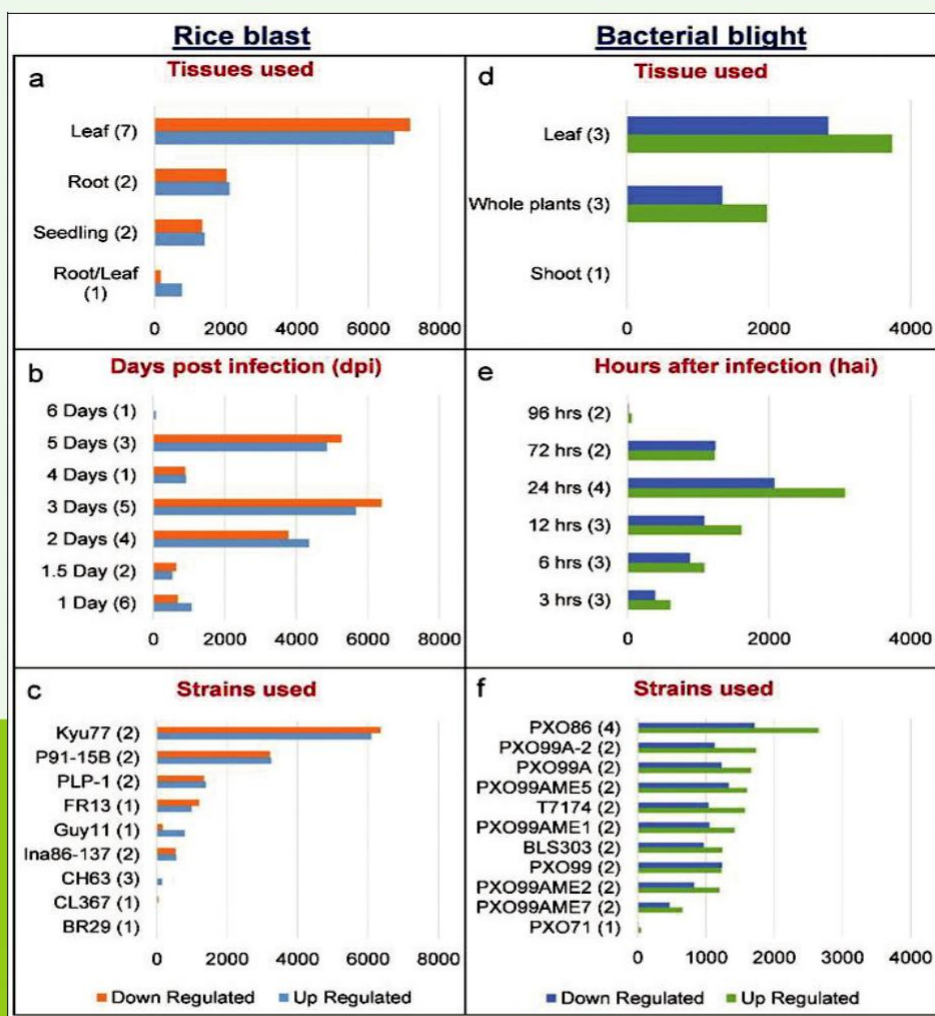
**Fig. 1.13:** Expression analysis of WRKY genes in blast infected panicle tissues at three-time intervals. T-Tetep, H-HP2216 with virulent and avirulent *M. oryzae* strains.

## RiceMetaSysB: Development of database of rice blast and bacterial leaf blight responsive genes

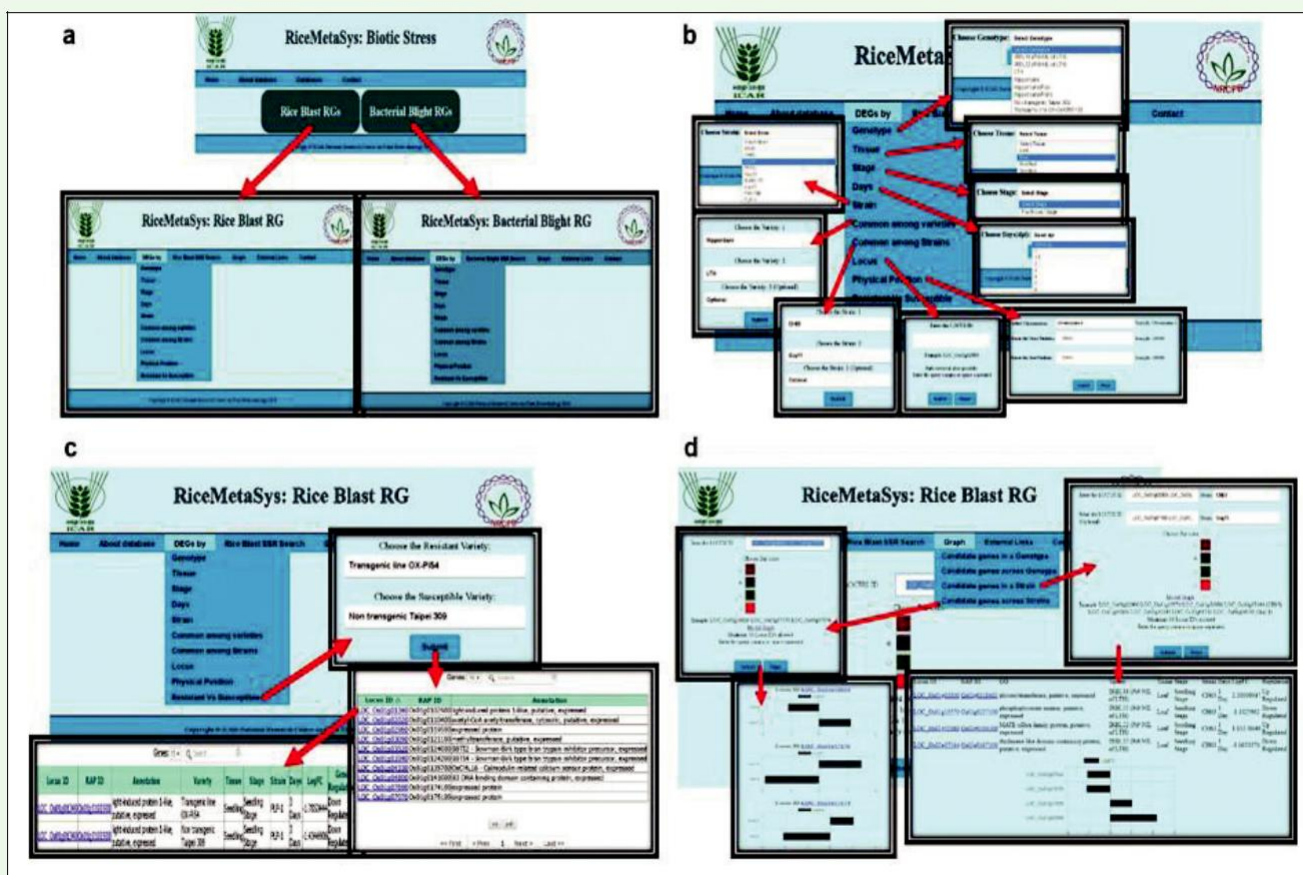
(Sureshkumar V, SV Amitha Mithra, Dwijesh C. Mishra, KK Chaturvedi, Anil Rai and Amolkumar U. Solanke)

Response to disease infestation depends upon the strain of the pathogen and the genetic background of the host. Two most important biotic stresses in rice are blast (caused by *Magnaporthe grisea*) and bacterial leaf blight (caused by *Xanthomonas oryzae* pv. *oryzae*). Currently there is no database which comprehensively covers gene expression profiles of these diseases across pathogen strains and host genotypes. Here, we developed RiceMetaSysB (Biotic Stress), an open source database which provides detailed information about rice blast and bacterial blight responsive genes. The database explores gene expression for blast and bacterial leaf blight disease across 241 and 186 samples, respectively. There are 15135 unique genes for rice blast and 7475 for bacterial leaf blight

in this database. Here, we facilitate users to retrieve responsive/candidate genes using different search options like genotypes, tissue, developmental stage of the host, strain, hours/days post inoculation as identified from publicly available microarray datasets (Fig. 1.14). ‘Common genes among varieties’ and ‘strains’ search options have been enabled to identify most robust candidate genes. User can query for physical position, simple sequence repeat (SSR) information, location of the SSRs like intron, exon, and UTRs from the disease responsive genes (Fig. 1.15). A two-dimensional representation of the data in the form of graph provides expression profiles across genes, genotypes, and strains. Batch retrieval facility has been enabled for locus id search and SSR search without any restriction on quantum of input entry. RiceMetaSysB (Biotic Stress) database will be helpful for crop improvement, in providing robust candidate genes responsible for rice blast and bacterial blight biotic stresses.



**Fig. 1.14:** Distribution of DEGs in RiceMetaSys (Biotic Stress) (A), (B) and (C) Distribution of Rice Blast responsive genes across tissues, Days post infection(dpi) and strains. (D), (E) and (F) Distribution of Bacterial leaf blight responsive genes across tissues, Hours after infection(hpi) and strains.



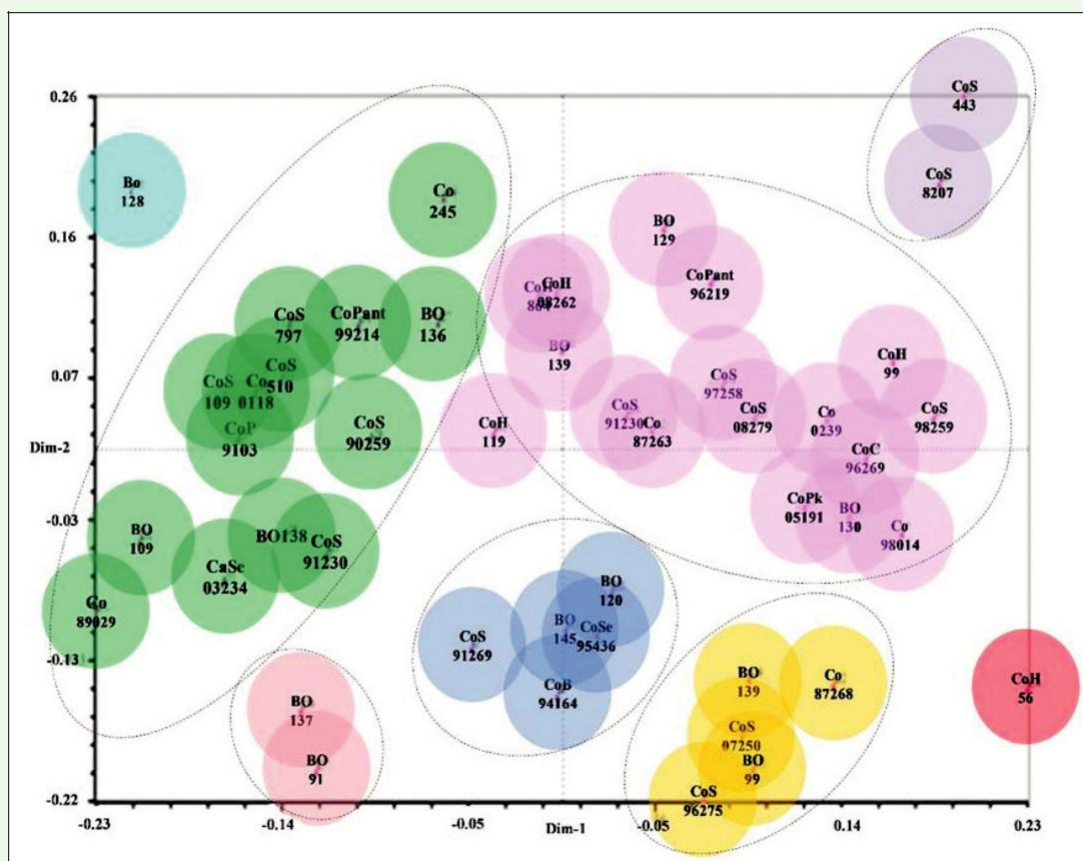
**Fig. 1.15:** A. Homepage of the RiceMetaSys Biotic Stress Database and two different links for Rice Blast and Bacterial Blight RGs Databases. B. Different Search option of in RiceMetaSys Biotic Stress Database like Genotype, Tissue, Stage, Days, Strain, Common among varieties/Strains, locus, Physical position and Resistant vs. Susceptible search. C. Resistant vs. Susceptible search input, its output and hyperlink of locus search. D. Interactive graph results of Candidate genes in and across strain.

### Assessment of IRAP and ISSR markers for resolving molecular phylogeny and detection of somaclonal variations in sugarcane

(Prashant Shingote, Pratima Sharma, Devanna, Kirti Arora, SV Amitha Mithra and Amolkumar U. Solanke)

Sugarcane is one of the major cash crops largely contributing to the sugar industry. Micropropagation is a commercially viable and widely used technique for seed production in sugarcane but is challenged by somaclonal variation. Evaluation of tissue culture raised plants using DNA molecular markers for testing their genetic fidelity, prior to commercialization, can assist in circumventing the probable losses caused by somaclonal variation. Since genome sequence of this complex polyploid is not yet available, multi-profiling anonymous DNA marker systems such as ISSR, AFLP and IRAP are routinely used for identification of informative markers in sugarcane. In this study, we assessed the utility of ISSR and IRAP markers in unravelling the molecular phylogenetic relationships and testing the genetic fidelity of tissue

culture raised plants in sugarcane. We identified 50 ISSR and 48 IRAP polymorphic loci from 12 ISSR and 8 IRAP markers in 47 Indian sugarcane genotypes, after screening 132 markers. IRAP was found to be superior to ISSR in terms of all parameters namely, marker index (2.68 against 1.76), resolving power (2.85 against 1.98), polymorphic loci per assay (6 against 4.1) except for mean polymorphic information content (0.31 and 0.34). The diversity clusters identified by the two marker systems were quite similar though they target different regions of the genome (Fig. 1.16). Testing the tissue culture raised plants with 20 polymorphic markers identified a clone to be a somaclonal variant. Single marker each from both the marker systems (UBC 40 and Gypsy and LTR6149+3'LTR) could detect this variant plant (Fig. 1.17). The unique amplicons were validated by Sanger sequencing. Thus, we are deploying both ISSR and IRAP markers system for detecting somaclonal variants in tissue culture industry through National Certification System for Tissue Culture-raised Plants (NCS-TCP).



**Fig. 1.16:** Integrated 2-D plot from IRAP and ISSR marker-based principal coordinate analysis (PCoA) of 47 sugarcane varieties. Similar colour represents related group.



**Fig. 1.17:** Genetic fidelity testing of tissue culture-raised sugarcane plantlets using ISSR marker UBC\_840 and IRAP marker LTR6149+3'LTR and sequences of unique polymorphic bands from variant clone. M: 1 kb DNA ladder, lanes 1-3 tissue culture-raised plantlet samples var. Co0118 sugarcane from batch I, and batch II (lanes 4 and 5), the arrow indicate genetically variable DNA fragment.

## Development of mapping population segregating for sheath blight resistance phenotype

(Richa Kamboj, Tapan Kumar Monda, Tilak Raj Sharma, Deepak Singh Bisht and Nagendra Kumar Singh)

Rice (*Oryza sativa* L) is a staple food for more than half of the world's population. Globally the major constrain in rice production is the damage caused by the attacking pathogens. Sheath Blight (ShB) is second most important disease of rice, next only to rice blast. ShB disease is caused by a necrotrophic fungus *Rhizoctonia solani*. Breeding for sheath blight resistance has been a futile exercise so far mainly due to the unavailability of a potent resistance

donor source. In the Kharif 2016 and 2017 we have screened the wild rice germplasm collected at our centre from 13 agro-climatic zones of India. The identified germplasm was used to the development of suitable mapping population (Fig. 1.18). We have developed a BC<sub>1</sub>F<sub>2</sub> mapping population consisting of 440 individuals, and a F<sub>2</sub> population consisting of 220 individuals. Leaf samples were collected from the mapping population and the genomic DNA was isolated (Fig. 1.19). In the present kharif season the mapping population will be scored for ShB resistance and susceptible phenotype at different locations in India.



Fig. 1.18: (a-e). Planting of BC<sub>1</sub>F<sub>2</sub> population in Kharif 2018.

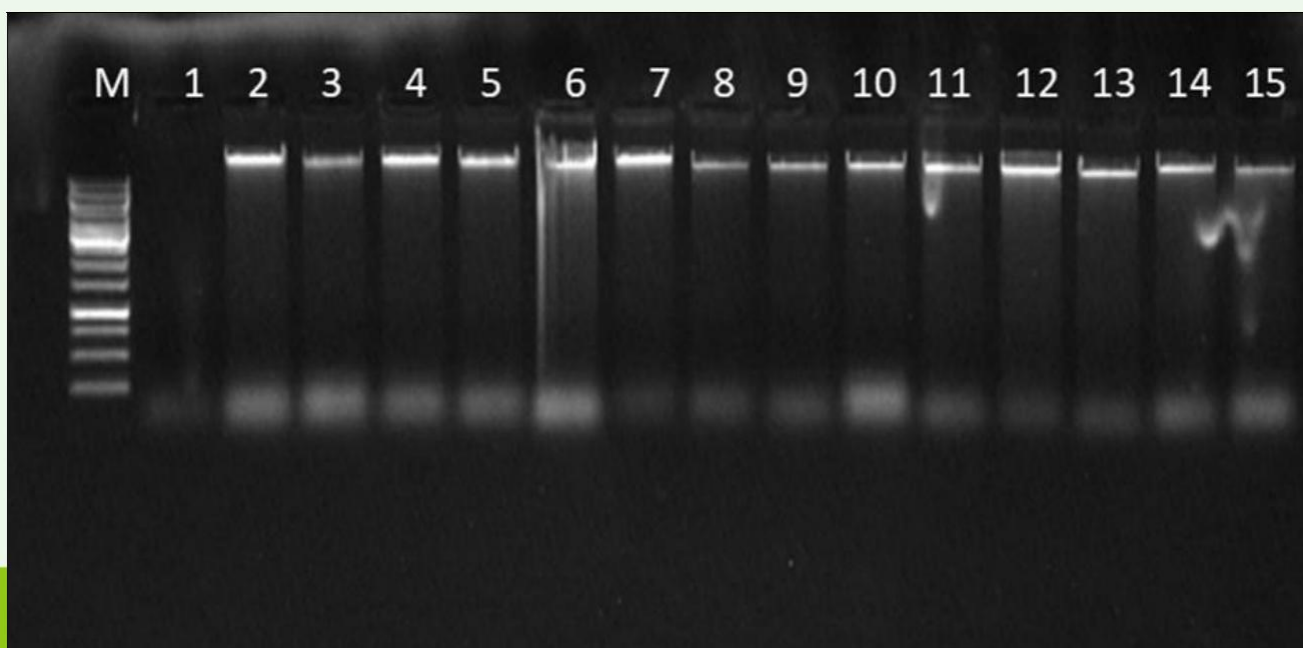


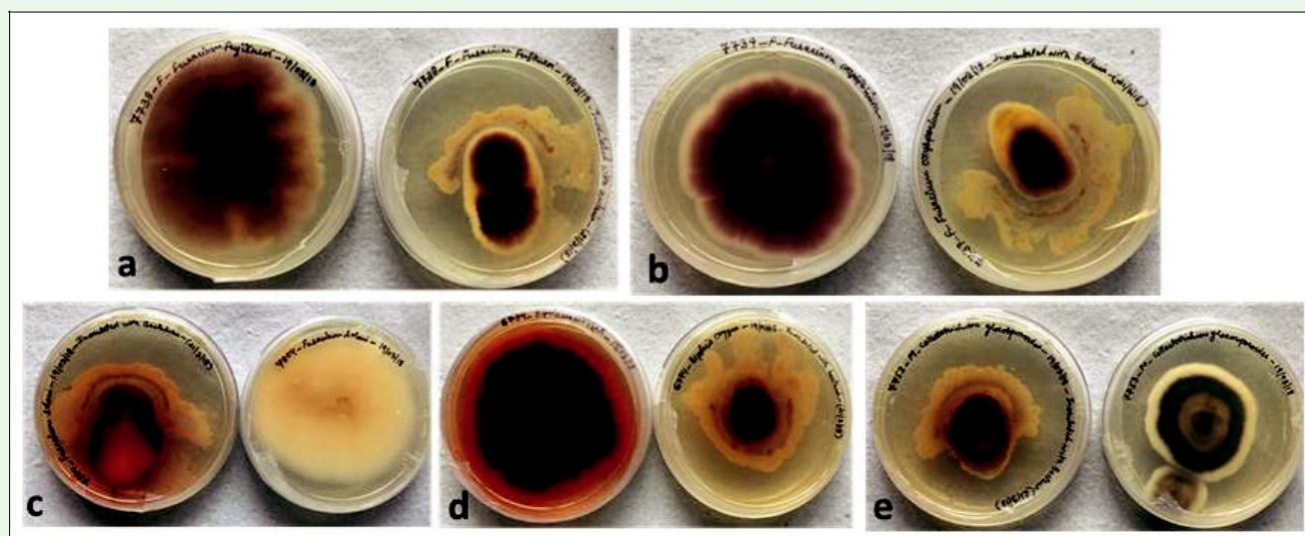
Fig. 1.19: A representative picture of genomic DNA isolated from F<sub>2</sub> mapping population segregating for sheath blight resistance phenotype.

### Characterisation of *Pseudomonas aeruginosa* strain against different plant pathogens

(Richa Kamboj, Tapan Kumar Monda, Tilak Raj Sharma and Deepak Singh Bisht)

At present the application of fungicide is the only known method deployed for the control of Sheath blight resistance in rice. However, recently we have identified *Pseudomonas* spp. acting antagonistically against *R. solani*, the causal agent of Sheath Blight disease in rice. Notably, our identified strain of

*Pseudomonas* showed a potent antagonistic activity against five different important plant pathogens including, *Fusarium oxysporium*, *Bipolaris oryzae*, *Fusarium fujikori*, *Fusarium solani* and *Colletotrichum gloeosporioides* (Figure 1.20). The genome sequence of our *Pseudomonas* strain revealed several niche specific genes and gene cluster encoding typical antibiotic metabolites which are yet to be validated for their antagonistic activity. After biochemical characterisation the strain will be deposited in Microbial Type Culture Collection.



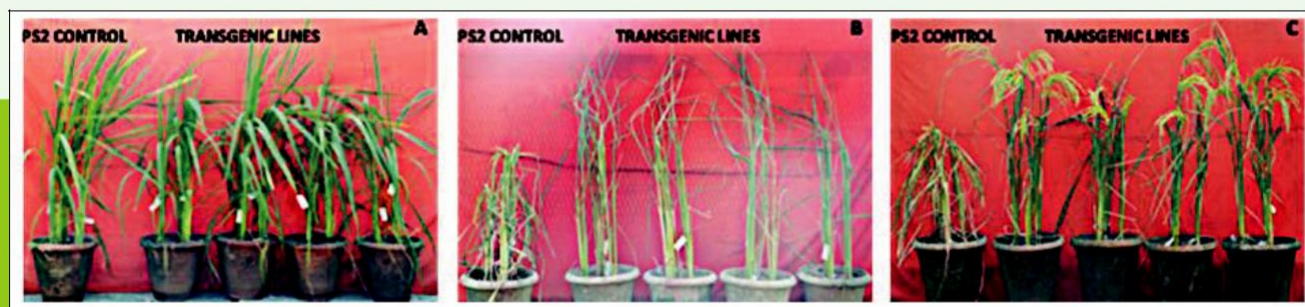
**Fig. 1.20:** Antagonistic activities of *Pseudomonas aeruginosa* strain against (a) *Fusarium fujikori*, (b) *Fusarium oxysporium*, (c) *Fusarium solani*, (d) *Bipolaris oryzae*, (e) *Colletotrichum gloeosporioides*.

### Transgenic rice with over expression of water stress tolerance genes

(Karikalan J, Amitha CR Mithra and Pranab Kumar Mandal)

Transgenic plants developed by overexpressing putative stress tolerance genes in the background of N22 and PSII for functional validation of the genes

are still being characterized. Two genes are found Southern positive with single copy number in each under PS-II background for the T2 and T3 plants. Two events in each case were confirmed. Water stress related physiological parameters are also being studied in these transgenic rice plants (Fig. 1.21).



**Fig. 1.21:** Evaluation of drought tolerance of pCambia-Os06g10210 transgenic rice lines. A, Non Transgenic (NT) and transgenic rice cv. Pusa Sugandh-II plants grown under non-stress conditions; B, plant after 10 days drought stress condition; C, plants after 8 days after recovery.

## Genetic modifications to improve biological nitrogen fixation for augmenting nitrogen needs of cereals

(Sapna Dhiman, Chetan Kumar Nagar, Subodh Kumar Sinha and Pranab Kumar Mandal)

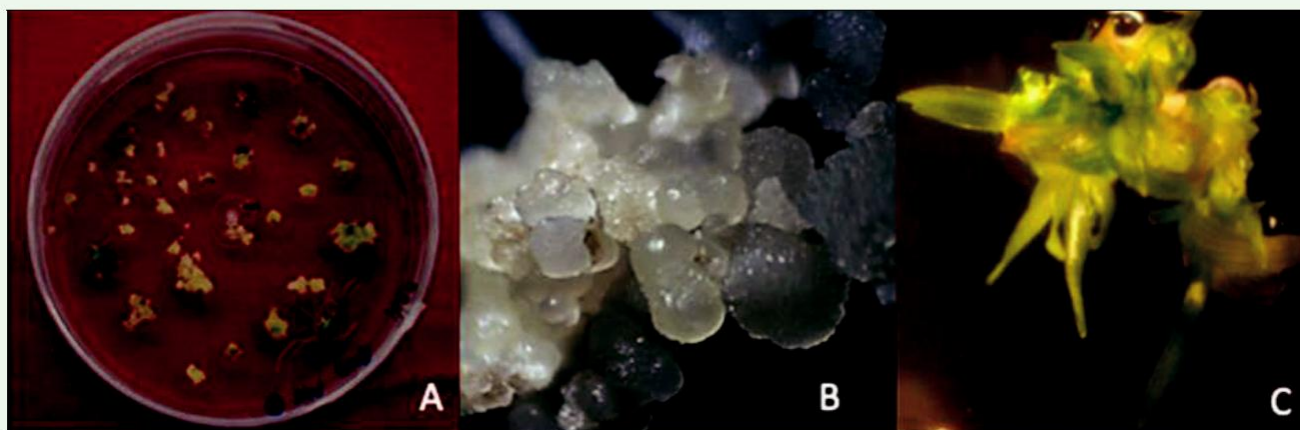
To develop transgenic rice plants that are able to support the formation of symbiosome compartments in rice roots for harboring/sequestering rhizobia and protecting them from plant defense system, two constructs developed at TERI (partner of an ICAR funded projet) were used for transforming rice cultivar Taipei 309. Both the constructs PA5 and PB6 were initially confirmed for the presence of transgene. The genes *MtN6*, *MtFL4*, *MtFL2*, *MtRPG*,

*MtVAPLjCER*, *LjNENA*, *LjNAP1*, *LjPIR1*, *LjARPC1* and *LjNPL* were shown to participate in rhizobial infection processes in legume roots. The vectors consist of the linked expression cassettes as shown below:

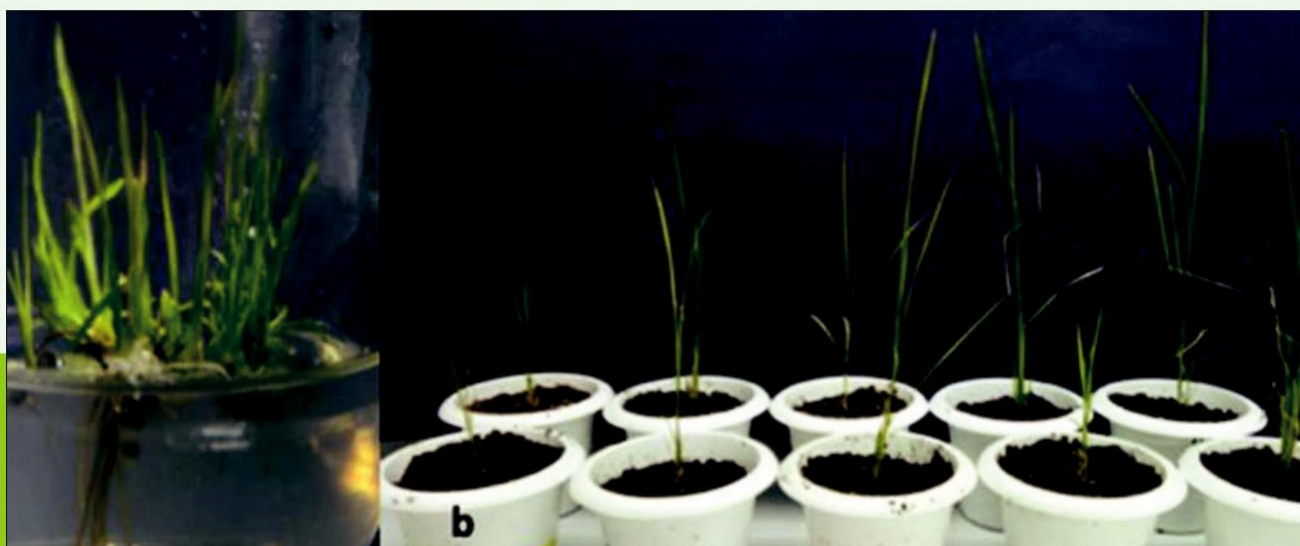
**PA5:** *OcsP:hpt:ocsT-OsCYCLOPSP:MtN6:masT-OsDMI2P:MtFL4:35T-OsPOLLUXP: MtRPG: nosT - OsUBI3P:MtFL2:Tm1 - OsDMI3P:LjCer:agsT*

**PB6:** *OcsP:hpt:ocsT-OsCYCLOPSP:LjNENA:masT-OsDMI2P:LjNAP1:35T-OsExpB5P: LjNPL:nosT-35SP:LjARPC1:E9T-OsDMI3P:MtVAP:35T-OsPOLLUXP:LjPIR1:agsT.*

During the reporting period we could transform Taipei 309 with PA5 construct through *Agrobacterium* infection (Fig. 1.22 and Fig. 1.23).



**Fig. 1.22:** Transformed calli on CHU-N<sub>6</sub> regeneration media. **(b):** Transformed calli after second round of regeneration; **(c):** Regenerated shoots after third phase of regeneration.



**Fig. 1.23:** Regeneration of putative transgenic plantlets.

## Low-light tolerance in rice

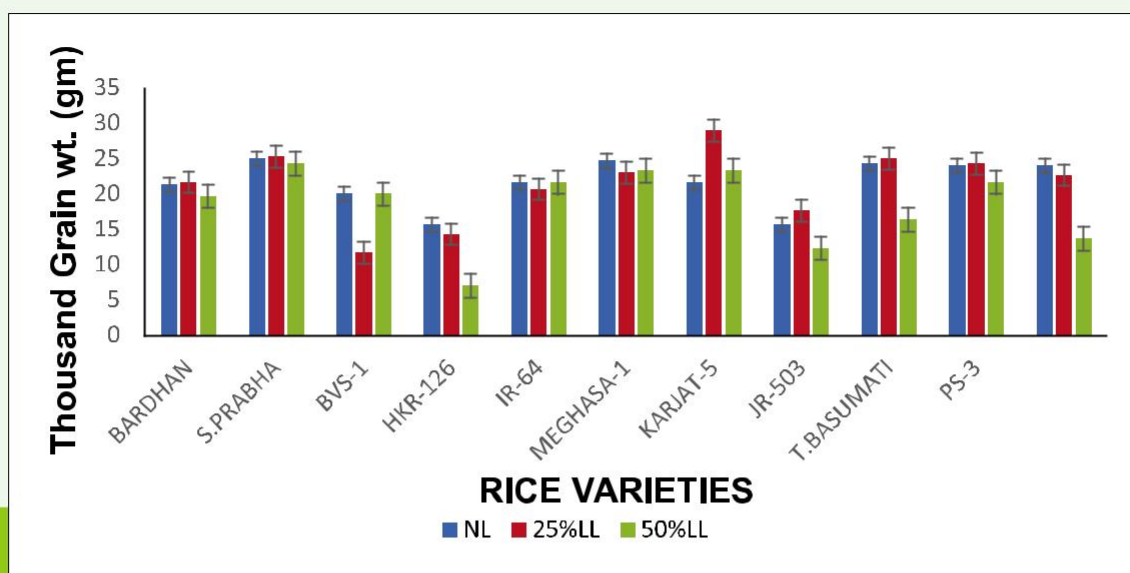
(Bharali B, Rai R, Baig MJ and Pandey R and Dash PK)

Eleven rice genotypes short-listed from one hundred rice genotypes screened in previous two growth seasons of *kharif* 2016 and 2017 were further screened for their response to light tolerance under 100% (NL) and 50% (LL) of light intensities during *kharif* 2018. All the genotypes exhibited reduction in yield under 75% and 50% light intensities. Tolerant genotypes compensated yield reduction under shade by improving the test weight (1000 grain weight) (Fig. 1.24) and grain yield per unit area (Fig. 1.25). Based on the pooled data of 2016-17 & 2017-18 on susceptibility index on yield and yield attributes; rice genotype Aki Sali was considered tolerant to low-light and used for crossing with the local popular variety viz., Ranjit (IET12504; cross between Pankaj x Mahsuri). Since the flowering time of Aki Sali coincided with the Ranjit, one hundred crosses (field and pots) were raised in *kharif*, 2018 (Fig. 1.26). Similarly, the NRRI variety: BVS-1 was also considered for making crosses with Ranjit following staggered transplanting. However, the cross using Aki Sali was successful, while it failed for BVS-1. The seeds from Aki Sali are preserved for multiplication

and phenotyping studies.

Physiological screening at NRRI, revealed increase in total chlorophyll content in most of the rice genotypes, most conspicuously in PS-3, T.Basumati and BVS-1 during *rabi* and Panindra and PS-3 during *kharif* (Fig. 1.27). Chlorophyll a/b ratio decreased under low light stress which is considered as a marker phenotype for the screening of low light tolerant genotypes. Among NRRI genotypes PS-3 and T.Basumati during *rabi* and Panindra and PS-3 during *kharif* showed a pronounced decrease in Chlorophyll a/b ratio.

The leaf anatomical structure was analysed in the contrasting genotypes. In lowlight (LL) grown leaf, the stomatal density was reduced as compared to normal light (NL) grown leaf on both surfaces. Also, the pore size (stoma) was smaller in lo light grown leaf compared to normal light grown leaf (Fig. 1.28). The dominant light receptor gene *phyA* from tolerant rice variety Aki-Sali has been cloned, and to make it constitutive photo receptor site-directed mutagenesis has been accomplished. Putative interacting factors of *phyA* from rice have been cloned (Fig. 1.29). and their role in stabilizing *phyA* by physical interaction is under investigation.



**Fig. 1.24:** Tolerant genotypes compensated yield reduction under lowlight (LL) and normal light (NL) by improving the test weight (1000 grain weight).

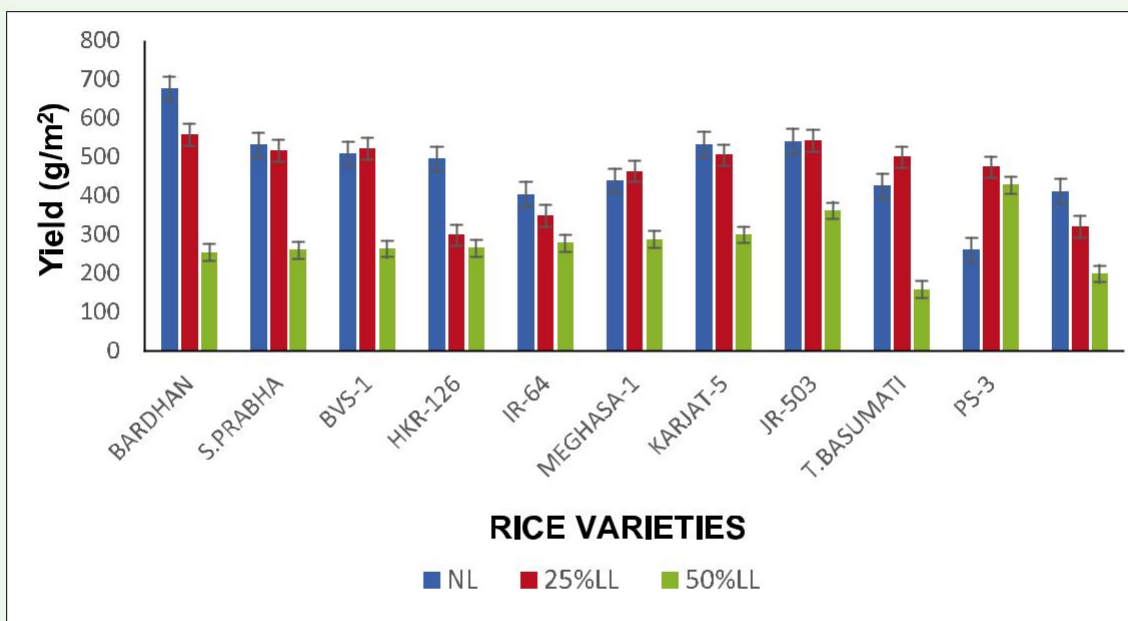


Fig. 1.25: Yield of some tolerant genotypes under lowlight condition.



Fig. 1.26: Generation of F1/mapping population using Aki-Sali X Ranjit and BVS1 X Ranjit.

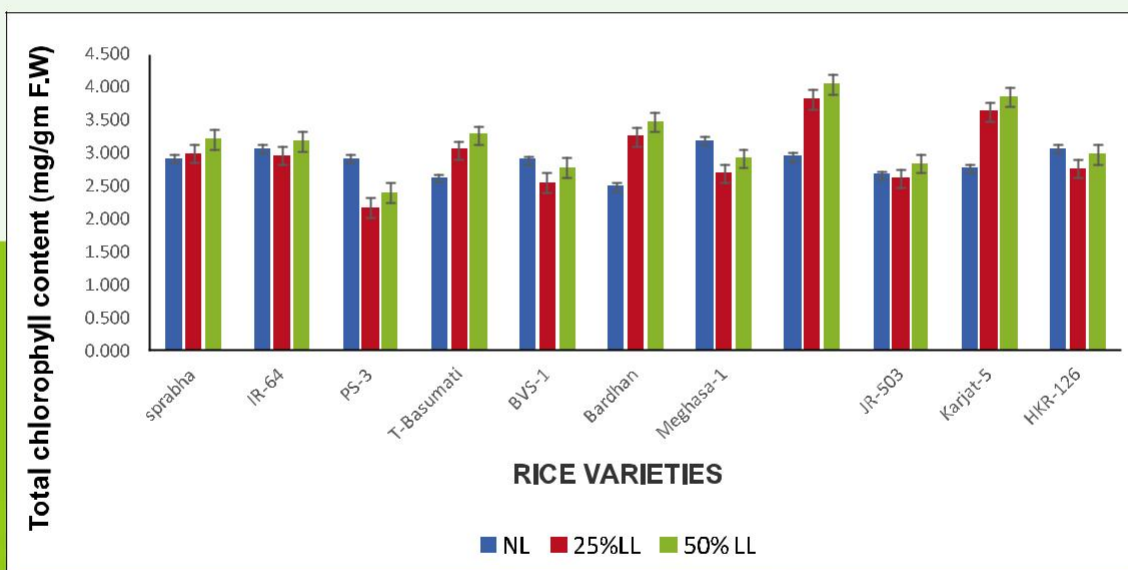
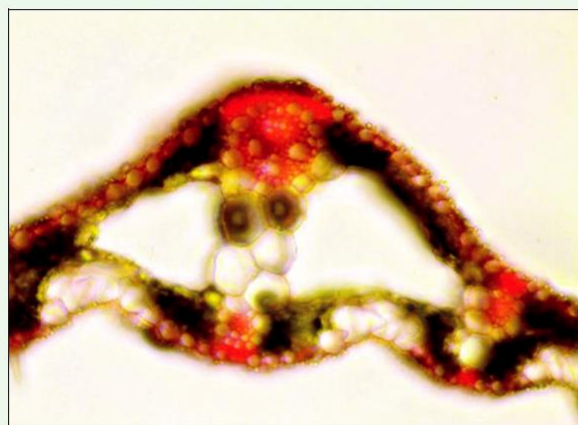
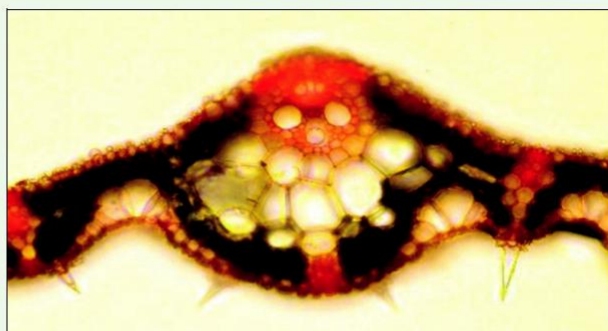


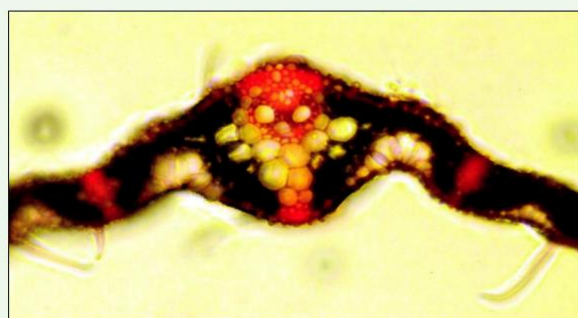
Fig. 1.27: Chlorophyll content in the screened rice genotypes under normal and lowlight conditions.



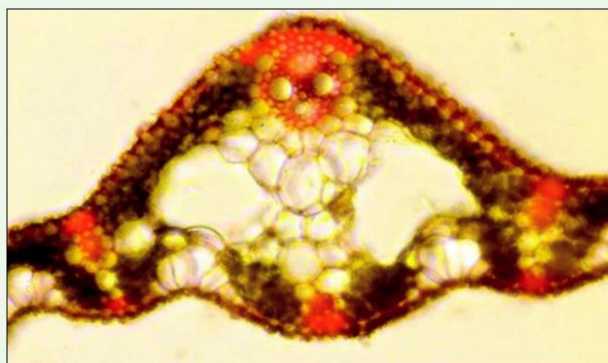
IR 64 NL



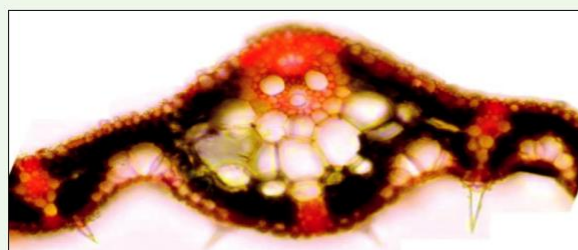
IR 64 LL



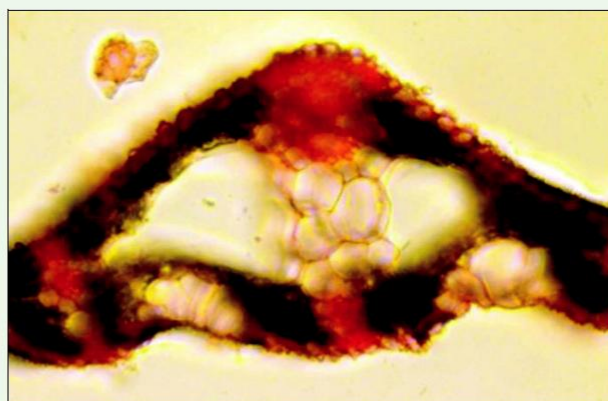
SP NL



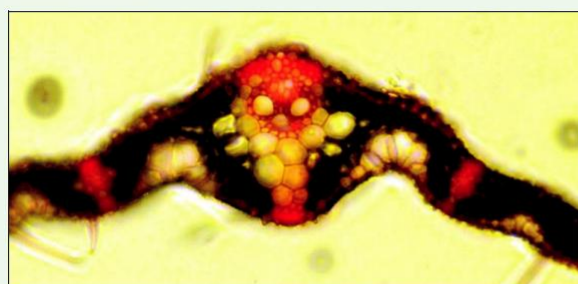
SP LL



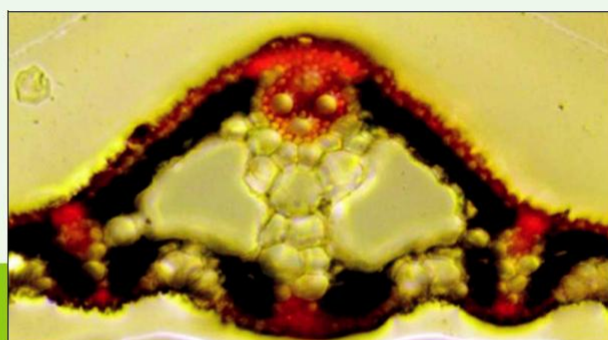
Assam 2 NL



Assam 2 LL

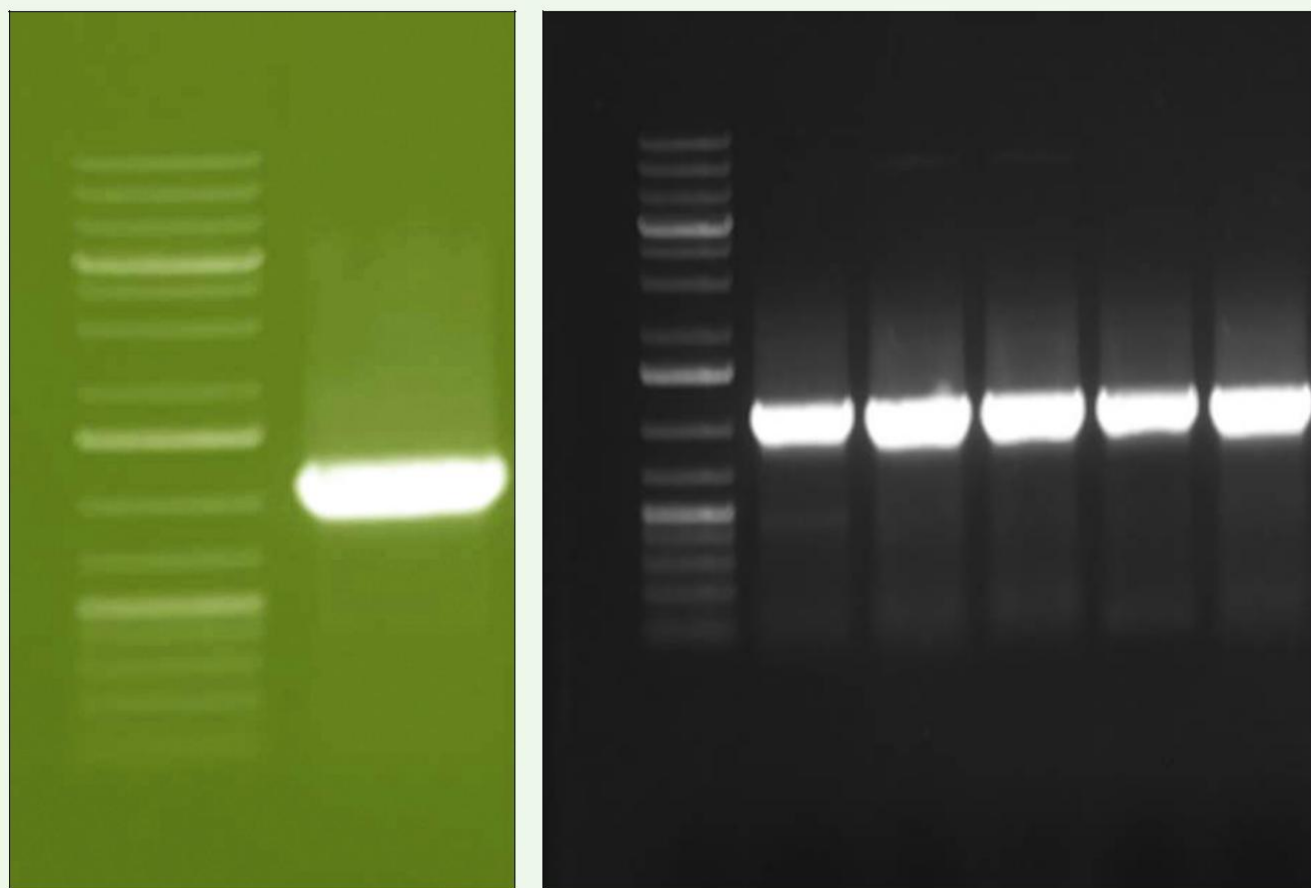


NRRI18 NL



NRRI18 LL

**Fig. 1.28:** Ultra-structure of rice leaf exposed to lowlight (LL) and normal light (NL) conditions. (IR- IR64 (Susceptible); SP-Swarnaprabha (Tolerant) and two tolerant genotypes (Assam 2 and NRRI18)).



A

B

C

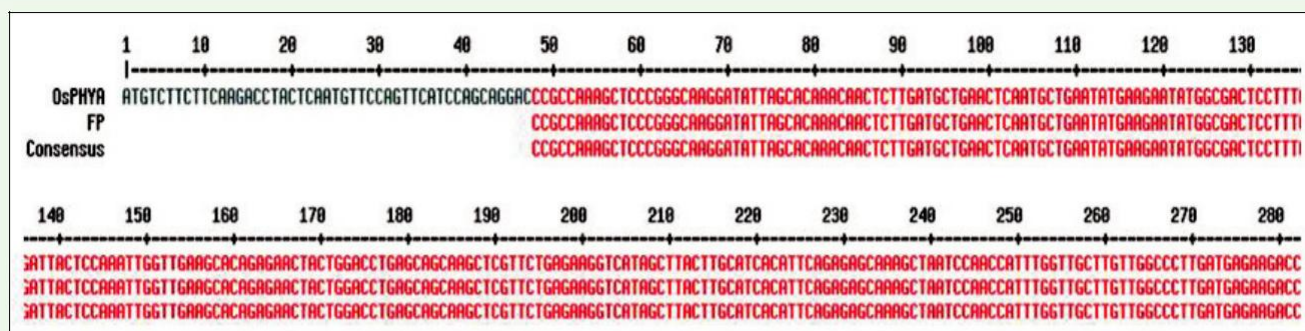


Fig. 1.29: Cloning *Phytochrome A* from rice. a) PCR amplification from rice, b) colony PCR, and c) sequence verification.

# Adaptation of Wheat to Climate

## Heat Induced

### Glycerol-3-phosphate acyl transferase (GPAT) -Molecular cloning and gene expression profiling under terminal heat stress in wheat (*Triticum aestivum* L.)

(Shaziya Sultana, Praful Jaiswal and Sharmistha Barthakur)

Wheat is a highly versatile and major cereal crop in terms of harvested area and adaptation to diverse regions. Terminal heat stress that is heat stress during the later or reproductive phases of wheat development from booting, heading, anthesis and grain filling stages is considered as one of the major environmental constraints in wheat breeding today. With rising global temperature most of the wheat cultivated regions throughout the world including India has been affected by terminal heat stress resulting in reduction in grain number per spike, grain weight and consequently significant reduction in grain yield. Towards elucidating terminal heat stress tolerance mechanisms in Indian bread wheat we are targeting various processes involved in plant stress tolerance and signaling.

Heat stress during the last phases of wheat development causes structural perturbations in the membrane leading to phase separation of the membrane glycerolipid constituents with characteristics disturbances of function such as loss of selective permeability and transport process.

GPATs a multi gene family member encodes enzymes involved in several lipid biosynthetic pathways and plays important biological roles in plant growth and development. A transcript expression profiling was carried out in two cultivars viz. HD 2967, a popular high yielding variety and Raj3765, a heat tolerant cultivar. Beginning with emerging flag leaf, four grain development stages were exposed to high temperature stress at 37°C for 2 h in temperature controlled growth chamber. Independent leaf and ear heads were utilized for gene expression profiling of GPAT gene using quantitative RT-PCR. Differential, abundant and positive modulation of GPAT was observed in both the cultivars under heat stress condition. To evaluate the effect of heat stress on developing wheat grain, certain physiological parameters like- Membrane Stability Index, Chlorophyll content, Relative Water Content, Hydrogen peroxide accumulation were also taken into consideration (Fig. 2.1). A full coding region of GPAT was cloned from variety Raj3765 and structurally confirmed by various *in silico* analyses (Fig. 2.2). As a whole, the results show positive modulation of GPAT gene under high temperature in wheat implicating its functional role in pollen development, gametogenesis, and early grain development vide induction of phospholipid and TAG synthesis.

**Fig. 2.1:** Real Time RT-PCR analysis of GPAT transcript levels under high temperature treatments at five developmental stages starting from flag leaf through 20 days post anthesis of cultivars HD2967 and Raj3765. Here relative fold change in expression is calculated by taking the lowest value obtained at 10 DAA stage of cultivar Raj3765 as 1 and calculating the other values respectively. FL-flag leaf, PA-pre anthesis, ANT-anthesis, 10DAA-10 days after anthesis, 20DAA-20 days after anthesis. The left panel shows membrane stability index (MSI) of the two wheat cultivars in both untreated control and heat treated condition at anthesis stage.



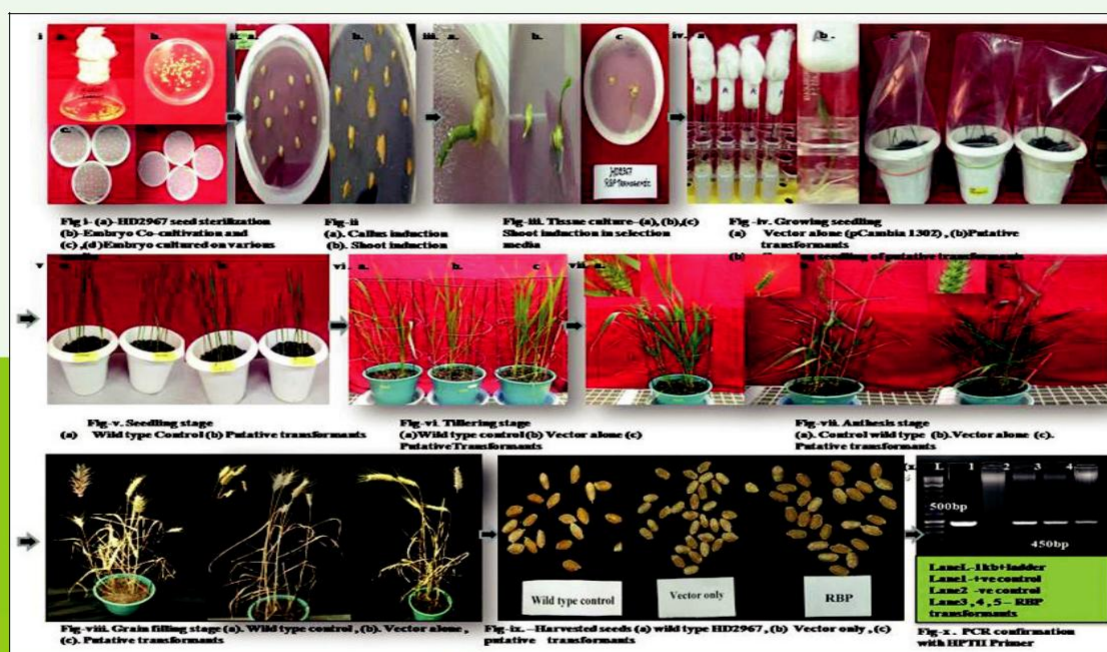
**Fig. 2.2:** Isolation of GPAT coding sequence from wheat cultivar Raj3785. RT-PCR amplification of approx 1.3 Kb fragment and cloning confirmation in pGEMT vector after blue white screening.

## Towards development of transgenic wheat using mature embryo as explants: optimization of an *in vitro* regeneration and transformation protocol using *TaRBP/ pCambia1302* construct

(Namrata Pandey and Sharmistha Barthakur)

Wheat is the most important staple crop grown all over the world. However, till date there is no single optimized protocol for *in vitro* regeneration and transformation of wheat. RBPs or RNA binding proteins are involved in post transcriptional modification that control mRNA metabolism allowing cells to rapidly adapt to changing physiological and environmental conditions. A protocol has been

standardized for wheat regeneration using mature seeds of a popular wheat cultivar HD2967. Different concentration of auxin, cytokinin, and gibberellin were used for optimization of wheat regeneration using mature embryos as explants. Number of seeds per spike varied in developed mature plants and next generation seeds were compared with tissue culture developed control HD2967 plants. This protocol was further used for wheat transformation using *RBP* gene identified and isolated from wheat in the lab earlier. The putative transformants after antibiotics screening, PCR analyses are now being advanced into next generation for southern analysis and functional characterization (Fig. 2.3).



**Fig. 2.3:** Representative flow chart showing various stages in developing transgenic bread wheat (cv HD2967) using mature embryo as explants and over expressing a RNA binding protein.

## Elucidating acquired and transgenerational thermotolerance in wheat vide hormonal and heat priming and clarifying the role of ubiquitin proteasome system

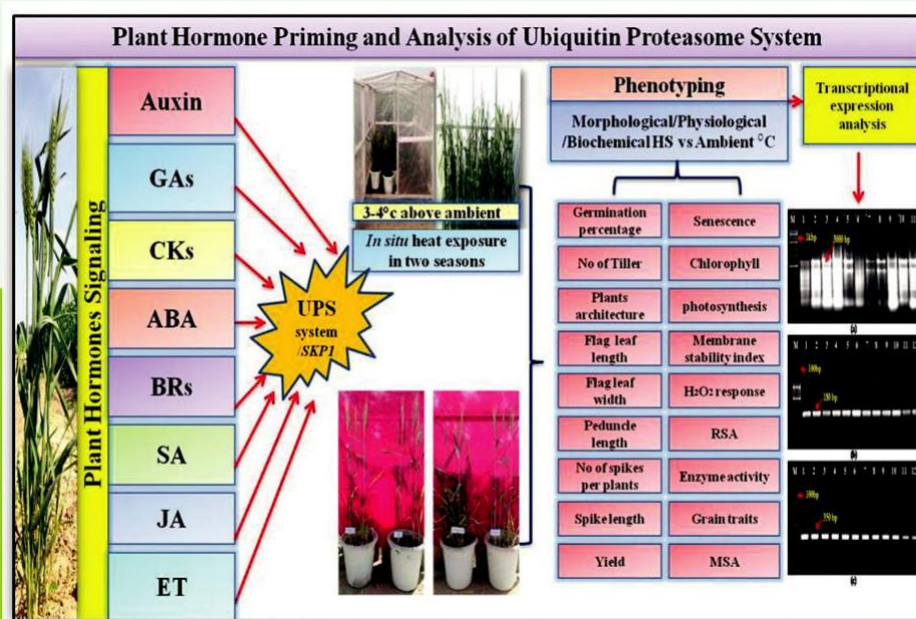
(Praful Jaiswal and Sharmistha Barthakur)

Ubiquitin proteasome system is intricately involved in protein turn over and signaling of various plant growth regulators vide post translational modifications. Towards isolating and functionally characterizing role of individual hormone in modulating heat stress response, we are studying their modulation and regulatory effect on wheat under heat stress. The *SKP1* gene involved in

protein turnover which is an essential component of SCF complex (Skp1-Cullin-F-box protein) was used as heat inducible transcript marker for molecular characterization. Differential modulation of *SKP1* gene was observed. Selected panel of plants in primed2 (P2) and primed3 (P3) generation are being further evaluated for elucidating molecular mechanisms of heat stress memory at the mechanistic level by extensive phenotyping as well as molecular characterization. In the Fig. 2.4, an outline of the experiment is shown and in Fig. 2.5, various characterizations that are being carried out are depicted.



**Fig. 2.4:** Outline of priming experiments carried out to evaluate acquired and transgenerational heat stress tolerance in bread wheat.



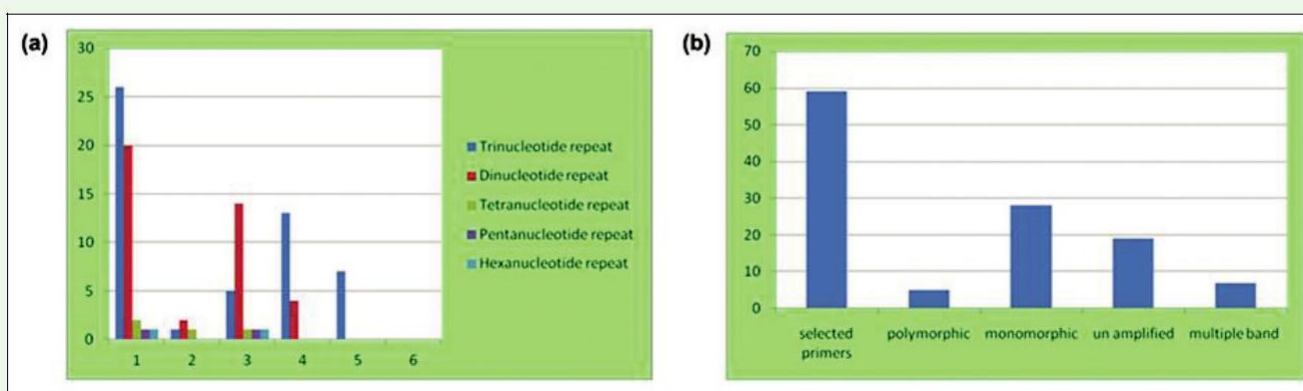
**Fig. 2.5:** Various plant hormonal and heat priming treatment and subsequent phenotyping and analyses.

## Microsatellite markers associated with heat stress tolerance in *Pennisetum glaucum* (L.) R.Br.

(Nigombam Sunil and Padaria Jasdeep Chatrath)

*Pennisetum glaucum* (L.) commonly known as Pearl millet is largely grown as a rainy season crop under rainfed conditions in Asia and Africa. To speed up the molecular breeding strategies for developing high temperature stress tolerant *P. glaucum*, there is an urgent need for a large number of markers associated with heat stress tolerance. Therefore, additional EST-SSR markers were developed for pearl millet research from the available heat stress responsive transcriptome pearl millet data (NCBI

SRA Accession No.: SRP151237). Fifty nine highly expressed unigenes with a threshold P-value of less than 0.005 and an expression upregulation of a fold change of greater than 2 were selected from the available heat stress responsive transcriptome data. Of these selected unigenes, 50 were the ones which had SSR motifs while 9 were used for development of STS markers. The EST-SSR markers selected consisted of 26 SSRs with trinucleotide motifs, 20 SSRs with di-nucleotide repeat motif, 2 SSRs with tetra-nucleotide repeat motif and 1 each from penta and hexanucleotide repeat motif. For development of STS markers, 9 highly expressed unigenes were randomly selected (Fig. 2.6).



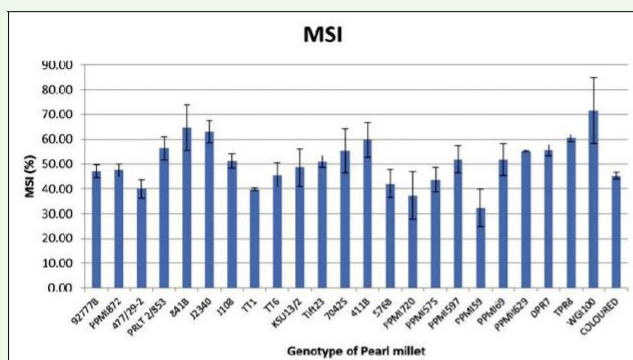
**Fig. 2.6:** a) Distribution of monomorphic, polymorphic, unamplified and multiple band over the tri, di, tetra, penta and hexa nucleotide repeats, b) Overall distribution of primers.

To validate the designed microsatellite markers, primers were designed for their amplification in 24 diverse genotypes of pearl millet (Table 2.1). The 24 genotypes were assessed for their high temperature stress tolerance ability by measuring membrane stability index (MSI). Variation in MSI (32.24% to 71.54%) was observed among the 24

genotypes, indicating variation in the sensitivity to high temperature stress. It was observed that the MSI value for the different genotypes ranged from 32.24% to 71.54% with genotype PPMI 59 having the lowest MSI (32.24%) and genotype WGI 100 having the highest MSI (71.54%) (Fig. 2.7).

**Table 2.1:** List of *P. glaucum* genotypes used in present study.

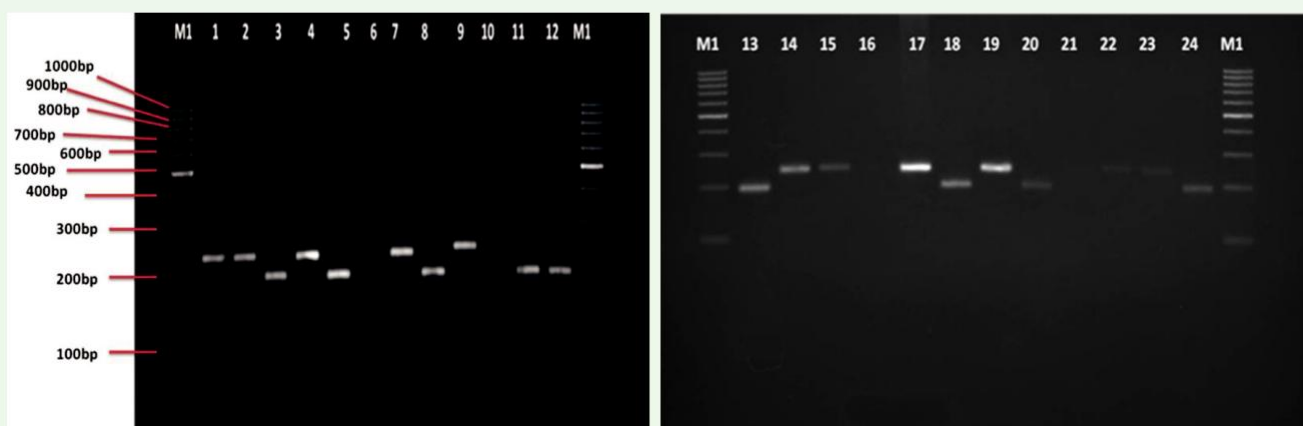
| S. No. | Genotype   | S. No. | Genotype | S. No. | Genotype | S. No. | Genotype |
|--------|------------|--------|----------|--------|----------|--------|----------|
| 1      | 92777B     | 7      | J108     | 13     | 411B     | 19     | PPMI69   |
| 2      | PPMI 872   | 8      | TT-1     | 14     | 576B     | 20     | PPMI627  |
| 3      | 477/29-2   | 9      | TT-6     | 15     | PPMI720  | 21     | DPR-7    |
| 4      | PRLT-2/853 | 10     | KSU 13/2 | 16     | PPMI575  | 22     | TPR-8    |
| 5      | 841B       | 11     | Tift-23  | 17     | PPMI597  | 23     | WGI-100  |
| 6      | J2340      | 12     | 7042S    | 18     | PPMI59   | 24     | Coloured |



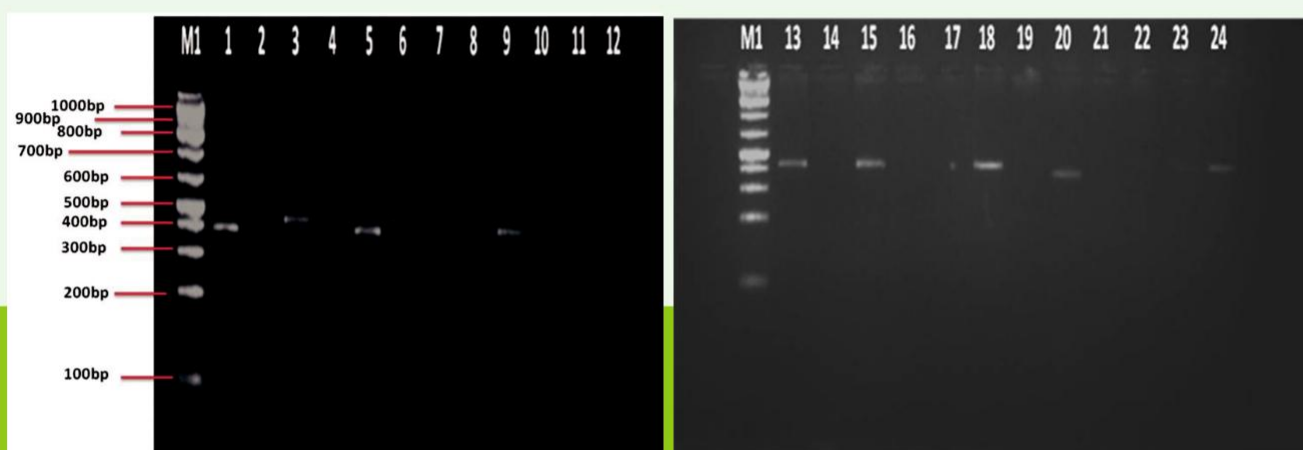
**Fig. 2.7:** Membrane stability index of different pearl millet genotypes.

Four EST-SSRs markers (15.38%) and one STS marker (16.66%) displayed polymorphism

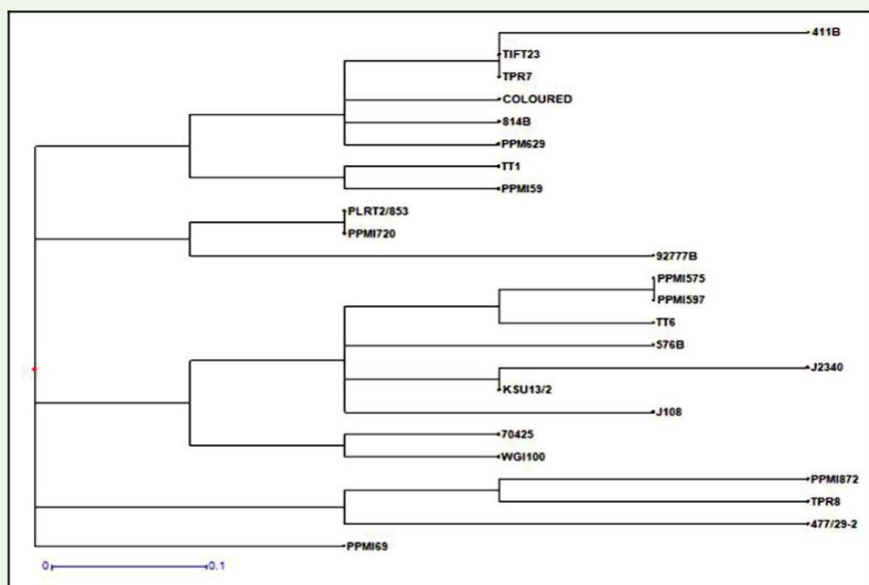
respectively (Fig. 2.8, Fig. 2.9). The PIC (Polymorphic Information Content) of newly developed EST-SSRs was in the range of 0.207 to 0.662 with an average value of 0.435. Based on the five polymorphic markers, the 24 pearl millet genotypes could be clustered into 5 main and 11 sub clusters (Fig. 2.10) Annotation of the five transcripts for which polymorphism was detected, were found to code for genes related to stress response and signaling pathway (Table 2.2). The EST-SSRs and STS markers thus analyzed would augment the existing SSR marker resource and find their applications in diversity assessment and marker-aided breeding programs for genetic improvement of pearl millet.



**Fig. 2.8:** Polymorphism resultant of EST-SSR marker (Primer 47SSRPgCL118Contig3) for 24 pearl millet genotypes run on 3% metaphor agarose. M1- 100 bp DNA ladder; lane 1 to 24 represents the genotypes of pearl millet:lane 1- 92777B; 2- PPMI872; 3- 477/29-2; 4- PLRT-2/853; 5-841B; 6-J2340 ; 7-J108; 8-TT-1; 9-TT-6; 10- KSU 13/2; 11- Tift-23; 12- 7042S; 13-411B; 14-576B; 15- PPMI720; 16-PPMI575; 17-PPMI597; 18- PPMI59; 19- PPMI69; 20-PPMI629; 21-DPR-7;22-TPR-8; 23-WGI-100; 24-PPMI683.



**Fig. 2.9:** Polymorphism resultant of 1 STS marker (Primre56STSPg CL4894Contig1) for 24 pearl millet genotypes run on 3% metaphor agarose. M1, 100 bp ladder, lane 1 to 24 represents the genotypes of pearl millet:lane. Lane 1- 92777B; 2- PPMI872; 3- 477/29-2; 4- PLRT-2/853; 5-841B; 6-J2340 ; 7-J108; 8-TT-1; 9-TT-6; 10- KSU 13/2; 11- Tift-23; 12- 7042S; 13-411B; 14-576B; 15- PPMI720; 16-PPMI575; 17-PPMI597; 18- PPMI59; 19- PPMI69; 20-PPMI629; 21-DPR-7;22-TPR-8; 23-WGI-100; 24-PPMI683.



**Fig. 2.10:** Diversity of the genotypes based on alleles of SSRs markers.

**Table 2.2:** Annotation of selected contigs showing polymorphism.

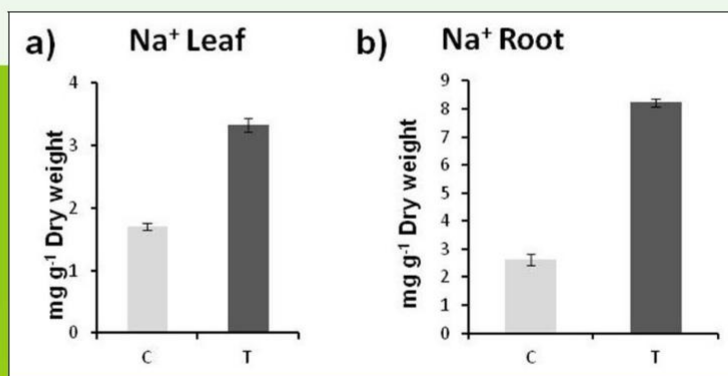
| Primer             | SSR/STS motif | No. of alleles | Product size (bp) | PIC value | Description                                                   | Functions                                                                            |
|--------------------|---------------|----------------|-------------------|-----------|---------------------------------------------------------------|--------------------------------------------------------------------------------------|
| SSRPgCL8220Contig1 | (CGC)5        | 3              | 195-225           | 0.4199    | cyanidin 3-O-rutinoside 5-O glucosyltransferase               | Having role in salt tolerance                                                        |
| SSRPgCL2813Contig2 | (GA)11        | 2              | 183-205           | 0.2078    | Zinc finger homeodomain 4 isoform                             | Role in abiotic stress tolerance via Na <sup>+</sup> /K <sup>+</sup> pump regulation |
| SSRPgCL4537Contig1 | (CT)7         | 3              | 280-308           | 0.5213    | No significant homology                                       | NA                                                                                   |
| SSRPgCL118Contig3  | (AGAT)6       | 4              | 196-268           | 0.6628    | putative CCA tRNA nucleotidyl transferase                     | Involved in small RNA metabolic pathways under various stress condition              |
| STSPgCL4894Contig1 | (A)11         | 2              | 402-424           | 0.3680    | probable LRR receptor-like serine threonine- kinase At1g34110 | Integral component of membrane, ATP binding; amino acid metabolic process            |

## Identification and cloning of genes for salinity tolerance in wheat

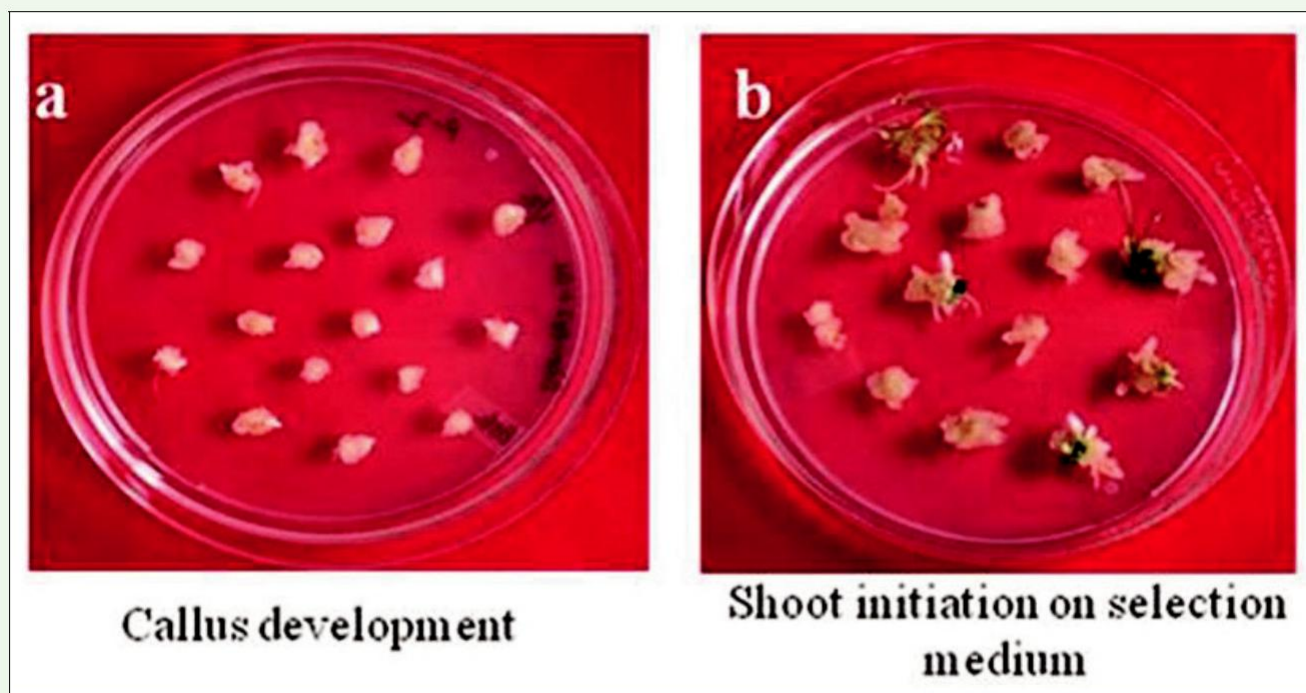
(Priyanka and Kanika)

Kharchia Local is a well known salt tolerant Indian land race. It is a very useful genetic resource which can be used to improve the salinity tolerance. This year salinity induced changes in various physiological parameters like chlorophyll, RWC, MSI and sodium and potassium content in Kharchia Local were evaluated. Nearly eight fold increase in the Na content of root was observed in Kharchia Local whereas this increase was three fold in the leaf tissue (Fig. 2.11). Expression analysis of *NHX1* gene in leaf and root tissues

following imposition of salinity stress was carried out. Transformation of wheat using *NHX1* gene is in progress (Fig. 2.12).



**Fig. 2.11:** Na<sup>+</sup> content in a) shoot and b) root tissue of Kharchia Local seedlings following stress treatment of 15 days.



**Fig. 2.12:** Development of transgenic wheat HD 2967 using mature embryo explants. a) Callus development from mature embryos b) Shoot initiation on selection medium.

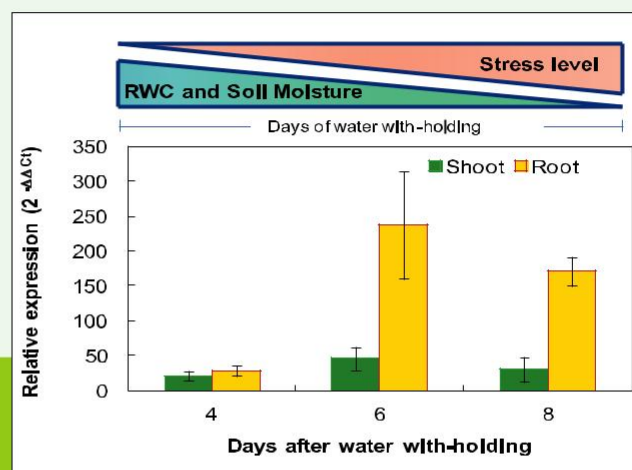
Soil salinity is a major constraint to wheat production around the world. Modern high yielding varieties are particularly sensitive to high salt stress. There are salt tolerant landraces and but there is limited information on genomic regions (QTLs) and genes responsible for their tolerance. QTLs are genomic regions containing genes controlling these quantitative traits, and QTL mapping is one of the most common approaches for the genetic dissection of quantitative traits, which provides the basis for map-based cloning of genes and marker-assisted selection (MAS) in crop breeding. In order to identify the (QTLs) development of mapping populations involving Kharchia local as one of the parent is going on.

### Cloning and characterization of *PM19* promoter from wheat

(Niladri Barman and Monika Dalal)

*PM19* is a highly conserved plasma membrane protein and gets induced by ABA, osmotic stress, cold, heat and salt stress. *PM19* gene was identified as one of the differentially expressed genes in root transcriptome study of two wheat genotypes Raj3765 (inductive root growth) and HD2329 (less or no inductive root growth) under drought stress. To

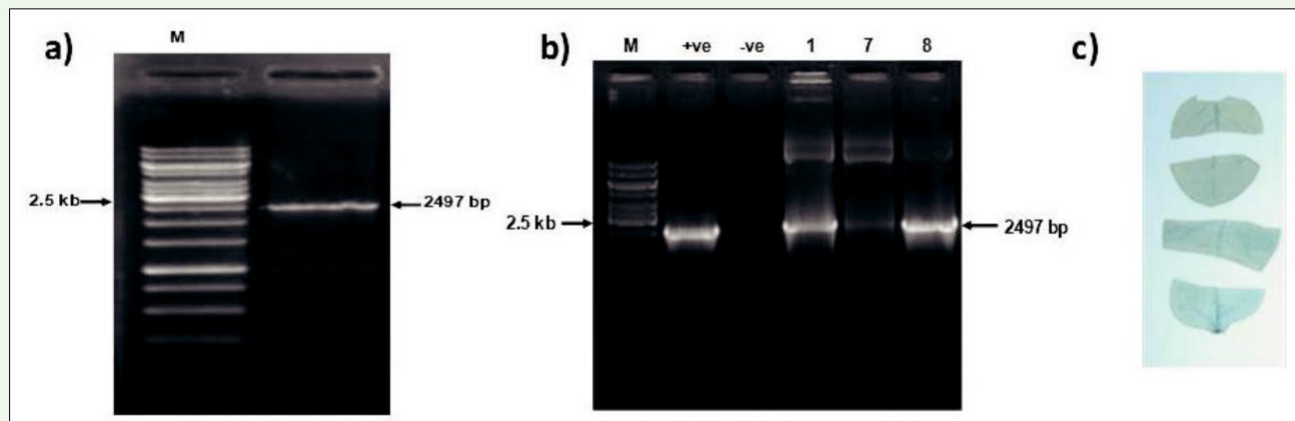
understand the role of *PM19* under drought, cloning and characterization of *PM19* promoter from wheat was carried out. The expression kinetics of *PM19* gene was analyzed in wheat genotype Raj3765 after 4, 6 and 8 days of moisture deficit stress by real-time RT-PCR analysis. *PM19* gene was significantly induced at 4 days after withholding irrigation (Soil moisture content 10.8 %) (Fig. 2.13).



**Fig. 2.13:** Relative expression analysis of *PM19* gene in shoot and root tissues of wheat genotypes under moisture deficit stress. Relative expression was calculated by  $2^{-\Delta\Delta C_t}$  method. Bars represent mean of three biological replicates.

The maximum expression was recorded after 6 days of stress with 45 fold and 236 fold increase in *PM19* expression in shoot and root, respectively. Though *PM19* expressed in both root and shoot under drought stress, yet level of induction was higher in

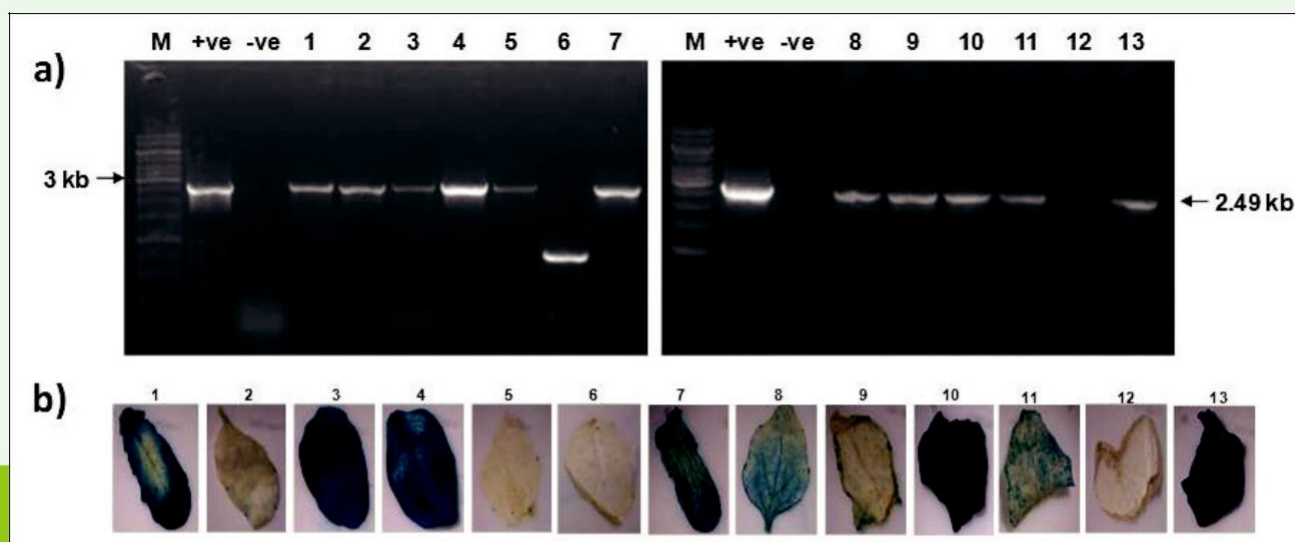
root than that of shoot. A 2.5 kb promoter fragment of *PM19* gene was isolated and cloned in to binary vector and functionality of *PM19* promoter (*PM19pRaj*) was analyzed by transient expression (Fig. 2.14).



**Fig. 2.14:** Isolation and cloning of *PM19* Promoter from wheat. a) Amplification of 2.5 kb promoter fragment by PCR; b) Confirmation of cloning in to binary vector by PCR; and c) Transient expression of *PM19* promoter in *T. benthamiana* leaves.

To elucidate its transcriptional regulation, *Arabidopsis* transgenics harbouring *PM19pRaj:GUS* were developed. The expression of *PM19pRaj* was confirmed in *Arabidopsis* T1 transgenic leaves under control condition by GUS histochemical assay (Fig. 2.15). The study revealed that *PM19* promoter is a

constitutively expressed gene with early and high level of upregulation under drought stress. The transgenic plants developed in the study will help in understanding transcriptional regulation and spatio-temporal expression of *PM19* gene in wheat.



**Fig. 2.15:** Analysis of *Arabidopsis* (T1) plants harbouring *PM19* promoter from wheat. a) Confirmation of transgenic plants by PCR, Lanes 1-13, PCR product from genomic DNA of T1 plants. +ve, plasmid DNA (*pCambia1300:PM19pRaj:GUS*) was used as template, -ve, wild type *Arabidopsis* genomic DNA was used as template, M: 1 Kb molecular weight marker, and b) Histochemical GUS staining of transgenic leaves from each event.

### Morphological and biochemical phenotyping of wheat genotypes and SNP genotyping

(Gayatri, Subodh Kumar Sinha, Nagendra Kumar Singh and Pranab Kumar Mandal)

Phenotyping of around 300 Indian and UK bread wheat genotypes has been completed at their seedling stage both under N-optimum as well as under N-stressed condition. Parameters measured are shoot and root length, Shoot and Root fresh and dry weight, chlorophyll and carotenoids, soluble protein content and assay of five candidate enzymes (nitrate reductase, glutamine synthetase, glutamate synthase, iso-citrate dehydrogenase, and Alanine amino transferase) related to nitrogen metabolism. We have also done SNP genotyping using 35K Breeder's array for around 360 Indian genotypes including the Indian genotypes used for seedling phenotyping under N-optimum and N-stress mentioned above and 208 RIL population of WL711 and C306. As the genotyping for the UK genotypes have already been conducted, these data would be used for GWAS at a later stage.

### Cloning and Expression analysis of AT1 (CM1, CM2, CM16 and 0.19) gene in different wheat genotypes (hexaploid, tetraploid and progenitors having A, B and D genome) at different developmental stages during grain filling

(Megha Kaushik, Subham Rai, Subodh Kumar Sinha and Pranab Kumar Mandal)

Wheat ATIs are considered causative agents that mediate intestinal inflammation by binding to toll-like receptor 4. During the reporting period seven ATIs (CM1, CM2, CM3, CM16, CMX2, 0.19, 0.53) are cloned from haploid bread wheat variety HD 2967. Sequences and their differences in protein structure are being analysed *in silico*. During the reporting period all seven ATIs (CM1, CM2, CM3, CM16, CMX2,

0.19, 0.53) expression profiling has been carried out 10 genotypes belonging to 2 genotypes each from AA, BB, DD, AABB and AABBDD genomes in the developing grains i.e. at 2 weeks after anthesis (WAA), 3WAA, 4WAA and 5WAA. It has been observed that the expression of most of the ATIs was several fold higher in hexaploid bread wheat as well as tetraploid durum wheat in comparison to diploid progenitors. Since these ATIs are there to protect protein and starch in the wheat grain from hydrolysis, we are estimating starch and protein in all the five genome species.

### Assessing nitrogen utilization efficiency in wheat - a different approach

(Alka Bharati, Subodh Kumar Sinha and Pranab Kumar Mandal)

Nitrogen utilization efficiency is directly related to yield because it depends on N assimilation, remobilization and finally efficient utilization of assimilated nitrogen for starch biosynthesis in the grain. Therefore, two genotypes have been selected, which are similar in uptake and biomass production during N limitation but shows contrasting response with respect to seed volume and seed weight reduction. It is expected that these genotypes will be contrasting with respect to starch biosynthesis or grain filling during nitrogen limitation rather contrasting for nitrogen uptake efficiency. The selected contrasting genotypes, HB208 and Chotilerma, were grown under controlled and N-stressed condition. Grain sample, collected during active filling stages last year, were subjected to RNA isolation and sequencing during this reporting period (Table 2.3). Carbon partitioning through <sup>14</sup>C-Sucrose feeding is also carried out during this reporting period and grain samples are collected after the sucrose feeding for further analysis. So far we have measured the radioactive count in the grain sample.

**Table 2.3:** Summary of 'Raw' reads generated trimming Alignment statistics.

| Wheat genotypes with their biological replicates | Total raw reads generated | Processed reads | Mapped reads |
|--------------------------------------------------|---------------------------|-----------------|--------------|
| HB_NC_1                                          | 117673438                 | 109960116       | 94771388     |
| HB_NC_2                                          | 128651446                 | 120885316       | 102082156    |
| HB_NC_3                                          | 145533890                 | 137137862       | 118946844    |
| CL_NC_1                                          | 158861718                 | 147780818       | 128371890    |
| CL_NC_2                                          | 136358676                 | 128141546       | 100885404    |

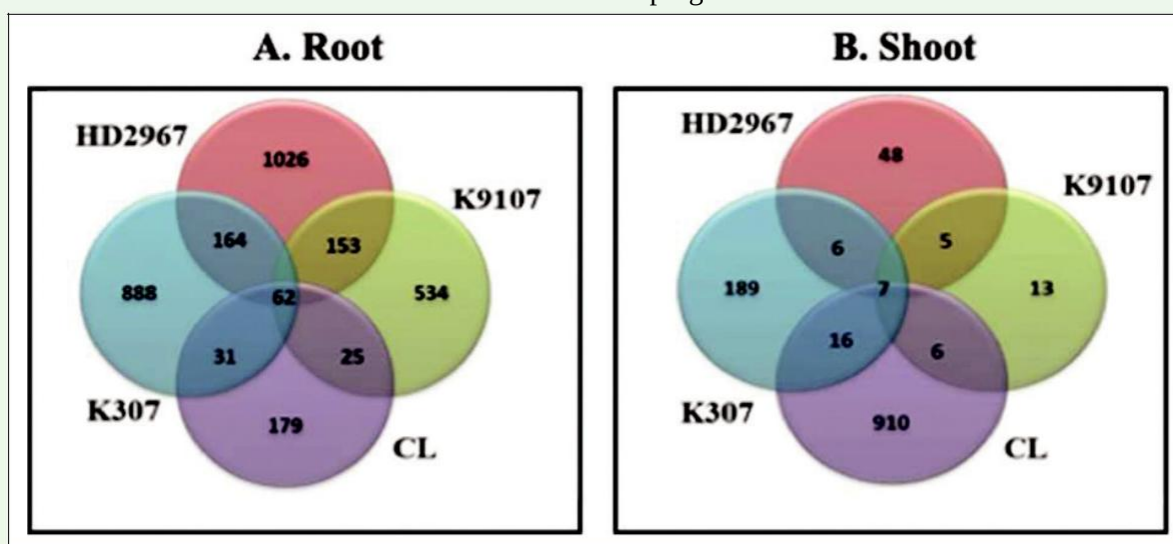
| Wheat genotypes with their biological replicates | Total raw reads generated | Processed reads | Mapped reads |
|--------------------------------------------------|---------------------------|-----------------|--------------|
| CL_NC_3                                          | 133738346                 | 125535472       | 110847490    |
| HB_NS_1                                          | 149985848                 | 141903752       | 123826562    |
| HB_NS_2                                          | 134722578                 | 126807078       | 97613512     |
| HB_NS_3                                          | 155650746                 | 147957844       | 131407936    |
| CL_NS_1                                          | 130559680                 | 123704866       | 107960456    |
| CL_NS_2                                          | 163556940                 | 154582526       | 134030140    |
| CL_NS_3                                          | 148410592                 | 140233908       | 124012938    |

### Transcriptome analysis and identification of nitrogen starvation responsive genes in wheat

(Subodh K Sinha, Akanksha Tyagi and Pranab Kumar Mandal)

Differential gene expression analysis was carried out to fetch out transcript IDs related to uptake and utilization of nitrogen (N) responsive genes in

different wheat genotypes (HD2967, K9107, K307 and Chotilerma under N stress conditions (Fig. 2.16). Gene ontology and enrichment analysis was done to address biological process, cellular component and molecular function of genes in each genotype. Validation of microarray data by quantitative real-time PCR. Further studies and analysis of the work is in progress



**Fig. 2.16:** Differentially expressed overlapped and unique genes revealed through Affymetrix 61K GeneChip wheat genome array analysis under N- optimal and starved conditions in four diverse wheat genotypes.

### Co-expression studies of *NAR2.1* and *NAR2.2* genes with high affinity nitrate transporter gene (*NRT2.1*) in root tissue of wheat (*Triticum aestivum* L.)

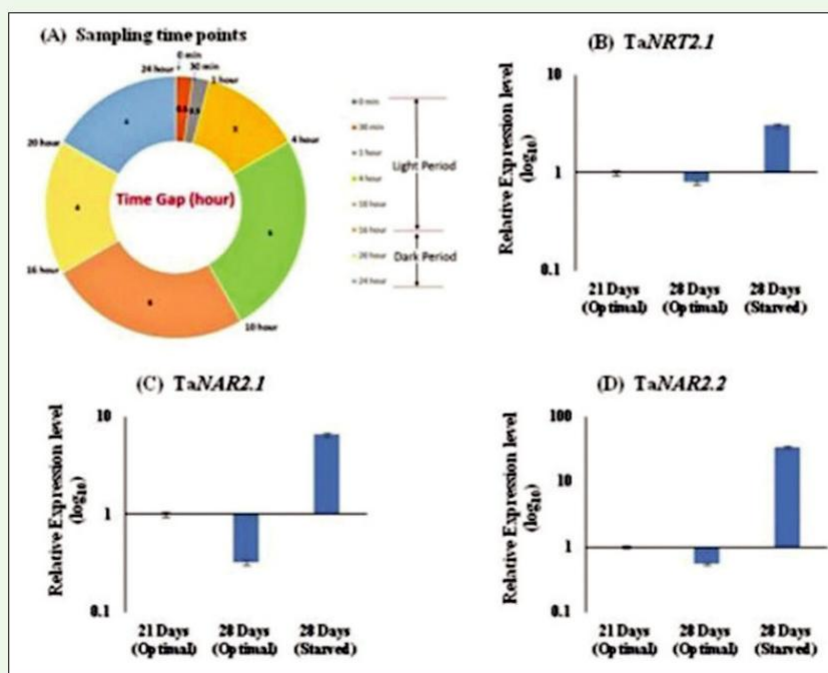
(Akash Paul, Pranab Kumar Mandal and Subodh Kumar Sinha)

Nitrate uptake is one of the important component traits that can be tapped for improvement of nitrogen use efficiency (NUE) in wheat. In case of wheat, NO<sub>3</sub><sup>-</sup> is the most preferred form of N which is transported through one of the most important nitrate transporter i.e. *NRT2.1*, among many under

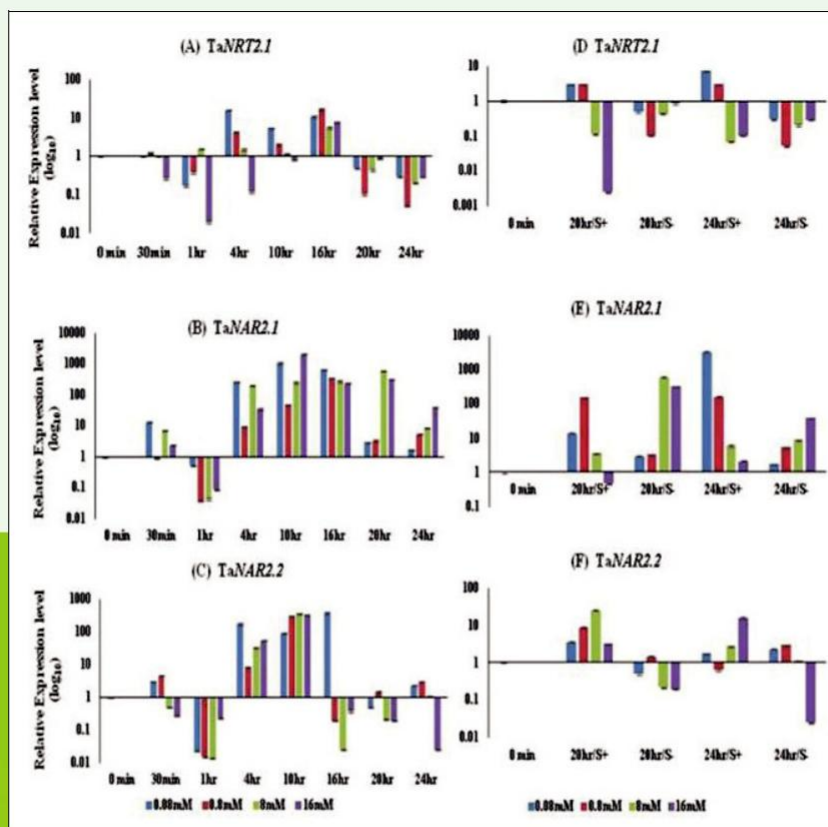
limited external N-availability, however it invariably requires additional accessory protein (*NAR2*) for its function which is not known in wheat. In order to identify the partner proteins (*NRT2.1*-*NAR2s*), expression studies of *NRT2.1* along with two *NAR2s* i.e. *NAR2.1* and *NAR2.2* genes have been studied in root tissues of 21 days old wheat seedling starved for a week and then subjected to different concentrations of nitrate treatments for different time intervals (Fig. 2.17). The transcript expression of all three genes (*NRT2.1*, *NAR2.1* and *NAR2.2*) gets up-regulated under nitrate starvation in root tissues as compared to optimally grown seedlings. The *NRT2.1* has been

found to be invariably up-regulated under lower nitrate concentration treatments. Similarly sucrose has been found to induce the expression of *NRT2.1* under lower nitrate concentration even during dark

period as compared to sucrose minus treatment, whereas it seems there has no such relation exist with *NAR2.1/2.2* expression (Fig. 2.18).



**Fig. 2.17:** Quantitative gene expression profile of *TaNRT2.1* (B), *TaNAR2.1* (C) and *TaNAR2.2* (D) genes in root tissue of 21 days (optimal), 28 days (optimal) and 28 days ( $\text{NO}_3^-$ -starved) old wheat seedlings. The time points of different treatments are shown in (A).



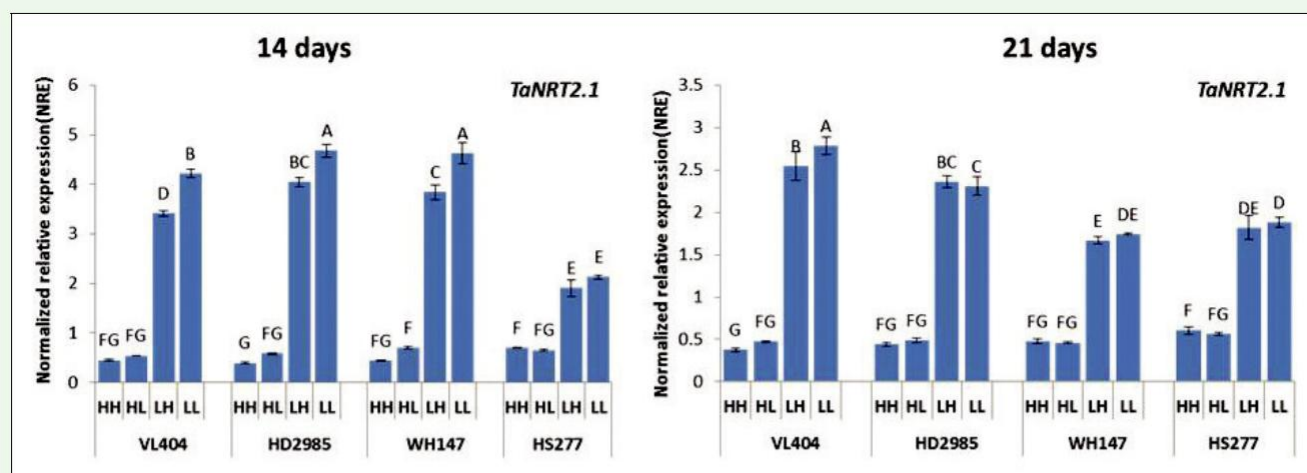
**Fig. 2.18:** Quantitative gene expression profile of *TaNRT2.1*, *TaNAR2.1* and *TaNAR2.2* genes in root tissue (28 days old) at different time points after 1 week N-starvation (A, B and C) and in the presence and absence of sucrose during dark period (20 and 24 hr).

## Expression of nitrate transporter (*TaNRT2.1* and *TaNRT1.1*) genes in root tissue under different external N conditions

(Amresh Kumar, Pranab Kumar Mandal and Subodh Kumar Sinha)

Previous year work involving measurement of  $\delta_{15}\text{N}$  of several diverse wheat genotypes grown under different external N concentration for two growth stages (14 and 21 days) and which were then subjected in N-uptake solution containing optimum and starved N concentration was reported. In order to understand expression profile of nitrate transporters genes in root tissue subjected to HH, HL, LH and LL nitrate treatments condition, the expression pattern of one HATS i.e. *TaNRT2.1*, and LATS i.e. *NRT1.1* (*TaNPF6.1*, *TaNPF6.2*, *TaNPF6.3* and *TaNPF6.4*) and *TaNRT1.5* (*TaNPF7.1* and *TaNPF7.2*) respectively was carried out in root tissue of 14 and 21 days old seedlings. Invariably the high affinity nitrate transporter gene i.e. *TaNRT2.1*, up-regulated under low exposure of nitrate conditions viz. LH and

LL, however more predominantly under LL condition than HH and HL conditions in all genotypes (Fig. 2.19). HD2985 and VL404 showed highest expression at 14 days and 21 days stage respectively whereas HS277 showed lowest expression at both the stages at LH and LL conditions. Among *NRT1.1* orthologues, *TaNPF6.3* showed typical low affinity nitrate transporter gene type expression pattern at 14 days stage at HH and HL conditions wherein VL404 and HS277 had highest and lowest expression respectively, but interestingly at 21 days the expression pattern was found to be reversed for each genotypes (Fig. 2.20), whereas *TaNPF6.4* showed this type of expression pattern at 21 days stage (Fig. 2.21) at which WH147 followed by HD2985 showed highest expression in HH and HL conditions respectively. As far as *AtNRT1.5* orthologue of wheat is concerned, *TaNPF7.2* expresses more in HH and HL conditions at both stages in all genotypes, whereas expression of *TaNPF7.1* was not observed to be specific to N-conditions, genotypes and growth stages (Fig. 2.22).



**Fig. 2.19:** Quantitative gene expression profile of *TaNRT2.1* at 14 and 21 days of growth stage in root tissue in different external N-conditions (HH: high-high; HL: high-low; LH: low-high and LL: low-low). Genotypes analysed are mentioned at the bottom.

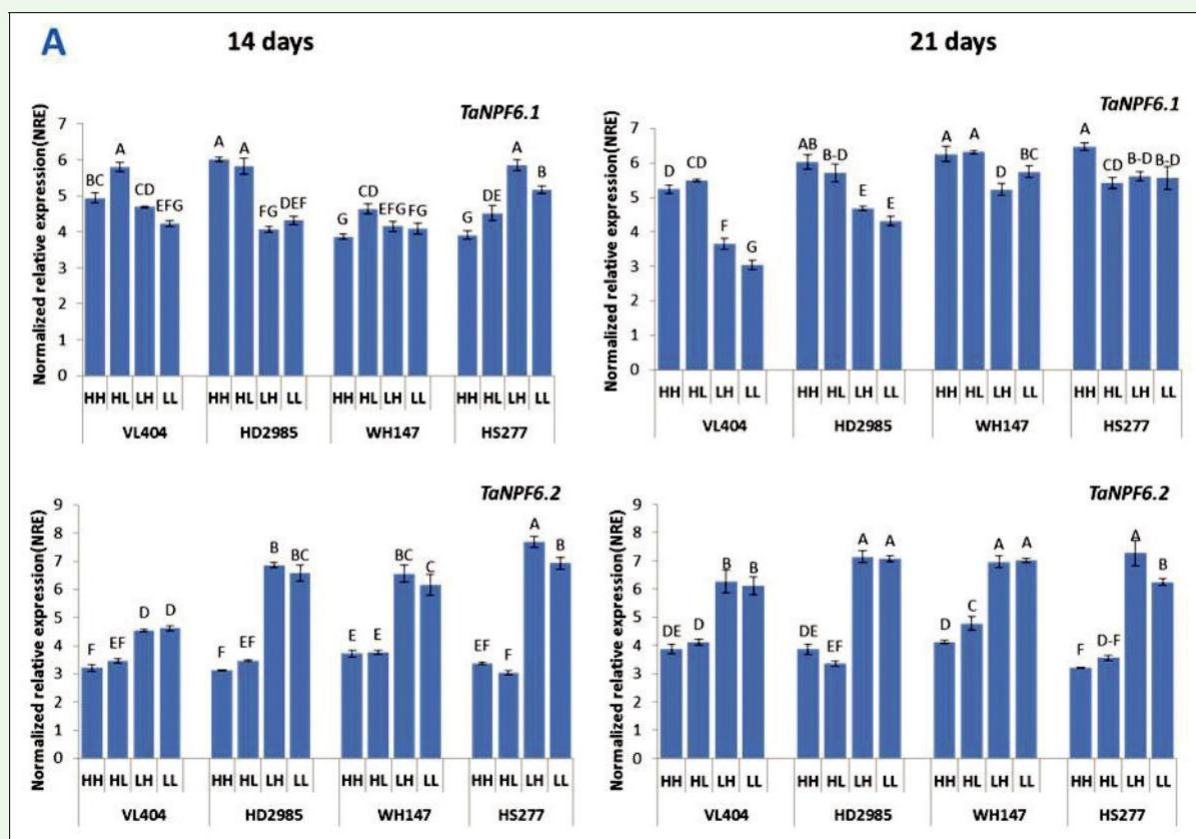


Fig. 2.20: Quantitative gene expression profile of *TaNP6.1* and *TaNP6.2* at 14 and 21 days of growth stage in root tissue in different external N-conditions (HH: high-high; HL: high-low; LH: low-high and LL: low-low). Genotypes analysed are mentioned at the bottom.

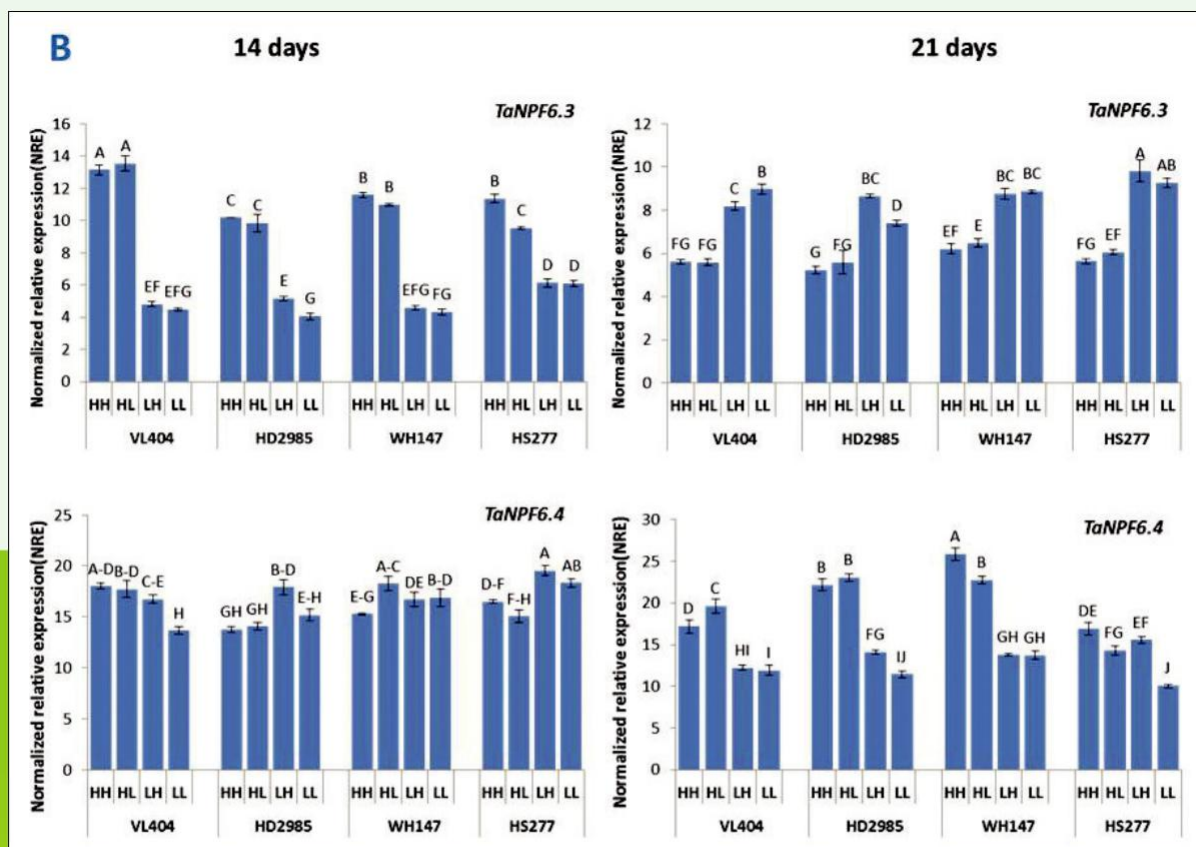


Fig. 2.21: Quantitative gene expression profile of *TaNP6.3* and *TaNP6.4* at 14 and 21 days of growth stage in root tissue in different external N-conditions (HH: high-high; HL: high-low; LH: low-high and LL: low-low). Genotypes analysed are mentioned at the bottom.

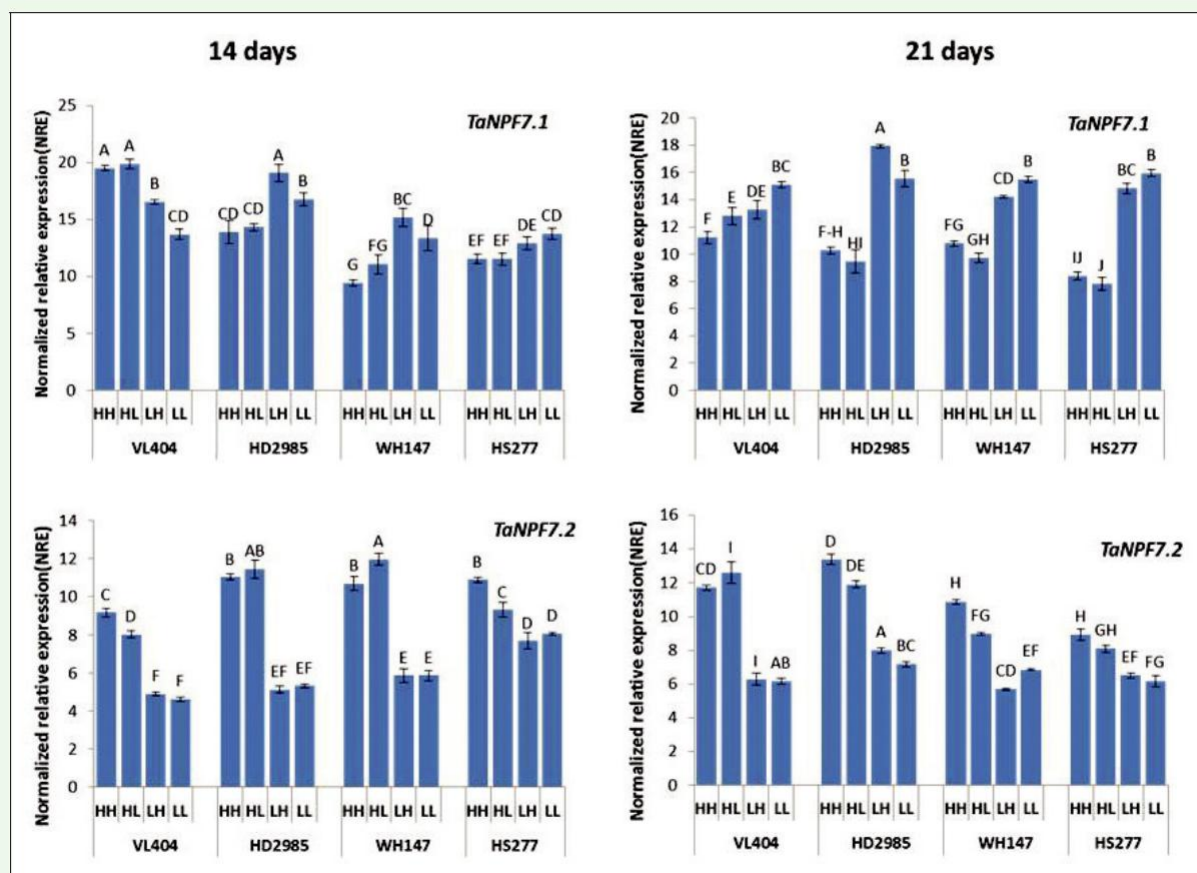


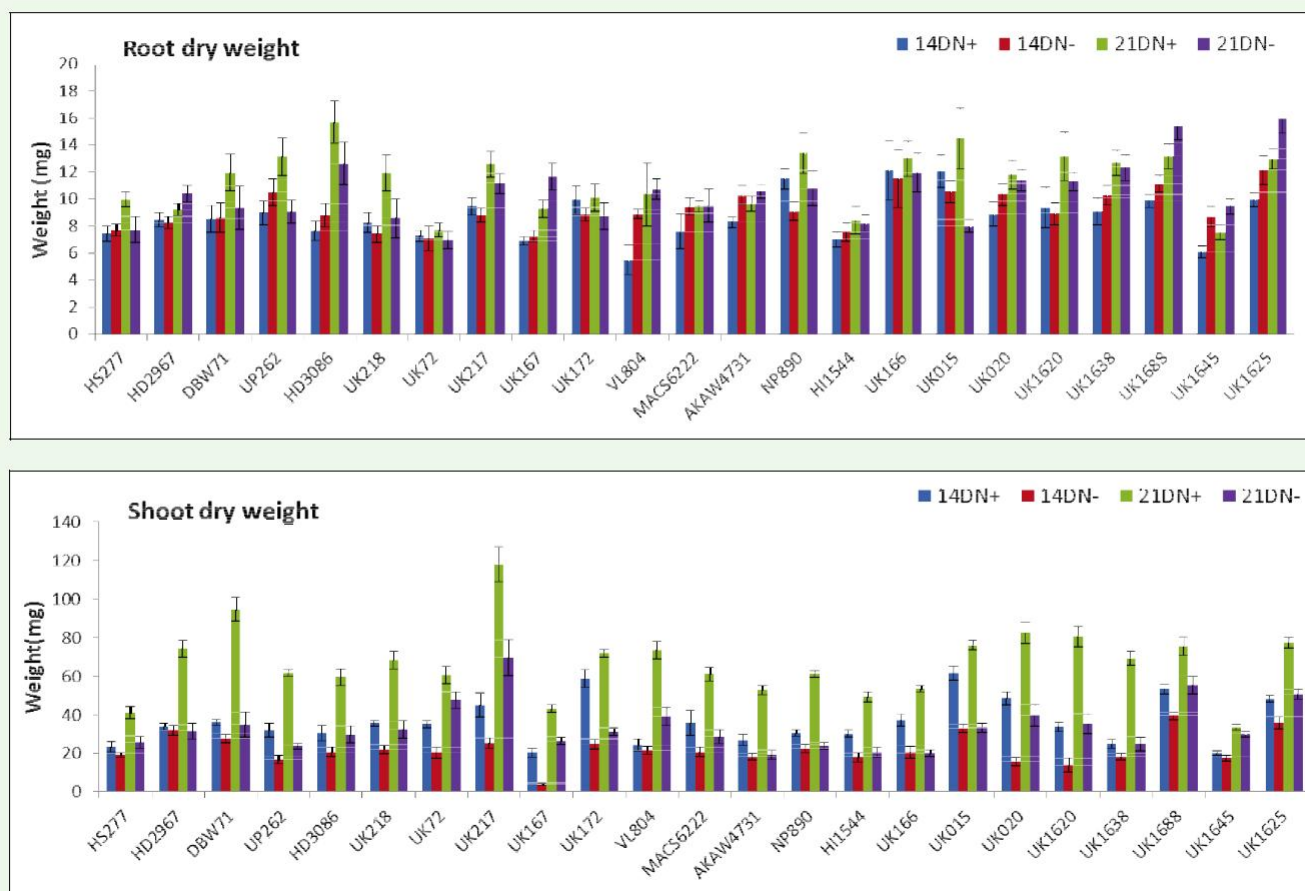
Fig. 2.22: Quantitative gene expression profile of *TaNPF7.1* and *TaNPF7.2* at 14 and 21 days of growth stage in root tissue in different external N-conditions (HH: high-high; HL: high-low; LH: low-high and LL: low-low). Genotypes analysed are mentioned at the bottom.

## Morphological and physiological characteri-zation of Indian and UK wheat genotypes for nitrate-uptake phenotype

(Amresh Kumar, Nagendra Kumar Singh, Pranab Kumar Mandal and Subodh Kumar Sinha)

Under Indo-UK Centre for Improvement of Nitrogen Use Efficiency in Wheat (INEW) project, a few (23) high and low yielding wheat genotypes from both the countries were selected for their characterization in terms of their nitrate-uptake capacity vis-à-vis root system architecture under optimum and starved -N conditions at two growth stages (14 and 21 days). In order to understand the biomass allocation in root and shoot tissues under these conditions, we measured dry weight of root and shoot of these wheat seedlings (Fig. 2.23). N-stress has been found to cause reduction in shoot biomass at both stages predominantly at later stage (21 days) than at beginning stage (14 days), whereas few genotypes have been observed to accumulate more biomass in their roots under N-stress condition at both the stages presumably demonstrating their increased N-foraging capacity of roots (Fig. 2.24). As

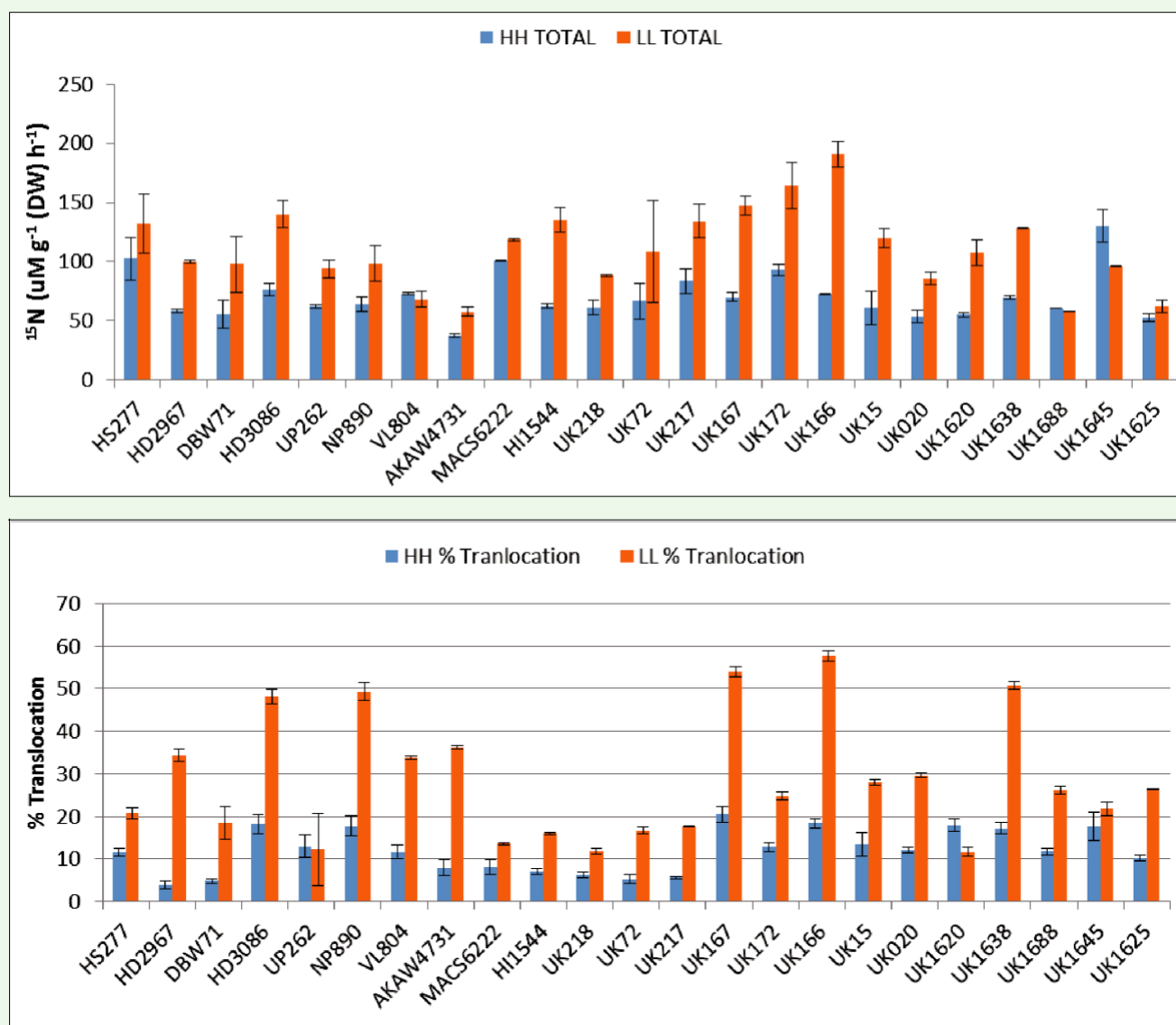
reported earlier five root system architecture traits (TRS, MRP, LRS, FOLRN and SOLRN) were measured for each of the genotypes under these conditions in order to understand their N-deprivation responsive modulation in root architecture. TRS in general has been found to be increased under N-stress condition depicting N-foraging ability of the genotypes. Other root traits were also observed to be varied under these N-conditions genotypically. In order to precisely phenotype the nitrate-uptake and translocation to above ground part traits of these genotypes, <sup>15</sup>N-labelled N-source was used to measure nitrate-influx using isotope ratio mass spectrometry in the wheat seedlings grown for 21 days both under optimal as well as starved-N conditions and then subjected to uptake solution containing same concentration of nitrate (10 atom% <sup>15</sup>N) for 10 minutes. Both <sup>15</sup>N-influx and its % -translocation to shoot were interestingly found to be more under lower external-N concentration than optimal condition, demonstrating the genetic variation in capacity of High Affinity Nitrate Transport System (HATS) and above ground translocation system of these genotypes (Fig. 2.25).



**Fig. 2.23:** Root and shoot dry weight of different Indian and UK genotypes under optimal (N+) and N-starved (N-) conditions at 14 and 21 days growth stages.



**Fig. 2.24:** 2D images showing root system architecture of few representative genotypes (Indian and UK) under optimal (N+) and N-starved (N-) conditions at 14 and 21 days growth stages.



**Fig. 2.25:**  $^{15}\text{N}$ -influx ( $\mu\text{M g}^{-1} (\text{DM}) \text{ h}^{-1}$ ) and % translocation of  $^{15}\text{N-NO}_3^-$  in shoot system at 21 days growth stage in HH and LL external N-concentration in different genotypes (Indian and UK).

## Leaf spot disease resistance management in wheat

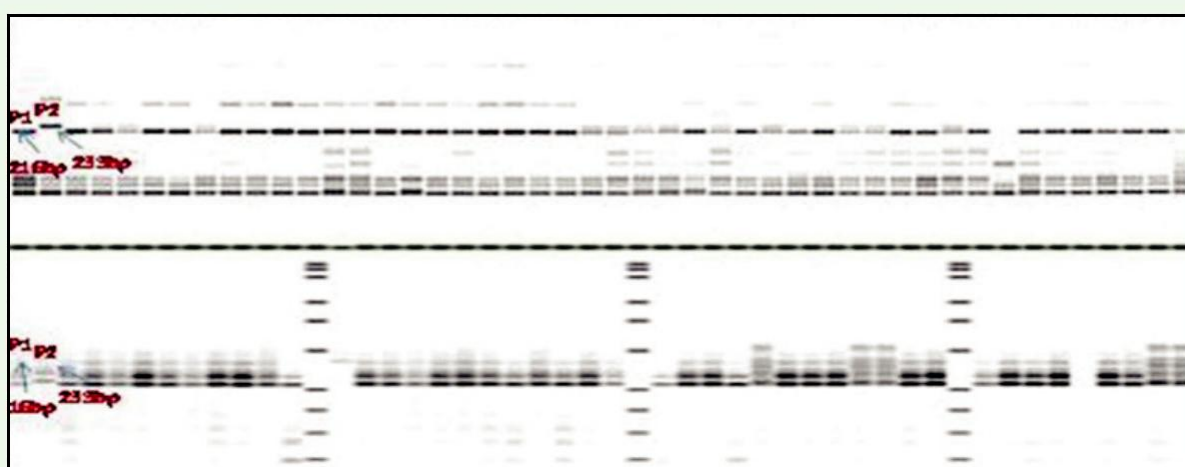
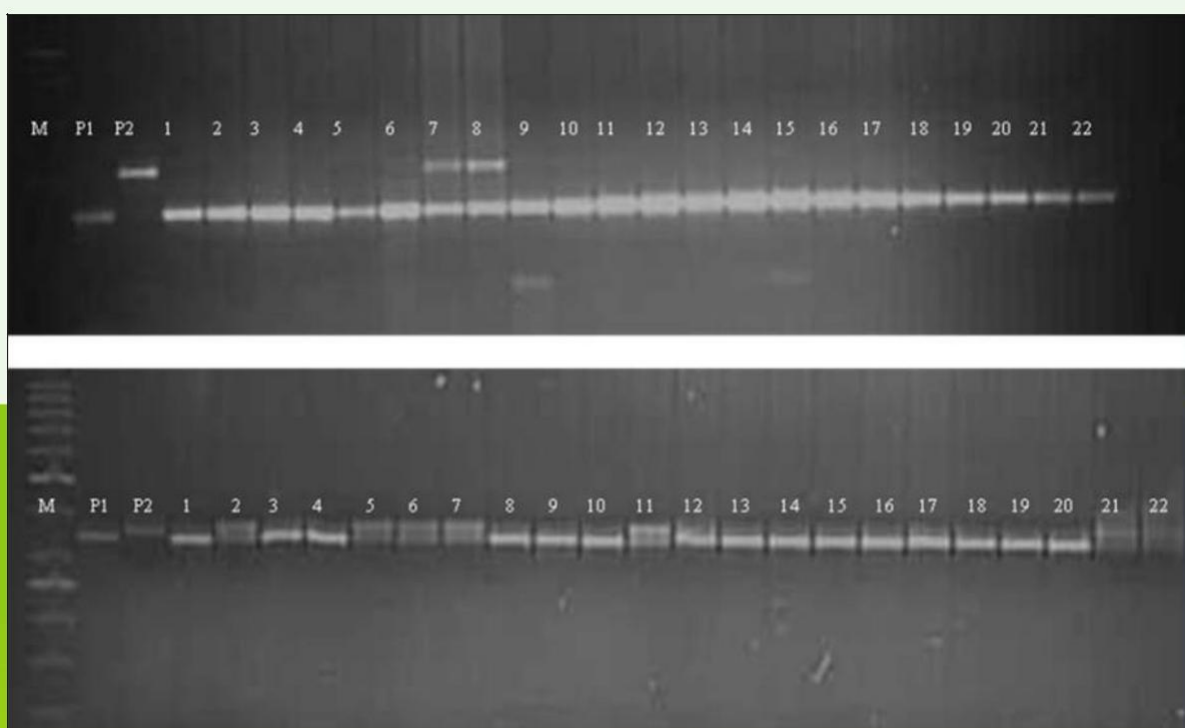
(Sanjay K Singh, Jasdeep C Padaria, PK Singh and Vaibhav K Singh)

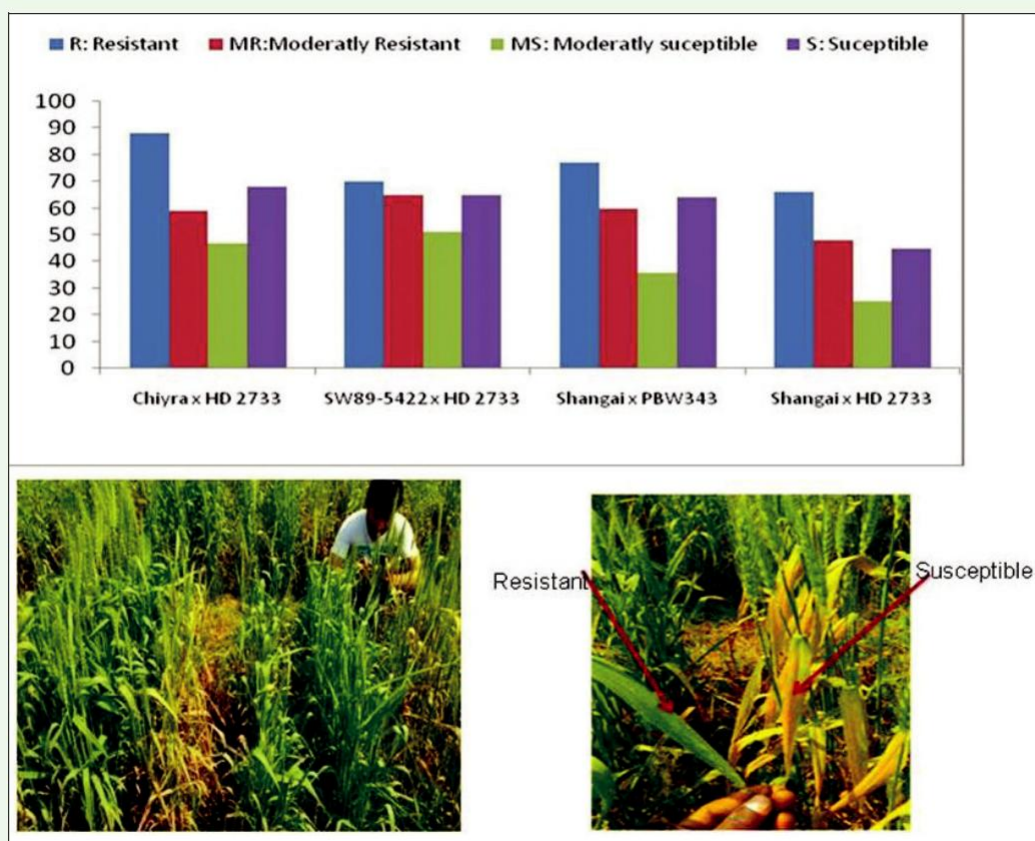
QTL (Qsb.bhu-2A-Xgwm425) was transferred from SW89-5422 while in the other; two resistant QTLs (Qsb.bhu-2A and Qsb.bhu-5B) from Chirya3 were transferred (Table 2.4). Foreground and whole genome background selection were done in BC3F1 generation. The foreground selection for spot blotch resistance QTLs was exercised using known SSR markers while background selection was performed

using 93 simple sequence repeat (SSRs) loci spread over the whole genome (Fig. 2.26, Fig. 2.27). In both cases the progenies selected showed up to 94.29% genetic similarity with the recipient parent. The derived lines exhibited significantly higher level of resistance against spot blotch under field condition (Fig. 2.28). 601 F1's (straight crosses including back crosses, three-way and double crosses) 407 F2, 266 F3, 376 F4, 03 F5, 182 F6 and 270 F7/Advance generation material was evaluated for initial vigour, crop establishment, physiological traits, disease resistance and plant ideotype (Fig. 2.29).

**Table 2.4:** Details of population and QTLs being introgressed.

| Populations         | QTLs       | Segregation in BC <sub>2</sub> F <sub>1</sub> |            |                |
|---------------------|------------|-----------------------------------------------|------------|----------------|
|                     |            | Heterozygous                                  | Homozygous | $\chi^2$ (1:1) |
| Chiya x HD 2733     | Qsb.bhu-5B | 195                                           | 233        | 3.36           |
|                     | Qsb.bhu-2A | 238                                           | 190        | 5.38*          |
| SW89-5422 x HD 2733 | Qsb.bhu-5B | 230                                           | 195        | 1.95           |
|                     | Qsb.bhu-2A | 160                                           | 140        | 4.00*          |
| Shangai x PBW343    | Qsb.bhu-5B | 181                                           | 161        | 1.17           |
|                     | Qsb.bhu-2A | 193                                           | 149        | 5.66*          |
| Shangai x HD 2733   | Qsb.bhu-5B | 230                                           | 195        | 1.95           |
|                     | Qsb.bhu-2A | 245                                           | 180        | 6.74**         |


**Fig 2.26:** Foreground selection in BC<sub>2</sub>F<sub>1</sub> progenies with QTL linked: Xgwm425 linked with (Qsb.bhu-2A) and another SSR marker (b) Xgwm67 (Qsb.bhu-5B) associated with spot blotch resistance QTL. P1= recurrent parent, P2= donor parent, M= 100bp ladder).

**Fig. 2.27:** Background selection in BC<sub>2</sub>F<sub>1</sub> progenies (P1= recurrent parent, P2= donor parent, M= 100 bp ladder).



**Fig. 2.28:** Screening of BC<sub>2</sub>F<sub>1</sub> generation plant of the four crosses of wheat varieties through artificial inoculation show the reaction of resistance and susceptible against *Bipolaris* leaf spot.



**Fig. 2.29:** Evaluation of advance generation material for initial vigour, crop establishment, physiological traits, disease resistance and plant ideotype.

Pipe line of entries Contributed in AICW&BIP Trials (NIVT): 2017-18

| Entry   | Trial  | Zone      | Agro. Condition |
|---------|--------|-----------|-----------------|
| HD 3289 | NIVT3A | All Zones | Irrigated       |
| HD3294  | NIVT5A | All Zones | RI TS           |
| HD3295  | NIVT5A | All Zones | RI TS           |
| HD3296  | NIVT5B | All Zones | RI TS           |

## Varieties released

Wheat Variety HD 3237 (Timely sown Restricted Irrigated condition (NWPZ) released and notified by CVRC 2018.

# Improvement of Stress Tolerance in Chickpea

## Genome-wide discovery and characterization of miRNAs and target genes under water-deficit stress conditions in chickpea

(Lal Bahadur Singh, Deshika Kohli and PK Jain)

Four small RNA libraries, from mature chickpea plants [drought tolerant (ICC4958) and drought-susceptible (ICC1882)] at flowering stage, under control and drought stress conditions were constructed and sequenced using Illumina Hiseq 2500 platform. A total of 146.7 million reads were generated. After discarding the low quality reads and further filtering with Rfam database putative unique small RNA reads were further used for bioinformatics

analysis. The sequence analyses revealed the presence of 658 miRNAs belonging to 37 conserved families and 186 novel miRNAs with MFEI  $\geq 0.85$  by mapping on chickpea genome. Analysis revealed 335 differentially expressed miRNAs under drought stress in both tolerant and susceptible genotypes. For validation of miRNAs obtained from high-throughput deep sequencing, poly (A)-tailed based quantitative real-time PCR (qRT-PCR) of miRNAs was performed. Seventeen miRNAs, including eleven conserved and six novel miRNAs were validated in root tissue of both genotypes (Fig. 3.1). Seven conserved and three novel miRNAs were found to be up-regulated under drought stress.

**Fig. 3.1:** Expression analysis of selected conserved miRNAs in root tissue of contrasting chickpea genotypes.

## Identification and validation of differentially expressed genes associated with seed protein content in chickpea

(Yashwant Yadav, Deshika Kohli and PK Jain)

Transcriptome analysis was performed in genotypes contrasting with respect to seed protein content in replicates. After extensive characterization by collaborators, the genotypes selected for transcriptome analysis were ICC 8397 (high protein content) and FG 212 for (low protein content). In all, four cDNA libraries were generated, in replicates, using genotypes contrasting for seed

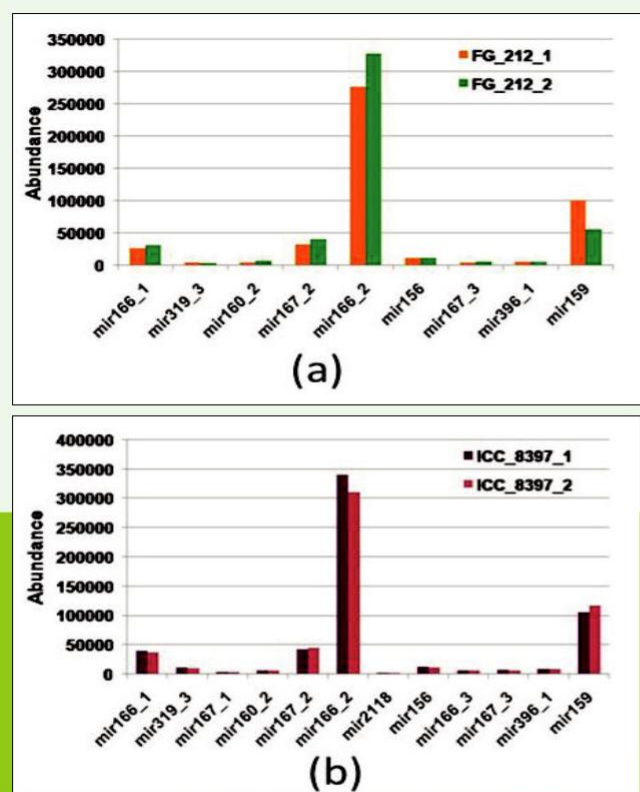
protein content. 88Mnreads with 150 bp read length were generated. Overall, 98.06% of the high-quality reads were mapped to the chickpea (kabuli) genome (Table 3.1). About 400 significantly differentially expressed genes were found in ICC 8397 and FG 212 genotypes. Six genes were short-listed based on their role in seed protein content and digital differential expression pattern for the real time PCR based expression analysis in both the contrasting cultivars (High and low protein content). The results obtained were consistent with the *in silico* expression profiles except UPBE3 gene, which encodes for ubiquitin protein ligase.

**Table 3.1: Generation of reads after RNA seq analysis in ICC8397 (high seed protein content) and FG 212 (Low seed protein content) genotypes**

| Samples    | Total reads | Reads after trimming | % reads mapped to genome |
|------------|-------------|----------------------|--------------------------|
| ICC 8397_1 | 21,121,315  | 21,121,315           | 99.31                    |
| ICC 8397_2 | 20,638,636  | 20,638,636           | 98.25                    |
| FG 212_1   | 25,340,469  | 25,340,469           | 98.73                    |
| FG 212_2   | 21,071,199  | 21,071,199           | 95.97                    |

#### Identification and validation of differentially expressed microRNAs associated with seed protein content

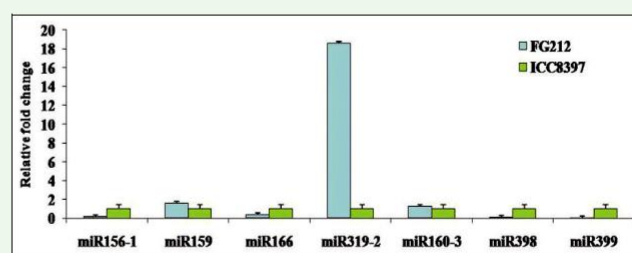
Two small RNA libraries using seeds sample from chickpea cultivars (FG 212 and ICC 8397) were constructed and sequenced in replicates using Illumina NextSeq small platform. 23 and 26 conserved miRNAs were identified in FG212 and ICC8397, respectively. The novel miRNAs identified in both the cultivars viz. FG212 and ICC8397 were 258 and 512, respectively. The most abundant miRNA was miR166 in both the cultivars (Fig. 3.2a and b). The log2 fold expression analysis revealed differential expression of known and conserved miRNAs between these cultivars.



**Fig. 3.2: Conserved miRNA families' abundance in (a) FG 212 and (b) ICC 8397.**

#### Validation of conserved miRNAs associated with seed protein content

Quantative real-time PCR based analysis of conserved miRNAs resulted in validation of seven miRNAs. Out of these 7 validated miRNAs, the real time based expression analysis of three miRNAs namely; miR160, miR398 and miR399 were in accordance with that obtained through digital expression analysis. However, other miRNAs namely, miR156, miR159, miR166 and miR319 exhibited opposite expression pattern (Fig. 3.3).



**Fig. 3.3: Real-time PCR based validation of conserved miRNAs in two chickpea cultivars with respect to seed protein content.**

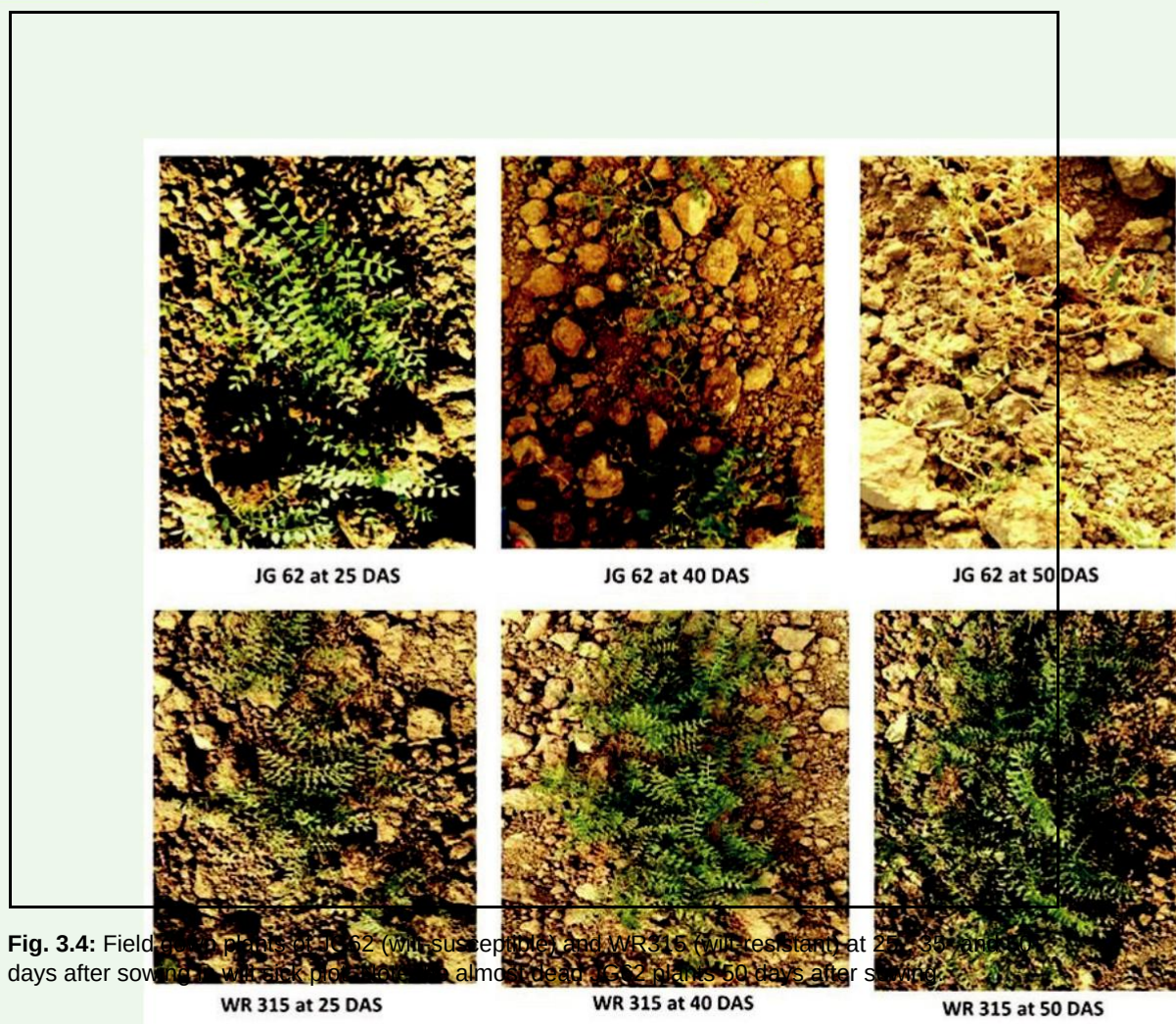
#### Quantification of *Fusarium oxysporum* f.sp. *ciceris* load in wilt-resistant and -susceptible genotypes of chickpea using intergenic spacer region-based marker

(Parichita Priyadarshini and PK Jain)

*Fusarium* wilt in chickpea is an important disease in chickpea and causes severe yield loss under favorable conditions. Quantitative detection of pathogen load is a crucial aspect in development of disease management strategies and breeding programs. The intergenic spacer region (IGS) is a stretch of non-coding DNA that separates tandemly repetitive ribosomal DNA units. Variation in length and nucleotide sequence of IGS among different species including *Fusarium* sp. has led to development of IGS based molecular markers for taxonomic studies. In the present study, we have utilized a IGS based marker to detect and quantify pathogen DNA by conventional PCR and real time PCR. The early wilting cultivar, JG 62 showed significant accumulation of *Fusarium oxysporum* f.sp. *ciceris* (Foc) DNA at 25 DAS (days after sowing) and 35 DAS. JG 62 plants were totally dead after 50 DAS whereas wilt resistant genotype, WR 315 showed no symptoms (Fig. 3.4). It can be inferred that there is a delay between pathogen multiplication inside host

and manifestation of symptoms. Thus present study implies that pathogen specific markers can be used

to confirm whether successful infection of host has occurred or not in time course experiments.



**Fig. 3.4:** Field growth of plants of JG 62 (wild susceptible) and WR 315 (wild resistant) at 25, 40 and 50 days after sowing. WR 315 plants show almost dead JG 62 plants 50 days after sowing.

### Identification and molecular cloning of a salt inducible MATE transporter gene from chickpea

(Pooja Kaur, Anjali Sharma, Vinod Kumar, Heena Saxena, Bharadwaj C, Jain PK and Nimmy MS)

Multidrug and toxic compound extrusion (MATE) proteins are the most recently identified family of multidrug transporters. In plants, this family is remarkably large compared to the human and bacteria counterpart, highlighting the importance of MATE proteins in plants. Salt responsive genes belonging to MATE family reportedly play a significant role imparting salt tolerance in plants. We have carried out a genome-wide survey of MATE family genes in the pulse crop chickpea and identified around 50 members in the family and according to the phylogenetic grouping selected a subgroup consisting of six genes for further study.

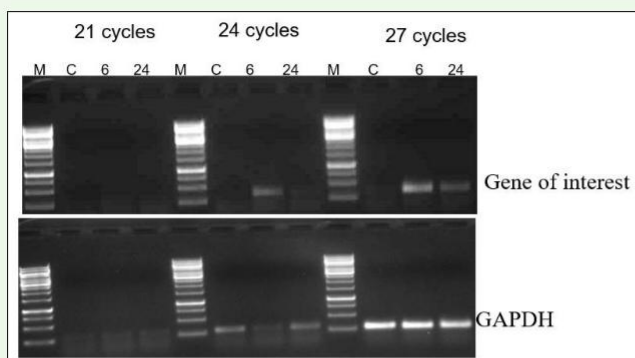
Initially a semi qRT PCR was performed for the gene selected for the study. Semi quantitative RT PCR method is based on the use of an internal control, which is a housekeeping gene (GAPDH) and hence on the assumption that the experimental conditions do not alter the expression levels of the selected control RNA (cDNA). The Semiquantitative RT PCR study shows an up-regulation of selected gene in stressed sample in contrast to wild type control sample, thereby, suggesting the salt responsive nature of gene (Table 3.2; Fig. 3.5 & 3.6). Further validation of the upregulation of the gene under salt stress was performed by a Real time quantitative RT PCR (Fig. 3.7). The gene showed upregulation of 7.5 fold at 6hpt which reduced to 1.8 fold at 24hpt which further confirmed the salt responsive nature of gene. Hence the differential gene expression studies revealed the role of *CarMATE6* as a candidate gene for salt tolerance (Fig. 3.5-3.7).

**Table 3.2:** Chickpea genotypes used for study on expression of MATE transporter gene

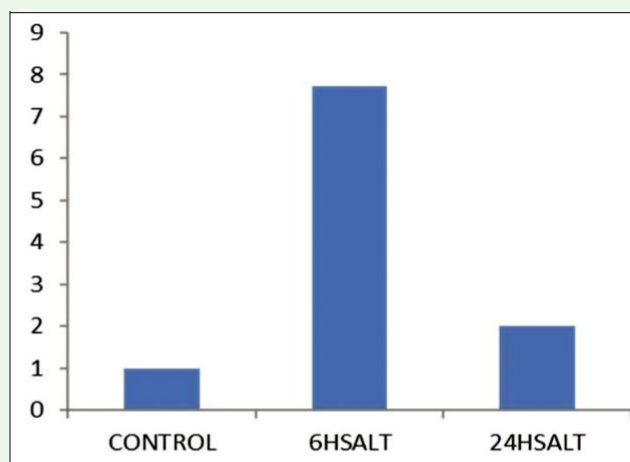
| Genotype | Response to salinity | Type   |
|----------|----------------------|--------|
| JG 11    | Tolerant             | Desi   |
| JG 62    | Tolerant             | Desi   |
| ICCV 2   | Susceptible          | Kabuli |
| DCP-92-3 | Susceptible          | Desi   |



**Fig. 3.5:** Seeds of both salt tolerant and susceptible chickpea genotypes were germinated in autoclaved bottles and shifted to solrite pots. 14-day old seedlings growing on solrite irrigated with Hoagland's solution were subjected to 80 mM NaCl stress and sampled at 6 and 24 hour post treatment (hpt). First row are the plants before stress; second row and third row are the plants at 6h & 24h salt stress respectively.

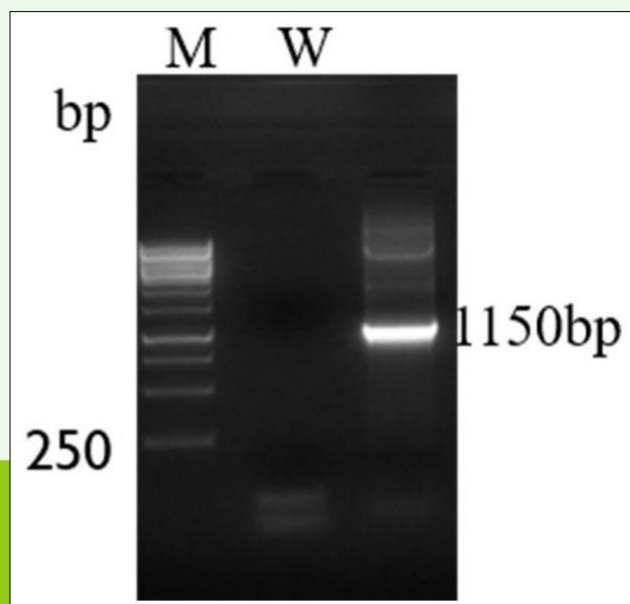


**Fig. 3.6:** Semi qRT-PCR showing up-regulated gene in response to salt stress in 6h and 24h corresponding to a house-keeping gene.



**Fig. 3.7:** Quantitative RT-PCR analysis of a putative salt responsive gene of chickpea in response to salt stress of 80 mM NaCl at 6 and 24h time interval. GAPDH gene was used for normalization. The expression level of genes in control sample was used as calibrator to calculate fold change.

Genes that are induced in response to salt stress are thought to play major role in conferring salt tolerance and this study provides a possible role of *CarMATE6* in stress adaptation and stress tolerance in chickpea. Hence this candidate salt inducible gene of ~1150bp (1.15kb) was further PCR amplified, cloned and sequence confirmed (Fig. 3.8). Detailed molecular, physiological and biochemical characterization is under progress.



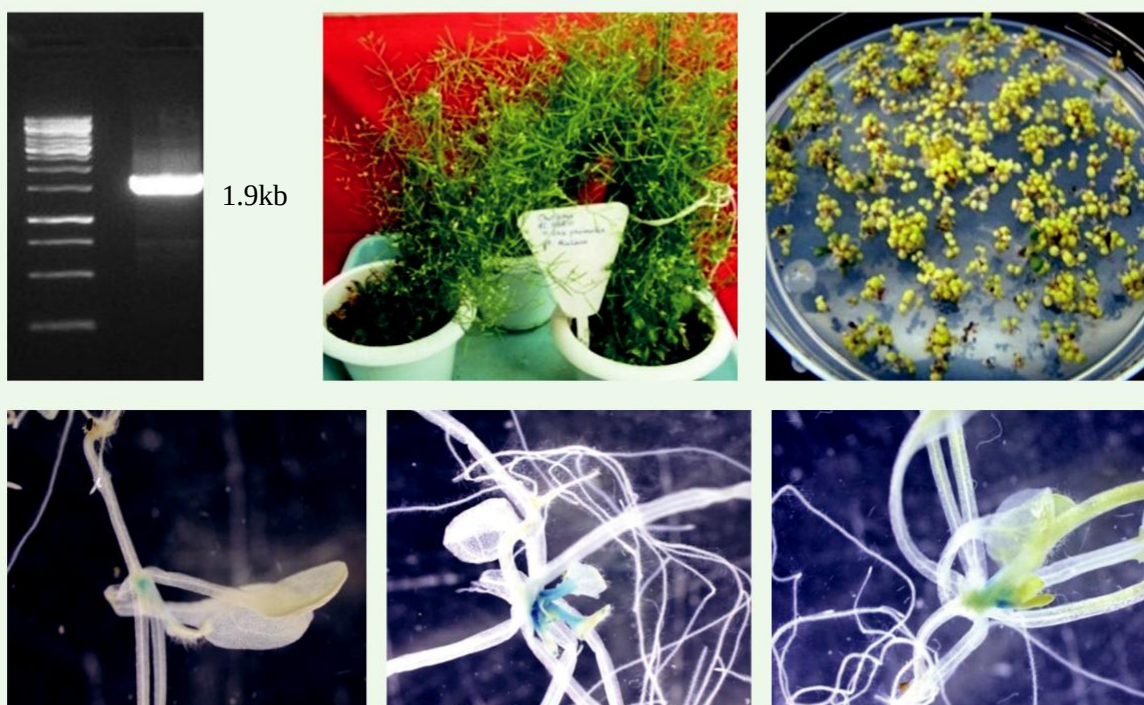
**Fig. 3.8:** PCR amplification of salt inducible *CarMATE6* gene of chickpea.

## Isolation and characterization of a salt stress inducible promoter of chickpea

(Nimmy MS, Vinod Kumar, Bharadwaj C and Jain PK)

Inducible promoters that drive expression of the transgene only upon exposure to a particular environmental stress are important in genetic engineering. In this study we have PCR amplified full length promoter of 1.9 kb of a previously characterized salt inducible *leucoanthocyanidin dioxygenase* gene of chickpea. The amplified upstream promoter fragment was cloned in pBI101 vector and transgenic Arabidopsis plants carrying the reporter gene *GUS* and promoter element fusion were generated. The T<sub>2</sub> progeny of 6 independent T1

lines were analyzed for determining the expression pattern of *GUS* gene. Analysis of the transgenic plants showed that *GUS* expression was located in the junction of petiole with the shoot and expression was more at 6h NaCl stress, but there was no expression with ABA and wound stress (Fig. 3.9). *In silico* analysis of sequence confirmed promoter fragment reveals presence of various *cis* elements. ABRE (abscisic acid response), MYB binding site having role in regulation flavonoid biosynthetic genes, HSE (heat stress response element) and TC-rich repeats (defense and stress responsiveness). Further analysis is required to identify the complete promoter region that would lead to the expression of the *uidA* gene on salt stress.



**Fig. 3.9:** PCR amplified promoter from genomic DNA of chickpea; Arabidopsis plants growing after floral dip transformation; T<sub>1</sub> primary transformants; GUS expression in two week old seedlings of T<sub>2</sub> lines, Basal, 6h NaCl, 24h NaCl.

## Isolation and characterization of insecticidal genes from native *Bacillus thuringiensis* isolates

The major insect pest of Chickpea, an important pulse crop, is *Helicoverpa armigera* (pod borer), which has increasingly developed resistance to chemical insecticides, necessitating alternative methods of control. Resistance towards this pest is largely not available in cultivated chickpea varieties. *Bacillus thuringiensis* (*Bt*), an aerobic, spore-forming Gram-positive bacterium that produces insecticidal crystal proteins (Cry) and vegetative insecticidal proteins

(Vip), which have specific toxicity towards insect pests of different insect orders while being safe for non-target species. Therefore, *Bt* has been extensively used as a biopesticide and for development of transgenic crops expressing these genes. However, in view of likelihood of development of resistance in target insects to the continuous deployment of these genes, isolation of novel genes is important. *Bt* isolates have been recovered from diverse habitats in India and *cry*- and *vip*-type genes have been isolated in our laboratory.

- 1) *Identification and characterization of novel genes:*  
NGS (Next Generation Sequencing)-based pool deconvolution technique had been developed to identify and clone potentially useful genes from native *Bt* isolates. Theoretical 3D structural modelling of a novel Cry **protein** (NCBI Accession No. KM053254) isolated from native *Bt* isolate SK711 had been done.
- 2) *Molecular modeling of a novel cyt1007 gene:*  
Three dimensional structure and putative binding sites of novel *cyt1007* gene isolated from native *Bt* isolate SK-1007 based on pool deconvolution technique, were predicted by molecular modelling.
- 3) *Evaluation of insecticidal activity of novel cry52Ca1 gene:* The novel *cry52Ca1* gene (NCBI Accession No. KM053253), identified by NGS and cloned from a native *B. thuringiensis* isolate, was expressed in *E. coli* for evaluation of its toxicity towards important lepidopteran pest, chickpea pod borer, *Helicoverpa armigera*.
- 4) *Evaluation of insecticidal and aphidicidal activity of novel cry4Ca2 gene:* The novel *cry4Ca2* gene (NCBI Accession No. KM053254) was expressed in *E. coli* for evaluation of its toxicity towards important lepidopteran and aphid pests, viz. *Earias vitella* (spiny bollworm), *Lipaphis erysimi* (mustard aphid), *Myzus persicae* (green peach aphid), *Brevicoryne brassicae* (cabbage aphid) in comparison in laboratory bioassays at Insect Culture Facility, in collaboration with Division of Entomology, ICAR-IARI (Table 3.3-3.6). Cry4Ca2 protein was found to have higher or comparable toxicity as compared with that of Cry1Ac1 used as reference at P .05 level of significance.

**Table 3.3: Efficacy of Cry4Ca2 protein towards *E. vittella* neonates**

| S. No | Toxin (10 ppm) | Mean $\pm$ standard error (%mortality) |
|-------|----------------|----------------------------------------|
| 1.    | Cry 4Ca2       | 55.17 $\pm$ 3.44 <sub>A</sub>          |
| 2.    | Cry1Ac1        | 44.82 $\pm$ 3.44 <sub>B</sub>          |

Different letter denotes significant difference at P 0.05

**Table 3.4: Efficacy of Cry4Ca2 protein towards adults of *L. erysimi***

| S. No | Toxin (10 ppm) | Mean $\pm$ standard error (%mortality) |
|-------|----------------|----------------------------------------|
| 1.    | Cry4Ca2        | 55.17 $\pm$ 3.44 <sub>A</sub>          |
| 2.    | Cry1Ac1        | 62.07 $\pm$ 6.89 <sub>A</sub>          |

Same letter denotes no significant difference at P 0.05

**Table 3.5: Efficacy of Cry4Ca2 protein towards adults of *M. persicae***

| S. No | Toxin (10 ppm) | Mean $\pm$ standard error (%mortality) |
|-------|----------------|----------------------------------------|
| 1.    | Cry 4Ca2       | 62.07 $\pm$ 3.44 <sub>A</sub>          |
| 2.    | Cry1Ac1        | 55.17 $\pm$ 9.12 <sub>A</sub>          |

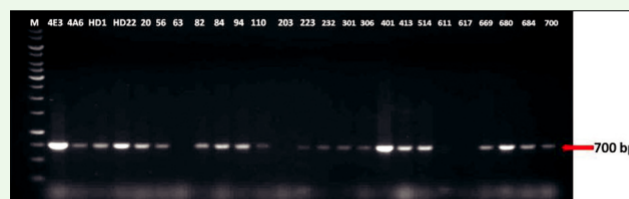
Same letter denotes no significant difference at P 0.05

**Table 3.6: Efficacy of Cry4Ca2 protein towards adults of *B. brassicae***

| S. No | Toxin (10 ppm) | Mean $\pm$ standard error (%mortality) |
|-------|----------------|----------------------------------------|
| 1.    | Cry4Ca2        | 32.14 $\pm$ 3.57 <sub>A</sub>          |
| 2.    | Cry1Ac1        | 46.43 $\pm$ 6.18 <sub>B</sub>          |

Different letter denotes significant difference at P 0.05

- 5) *Identification of vip3A-type genes from native Bt isolates:* Ninety four native *Bt* isolates were screened for the presence of partial *vip3A*-type genes by PCR amplification using gene-primers. Presence of *vip3A*-type genes was observed as a 700 bp PCR product (Fig. 3.10). The positive isolates have been selected for amplification of full length 2.3 kb *vip3A*-type genes.



**Fig. 3.10: Screening of native *Bt* isolates for the presence of *vip3A*-type genes by PCR amplification with gene-specific primers.**

# Stress Tolerance and Quality Improvement in

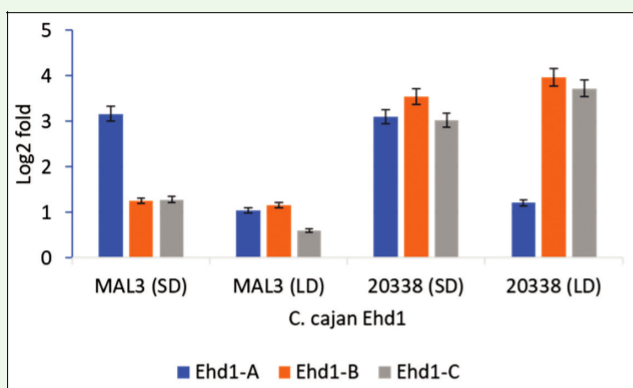
## Genomics of flowering traits in pigeonpea for identification of genes with emphasis on photoperiod related pathways

(K Tribhuwan, K Durgesh, Sandhya and Kishor Gaikwad)

Understanding the molecular mechanism underlying photoperiod sensitivity is crucial to extend the cropping area of *Cajanus cajan*, a photoperiod sensitive major grain legume of India and Africa. For expanding the scope of investigations, in the present study, we screened 12 accessions of *C. cajan* under both long day (LD) and short day (SD) photoperiods and identified three of them to be photoperiod-insensitive while the remaining genotypes were observed to be photoperiod sensitive. Further a Genome-wide survey for FT and CCT domain family genes [*CONSTANS*, *CO-like (COL)*, and *TOC1*], known to be involved in photoperiod response and flowering in many plants, revealed that 13 FT and 33 CCT family genes are present in the *C. Cajan* genome (Fig. 4.1). Gene expression pattern across 30 different tissues of *C. Cajan* identified two FT genes, *CcFT-09* and *CcFT12* particularly up-regulated in reproductive leaf tissues. Similarly, among the CCT genes, *CcCCT-23* (a *COL* gene) was

identified as a probable activator of FT. Functional protein association network analysis suggested that *CcCCT-23* interacted with many other proteins and signalling molecules like *CcGIGANTEA (CcGI)*, *CcFKF1*, *CcELF3*, *CcCOP1* and *CcPHYB*. Expression analysis of these genes in a photoperiod sensitive (MAL-3) genotype revealed *CcFT-09* and *CcCCT-23* as the major FT and FT-activator genes. However, in a photoperiod insensitive genotype (ICP20338) under SD condition flowering induction was found to be regulated by two independent pathways working simultaneously. One of the pathways utilized photoperiod dependent induction of flowering with involvement of *CcFT-09*, *CcCCT-23*, *CcGI*, *CcFKF1* etc. proteins, while the other pathway was photoperiod independent comprising of *CcEhd1*, *CcFT-12* and *CcCCT-04* proteins (Fig. 4.2). Interestingly, photoperiod dependent pathway of flowering appears to be shut down or not effective in the plants grown under LD conditions. This study will pave way for detailed characterization of the candidate genes involved in photoperiodic regulation of flowering in *C. cajan* and their subsequent utilization in developing photoperiod insensitive varieties

**Fig. 4.1:** A. Phylogenetic analysis and protein structure of CcFT family genes in *C. Cajan* (The classification of each CcFT into sub-group mentioned in the bracket).



**Fig. 4.2:** Expression (qRT-PCR) analysis *CcEhd1*, photoperiod sensitivity related gene in the reproductive leaf (RL) tissue, compared to the vegetative leaf (VL), under SD and LD photoperiod condition in two genotypes (MAL-3 and ICP-20338) of *C. cajan*.

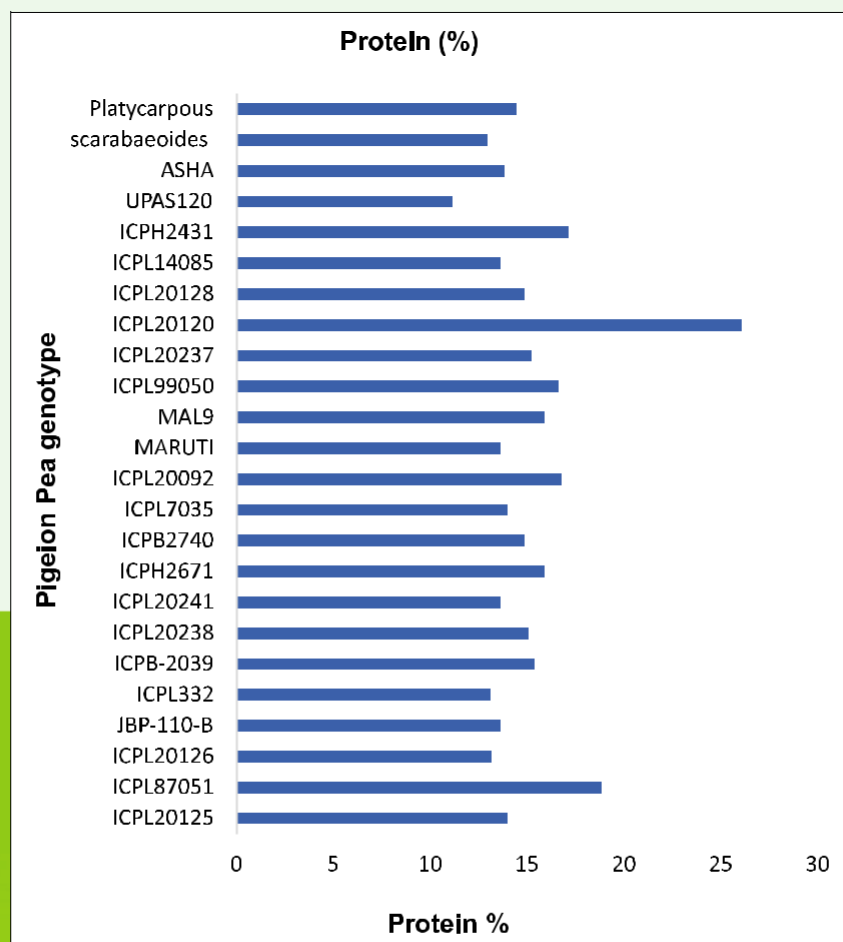
## Screening of various pigeonpea genotype for quality traits (protein content and cooking quality)

(Kumar Durgesh, Kishor Gaikwad and Sandhya)

Under quality traits subproject, a total of 24 pigeonpea genotypes were screened for high, medium and low protein content using both Bradford method

as well as Kjeldahl methods. The Kjeldahl method was used to determine nitrogen content which was then multiplied by a factor to obtain the protein content. Standard curve was made using BSA. Protein content varies from 11 to 24.7 % using Bradford assay and 11-26 % using Kjeldahl method (Fig. 4.3) in tested pigeonpea genotype. Seed samples of 18 pigeonpea genotypes were digested for estimating mineral content particularly Fe and Zn analysis. Fe content varied from 21 to 64  $\mu\text{g/g}$  while Zn content was 15-38  $\mu\text{g/g}$  among 18 genotypes. Optimization of the protocol for determination of other (ANF) anti-nutritional factors like phytic acid total phenol and tannin are under process.

Cooking process is the combination of heating and hydration. Under cooking quality, we have preliminary estimates for the cooking quality related parameters like swelling capacity/ swelling index and hydration capacity/hydration index etc. (Table 4.1) however these experiments were performed with less no. of seeds because of limited seed availability. We are still standardizing the protocol for estimating cooking time.



**Fig. 4.3:** Protein content in different pigeonpea genotypes.

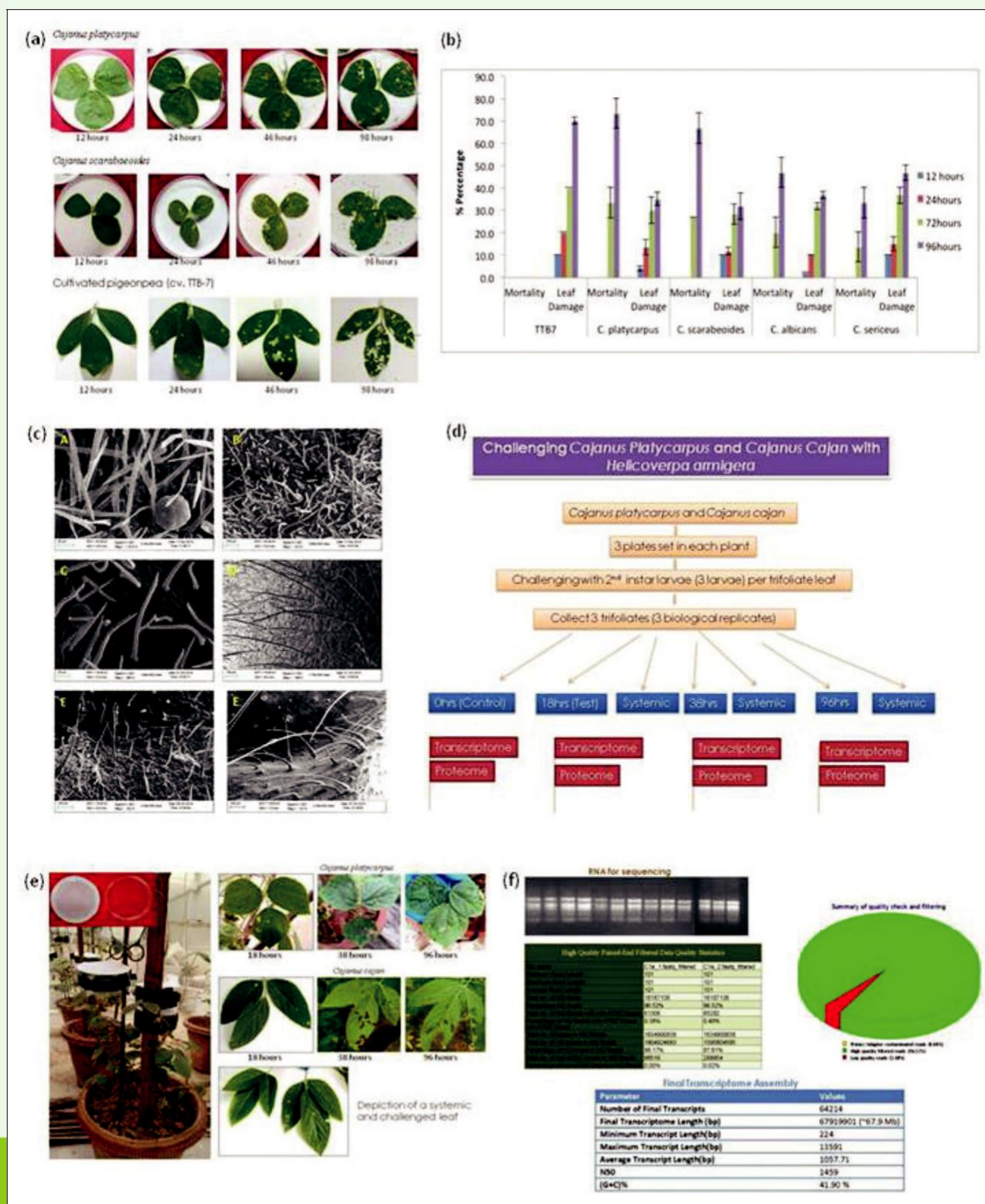
**Table 4.1: Cooking characteristics of various pigeonpea genotypes**

| S. No | Genotype         | H. Capacity | H. Index | S. Capacity | S. Index |
|-------|------------------|-------------|----------|-------------|----------|
| 1     | ICPL20125        | 0.133       | 1.455    | 1.5         | 10       |
| 2     | ICPL87051        | 0.162       | 1.276    | 1.7         | 7.29     |
| 3     | ICPL20126        | 0.14        | 1.2      | 1.7         | 8.5      |
| 4     | JBP-110-B        | 0.125       | 1.415    | 0.9         | 3.86     |
| 5     | ICPL332          | 0.113       | 1.172    | 1.2         | 6.55     |
| 6     | ICPB-2039        | 0.112       | 1.264    | 1.4         | 9.03     |
| 7     | ICPL20238        | 0.113       | 1.478    | 1.2         | 9.34     |
| 8     | ICPL20241        | 0.125       | 1.21     | 1.4         | 7.64     |
| 9     | ICPH2671         | 0.202       | 1.287    | 2.7         | 9        |
| 10    | ICPB2740         | 0.107       | 1.103    | 1.4         | 8.4      |
| 11    | ICPL7035         | 0.232       | 1.252    | 2.4         | 7.58     |
| 12    | ICPL20092        | 0.093       | 0.966    | 1.6         | 10.67    |
| 13    | MARUTI           | 0.133       | 1.311    | 1.2         | 7.2      |
| 14    | MAL9             | 0.105       | 0.875    | 2.7         | 11.57    |
| 15    | ICPL99050        | 0.15        | 1.475    | 1.7         | 10.2     |
| 16    | ICPL20237        | 0.102       | 1.371    | 1.6         | 12       |
| 17    | ICPL20120        | 0.14        | 1.12     | 2.6         | 7.38     |
| 18    | ICPL20128        | 0.143       | 1.162    | 1.5         | 6.92     |
| 19    | ICPL14085        | 0.095       | 0.77     | 1.5         | 9        |
| 20    | ICPH2431         | 0.142       | 1.441    | 1.8         | 10.8     |
| 21    | UPAS120          | 0.152       | 1.625    | 1.6         | 10.73    |
| 22    | ASHA             | 0.112       | 1.155    | 1.4         | 7.64     |
| 23    | C. Scarabaeoides | 0.007       | 0.125    | 0.1         | 1        |
| 24    | C. Platycarpus   | 0.002       | 0.071    | 0           | 0        |

### Understanding the mechanism of resistance to *Helicoverpa armigera* in pigeonpea wild relative, *Cajanus platycarpus* (Rohini)

Gram pod borer or *Helicoverpa armigera* is one of the major pests of pigeonpea. Development of resistance in pigeonpea to this herbivore has been a daunting task due to non-availability of resistant resources in the cultivated germplasm. However, it is a well-known fact that the wild relatives of pigeonpea possess the ability to resist the herbivore. In this direction, wild relatives of pigeonpea were used for understanding the mechanism of resistance to *Helicoverpa armigera* and identification of candidate genes to mitigate the herbivore. For this

four wild pigeonpea species, viz. *C. scarabaeoides*, *C. albicans*, *C. platycarpus* and *C. sericeus*, were screened and *C. scarabaeoides* and *C. platycarpus* showed higher resistance to pod borer than *C. cajan* cultivars. RNA-seq and differential gene expression analysis were carried out between *C. platycarpus* and cultivar TTB7 at different time points after challenge with the insect larvae. Herbivory-specific genes with >2-fold differential expressions were identified. Genes with probable role in (i) insect structural destruction; (ii) interference in digestion; (iii) reduction in availability of nutrients; and (iv) insect toxic products have been shortlisted for validation. Integration of this information with the transcriptome data will aid in better understanding of the resistance phenomena existing in the wild relative (Fig. 4.4).



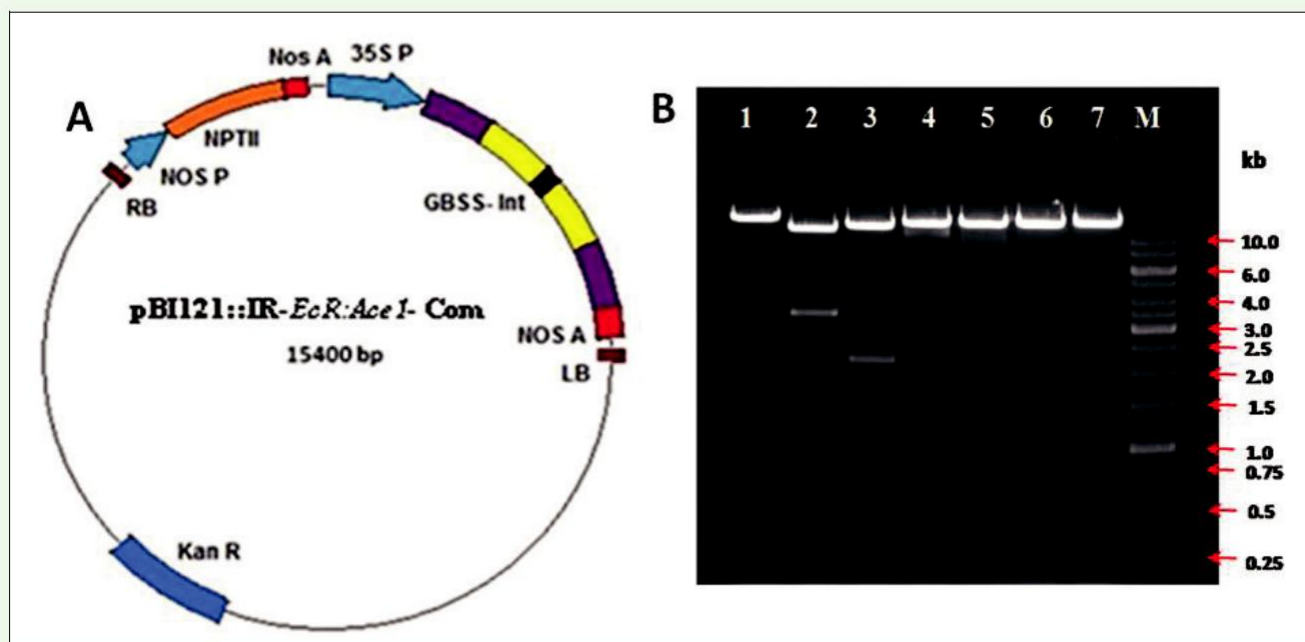
**Fig. 4.4:** Detached leaf assay for checking resistance levels of *Cajanus* species to *H. Armigera*; (b) Percent mortality of the *Helicoverpa* larvae fed on the leaves of cultivated and wild pigeonpea species; (c) Electron micrographs of trichomes on the leaves and pods of cultivated and wild pigeonpea species; (d) Schematic representation of deliberate challenging of *Cajanus platycarpus* and *Cajanus cajan* with *Helicoverpa armigera* at different time points; (e) Experimental set up for simultaneous challenging of *Cajanus platycarpus* and *Cajanus cajan* with *Helicoverpa armigera*; (f) Snapshot of transcriptome sequencing and assembly of the challenged samples of wild relative *Cajanus platycarpus* and cultivated pigeonpea.

## Host-delivered RNAi-mediated combinatorial silencing of acetylcholinesterase1 and 20-hydroxyecdysone receptor genes for *Helicoverpa armigera* resistance

(Jyotsana Tilgam, Venkat Raman, Saurav Tyagi, Rohini Sreevathsa and Debasis Pattanayak)

A combinatorial approach for silencing of two important genes encoding 20- hydroxyecdysone receptor (*HaEcR*) and acetylcholinesterase 1 (*HaAce1*) of *H. armigera* through transgenic tobacco delivered RNAi was employed. A 533 bp long *HaAce1* target region spanning from 1251 to 1783 nt position of *HaAce1* partial cDNA sequence (GenBank Accession No. DQ064790) was selected that shared the least homology to the sequences of various other non-

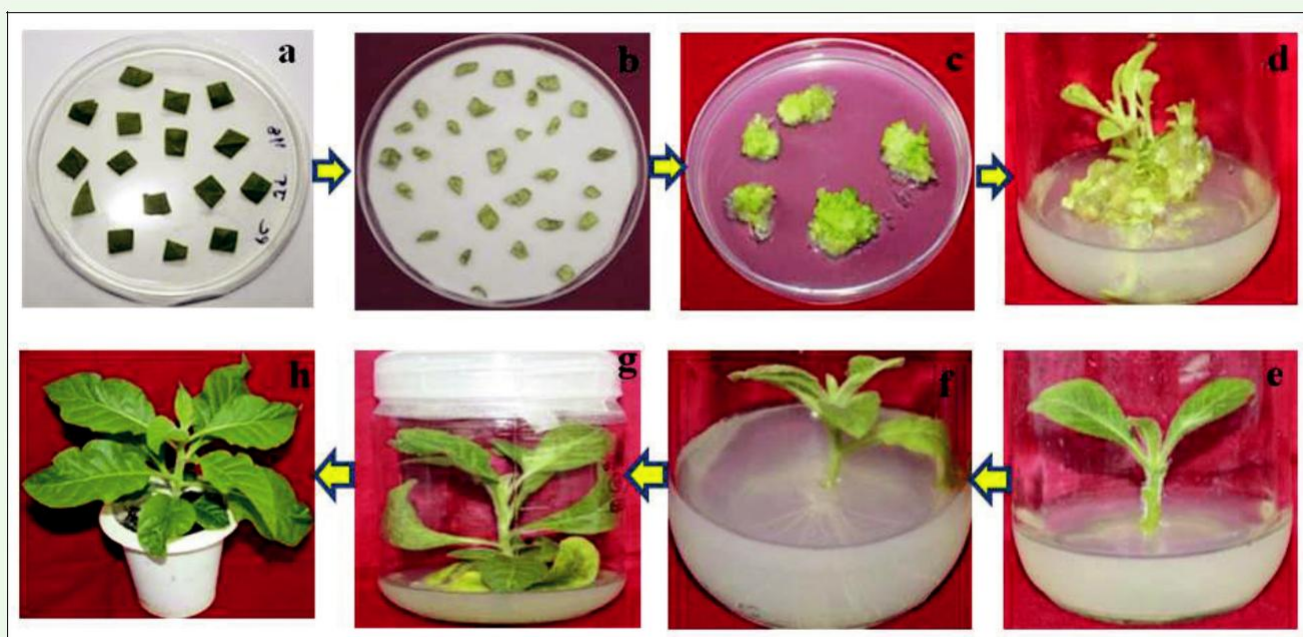
target organisms. For *HaEcR*, a 482 bp sequence spanning from 245 to 726 *HaEcR* cDNA sequence (GenBank accession No. EU526831) has chosen based on previous published report. An inverted repeat construct was designed by placing these two cDNA sequences in sense and antisense orientation intervened by potato granule bound starch synthase (*GBSS*) intron. The synthesized inverted repeat construct was cloned under the transcriptional control of CaMV 35S promoter and NOS terminator on the backbone of plant binary vector, pBI121 (Fig. 4.5) The recombinant binary vector cassette, pBI121::IR-*EcR:Ace1*-Com, was used for *Agrobacterium* mediated tobacco (*Nicotiana tabacum* L.; var. Petit Havana) transformation.



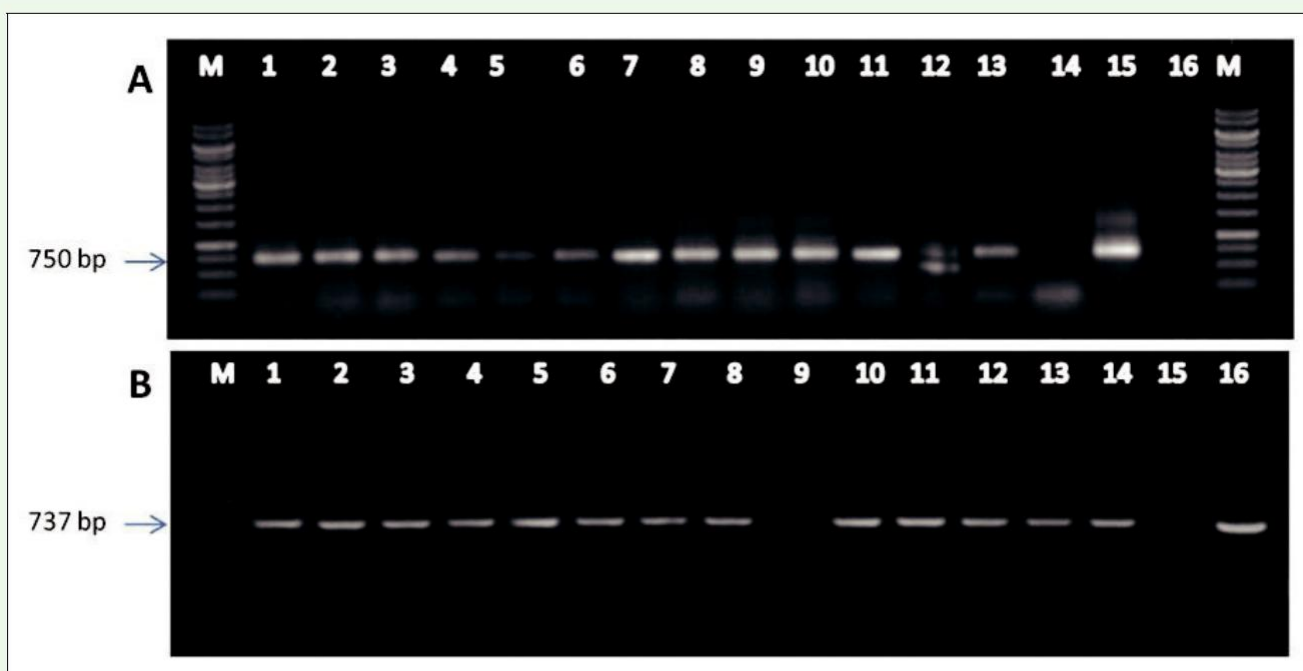
**Fig. 4.5:** Construction of RNAi combinatorial binary vector. (A) Vector map of pBI121::IR-*EcR:Ace1*- Com. (B) Restriction analysis of pBI121::IR-*EcR:Ace1*-Com. Lanes: 1, *Hind*III digestion linearized the vector; 2, *Hind*III and *Eco*RI digestion released 3.3 kb RNAi cassette; 3, *Hind*III and *Sac*I digestion released 2.3 kb fragment of 35S promoter and inverted repeat fusion; 4, *Xba*I and *Bam*HI digestion released 1.2 kb *EcR-Ace1* sense fragment alongwith *GBSS* intron; 5, *Bam*HI and *Sac*I digestion released 1.1 kb *EcR-Ace1* antisense fragment; 6, *Hind*III and *Xba*I digestion released 800 bp promoter fragment; 7, *Sac*I and *Eco*RI digestion released 250 bp terminator fragment; M, 1 kb DNA ladder.

The recombinant binary vector, pBI121::IR-*EcR:Ace1*-Com, was used for *Agrobacterium* mediated tobacco (*Nicotiana tabacum* L.; var. Petit Havana) transformation. A total of 48 putative transformants were generated from 360 tobacco leaf disc explants co-cultivated (Fig. 4.6). Thirty two transgenic tobacco lines were obtained after screening of 48 putative transformants by PCR and RT-PCR using gene specific and *nptII*-specific primers (Fig. 4.7). These transgenic lines were subjected to detached leaf insect bioassay using 2<sup>nd</sup> instar larvae of *H. armigera*. Eleven out of 32

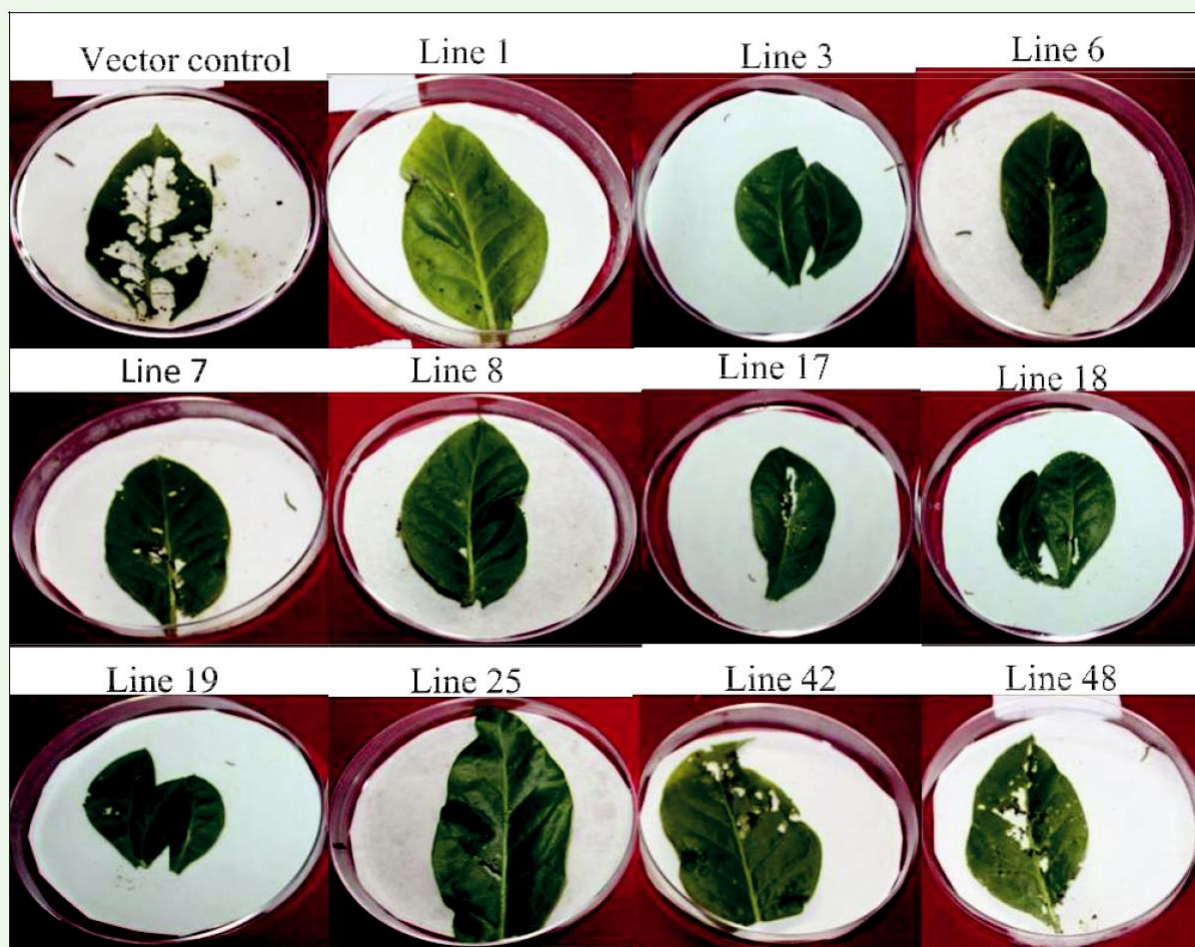
transgenic tobacco lines demonstrated moderate to high level of resistance to *H. armigera* as evidenced from less leaf damage (Fig. 4.8). Leaf damage to the extent of 5-40% was observed in eleven tobacco transgenic lines compared to 90% leaf damage in control leaves. Less than 10% leaf damage was observed in five of these 11 RNAi tobacco lines (Fig. 4.9). *H. armigera* larvae reared on leaves of these 11 transgenic tobacco lines showed retarded growth and attained almost half of the length compared to larvae fed on control tobacco leaves (Fig. 4.10A & B).



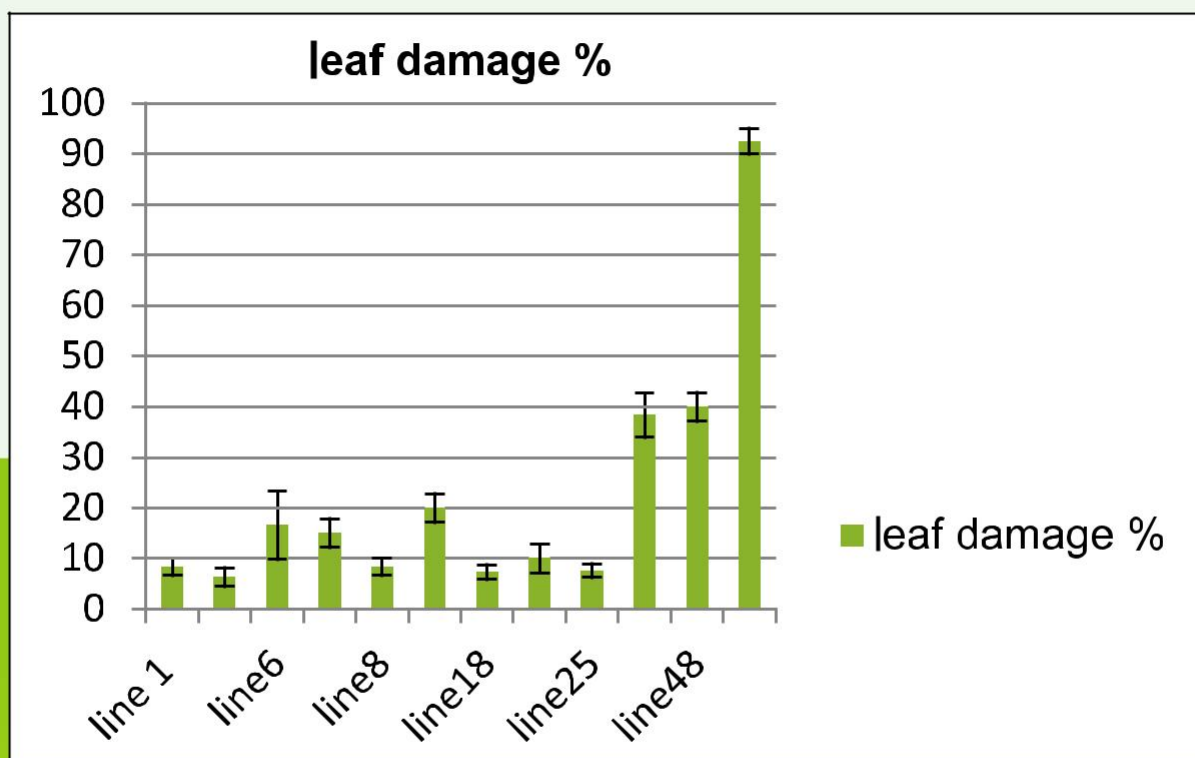
**Fig. 4.6:** *Agrobacterium*-mediated transformation of tobacco with pBI121::IR-*EcR:Ace1*-Com gene construct. a. Pre-culture; b. Co-cultivation; c. Selection; d. Adventitious Shooting; e. Shoot elongation; f. Rooting; g. Fully developed plantlet; and h. Plant transferred to pot.



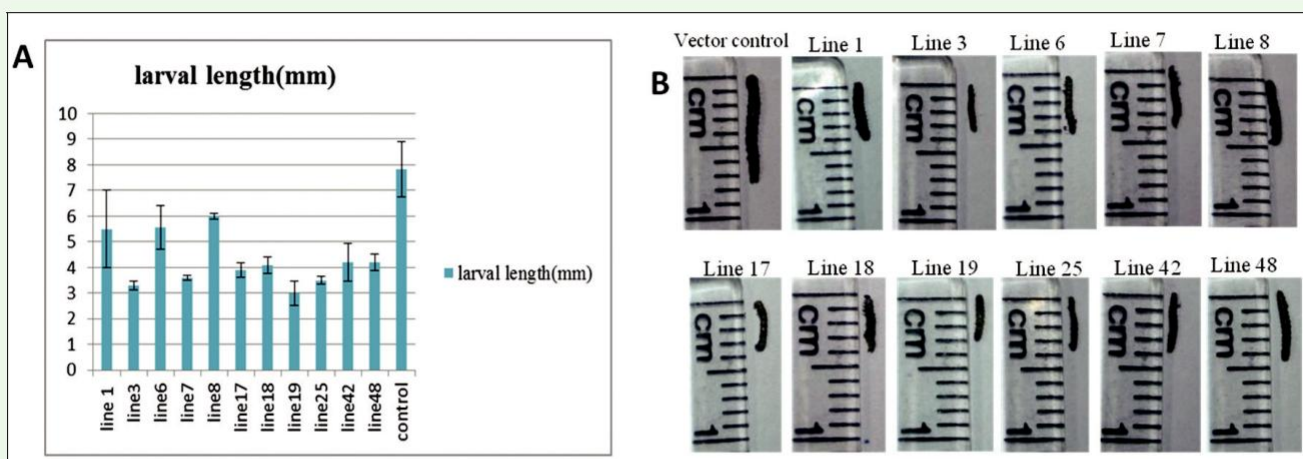
**Fig. 4.7:** Screening of putative tobacco transformants. (A) PCR confirmation of IR-*EcR:Ace1*-Com tobacco lines using gene specific primers. Lanes: M, 1 kb ladder; 1- 13, transgenic tobacco lines; 14, pBI121 tobacco line; 15, positive control; and 16, water control & (B) RT- PCR confirmation of transgene expression in IR-*EcR:Ace1*-Com tobacco lines. Lanes: M, 1 kb ladder; 1-13, transgenic tobacco lines; 14, pBI121 tobacco line; 15, non-transgenic tobacco line; and 16, positive control.



**Fig. 4.8:** Detached leaf insect bioassay of tobacco IR-Ace1:EcR Com RNAi transgenic lines with 2nd instar larvae of *H. armigera* after three days of feeding.



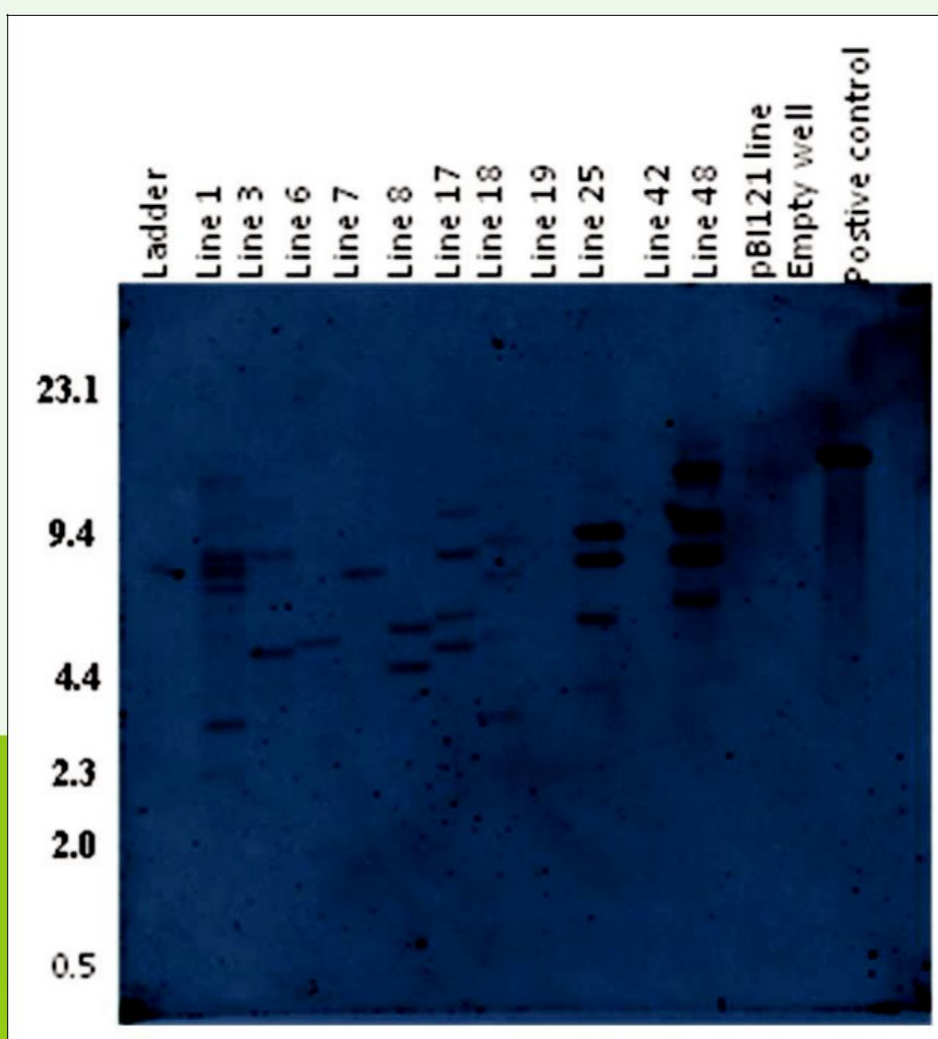
**Fig. 4.9:** Per cent leaf damage of tobacco transgenic lines by *H. armigera*. Transgenic tobacco lines 1, 3, 8, 18 and 25 showed less than 10% leaf damage while the vector control tobacco leaf showed 90% leaf damage.



**Fig. 4.10: (A)** Growth retardation of larvae fed on IR-*EcR:Ace1*-Com transgenic tobacco lines **(B)** Growth comparison of larvae fed on control and transgenic tobacco leaves.

The 11 transgenic tobacco lines demonstrating resistance against *H. armigera* in insect bioassay were characterized for the gene integration and copy number through Southern hybridization.

Two IR-*EcR:Ace1*-Com lines, line 6 and line 7, carried single copy of the transgene, while the rest of nine RNAi transgenic lines contained 2-4 copies of the transgene (Fig. 4.11).



**Fig. 4.11:** Southern hybridization analysis of IR-*Ace1:EcR*-Com tobacco transgenic lines. Lines 1, 3, 6, 7, 8, 17, 18, 19, 25, 42 and 48 are transgenic tobacco lines. pBI121 line is tobacco line transformed with pBI121. Numbers on the left indicates size of the band in kb.

# Biotechnological Approaches for Mustard Improvement

On the oilseeds map of the world, India occupies a prominent position, both in regard to acreage and production. India is the 4<sup>th</sup> largest edible oil economy in the world and contributes about 10% of the world oilseeds production.

## Development of aphid resistance in Indian mustard

(Deepa Bhatt, Muthuganeshan Annamalai, Rohit Chamola and Ramcharan Bhattacharya)

Hemipteran insect-pest mustard aphid (*Lipaphis erysimi*) has been identified as a major insect-pest of a rapeseed-mustard group of crops because of its widespread occurrence and devastating damaging effect. Breeding for aphid resistance has been difficult primarily because of lack of resistance source within the crossable germplasms. Therefore, biotechnological intervention for devising strategies and accessing distantly related genes for developing aphid resistance has remained a priority in the mustard improvement program.

## Host delivered RNAi based gene silencing for developing aphid resistance in mustard

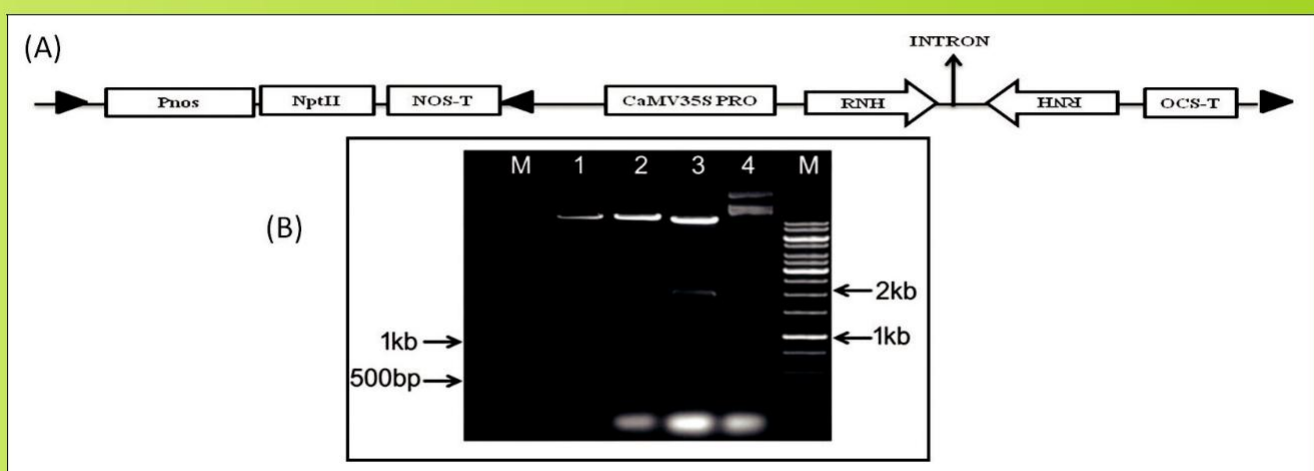
Once the ubiquitous nature of RNAi based gene

silencing was established, several studies indicated its potential use in developing insect resistance in host plants. We had conducted several insect bioassay experiments for analyzing the effect of selected dsRNAs, homologous to important aphid genes, on fecundity and survival of the feeding aphid-nymphs.

Based on the results of diet bioassay, three potential targets *LePG1*, *LePG2*, and *LePG3* were identified. Feeding of dsRNA homologous to these target aphid genes resulted in ~50-60% reduction in nymph production as well as induced ~65-75% aphid mortality. Ingestion of dsRNA also resulted in significant down-regulation of the target genes in aphid nymphs.

For a host-mediated generation of these selected dsRNAs the *LePG1* coding sequence (RNH) was amplified and cloned in sense and antisense direction in dsRNA expression cassette in plant transformation vector pBinAR (Fig. 5.1). Several independent transgenic *B. juncea* lines were generated through *Agrobacterium*-mediated floral dip transformation method. Initially, the putative transgenic lines were

scored by gene-specific PCR analysis.



**Fig. 5.1:** (A) Schematic view of the T-DNA region of the binary vector pBinAR having expression cassette *LePG1*-dsRNA under *CaMV35S* promoter. (B) Restriction based confirmation of gene cloning in binary vector pBinAR. Lanes: M (LHS), 100 bp DNA ladder; 1, *EcoRI*+*Bam*HI restricted rpBinAR plasmid showed 1 kb insert released; 2, *Kpn*I+ *Bam*HI restricted rpBinAR plasmid; 3, *Pac*I+*Hind*III restricted rpBinAR plasmid; 4, Uncut rpBinAR plasmid, M (RHS), 1kb DNA ladder.

RNA samples isolated from the transgenic plants were analyzed for assessing the expression of the dsRNA. In qRT-PCR, the transcript level of the *LePG1*-dsRNA was compared in quantitative terms on the basis of normalized Ct values. The expression level was represented as fold ratios of the transcript level using the least value as the common denominator (Fig. 5.2). The qRT-PCR results showed significant variation in the expression of *LePG1*-dsRNA among the independent transgenic lines. Two lines LP1 and LP2 showed relatively high expression of *LePG1* dsRNA.

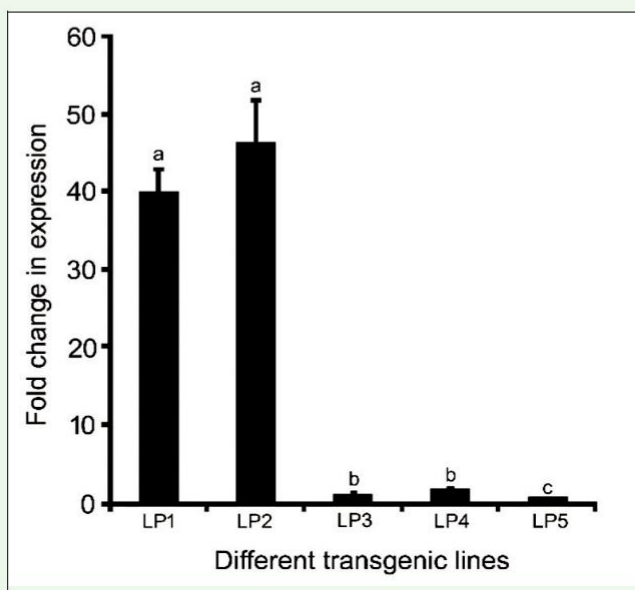


Fig.5.2: Fold change expression of *LePG1* in the *B. juncea* transgenic lines.

### Aphid bioassay

Five PCR positive transgenic lines along with the control line were used for aphid bioassay. Ten 3<sup>rd</sup> instar aphids were placed on each leaf in planta and caged (Fig. 5.3). A same number of aphids were also released on detached leaf discs.

After 7 days of inoculation majority of the adult aphids and nymphs on transgenic leaves were found dead. The highest mortality of 50.5% and 49% was obtained in case of transgenic line LP5 LP4 respectively (Fig. 5.4A). It was also observed that the production of the nymph was also affected due to feeding on the transgenic plants. The fecundity of nymphs was reduced up to 36% in LP2 compared to 82% on the control plant (Fig. 5.4B). Whereas in case of other transgenic lines the reduction of aphid fecundity varied from 4% to 36% at the end of the seven-day bioassay period. The mean mortality and fecundity values recorded for transgenic plants were

significantly different compared to the respective values obtained from control plants when analyzed by student's t-test ( $P < 0.05$ ).

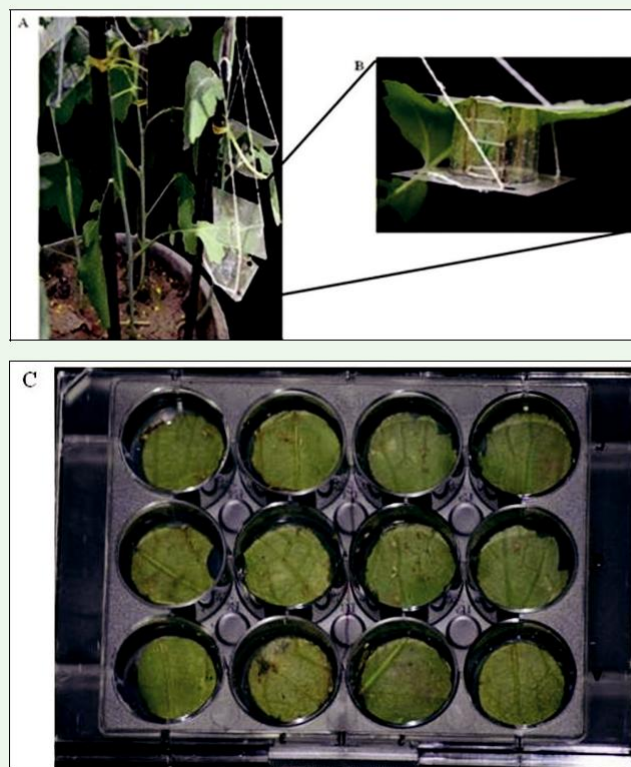


Fig. 5.3: Bioassay setup in mustard plants. (A) Custom-made clip cages used for in planta bioassay of transgenic lines expressing *LePG1*-dsRNA (B) clip cage attached to the leaf of the transgenic plants (C) leaf disc assay of the transgenic leaf.

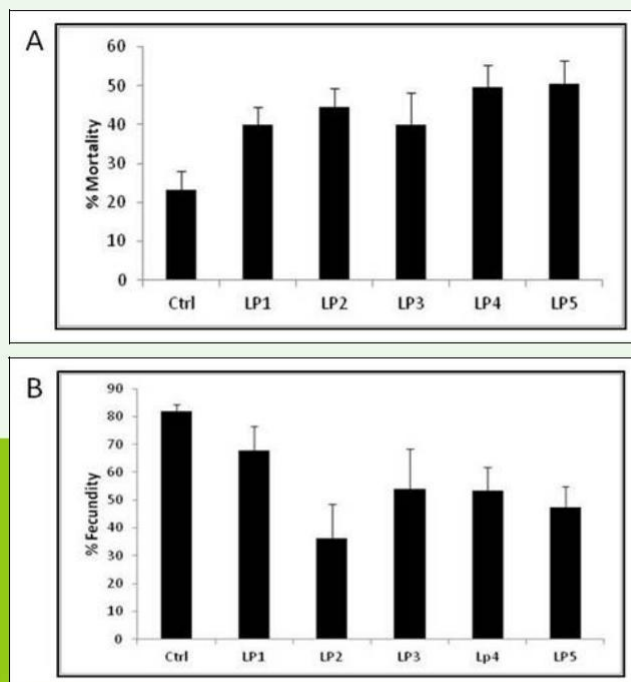


Fig. 5.4: (A) Mortality of *L. erysimi* was calculated on *B. juncea* transgenic leaves and control leaves at 7 days post inoculation. (B) Calculation of aphid fecundity on *B. juncea* transgenic leaves along with control nontransgenic leaves at 7 day post inoculation.

### Functional evaluation of aphid inducible promoter

Aphid probing is limited to phloem tissues. Phloem-specific promoters are appropriate for the transgene expression. Expressing the transgene under tissue-specific aphid inducible promoter will avoid unnecessary background expression of the transgene which will reduce the chance of pleiotropic effects and metabolic payoffs. *Arabidopsis* *PR1* promoter is known to be induced by pathogen attack. We evaluated *B. juncea* *PR1* promoter for its activation by aphids. The promoter was isolated by PCR and fused with a *gus* reporter gene in pCambia1305.1. The clone was confirmed by *EcoRV* digestion which released 0.9 and 1.5 kb fragments. The binary plasmid *PR1:gus* was mobilized into the *Agrobacterium tumefaciens* GVK3101 strain. Transient expression of the construct was evaluated in *Nicotiana benthamiana*. Several transgenics in tobacco were developed and in transgenic plants, leaves were analyzed for the *gus* expression by GUS histochemical staining. Molecular confirmation is done by PCR amplification of plant selectable marker (986 bp). Further molecular analysis and evaluation of promoter activation by aphids are in progress.

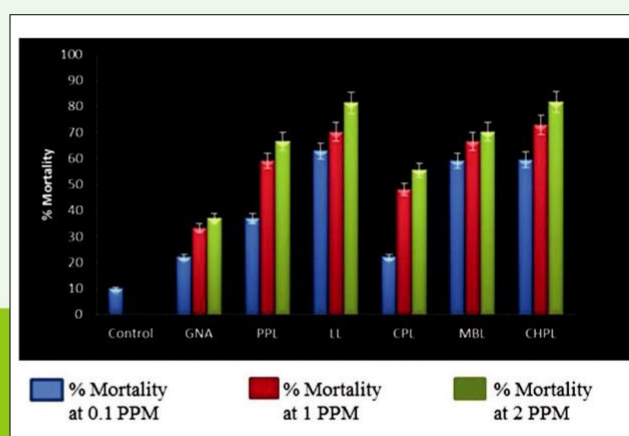
### Use of insecticidal genes from legumes for developing aphid resistant mustard (Rakesh Kumar Prajapat and Rekha Kansal)

Aphids are the major agricultural pests that cause significant yield losses each year. Excessive dependence on insecticides for aphid control is undesirable because of the development of insecticide resistance, the potential negative effects on non-target organisms and environmental pollution. Transgenic strategies expressing insecticidal *Bacillus thuringiensis* toxin, have been found to be effective against many insect pests belonging to the order Lepidoptera and Coleoptera, but for sap-sucking hemipteran aphids, it is ineffective. Therefore, transgenic plants engineered for resistance to aphids via a non-toxic mode of action could be an efficient alternative strategy using other insecticidal proteins such as protease inhibitors, lectins, amylase inhibitors. Lectins are the carbohydrate-binding proteins that play a defensive role in plants in response to an attack by herbivores or pathogens.

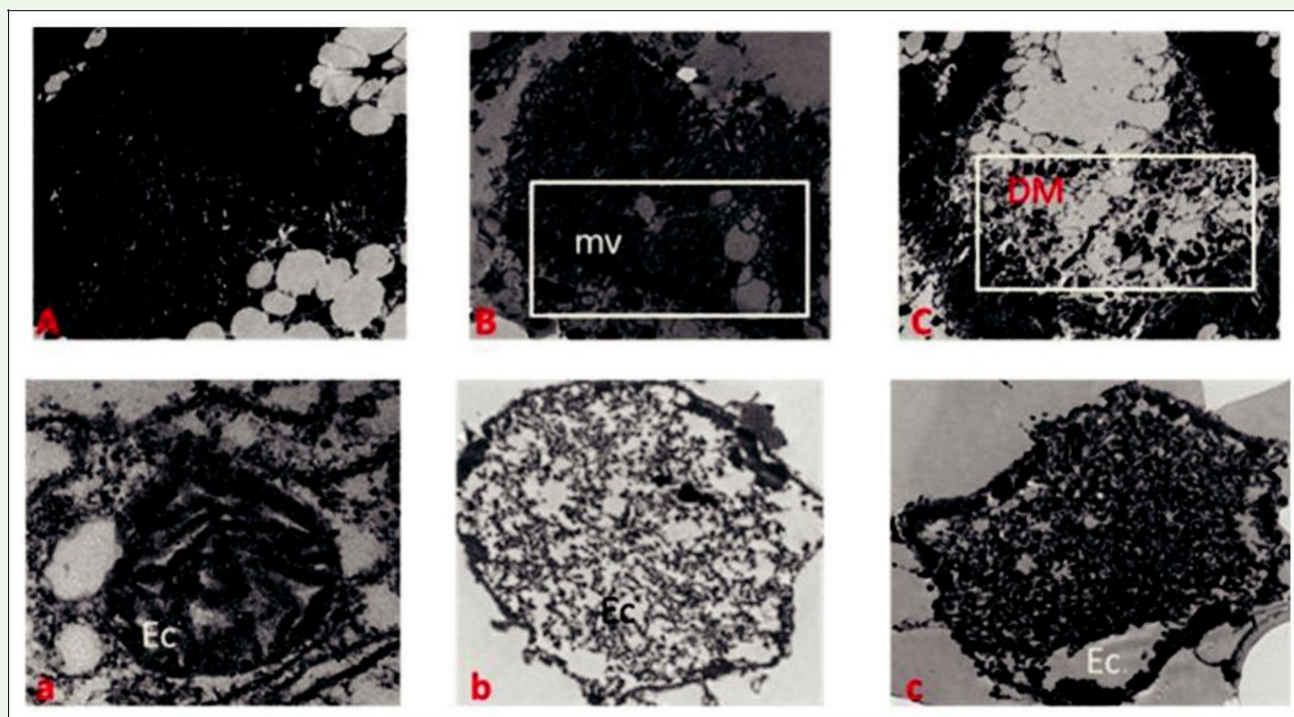
The lectin genes from five legumes viz. pigeonpea (PPL), lentil (LL), mothbean (MBL), cowpea (CWPL) and chickpea (CHPL) were cloned in the pET expression system. The proteins were expressed *in vitro* and the quantity of protein recovered from the total sample was 443, 137.74, 46.04, 107.18 and 78.92 µg/ml for PPL, LL, MBL, CWPL, and CHPL respectively. The purified protein was used to perform insect bioassay to evaluate the toxicity of legume lectins against hemipteran (*L. erysimi*) and lepidopteran (*H. armigera*) by artificial diet incorporation.

### Insecticidal potential of legume lectin proteins against aphids

The bioassay was carried out at three different concentrations of 1, 10, 20 µg g<sup>-1</sup> of diet on the basis of total protein concentration under controlled conditions of 18 ± 2 °C, 70 ± 10% RH, and 16:8 L:D against adult aphids. LL and CHPL were found to be more effective at all three concentrations (59.25 to 81.47% mortality) than PPL, MBL and CWPL (Fig. 5.5). The efficacy of all the extracted legume lectins was compared with GNA lectin against *L. erysimi*. The legume lectin proteins were found to be more or equally effective than the GNA. The transmission electron microscopy also showed that the foregut of aphid fed with a diet including legume lectin proteins was damaged (Fig. 5.6) compared to control. This clearly implied that these lectins are toxic proteins for aphid and can be used to generate transgenic plant with increased Aphid resistance.



**Fig. 5.5:** Graph showing mortality percent of aphids (*Lipaphis erysimi*) after 96 hr on an artificial diet supplemented with legume lectin proteins at different concentration.



**Fig. 5.6:** Transmission electron micrograph showing ultrastructure of *L.erysimiforgut* cells (A & a-Control aphid gut structure; B&b showing changes in the morphology of MBL treated aphid gut; C&c representing deformed microvilli in CHPL treated aphid gut. Mv (Microvilli), DM (Descending midgut), Ec (epithelium cell).

The LC<sub>50</sub> values for all these lectins against *L. erysimi* viz., GNA, CwPL, PPL, MBL, CHPL and LL were 7.102, 0.556, 0.376, 0.015, 0.013 and 0.004 µg/g of diet

respectively on 4th day (Table 5.1), indicating that LL was comparatively more toxic than the remaining lectins using artificial diet assay.

**Table 5.1:** Toxicity of selected legume lectins against adults of aphid (*L. erysimi*)

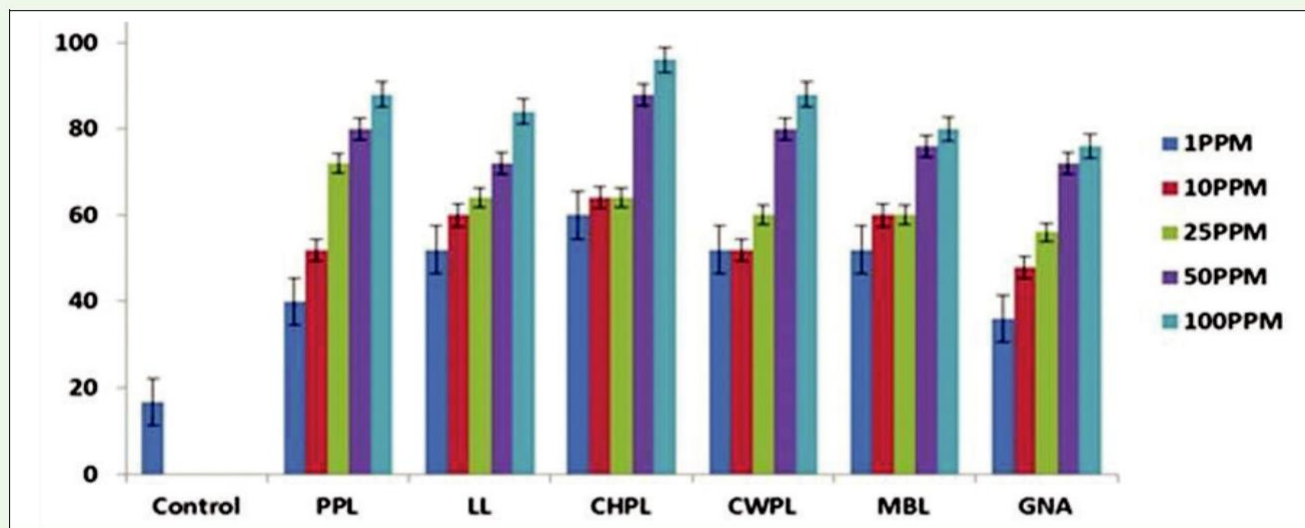
| Legume lectins | LC <sub>50</sub> µg/g of diet on the 4 <sup>th</sup> day | 95% Fiducial limit |        | Slope ± S.E | Chi-square value | DF |
|----------------|----------------------------------------------------------|--------------------|--------|-------------|------------------|----|
|                |                                                          | Lower              | Upper  |             |                  |    |
| GNA            | 7.102                                                    | -----              | -----  | 0.37±0.26   | 0.005            | 1  |
| PPL            | 0.376                                                    | 0.0007             | 1.7163 | 0.53±0.25   | 0.003            | 1  |
| MBL            | 0.015                                                    | -----              | -----  | 0.23±0.25   | 0.415            | 1  |
| LL             | 0.004                                                    | -----              | -----  | 0.25±0.25   | 0.487            | 1  |
| CWPL           | 0.556                                                    | 0.0412             | 4.0646 | 0.57±0.25   | 0.314            | 1  |
| CHPL           | 0.013                                                    | -----              | -----  | 0.30±0.25   | 0.878            | 1  |

### Insecticidal potential of legume lectin proteins against *Helicoverpa armigera*

The artificial diet prepared for the *Helicoverpa armigera* bioassay was mixed with five different concentrations of each lectin proteins i.e.: 1 ppm, 10 ppm, 25ppm, 50 ppm, and 100 ppm. From the experiments, LL was found to be the most toxic protein against *H. armigera* at all concentration. At

25 ppm, PPL protein was most effective compared to other lectin proteins (Fig. 5.7).

Against neonates of *H. armigera*, based on the LC<sub>50</sub> value on the 8th day after treatment LL (0.580µg/g of diet) and CHPL (0.895 µg/g of diet) were found to be more toxic followed by MBL (1.329 µg/g of diet) (Table 5.2). LL and CHPL protein were found to be 13.69 and 8.08 folds more effective than GNA lectin.



**Fig. 5.7:** Graph showing mortality (%) of *H. armigera* on an artificial diet supplemented with legume lectin proteins at different concentration after the 8th day.

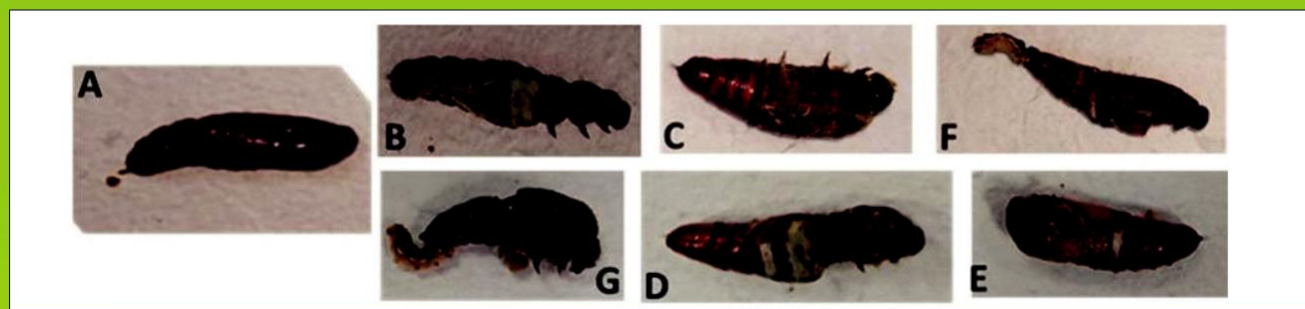
**Table 5.2:** Toxicity of selected legume lectins against *H. armigera* on the 8th day

| Legume lectins | LC <sub>50</sub> µg/g of diet on the 8 <sup>th</sup> day | 95% Fiducial limit |         | Slope ± S.E | Chi-square value | DF |
|----------------|----------------------------------------------------------|--------------------|---------|-------------|------------------|----|
|                |                                                          | Lower              | Upper   |             |                  |    |
| GNA            | 7.231                                                    | 0.5165             | 19.9901 | 0.55±0.19   | 0.942            | 3  |
| PPL            | 3.726                                                    | 0.4676             | 8.9273  | 0.71±0.19   | 1.459            | 3  |
| MBL            | 1.329                                                    | 0.0000             | 6.9722  | 0.37±0.18   | 1.315            | 3  |
| LL             | 0.580                                                    | 0.0000             | 3.6619  | 0.37±0.16   | 1.312            | 3  |
| CWPL           | 2.235                                                    | 0.0107             | 7.5686  | 0.51±0.18   | 4.315            | 3  |
| CHPL           | 0.895                                                    | 0.0022             | 3.6000  | 0.55±0.19   | 6.801            | 3  |

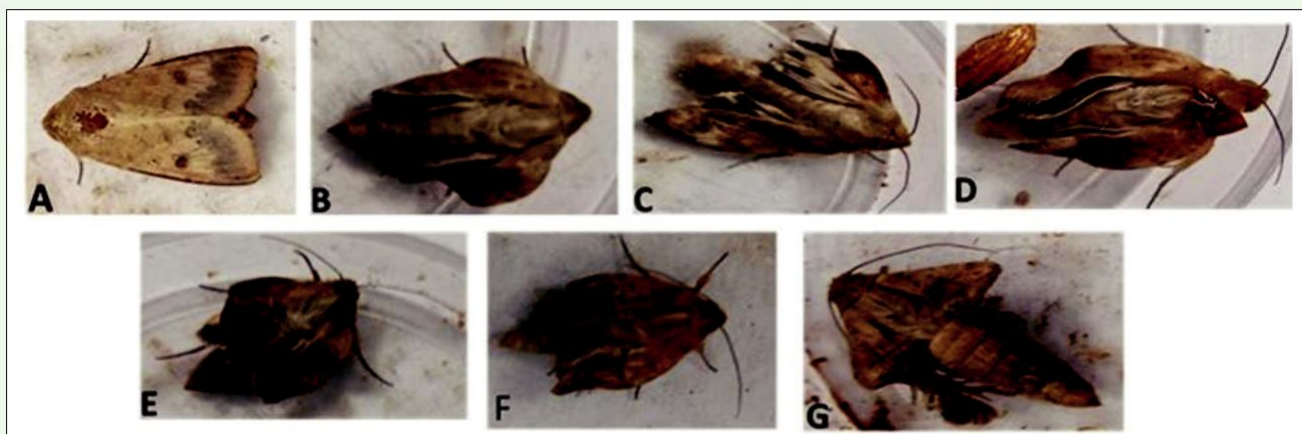
In another assay of *H. armigera*, 8 days old larvae were reared on an artificial diet supplemented with different lectins to see the effect on next-generation growth and development. Lectin proteins exhibited the effect on different stages of development like larval growth, pupal development and emergence of the adult from pupa (Fig.5.8 and 5.9).The moths emerging from lectin protein fed larvae had a sex ratio favoring females and Laid lower number of fertile eggs than those from the untreated diet. Similarly, the pupal mortality and malformation

were significantly high for LL protein and L-P% mortality was higher for PPL, CWPL, and CHPL as compared to GNA for 8 days old larvae of *H. armigera* at 50 ppm.

In contrast to *Bt*, alternative sources of potential insecticidal gene products including several different classes of proteins like lectins have been shown to be insecticidal towards a range of economically important insect pests by direct assay. These genes can be expressed in developing transgenic plants.



**Fig. 5.8:** Effect of legume lectins on pupation of *H. armigera*. (A- Control; B-PPL; C-CwPL; D-CHPL; E-MBL; F-LL; G-GNA).



**Fig. 5.9:** Effect of legume lectins on growth and winged development of *H. armigera* during emergence (A- Control; B-PPL; C-CwPL; D-CHPL; E-MBL; F-LL; G-GNA).

### Genetic Engineering to develop Alternaria-resistant mustard

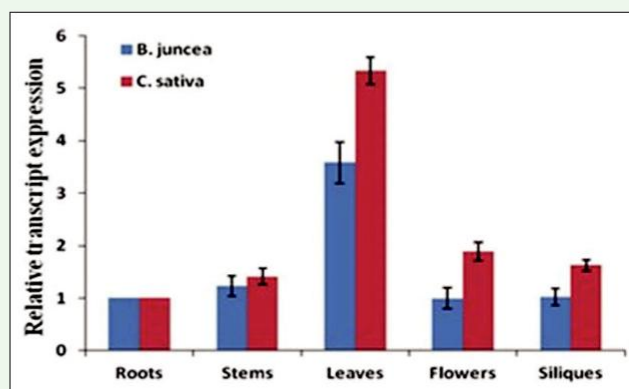
*Alternaria* causing Alternaria blight or ringspot is an important pathogen of *B. juncea* and leads to a great yield loss of this crop. To protect themselves, plants use multidimensional approaches such as morphological, biochemical and molecular defense responses that allow them to survive under such conditions. We are trying to understand these mechanisms for the purpose of developing Alternaria-resistant mustard.

### Transcript profiling of wrky33 in different tissues in *B. juncea* and *Camelina sativa* (Vinod Jangid and Anita Grover)

The WRKY proteins comprise a major family of transcription factors that are essential in pathogen and hormonal responses (such as JA and SA) of higher plants and a number of plant-specific reactions. They share a DNA binding domain, designated as the WRKY domain, which contains an invariant WRKYGQK sequence and a CX<sub>4</sub>-5CX<sub>22</sub>-23HXX zinc-binding motif. The WRKY33 was shown to be required for resistance toward the necrotrophic pathogens such as *A. brassicicola* and *B. cinerea*. WRKY33 regulates negatively with the SA pathway and positively with the JA pathway.

The expression of WRKY33 transcript was found to be higher in leaf tissues compared to other tissues in both the plant species. The transcript expression level of CsWRKY33 gene was found to be higher in

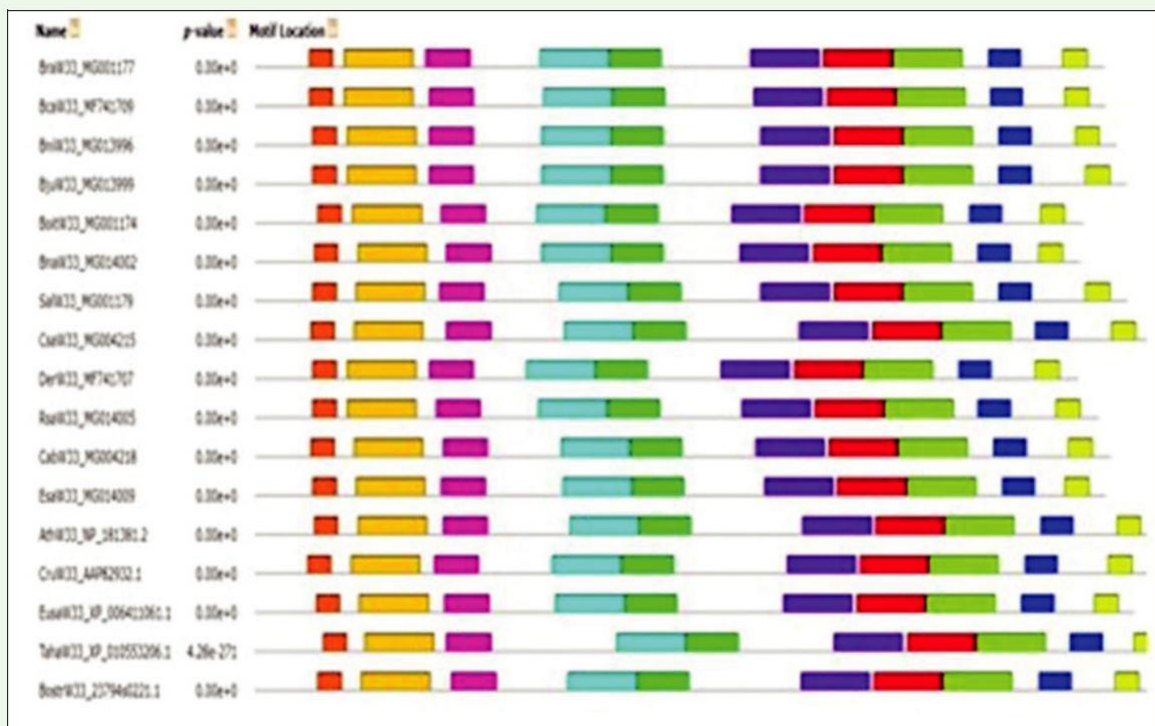
different tissues as compared to BjWRKY33 transcripts (more than 1.8 folds in leaf tissues) (Fig. 5.10)



**Fig. 5.10:** Relative transcript expression profiling of BjWRKY33 and CsWRKY33 genes in different plant tissues. The qPCR result demonstrating the differential expression of WRKY33 gene from different tissues of *B. juncea* and *C. sativa*. For the normalization, the  $\alpha$ -tubulin gene was used as an internal control. Three experimental replicates were used for calculation of the mean values and standard errors with three technical replicates.

### In silico analysis of WRKY33 proteins in different Brassica coenospecies

Different WRKY33 protein sequences of various members of the Brassicaceae family were analyzed *in silico* (Fig.5.11). Online-based MEME motif search analysis showed the presence of significant motifs which participate in forming functional WRKY domains. The amino acid residues in the motif were found to be conserved across the Brassica WRKY33 homologs which support in favor of their crucial role in effective DNA protein interaction as a transcription factor.



**Fig. 5.11:** The motif scan analysis of WRKY33 protein of different *Brassica* species using MEME and MAST suite. The results of motif scan analysis of WRKY33 protein from different *Brassica* species revealing the presence or absence of common and uncommon motifs and their distribution in protein.

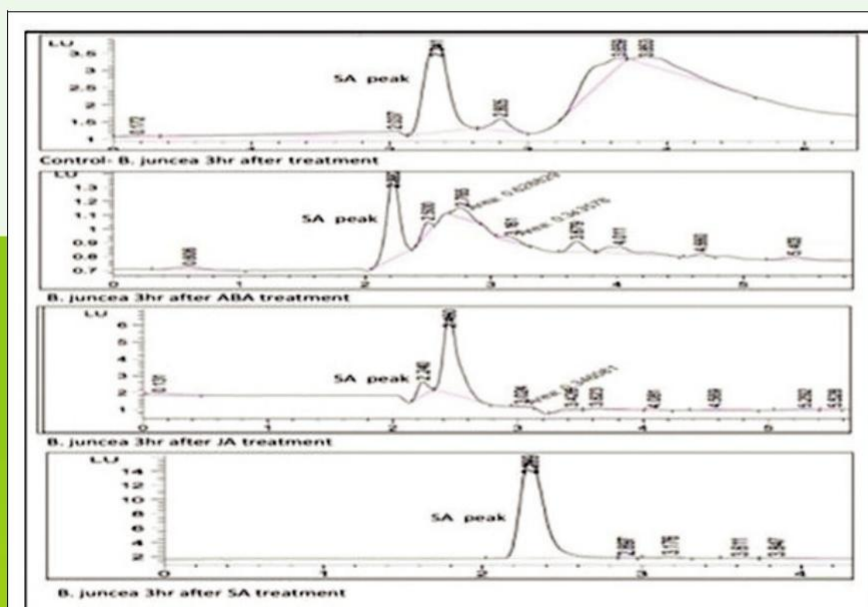
## Interaction of SA, JA, and ABA defense signaling pathways in *Brassica juncea*

(Shikha Dixit and Anita Grover)

### Quantification of Salicylic acid in hormone-treated samples of *Brassica juncea*, *Camelina sativa* and *Sinapis alba* using HPLC

Elevation in SA levels was observed in SA treated samples 3hr post-treatment on the other hand low SA levels were observed in JA and ABA-treated

samples as compared to mock plants. Same can be observed from the following figures (Fig. 5.12) where JA and ABA-treated samples show lower peak area as compared to control whereas, SA treated samples show higher peak area as compared to control samples in both the genotypes. It can also be understood from the chromatogram that the level of SA in JA and ABA-treated samples of *C. sativa* was lower as compared to *B. juncea* subjected to the same treatment.



Level of SA estimated in Hormone treated samples of *B. juncea* using HPLC

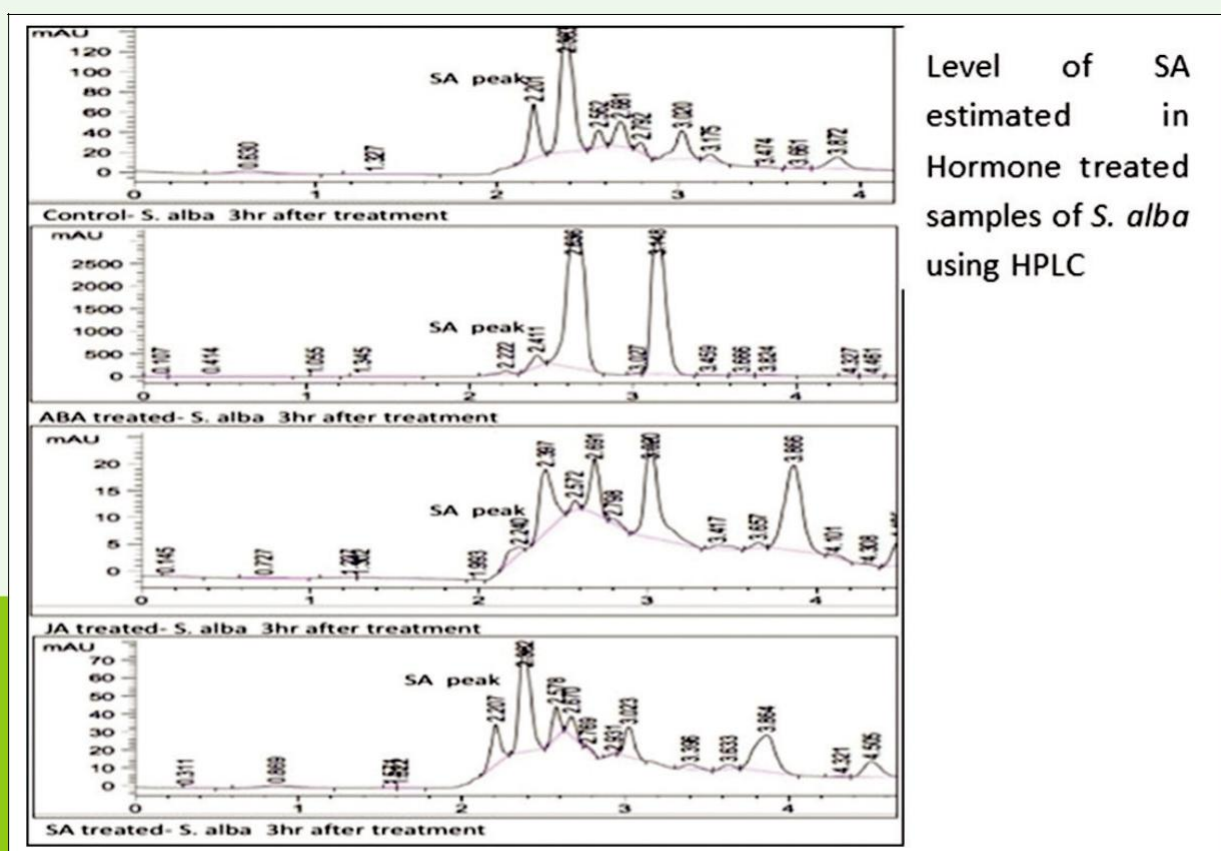
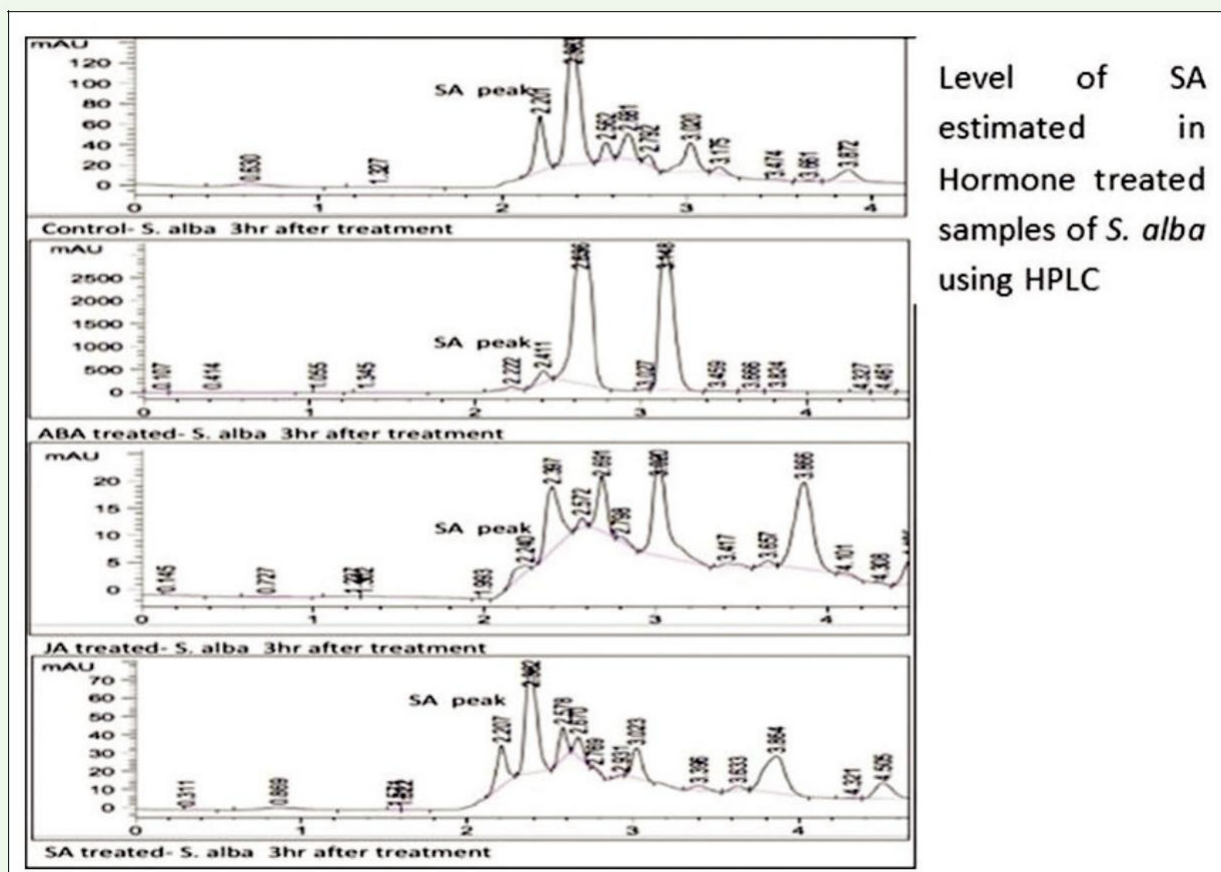


Fig. 5.12: Quantification of Salicylic acid in hormone-treated samples of *B. juncea*, *C. sativa*, and *S. alba*.

## Analysis of transgenic *B. juncea* plants overexpressing chitinase gene

(Zahoor M and Anita Grover)

Chitinase is an antifungal PR protein (belonging to PR3 class) by virtue of hydrolyzing chitin of fungal cell wall. We have developed transgenic *B. juncea* plants overexpressing chitinase gene. Transgenic plants are at the T<sub>2</sub> stage and have been analyzed by RT-PCR. The fold change of the BjChi-IV gene in

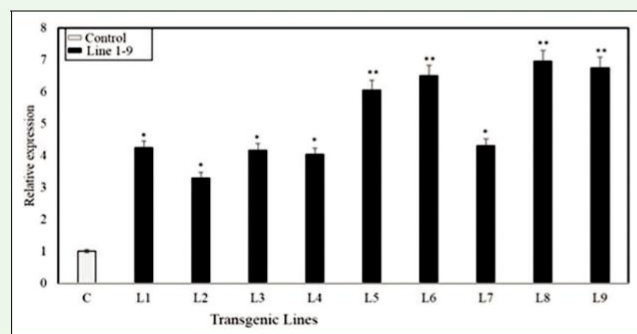


Fig. 5.13: RT-PCR analysis of transgenic T<sub>2</sub> *B. juncea* (chitinase-IV gene) plants. (\*P < 0.05; \*\*P < 0.01).

*B. juncea* wild type and 35s-BjChi-IV overexpressed *B. juncea* plants, high accumulation was observed in transgenic lines 5, 6, 8, and 9 while remaining lines (1, 2, 3, 4 and 7) revealed relatively low expression levels of BjChi-IV (Fig. 5.13).

## Enzyme assays of T<sub>2</sub> Transgenic *B. juncea* (chitinase-IV gene) Plants

Four transgenic lines were selected on RT-PCR expression results to carry out the activity of different enzymes after *A. brassicae* infection. At the time of infection by necrotrophic fungal pathogens plants activate different enzymes to counteract the incoming pathogen, such as Superoxide Dismutase (SOD), Glutathione Reductase (GR), Catalase (CAT) and Peroxidase (POD) are the most important in neutralizing the effect of superoxide (O<sub>2</sub><sup>-</sup>), or its dismutation product hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>). The enzyme activity was higher in Line 5, 6, 8 and 9 at 24 hr hpi and there was a decline in the activity at 48hr hpi (Fig.5.14).

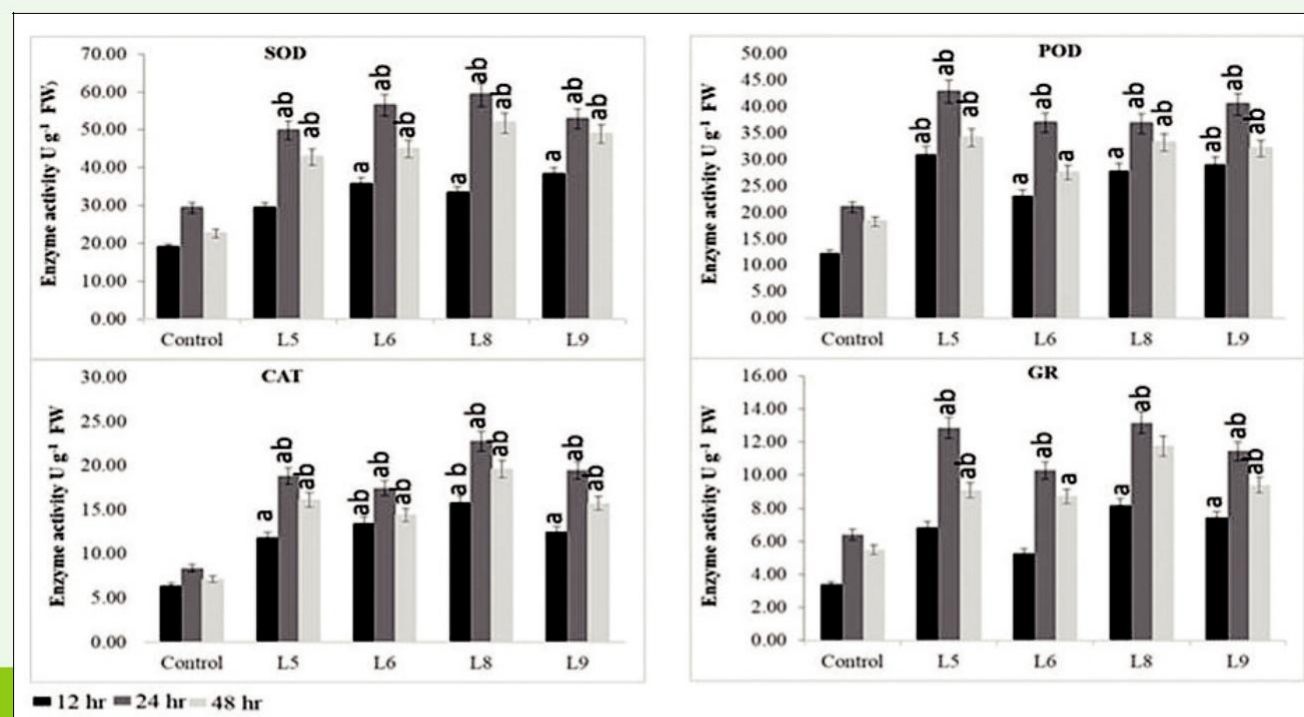


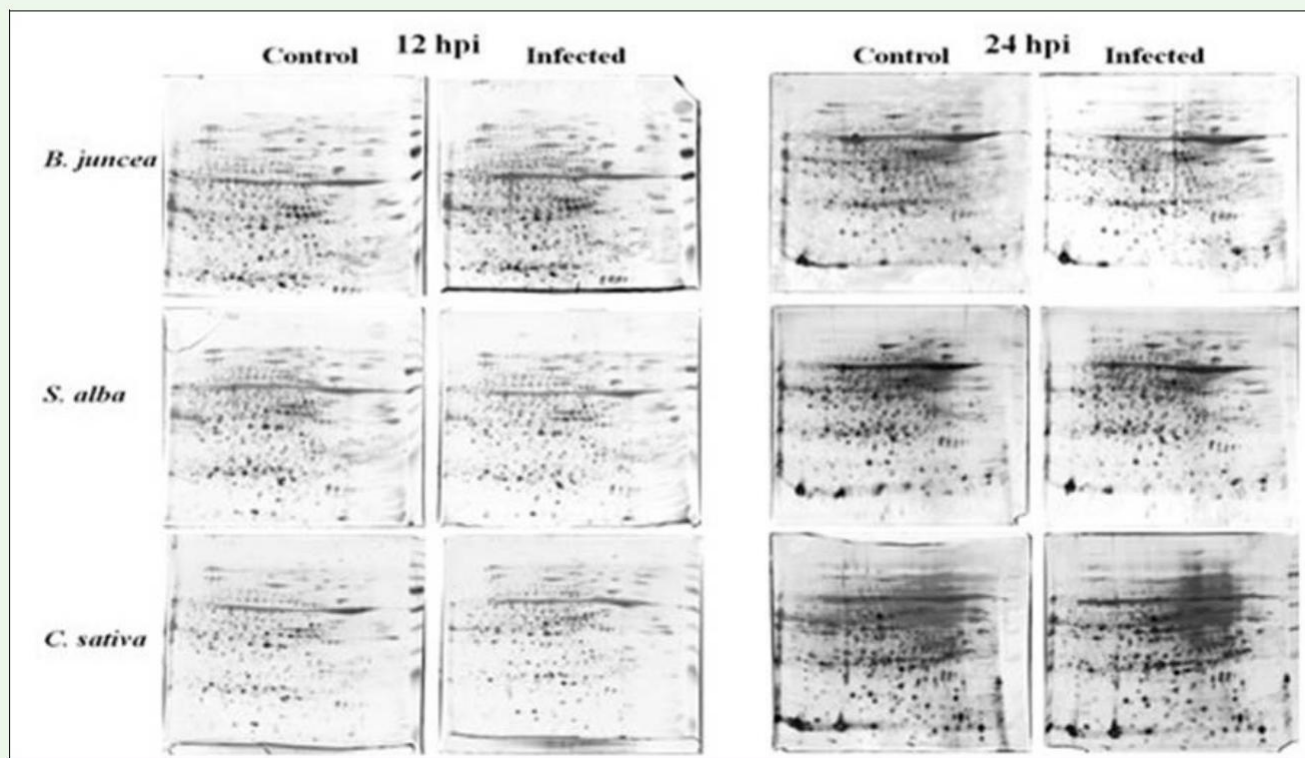
Fig. 5.14: Enzyme assays of T<sub>2</sub> Transgenic *B. juncea* (chitinase-IV gene) Plants (SOD, POD, CAT & GR).

## Protein expression analysis of *Alternaria* infected samples from *B. juncea*, *S. alba*, and *C. sativa*

(Prashant Yadav and Anita Grover)

Proteins were isolated from the *Alternaria* infected leaves of the *B. juncea*, *S. alba*, and *C. sativa*.

2-D gel electrophoresis was performed by first separating the proteins according to their isoelectric point (IPG; pH=4-7) and then in the second dimension by SDS-PAGE. Gels were stained by MS-compatible silver staining. Significant spots will be analyzed by MALDI-TOF (Fig. 5.15).



**Fig. 5.15:** Protein expression study through 2D profiling of proteins from 12 hpi (left) and 24 hpi (right) isolated from the leaves of *B. juncea*, *S. alba*, and *C. sativa* after *Alternaria* infection.

## Identification of the genes and pathways introgressed from wild species *Diplotaxis eruroides* in *Brassica juncea* for developing *Alternaria* resistance

(Naresh Vasupalli, Sharani Choudhary, RC Bhattacharya and Mahesh Rao)

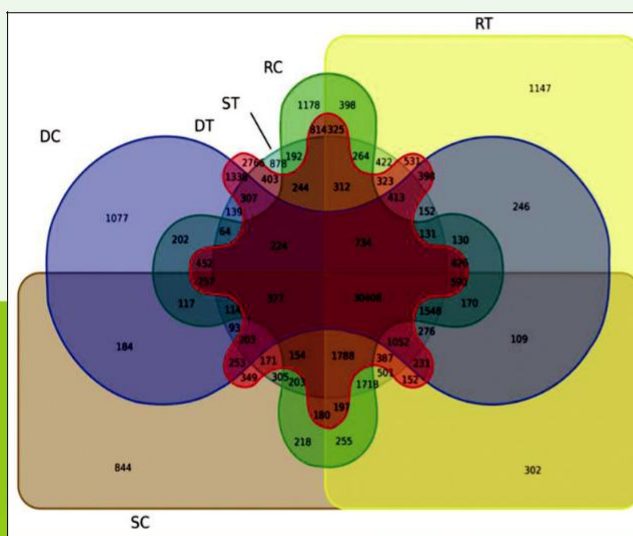
*Alternaria* leaf spot, caused by *A. Brassicae*, is widespread among the major diseases of *B. juncea* and economically very significant inflicting 35-47% yield loss. The wild species *Diplotaxis eruroides* has a high level of resistance to *A. brassicae*. For introgressing resistance genes from *D. eruroides* to *B. juncea*, an advanced backcross population (BC2-F6-7) was developed from interspecific hybrids involving *D. eruroides* and *B. juncea*. This introgression population has been subjected to extensive phenotyping for the *Alternaria* resistance under natural infection as well as artificial epiphytotic condition. As a result,

promising introgression lines (ILs) with a high level of *Alternaria* resistance have been identified. Having these extremely potential ILs, it will be pivotal to generate molecular markers linked to *D. eruroides* chromosomal segments and the resistance trait in these ILs for employing MAS. For generating markers low coverage whole genome sequence data was developed for NILs (stable resistance (R) line and susceptible (S) line) segregating only for the resistance trait. A total of 490,297,347 & 454,773,567 quality reads were developed with an average length of 114 & 134.1 for R and S line respectively. The *de novo* assembly of these reads developed a total of 670,851 and 343,363 contigs for R and S line respectively. In these contigs, a total of 268,156 and 196,536 genes were predicted in the R and S line contigs, respectively. Subsequently, a total of 53,282 and 26,598 contigs were identified as specific to R

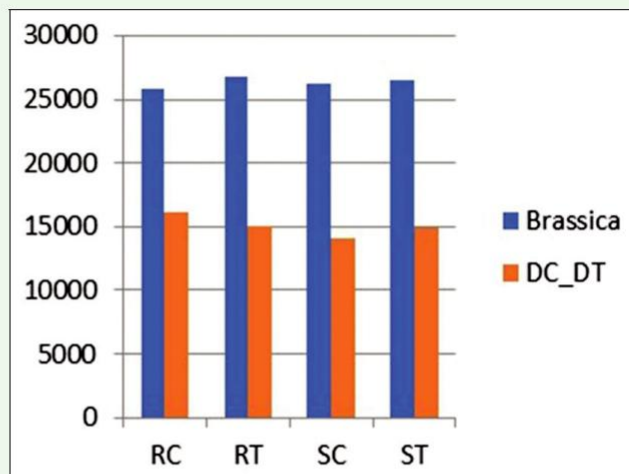
and S line, respectively. In these contigs, a total of 1,424 genes were predicted specific to the R line compared to S line and 1,051 genes were specific to S line in comparison to R line. The *de novo* assembly of *D. erucoides* RNA-Seq data was used to identify the expressed genes in 1,424 R line-specific genes. A total of 64 genes were identified as common genes between these two. Gene-specific primers will be synthesized for these 64 genes and will be used for the identification of markers associated with *Alternaria* resistance.

Transcriptome data generation from those stable introgressed line along with resistant parent (*D. erucoides*) followed by analysis has been undertaken with an aim to identify genes and pathways involved in resistance (Fig. 5.16). A *Diplotaxis* (the resistant parent) reference transcriptome was prepared from *Diplotaxis* control and *Alternaria* infected plants. Contigs from resistant introgressed line (RC & RT) and susceptible introgressed line (SC & ST) were mapped to both *Diplotaxis* transcriptome reference and Brassica database to categorize them into *Diplotaxis erucoides* origin and *Brassica juncea* origin (Fig.5.17).

Differentially expressed genes (fold change >2, p-value>0.05) were mined from *Diplotaxis*-originated genes. Each of them was assigned to a particular biological process and network analysis was carried out. Final selection of genes based on in silico analysis and validation through molecular analysis is well underway.



**Fig. 5.16:** Pictorial representation of common and unique genes present in the following 6 samples: RT: Resistant treated, RC: Resistant control, ST: Susceptible treated, SC: Susceptible control, DT: *Diplotaxis* Treated, DC: *Diplotaxis* control.



**Fig 5.17:** Categorization of genes among *B. juncea* and *D. erucoides* origin in 2 introgressed lines in both control and *Alternaria* infected condition

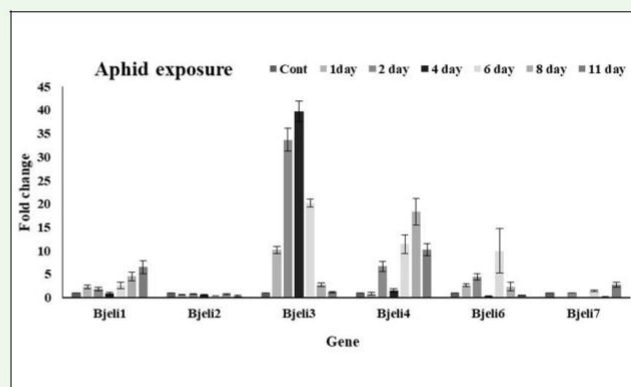
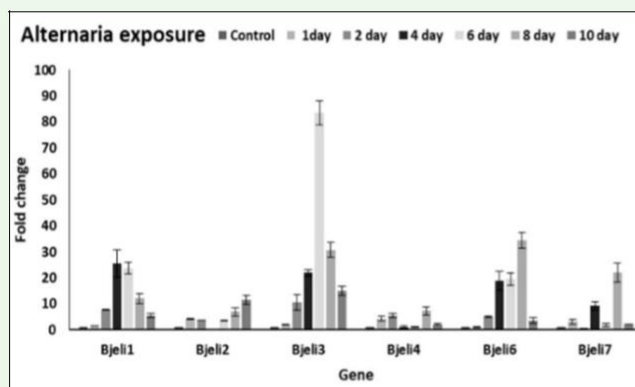
### Peptide elicitors of plant immunity in *B. juncea*

(Chetna Agarwal, Murali Krishna Koramutla and RC Bhattacharya)

Plant elicitor peptides constitute primary defense signal that prompt and intensify the first line of inducible plant defense. Endogenous elicitors initiate a cascade of defense signaling which culminates into the production of defense proteins and secondary metabolites, leading to resistance. Therefore, this host-derived peptide elicitors, if identified could be deployed for activation of quantitative defense against pest and pathogens in crop plants. Elicitor peptides were mined from *B. juncea* genome through bioinformatics tools by using known elicitor sequences from *Arabidopsis*. A total of 14 elicitor peptides were identified based on a sequence similarity matrix. On the basis of their amino acid identities and similarities, all the predicted fourteen paralogues encoding elicitors were divided into different groups naming Bjeli1 to Bjeli7. Most of the defense-related genes are induced by the activation of different signaling molecules. The activation pattern of these Bjelis was studied in response to the application of MeJ, SA, ABA and wounding in *B. juncea* in a time course manner. All the paralogues of Bjeli were induced by application of above-mentioned treatments and conditions except the paralogue Bjeli5 which was not inducible. In the case of wounding activation was more prominent in Bjeli3. Overall induction of Bjeli3 was found to be more among all the Bjelis in all the treatments. Further transcriptional activation of Bjeli3 was studied under *Alternaria* infection and aphid infestation in a

time course manner. Interestingly, among all Bjelis, Bjeli3 was found to be most induced on the sixth day in *Alternaria* treatment while maximum induction

was found on the fourth day in aphid infestation (Fig. 5.18). Our results indicating a relationship of this gene and its encoded peptide in plant defenses.



**Fig. 5.18:** Time course induction of elicitor paralogs in A) *Alternaria* treatment B) Aphid treatment. *Brassica juncea* leaves were harvested at 1, 2, 4, 6, 8 and 10 days after *Alternaria* treatment and at 1, 2, 4, 6, 8 and 11 days after Aphid infestation.

### Introgression of gene (s) for *Alternaria* tolerance from wild species into *Brassica juncea*

(Anamika Kashyap, NC Gupta, Ashish Kumar, KK Singh, Naveen Singh, Rohit Chamola, Lakshman Prasad, Usha Pant, RC Bhattacharya and Mahesh Rao)

The introgression line (BC2F9) developed in *Brassica juncea* for *alternaria* resistance from

wild species *Diplotaxis erucoides* were subjected for screening at ICAR-NIPB, Delhi under control condition and at GBPUAT, Pantnagar and IARI-RS PUSA Bihar under natural condition (Fig. 5.19). The scoring was done and the resistant lines/plants were identified from all the locations. The stable lines for the resistance were identified for the further studies.



**Fig. 5.19:** screening of the *Brassica juncea* introgression lines at GBPUAT, Pantnagar. C: control *B. juncea*; S: Susceptible ILs; R: Resistant ILs.

### Enlarging gene pool of *Brassica juncea* through wide hybridization

(Mahesh Rao, Mridula Pandey, NC Gupta, Rohit Chamola, Naveen Singh and RC Bhattacharya)

The genetic diversity is always mandatory for the improvement of crops and in case of *Brassica juncea* the genetic diversity in the native gene pool is very low. Total 92 resynthesized *Brassica juncea*

lines (RBJ-1 to RBJ-92) were resynthesized using parental diploid species *B. rapa* and *B. nigra*. Also, one population consisting 156 lines of introgression (ERJ-1 to ERJ-156) were developed using wild species *Diplotaxis eruroides*. These two materials are under phenotypic and genotypic evaluation in the subsequent generation. The genetic diversity for the resynthesized *Brassica juncea* lines (RBJ-1 to

RBJ-92) along with *B. rapa* (12 accessions), *B. nigra* (6 accessions) and ten popular Indian mustard varieties was done using SSR markers. The unrooted *neighbor-joining* dendrogram (Fig. 5.20) depict that the resynthesized *B. juncea* lines are having good genetic diversity and it will help in the improvement of the Indian mustard varieties.

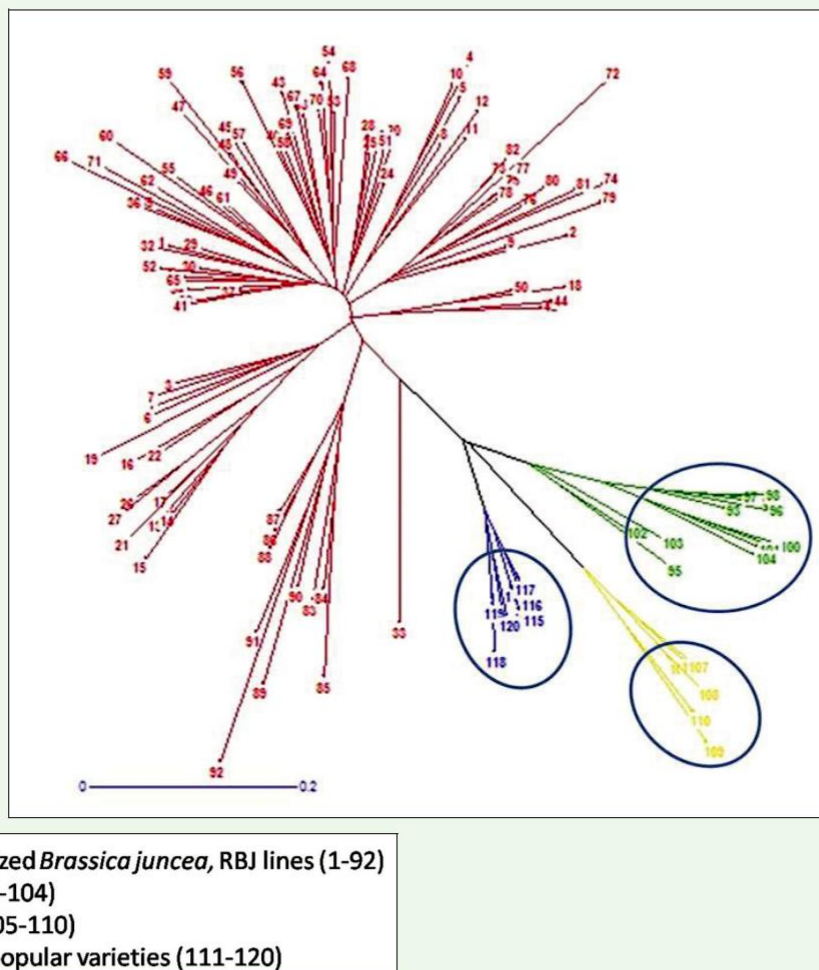


Fig. 5.20: Unrooted *neighbor-joining* dendrogram among 120 *Brassica juncea* lines.

The exploitation of progenitor species and wild relative will always help in the enhancing the genetic diversity of Indian mustard. During 2018-19,  $F_1$  derived from *B. rapa* x wild relatives (9 crosses); *B. rapa* x *B. nigra* (37 crosses) from 2017-2018 were subjected for colchicine treatment for development of synthetic amphidiploids. The new crosses were also attempted this year between *B. rapa* x *B. nigra* (50 crosses), *B. rapa* x wild relatives (11 crosses), *B. rapa* x *B. oleracea* (17 crosses) (reciprocal also in each cross combinations) with the objective of development of synthetic *B. juncea*, introgression from wild species and synthesis of *B. napus*.

Apart from these, important genetic stocks of *Brassica spp.* viz., forty six accessions of *B. nigra*, 68 accessions of *B. rapa*, twenty six species of *Brassica* wild species with different no of accessions in each species, two synthetic amphidiploids developed using wild species and *Brassica rapa* and thirteen CMS lines of *Brassica spp.* (twelve in *Brassica juncea* and one in *Brassica oleracea* background) are maintained every year and using these genetic stocks as and when required according to the experiment (the detailed information is available with ICAR-NIPB, Delhi).

## Novel germplasm registered

(Rashmi Yadav, Nanjundan J, AK Tewari, PS Sandhu, Mahesh Rao, Laxman Prasad, Sandeep Kumar, Vikendra Kaur, Ashok Kumar, JC Rana and Radhamani J)

Two germplasm of *B. juncea* IC265495 and IC313380 were registered with registration no. INGR18032 and registration no. INGR18033 respectively for white rust resistance. These germplasms gave resistant response against the different isolates of white rust tested. These can be used as the potential source for the development of the resistant cultivar in the white rust affected areas. The scientist from ICAR-NIPB was a collaborator in this novel germplasm.

## Pathotyping of Delhi isolate of *Albugo candida* causing white rust disease on rapeseed-mustard

(Mahesh Rao, Jameel Akhtar, Arun Kumar, RC Bhattacharya and Ashish Kumar)

The white rust pathogen, *Albugo candida* (Delhi isolate) was purified using single pustule isolate and maintained on seedlings of susceptible cv. 'Varuna' under controlled environmental conditions at National Phytotron Facility, IARI, New Delhi. Inoculation method was optimized and incubation period for resulted infection with 8-9 days, 11-15 pinheads/100 mm<sup>2</sup> area and 30-80% disease severity was observed on the inoculated plants. The pustules were observed as white smooth and rough surface, circular to oval in shape, coalesced and depressed at the center. All the plants inoculated at cotyledonary and true leaf stages were found infected having 40-80% leaf area covered with rusty pustules. Under the polyhouse condition, infection initiated on 8-9 days of pathogen inoculation along with 40-60% disease severity. However, 13.5% of plants were showed systemic infection and resulted in stag head formation.

## Evaluation of wild and released cultivars/ germplasm/introgressed lines of *Brassica* species

(Ashish Kumar, Mahesh Rao, Anamika Kashyap, NC Gupta, Jameel Akhtar and RC Bhattacharya)

*Brassica* wild relatives *Brassica fruticulosa*, *B. tournefortii*, *Camelina sativa*, *Diplotaxis assurgens*,

*D. catholica*, *D. cretacia*, *D. eruroides*, *D. muralis*, *D. tenuisiliqua*, *D. viminea*, *Erucastrum canariense*, *E. abyssinicum*, *Lepidium sativum* were found resistant against Delhi isolate of *A. candida*. Moreover, these *Brassica* wild relatives will further be tested against other prevalent isolates of *A. candida*.

Among various released cultivars (35), GSC-7 of *B. napus* and PC-6 of *B. carinata* were found resistant at both cotyledonary and true leaf stages of the plants against Delhi isolate of *Albugo candida*, while none of the *B. juncea* released cultivars found resistant against the tested isolate of the pathogen.

A total of 35 *Brassica* species germplasm obtained from NBPGR were tested for resistance against white rust pathogen (Delhi isolate). Among them, 10 accessions of *B. juncea* viz., EC 766272, 766230, 766164, 766152, 766145, 766133, 766134, 766136, IC 313380 and 597932 and four accessions of *B. carinata* viz., EC 206651, IC 341170, 555891 and EC 206642 found resistant at true leaf stage against the white rust pathogen.

Among 26 accessions of *Lepidium* and five *B. tournefortii* species tested against the white rust, three accessions, IC 508401, 508406 and 605224 of *Lepidium* Species found completely resistant at both cotyledonary and true leaf stages. However, none of the *B. tournefortii* accessions found resistant at any of the two growth stages.

Advanced germplasm/introgressed *Brassica* lines (760) were evaluated against *A. candida* Delhi isolate under controlled environmental condition (Phytotron chamber). Among 760 tested germplasm lines, 11 lines namely, RBJ 37, 40, 56; BR 4; IL 18; ERJS 92, ERJ 2, 6, 33, 37, 110 were resistant at cotyledonary leaf stage whereas, RBJ 41, 42, 60; WRR 1, 2, 26, 43; ERJS 4, 34, 37, 44, 45, 54, 67, ERJ 3, 4, 18, 44, 52, 63, 72, 80, 90, 94, 109, 112, 117, 137, 138, 142, 145, 152, 157, 167 found resistant only at true leaf stage of the plants. However, germplasm lines; ERJS 30, 35, 38, 39 and 78 were found completely resistant at both cotyledonary and true leaf stages of the plants (Fig. 5.21). Rest of the germplasm/advanced introgressed lines showed susceptible reaction with varying level of disease severity ranging from 30-100%.

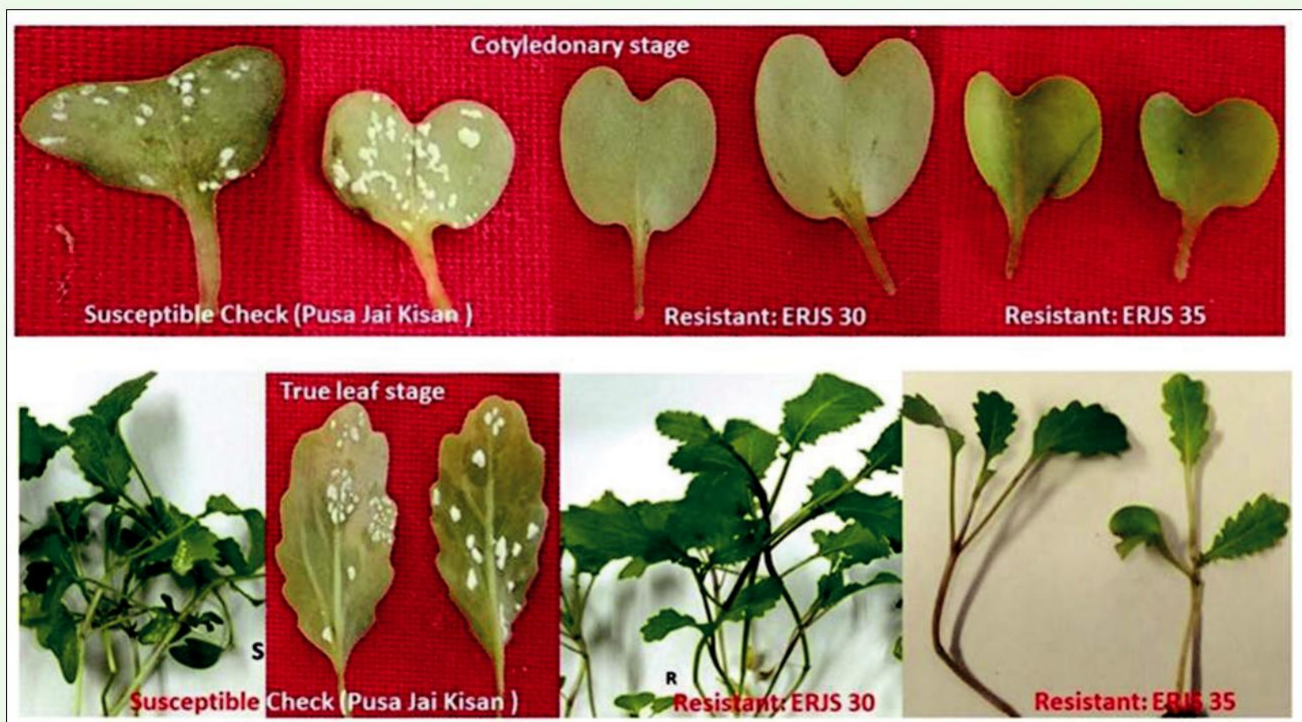


Fig. 5.21: Screening of introgressed lines of *Brassica* species

#### Quantification of defense/antioxidant pathway enzymes present in resistant/tolerant and susceptible germplasm of *Brassica* species (Biochemical Analysis)

(Ashish Kumar, Mahesh Rao, Sandip Kumar and Jameel Akhtar)

Biochemical analysis viz., catalase (CAT), ascorbate peroxidase (APOX), glutathione reductase (GR), Guaiacol peroxidase and SOD were carried out to gather information on the changes in defense/antioxidant pathway enzymes in different resistant/tolerant and susceptible germplasm/introgressed lines of *Brassica* species in response to infection with Delhi isolate of *A. candida*.

Inoculation of *A. candida* on *Brassica* leaves resulted in higher accumulation of defense enzymes viz., CAT, APOX, GR and peroxidase in healthy leaves of resistant genotypes as compared to susceptible genotypes which probably play a greater role in inbuilt resistance system in plants. The general positive correlation was observed between increasing levels of these enzymes in host tissues with higher resistance and pathogen incompatibility.

#### Development of *Sclerotinia* stem rot disease resistance in Indian mustard (*Brassica juncea*)

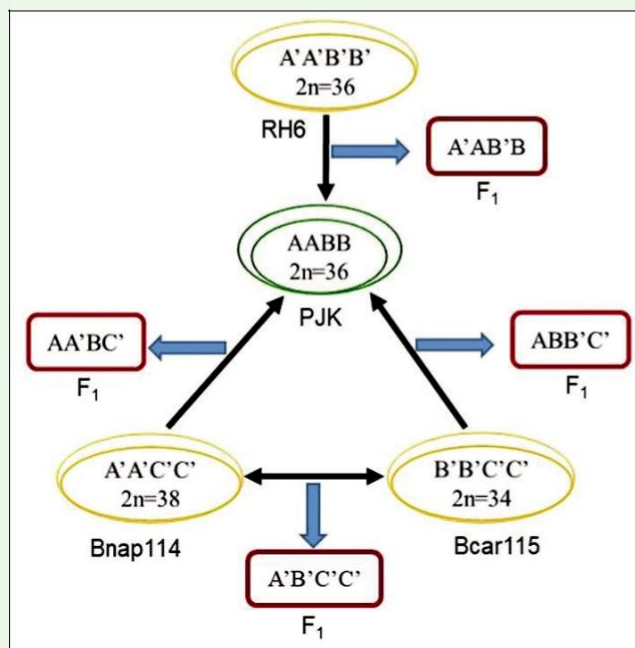
(Navin C Gupta, Rajesh Kumar, Mahesh Rao, Rohit Chamola, Aashish Kumar and RC Bhattacharya)

Stem rot disease caused by *Sclerotinia sclerotiorum* (Lib.) de Bary is one of the most devastating diseases of dicotyledonous species including oilseed *Brassica* and causes severe yield loss annually. Currently breeding for *Sclerotinia* stem rot disease resistance in oilseed rely solely on the availability of germplasm. But the lack of resistance in cultivated species and narrow range of available *Brassica* germplasm prompted to transfer the trait of interest and widen the genetic variability through the intergeneric and interspecific hybridization approaches.

#### Interspecific crosses for transferring the *Sclerotinia* stem rot disease resistance in rapeseed-mustard (*B. juncea*)

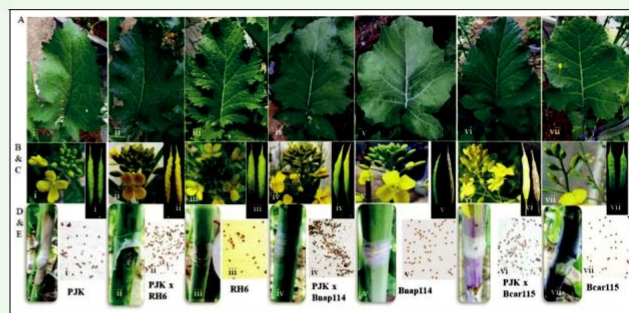
The previously identified resistance source for *Sclerotinia* stem rot disease in one line from each *B. carinata* (Bcar115), *B. napus* (Bnap114) and *B. juncea* (RH1222-28/RH-6), an allotetraploid with B and C genomes (BBCC, 2n=34), with A and C genomes

(AACC,  $2n = 38$ ), and with A and B genomes (AABB,  $2n = 36$ ), respectively, were used as donor parent to transfer the resistance trait into one of the elite and susceptible cultivar of Indian mustard, Pusa Jaikisan (PJK) (Fig. 5.22). In the other effort, both the resistant parent *B. napus* and *B. carinata* were used in reciprocal crosses to combine the resistance trait for Sclerotinia stem rot disease into  $F_1$  (Fig. 5.22).



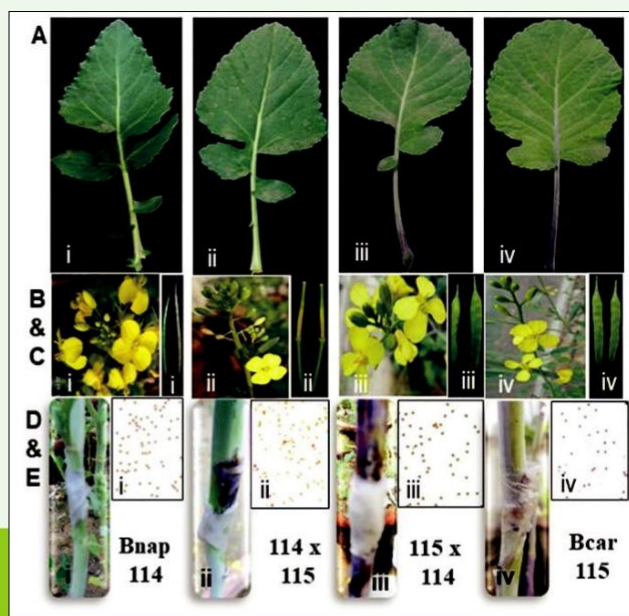
**Fig. 5.22:** Schematic diagram of the interspecific hybridization between *B. juncea* x *B. napus*, *B. juncea* x *B. carinata*, *B. juncea* x *B. juncea* (RH6) and reciprocal crosses between *B. napus* & *B. carinata*. Black arrows depict the hybridization between pollen (resistant/ tolerant) and pistillate parents (recessive) and blue arrow depicts the outcome of the interspecific crosses.

In the crosses of PJK x *B. napus*, PJK x *B. carinata*, PJK x RH6, 22, 25, and 34, hybrid lines respectively, were obtained and grew up to maturity. Similarly in case of reciprocal crosses between *B. napus* and *B. carinata* 12 and 01 hybrid lines were grew till maturity. All these interspecific hybrids produced were analyzed for morphological variation and viability of the pollens. The  $F_1$  progenies obtained from different crosses were found intermediate to the parents in many of the morphological characters such as leaf shape, leaf margin, flower color, stalk and silique length, etc. (Fig. 5.23 and 5.24). The pollen viability test as shown in Fig. 5.23E-i, ii, iii, iv, v and vii has 100% fertility except in case of PJK X *B. carinata* where ~15% sterility was observed (Fig 5.23E-vi).



**Fig. 5.23:** Morphological variation and stem rot disease assay of the hybrids and their parental lines. (A) Leaf, (B) Inflorescence and flower, (C) Siliques, (D) Sclerotinia stem rot disease assessment 21 dai, and (E) Pollen viability. Numerals represents (i) Pusa Jaikisan, (ii) PJK x RH6, (iii) RH6, (iv) PJK x *B. napus*, (v) *B. napus*, (vi) PJK x *B. carinata*, and (vii) *B. carinata*.

The crossability between *B. napus* and *B. carinata* and vice versa were found normal when *B. carinata* was taken as the female parent and pollen fertility is also remain intact in this case but 100% pollen sterility was observed when *B. napus* were taken as a female parent (Fig. 5.24E-iii). Additionally, the silique morphology as shown in Fig. 5.24C-ii, due to self-incompatibility the siliques remained rudimentary in the case where *B. carinata* was used as the male parent.



**Fig. 5.24:** Morphological variation and stem rot disease assessment in the reciprocal hybrids between *B. napus* and *B. carinata*. (A) Leaf, (B) Inflorescence and flower, (C) Siliques, (D) Assessment of Sclerotinia stem rot disease (21 dai), and (E) Pollen viability. Numerals represent (i) *B. napus*, (ii) *B. napus* x *B. carinata*, (iii) *B. carinata* x *B. napus*, and (iv) *B. carinata*.

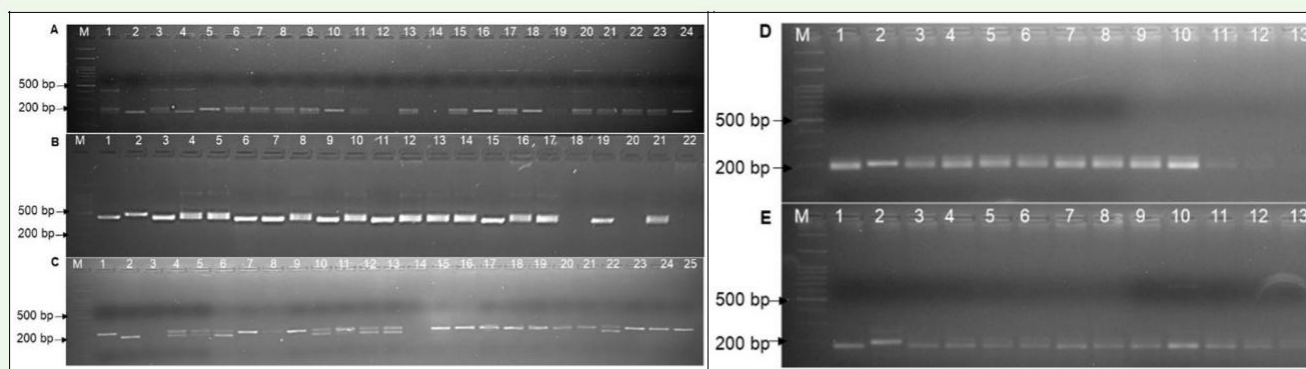
## Sclerotinia stem rot disease assessment

The parental lines along with their hybrids in F1 were evaluated for *Sclerotinia* stem rot disease by the non-injury method of stem inoculation assay under field condition and recorded the disease development in terms of lesion length after 21 days of inoculation (dai). The resistant lines RH6 (Fig. 5.24E-iii), Bnap114 (Fig. 5.24E-v), and Bcar115 (Fig. 5.24E-vii) have developed either scar marks or lesion length up to 5 cm at the site of infection whereas in susceptible PJK line the lesion developed throughout the stem (Fig. 5.24E-i) even after 7 days of inoculation. In all the hybrid progenies, the disease resistance was observed successfully transferred from the donor parent as the lesion length measured was found 0.5 cm to 5 cm. In reciprocal crosses between Bnap114 and Bcar115, the diseased lesion measured in parents and their hybrids were just a scar mark over the site of infection (Fig. 5.24E-i-iv). The stem rot disease evaluation in hybrids corroborates the resistance pattern as of their parental lines. Hence, the

present work reinforces the facts that interspecific transferability of the disease resistance is feasible through the normal breeding procedure.

## Hybridity confirmation by SSR markers

Morphological studies followed by SSR fingerprinting of all the interspecific hybrids along with their parents have confirmed the hybridity in the derived population. Out of 369 markers tested, two sets for each cross combinations the polymorphic simple sequence repeat (SSR) markers were identified and used in fingerprinting of all the respective lines. The SSR markers 610, 7223 and 821 were identified as biallelic and utilized for confirmation of the hybrids generated from PJK x RH6, PJK x Bnap114, and PJK x Bcar115, respectively (Fig. 5.25A, B, & C). The SSR primers 160 and 610 were also observed biallelic and The F1 population of reciprocal crosses 12 in case of *B. napus* x *B. carinata* and 01 lines from *B. carinata* x *B. napus* were observed carrying both the unique parental alleles as depicted in Fig. 5.25D & E.



**Fig. 5.25:** Hybridity confirmation in F1 lines of the interspecific crosses through SSR markers. (A) SSR marker '610' polymorphism in population of PJK X RH6; Lane 1-PJK, Lane 2-RH6, and Lane 3-24-F1 hybrids, (B) SSR marker '7223' polymorphism in population of PJK X Bnap114; Lane 1-PJK, Lane 2-Bnap 114, and Lane 3-22-F1 hybrids, (C) SSR marker '821' polymorphism in population of PJK X Bcar115; Lane 1-PJK, Lane 2- Bcar115, and Lane 3-25-F1 hybrids, (D) SSR marker '160', and (E) SSR marker '610' polymorphism in population of reciprocal crosses of Bnap114 x Bcar115; Lane 1-Bnap114, Lane 2-Bcar115, and Lane 3-12 F1 hybrids of *B. napus* x *B. carinata* and Lane 13: one hybrid line of *B. carinata* x *B. napus*. Lane M: 100 bp DNA ladder.

The transfer of resistance trait has been found in all the crosses and thus the *in vivo* stem inoculation of the hybrids showed resistance/tolerance to the *S. sclerotiorum* pathogen. The observation of the silique development and seed setting indicated the bilateral interspecific compatibility in case of *B. juncea* with *B. juncea*, *B. napus* and *B. carinata* whereas unilateral interspecific compatibility in the crosses of *B. carinata* with *B. napus*. The number of the hybridembryos

obtained has varied in different combination of crosses and that depends on the subspecies used and the direction of the crossing. The Interspecific hybrids developed and confirmed above in case of RH6 x PJK, PJK x *B. napus* and PJK x *B. carinata* were used to develop the F2 recombinant lines and the hybrids were also utilized in test crosses to advances the back cross lines towards near-isogenic lines (NILs) development.

## Construction of CRISPR/Cas9 and GFP-tailswap vectors for developing haploid inducer line(s) in *B. juncea* and *B. oleracea*

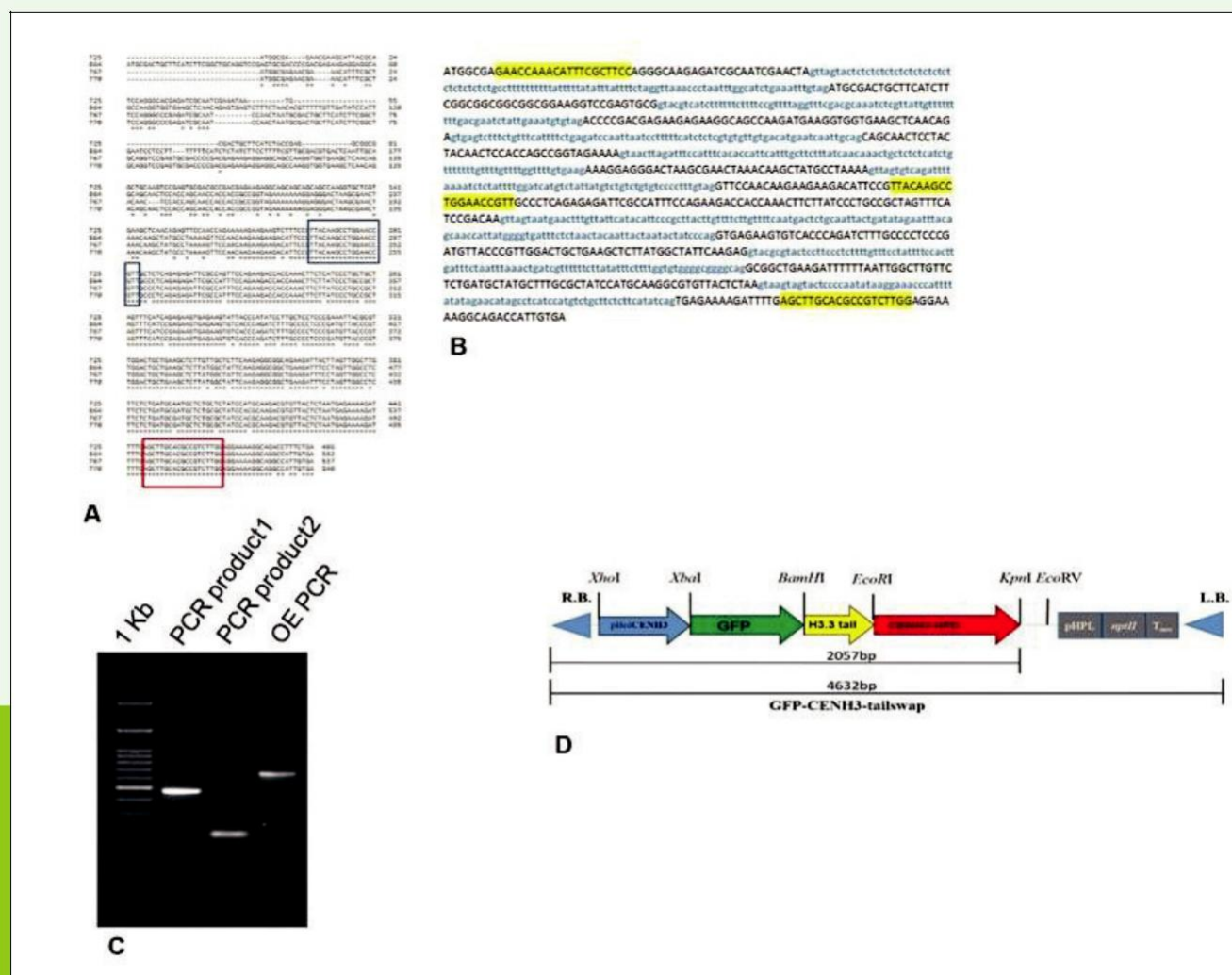
(Ritesh Kumar Raipuria, Ashu, Ramcharan Bhattacharya and Anshul Watts)

Since *B. juncea* contains three *CENH3* copies (KR676378, KU837267, and KR676374) and four transcripts (KU837265, KU837264, KU837266, and KR676381), therefore sgRNAs targeting each copy were predicted. Further, two sgRNAs were selected targeting all three *CENH3* copies (Fig. 5.24A). Similarly, three different sgRNAs targeting different regions (5', mid and 3' end) of *B. oleracea* (Pusa Meghna) *CENH3* gene (Bol018461) were designed (Fig. 5.26B).

The binary vector pORE:O4 was modified by cloning the CaMV35S promoter and Cas9 gene between *EcoRI*-*Clal* and *Clal*-*SpeI* sites, respectively. One more cassette named guide-cassette containing

AtU6 promoter, SgRNA, scaffold, and terminator were synthesized for the transcription of sgRNA and cloned between *SacII*-*BamHI* restriction sites. Now, this modified vector can be used to knockout different genes by simply replacing the sgRNA of targeting gene in the existing guide-cassette. Overlap extension PCR strategy was used to clone the sgRNA for *CENH3* gene in this vector. Additionally, two CRISPR/Cas9 vectors which have two different sgRNA were synthesized for knockout of *CENH3* gene in *B. juncea* and *B. oleracea*.

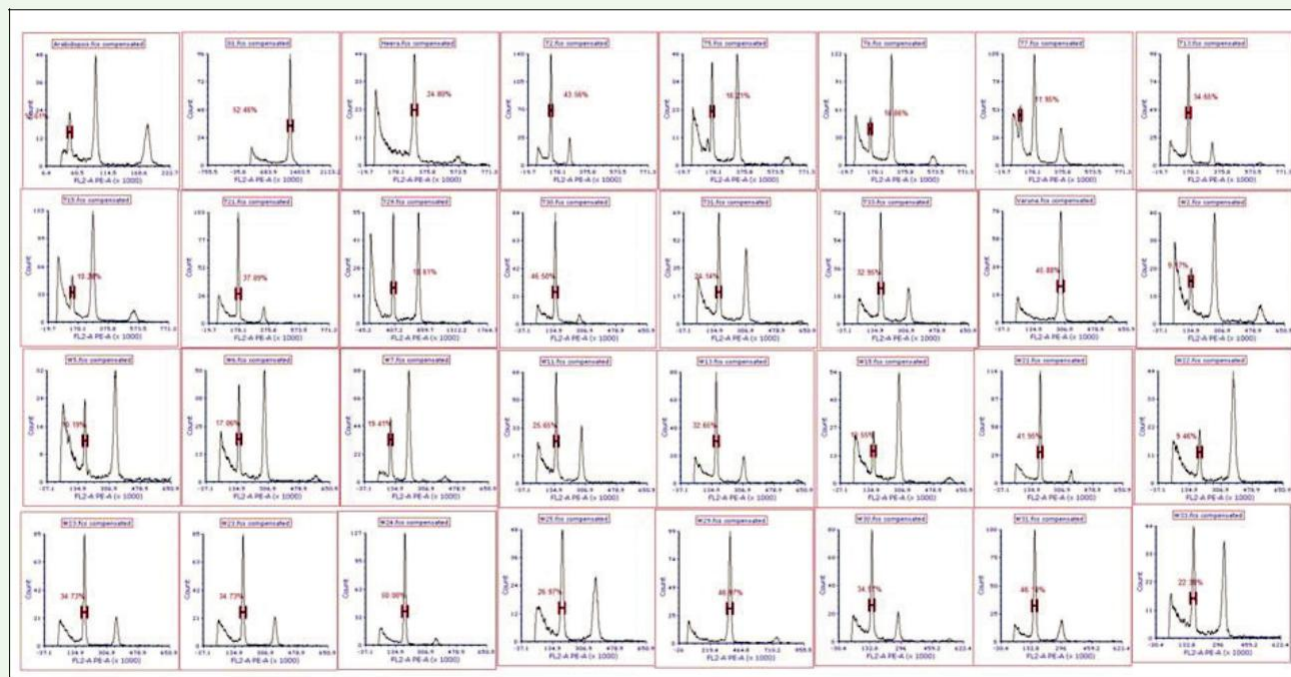
Since *CENH3* is essential and knockout of this gene is lethal. Therefore, a GFP-tail swap construct was made by cloning four different fragments; *B. oleracea* *CENH3* promoter, a Green fluorescent protein (GFP), histone H3.3 tail, *CENH3* histone fold domain (Fig. 5.26D), sequentially in pORE:O4 binary vector.



**Fig. 5.26:** A) Position of two sgRNA which used to target the *B. juncea* *CENH3* gene. B) Position of three sgRNAs in *B. oleracea* *CENH3* gene. C) Overlap extension PCR used cloning of sgRNA. D) GFP-tailswap contain *B. oleracea* *CENH3* promoter+GFP+H3.3 tail+CENH3 HFD.

## Estimation of various wild Brassica genome size through flow cytometry

The genome size of various wild species was estimated using flow cytometry technique (Fig. 5.27 & Table 5.3).



**Fig. 5.27:** Estimation of genome size in various Brassica species through FACS. Where W2 and T2 are *B. fruticulosa*; W5 and T5 are *B. oxyrrhina*; W6 and T6 are *B. spinensis*; W7 and T7 are *B. tournefortii*; W11 is *Didesmus bipinnatus*; W13 and T13 are *Diplotaxis catholica*; W15 and T15 are *Diplotaxis eruroides*; W21 and T21 are *Enarthocarpus lyratus*; W22 is *Erucastum abyssinicum*; W23 is *Erucastum canariensis*; W24 is *Erucastum cardaminoides*; W25 is *Erucastum gallicum*; W29 and T29 are *Orychophragmus violaceus*; W30 and T30 are *Sinapis alba*; W31 and T31 are *S. arvensis*; W33 and T33 are *Trachystomaballi*.

**Table 5.3:** List of estimated genome size of the different *Brassica* species with fluorescence-activated cell sorting (FACS) technique.

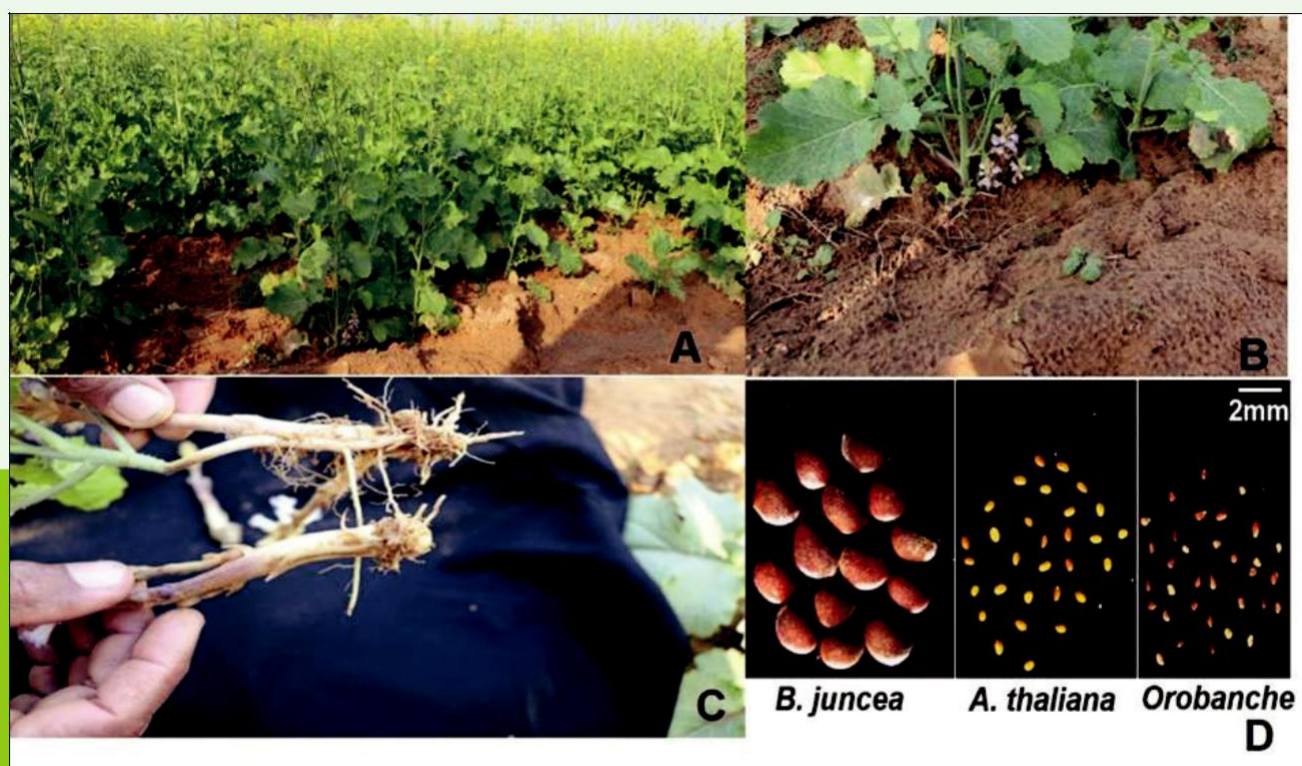
| Species name                | Median     | CV   | Genome size (Mb) | 2C Value (pg) |
|-----------------------------|------------|------|------------------|---------------|
| <i>Arabidopsis thaliana</i> | 45960.15   | 4.35 | 135              | 0.276         |
| <i>Orobancha ageyptica</i>  | 1161188.38 | 1.97 | 3411             | 6.97          |
| <i>Orobancha ageyptica</i>  | 1166761.88 | 1.61 | 3426             | 7.00          |
| <i>B. fruticulosa</i>       | 129823.90  | 4.24 | 381              | 0.78          |
| <i>B. fruticulosa</i>       | 125616.60  | 3.37 | 367              | 0.75          |
| <i>B. oxyrrhina</i>         | 168764.95  | 3.08 | 489              | 1.01          |
| <i>B. oxyrrhina</i>         | 162272.70  | 3.16 | 474              | 0.97          |
| <i>B. spinensis</i>         | 142718.20  | 3.49 | 421              | 0.86          |
| <i>B. spinensis</i>         | 137986.00  | 4.03 | 406              | 0.83          |
| <i>B. tournefortii</i>      | 105025.25  | 4.41 | 308              | 0.63          |
| <i>B. tournefortii</i>      | 90495.60   | 5.82 | 264              | 0.54          |
| <i>Didesmus bipinnatus</i>  | 141984.91  | 3.74 | 416              | 0.85          |
| <i>Diplotaxis catholica</i> | 156086.65  | 3.33 | 460              | 0.94          |
| <i>Diplotaxis catholica</i> | 155849.80  | 3.50 | 460              | 0.94          |
| <i>Diplotaxis eruroides</i> | 142817.70  | 3.93 | 421              | 0.86          |
| <i>Diplotaxis eruroides</i> | 134850.80  | 4.89 | 396              | 0.81          |

| Species name             | Median    | CV   | Genome size (Mb) | 2C Value (pg) |
|--------------------------|-----------|------|------------------|---------------|
| Enarthocarpuslyratus     | 177284.85 | 2.48 | 518              | 1.06          |
| Enarthocarpuslyratus     | 169760.95 | 3.29 | 499              | 1.02          |
| Erucastum abysinicum     | 187578.3  | 2.95 | 553              | 1.13          |
| Erucastum canariens      | 178868.59 | 2.90 | 523              | 1.07          |
| Erucastum cardaminoides  | 182347.80 | 2.85 | 538              | 1.1           |
| Erucastum gallicum       | 184266.70 | 3.57 | 543              | 1.11          |
| Orychophragmus violaceus | 368639.45 | 2.36 | 1081             | 2.21          |
| Orychophragmus violaceus | 375610.91 | 3.11 | 1105             | 2.26          |
| Sinapis alba             | 139331.80 | 3.89 | 411              | 0.84          |
| Sinapis alba             | 138273.40 | 3.70 | 406              | 0.83          |
| S. arvensis              | 147522.55 | 3.73 | 435              | 0.89          |
| S. arvensis              | 153810.85 | 3.71 | 450              | 0.92          |
| Trachystomaballi         | 162110.40 | 3.94 | 474              | 0.97          |
| Trachystomaballi         | 161162.80 | 3.63 | 474              | 0.97          |

## Understanding the interaction of *B. juncea* and *Orobanche ageyptica*

*Orobanche ageyptica* is a complete root parasite which can attack Cruciferae, Asteraceae, Fabaceae, and Solanaceae family. In India, *B. juncea* is the major crop which is highly affected due to its infestation.

So, we have collected *O. ageyptica* seed and *Orobanche* plants from the *B. juncea* infected plants from the farmer's field (Fig. 5.28) in the Charkhidadri district of Haryana. The seeds of *Orobanche* are smaller than *A. thaliana* seeds.



**Fig. 5.28:** *Orobanche* and *Brassica* interaction. A) *O. ageyptica* found in the *B. juncea* field; B) *O. ageyptica* near the *B. juncea* plant; C) *O. ageyptica* attached to *B. juncea* roots; D) Comparison of seed size of *O. ageyptica*, *A. thaliana* and *B. juncea*.

# **Human Resource Development**

**Post-graduate Teaching Programme**

**Training and Capacity Building**

# Post-graduate Teaching

NIPB has been actively engaged in human resource development in the area of plant Molecular Biology and Biotechnology since its inception. Currently 38 Ph. D. and 14 M. Sc. students are registered in the discipline of Molecular Biology and Biotechnology at the Centre. Four Ph. D. and seven M. Sc. students were awarded with doctoral and master's degrees, respectively, in February 2019.

## Students on roll in the discipline of Molecular Biology and Biotechnology during the Academic Session 2018-19

| S. No                  | Name of the student                    | Chairperson             |
|------------------------|----------------------------------------|-------------------------|
| <b>Ph. D. Students</b> |                                        |                         |
| 1.                     | Mr. Rakesh Bhowmick (10152)            | Prof. N. K. Singh       |
| 2.                     | Ms. Antara Das (10154)                 | Dr. Kishor Gaikwad      |
| 3.                     | Mr. Prashant Yadav (10155)             | Dr. Anita Grover        |
| 4.                     | Ms. Mamta Gupta (10217)                | Dr. Sarvjeet Kaur       |
| 5.                     | Mr. Albert Maibam (10315)              | Dr. Jasdeep C. Padaria  |
| 6.                     | Mr. Vinod Kumar Jangid (10318)         | Dr. Anita Grover        |
| 7.                     | Mr. Manoj M. L. (10495)                | Prof. N. K. Singh       |
| 8.                     | Ms. Shikha Dixit (10498)               | Dr. Anita Grover        |
| 9.                     | Mr. Panazade Kishor Prabhakar (10500)  | Dr. Jasdeep Padaria     |
| 10.                    | Ms. Priyanka (10501)                   | Dr. Kanika              |
| 11.                    | Mr. Pawan Sitaram Mainkar (10549)      | Dr. Rekha Kansal        |
| 12.                    | Mr. Lalbahadur Singh (10696)           | Dr. P. K. Jain          |
| 13.                    | Ms. Priyanka Singh (10697)             | Prof. N. K. Singh       |
| 14.                    | Mr. Chetan Kumar Nagar (10720)         | Dr. Pranab Kumar Mandal |
| 15.                    | Ms. Sharani Choudhury (10836)          | Dr. R. C. Bhattacharya  |
| 16.                    | Ms. Sreeshma N (10839)                 | Prof. N. K. Singh       |
| 17.                    | Mr. Kishor Uttamrao Tribhuvan (10841)  | Dr. Kishor Gaikwad      |
| 18.                    | Mr. Kuldeep Kumar (10842)              | Dr. Kishor Gaikwad      |
| 19.                    | Mr. Deepanshu Jayaswal (10843)         | Dr. Rekha Kansal        |
| 20.                    | Mr. Mahendra C (10844)                 | Dr. Kanika              |
| 21.                    | Ms. Gamage Dona Gaya Chaturani (10918) | Dr. Anita Grover        |
| 22.                    | Mr. Soham Choudhury (11068)            | Dr. Monika Dalal        |



| S. No                  | Name of the student                | Chairperson               |
|------------------------|------------------------------------|---------------------------|
| <b>Ph. D. Students</b> |                                    |                           |
| 23.                    | Mr. Sougata Bhattacharjee (11069)  | Dr. Debasis Pattanayak    |
| 24.                    | Ms. L. Ashakiran Devi (11070)      | Dr. R. C. Bhattacharya    |
| 25.                    | Ms. Bablee Kumari Singh (11071)    | Dr. S.V.A.C.R. Mithra     |
| 26.                    | Mr. Abinash Biswajit Sathy (11072) | Dr. Amolkumar U. Solanke  |
| 27.                    | Mr. Prashant R. Shingote (11073)   | Dr. Kanika                |
| 28.                    | Mr. Sachin (11074)                 | Dr. Pranab Kumar Mandal   |
| 29.                    | Mr. Suhas Karkute (11288)          | Dr. Amolkumar U. Solanke  |
| 30.                    | Mr. Niladri Barman (11289)         | Dr. Monika Dalal          |
| 31.                    | Mr. Bipratip Dutta (11290)         | Dr. S. V. A. C. R. Mithra |
| 32.                    | Ms. Sheel Yadav (11291)            | Dr. Pradeep Kumar Jain    |
| 33.                    | Mr. Akash Paul (11293)             | Dr. Kishor Gaikwad        |
| 34.                    | Mr. Saakre Manjesh (11294)         | Dr. Debasis Pattanayak    |
| 35.                    | Ms. Taku Monya (11295)             | Dr. Rhitu Rai             |
| 36.                    | Mr. Mawuli Kwamla Azameti (11362)  | Dr. Jasdeep C. Padaria    |
| 37.                    | Ms. Shaziya Sultana (11367)        | Dr. Sharmistha Barthakur  |
| 38.                    | Mr. Sandesh Tulsiram (11370)       | Dr. Anita Grover          |
| <b>M. Sc. Students</b> |                                    |                           |
| 1.                     | Mr. Muhammed Shamnas V (20994)     | Dr. Subodh Kumar Sinha    |
| 2.                     | Mr. Johan Ajnabi (20995)           | Dr. Monika Dalal          |
| 3.                     | Mr. Krishnayan Paul (20996)        | Dr. Debasis Pattanayak    |
| 4.                     | Mr. Naresh Kumar Samal (20997)     | Dr. R. C. Bhattacharya    |
| 5.                     | Mr. Megavath Ravi Naik (20998)     | Dr. Pranab Kumar Mandal   |
| 6.                     | Mr. Zaherul Islam (20999)          | Dr. Tapan Kr. Mondal      |
| 7.                     | Mr. Somya Gupta (21043)            | Dr. Sarvjeet Kaur         |
| 8.                     | Mr. Ankan Chakraborty (21148)      | Dr. Monika Dalal          |
| 9.                     | Mr. Samar Deb (21149)              | Dr. Tapan K. Mondal       |
| 10.                    | Ms. Priyanka Kumari (21150)        | Dr. Subodh Kumar Sinha    |
| 11.                    | Mr. Jeet Roy (21151)               | Dr. S. V. A. C. R. Mithra |
| 12.                    | Ms. Ashika Debbarma (21152)        | Dr. Sarvjeet Kaur         |
| 13.                    | Ms. Tamilarasi M. (21153)          | Dr. Kanika                |
| 14.                    | Mr. Sovanlal Sahu (21154)          | Dr. P. K. Dash            |

## Degrees awarded in the discipline of Molecular Biology and Biotechnology during the Convocation held in February, 2019

| S. No.                 | Name of the Student and Roll No. | Chairperson, Advisory Committee | Thesis Title                                                                                                                                                                          |
|------------------------|----------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ph. D.</b>          |                                  |                                 |                                                                                                                                                                                       |
|                        | Mr. H. H. Kumaraswamy (9864)     | Prof. N. K. Singh               | Study of population structure and phylogeny in Indian wheat landraces using DNA markers                                                                                               |
|                        | Mr. Deepak V. Pawar (10153)      | Prof. N. K. Singh               | Functional analysis of blast resistance gene Pi56or(t) from <i>Oryza rufipogon</i>                                                                                                    |
|                        | Mr. Alim Junaid (10314)          | Dr. Kishor Gaikwad              | Understanding the epigenetic regulatory networks in CMS based pigeonpea hybrid.                                                                                                       |
|                        | Mr. Rakesh K. Prajapat (10317)   | Dr. Rekha Kansal                | Comparative <i>in silico</i> analysis and functional validation of legume lectins for insecticidal activity                                                                           |
| <b>M. Sc. Students</b> |                                  |                                 |                                                                                                                                                                                       |
|                        | Mr. Niladri Barman (20837)       | Dr. Monika Dalal                | Cloning and characterization of drought inducible PM19 promoter from wheat                                                                                                            |
|                        | Mr. Sourav K. Das (20838)        | Dr. S.V.A.C.R. Mithra           | Marker development for major drought and heat stress tolerance QTLs and their validation in rice ( <i>Oryza sativa</i> L.)                                                            |
|                        | Mr. Bipratip Dutta (20839)       | Dr. Amolkumar U. Solanke        | Characterization of <i>Magnaporthe</i> -responsive <i>WRKY</i> genes in contrasting rice genotypes for panicle blast resistance                                                       |
|                        | Ms. Shaziya Sultana (20840)      | Dr. Sharmistha Barthakur        | Molecular cloning and gene expression profiling of (GPAT) (Glycerol-3-phosphate acyl transferase) under terminal heat stress in wheat ( <i>Triticum aestivum</i> L.)                  |
|                        | Mr. Akash Paul (20841)           | Dr. Subodh K. Sinha             | Co-expression studies of <i>NAR2.1</i> and <i>NAR 2.2</i> genes with high affinity nitrate transporter gene ( <i>NRT2.1</i> ) in root tissues of wheat ( <i>Triticum aestivum</i> L.) |
|                        | Mr. Sunil Ningombam (20842)      | Dr. Jasdeep C. Padaria          | Development of microsatellite marker associated with heat stress tolerance in <i>Pennisetum glaucum</i> (L.) R.BR                                                                     |
|                        | Ms. Taku Monya (20843)           | Dr. Rhitu Rai                   | Isolation, cloning and functional validation of a TAL like effector gene from <i>Xanthomonas oryzae</i> pv. <i>oryzae</i>                                                             |

## Training and Capacity Building

### Details of training by ICAR-NIPB staff during 2018-19

| S. No.                      | Name              | Subject Area                                       | Duration (Month of the year)  | Host Institute                 |
|-----------------------------|-------------------|----------------------------------------------------|-------------------------------|--------------------------------|
| <b>Scientific Staff</b>     |                   |                                                    |                               |                                |
| 1.                          | Dr. Anita Grover  | Advances in Microscopy                             | 3 days (January 17-19,2019)   | CIRCOT, Mumbai                 |
| 2.                          | Dr. Monika Dalal  | High Throughput Wheat Phenotyping                  | 2 days (September17-19, 2019) | University of of BOLOGNA Italy |
| 3.                          | Dr. Sandhya       | SHARED Communication Science, Health, Agriculture, | 5 days (January 14-18, 2019)  | NISCAIR, New Delhi             |
| <b>Technical Staff: Nil</b> |                   |                                                    |                               |                                |
| <b>Administrative Staff</b> |                   |                                                    |                               |                                |
| 1                           | Ms. Nidhi Nailwal | Assisstant                                         | NIPB                          | ISCM                           |
| 2                           | Mr. Mohit Sikka   | Assisstant                                         | NIPB                          | ISCM                           |

### HRD Fund Allocation and Utilization

| Total HRD fund allocation for 2018-19 (Rs. in Lakh) | Actual Expenditure for 2018-19 (Rs. In Lakh) | % utilization |
|-----------------------------------------------------|----------------------------------------------|---------------|
| 1.055                                               | 1.054                                        | 99.90         |



# **Extension and Outreach**

- 1. Mera Gaon Mera Gaurav**
- 2. Krishi Unnati Mela**

# 1. Mera Gaon Mera Gaurav

## Farmers Day:

The ICAR-National Institute for Plant Biotechnology, New Delhi organized Farmers Day under the innovative initiative “Mera Gaon Mera Gaurav” programme at Nekpur village, District Bulandshar, Uttar Pradesh on 6th October, 2018 in order to address the farmers need, their problems, and biotechnological intervention in agricultural research. About two hundred fifteen (215) farmers of MGMG adapted villages, viz. Nekpur, Khajpur, Nar-Mohammad, Agauta, Kakurghadhi, Badauth of

Mathura and Bulandshar districts of Uttar Pradesh, along with 36 scientific and technical staff had participated in the programme. The participating farmers discussed their day to day farming related difficulties in the meeting. Ten farmers were felicitated with Krishak Mitra Samman award by the Director, NIPB. He also distributed seed packets of the mustard variety, “Pusa Mustard 28”, along with bags and some literature materials related to NIPB technologies to the farmers.



## Kishan Goshties:

The ICAR-NIPB, New Delhi and Nand Educational Foundation for Rural Development (NEFORD), Lucknow, jointly organized National Kishan Goshti and seed distribution under Mera Gaov Mera Gaurav Programme on 13<sup>th</sup> October, 2018, at 10.30 AM in Dumraov village, Mau District, Uttar Pradesh in order to address the farmers' need, their problems, and biotechnological intervention in agricultural research of eastern Uttar Pradesh. Three hundred seventy (370) farmers of Mau, Azamgarh, Ghazipur,

Ballia, Gorakhpur districts of Uttar Pradesh along with four Directors of different agricultural institutes and scientists from ICAR-IARI, ICAR-NIPB, New Delhi and ICAR-IISS and ICAR-NBAIM, Mau participated in the programme. The dignitaries honoured twenty two farmers and two scientists, Dr. A. K. Singh and Dr. Mahesh Rao, with Krishak Mitra Samman, and distributed seed packets of mustard varieties, "Pusa Mustard 28" and "Pusa Jai Kishan", along with bags and literature related to NIPB technologies, to the farmers.



NIPB staff participated in Krishi Unnati Mela held at IARI, New Delhi on 05 to 07 March 2019. NIPB staff and students actively participated in stall arrangement, poster presentation and distributed

newsletter, pamphlets of different crop varieties developed at NIPB. The NIPB stall was visited by a large number of farmers and school students.

# **Other Institutional Activities**

**Institutional Projects**

**Externally Funded Projects**

**Technology Commercialization and IPR**

**Linkages and Collaborations**

**Awards and Honours**

**Visits Abroad**

**Recruitments/Promotions/Retirements**

**Other Activities**

**Participation in Conferences/Seminars/Symposia**

**List of Publications**

**Important Committees**

# Institutional Projects

| Project Title                                                                 | Date of Start               | Date of Completion/<br>ongoing | Principal Investigator | Name of Associates                   |
|-------------------------------------------------------------------------------|-----------------------------|--------------------------------|------------------------|--------------------------------------|
| Search and deployment of genes for stress tolerance and grain quality in rice | 1 <sup>st</sup> April, 2017 | 31 <sup>st</sup> March, 2020   | Prof. N. K. Singh      | Dr. T. K. Mandal                     |
|                                                                               |                             |                                |                        | Dr. Vandna Rai                       |
|                                                                               |                             |                                |                        | Dr. P. K. Dash                       |
|                                                                               |                             |                                |                        | Dr. Rhitu Rai                        |
|                                                                               |                             |                                |                        | Dr. S. V. A. C. R. Mithra            |
|                                                                               |                             |                                |                        | Dr. Amolkumar U. Solanke             |
|                                                                               |                             |                                |                        | Dr. Deepak S. Bisht                  |
|                                                                               |                             |                                |                        | Dr. Pankaj Kumar                     |
|                                                                               |                             |                                |                        | Ms. Rita                             |
| Adaptation of wheat to climate change induced stresses                        | 1 <sup>st</sup> April, 2017 | 31 <sup>st</sup> March, 2020   | Dr. Sanjay Singh       | Dr. J. C. Padaria                    |
|                                                                               |                             |                                |                        | Dr P. K. Mandal                      |
|                                                                               |                             |                                |                        | Dr. Kanika                           |
|                                                                               |                             |                                |                        | Dr. S. Barthakur                     |
|                                                                               |                             |                                |                        | Dr. Monika Dalal<br>Dr. Subodh Sinha |
|                                                                               |                             |                                |                        | Mr. R. S. Jaat                       |
| Improvement of stress tolerance in chickpea                                   | 1 <sup>st</sup> April, 2017 | 31 <sup>st</sup> March, 2020   | Dr. Sarvjeet Kaur      | Dr. P. K. Jain                       |
|                                                                               |                             |                                |                        | Mr. R. A. Nagar                      |
|                                                                               |                             |                                |                        | Dr. M. S. Nimmy                      |
|                                                                               |                             |                                |                        | Mr. Rakesh Narula                    |
| Stress tolerance and quality improvement in pigeonpea                         | 1 <sup>st</sup> April, 2017 | 31 <sup>st</sup> March, 2020   | Dr. Debasis Pattanayak | Dr. Kishor Gaikwad                   |
|                                                                               |                             |                                |                        | Dr. Rohini Sreevathsa                |
|                                                                               |                             |                                |                        | Dr. A. Dinabandhu                    |
|                                                                               |                             |                                |                        | Dr. Sandhya                          |
|                                                                               |                             |                                |                        | Dr. K.P. Singh                       |
|                                                                               |                             |                                |                        | Dr. R.S. Niranjana                   |
|                                                                               |                             |                                |                        | Ms. Megha                            |
| Biotechnological approaches for Brassica improvement                          | 1 <sup>st</sup> April, 2017 | 31 <sup>st</sup> March, 2020   | Dr. R. C. Bhattacharya | Dr. Anita Grover                     |
|                                                                               |                             |                                |                        | Dr. Rekha Kansal                     |
|                                                                               |                             |                                |                        | Dr. Ashish Kumar                     |
|                                                                               |                             |                                |                        | Dr. Navin C. Gupta                   |
|                                                                               |                             |                                |                        | Dr. Mahesh Rao                       |
|                                                                               |                             |                                |                        | Mr. Anshul Watts                     |
|                                                                               |                             |                                |                        | Mr. R. N. Gupta                      |
|                                                                               |                             |                                |                        | Mrs. Seema Dargan                    |
|                                                                               |                             |                                |                        | Dr. Rohit Chamola                    |
|                                                                               |                             |                                |                        | Mrs. Sandhya Rawat                   |

# Externally Funded Projects

| S. No. | Funding agency                              | Project Title                                                                                                                                         | Principal Investigator | Budget<br>(Rs. in lakh) |
|--------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------|
| 1.     | ICAR-NPTC                                   | Functional genomics of rice, wheat                                                                                                                    | N. K. Singh            | 500.00                  |
| 2.     | DBT-ISCB                                    | Improvement of pigeonpea for plant type, early maturity, pod borer resistance and moisture stress tolerance (Network programme)                       | N. K. Singh            | 640.00                  |
| 3.     | DBT                                         | From QTL to Variety: breeding for abiotic stress tolerance in rice-Phase-II                                                                           | N. K. Singh            | 1000.00                 |
| 4.     | ICAR- NPTC                                  | Development of Alternaria-resistant mustard                                                                                                           | Anita Grover           | 50.00                   |
| 5.     | ICAR- NPTC-                                 | Development of aphid resistant transgenic Brassica                                                                                                    | Rekha Kansal           | 58.33                   |
| 6.     | ICAR-NICRA                                  | National innovations for climate resilient agriculture (NICRA)                                                                                        | J. C. Padaria          | 54.02                   |
| 7.     | ICAR                                        | Exploiting alien genetic resources for developing climate resilient wheat and understanding mechanism of heat tolerance                               | J. C. Padaria          | 25.65                   |
| 8.     | ICAR-NFBSFARA                               | Phenomics of moisture deficit and low temperature tolerance in rice                                                                                   | P. K. Mandal           | 405.00                  |
| 9.     | CIMMYT                                      | Improving productivity of wheat through enhanced nitrogen use efficiency                                                                              | P. K. Mandal           | 26.60                   |
| 10.    | DBT                                         | Identification and characterization of peptides in selected wheat cultivars which are less immunogenic to patients with celiac disease                | P. K. Mandal           | 24.508                  |
| 11.    | ICAR- CRP on Biofortification               | Exploration for development of a wheat with low immunogenic gluten                                                                                    | P. K. Mandal           | 138.00                  |
| 12.    | ICAR- Incentivizing Research in Agriculture | Genetic modification to improve biological nitrogen fixation                                                                                          | P. K. Mandal           | 1290.00                 |
| 13.    | ICAR-NPTC                                   | Amelioration of cold induced sweetening in potato                                                                                                     | D. Pattanayak          | 4.50                    |
| 14.    | DBT                                         | Development of haploid inducer line and enhancement of seed meal quality in Brassica juncea through CRISPR/Cas mediated genome editing                | R. C. Bhattacharya     | 50.40                   |
| 15.    | ICAR-NASF                                   | Characterization, mapping and transcriptome analysis of seed protein, $\beta$ -carotene and mineral contents in chickpea ( <i>Cicer arietinum</i> L.) | P. K. Jain             | 85.00                   |
| 16.    | ICAR-NASF                                   | Identification and molecular tagging of gene(s) controlling resistance to chilli leaf curl virus infection in chilli ( <i>Capsicum annum</i> L.)      | P. K. Jain             | 39.00                   |
| 17.    | ICAR-NPTC                                   | Development of transgenic wheat for heat stress tolerance                                                                                             | Sharmishtha Barthakur  | 4.50                    |
| 18.    | ICAR-NPTC                                   | Finishing of pigeonpea genome and functional genomics of fertility restoration                                                                        | Kishor Gaikwad         | 10.00                   |
| 19.    | ICAR-CRP on Genomics                        | Decoding the genomes of guar and black gram for the identification of important genes                                                                 | Kishor Gaikwad         | 110.00                  |

| S. No. | Funding agency                                       | Project Title                                                                                                                                                                                       | Principal Investigator    | Budget<br>(Rs. in lakh) |
|--------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------|
| 20.    | ICAR-NASF                                            | Understanding cellular and genetic mechanisms and identifying molecular markers for seed viability in soybean                                                                                       | Kishor Gaikwad            | 59.84                   |
| 21.    | ICAR-NPTC                                            | Development of wheat transgenics with enhanced tolerance to drought                                                                                                                                 | Monica Dalal              | 4.50                    |
|        | DST-SERB                                             | Deciphering the transcriptional regulator(s) of hydrotropism in wheat                                                                                                                               | Monica Dalal              | 40.70                   |
| 22.    | ICAR-NPTC                                            | Functional genomics of terminal heat stress and drought tolerance in wheat                                                                                                                          | Vandna Rai                |                         |
| 23.    | DBT                                                  | Structural and functional genomics study of deepwater adaptations of local rice land races of Assam                                                                                                 | T. K. Mondal              | 24.47                   |
| 24.    | DBT                                                  | Generation and comparative analysis of salinity responsive miRNAs and miRNA-mediated pathways of halophyte ( <i>Oryza coarctata</i> ) and a tolerant glycophyte ( <i>Oryza sativa</i> cv Nonabokra) | T. K. Mondal              | 60.60                   |
| 25.    | National Tea Research Foundation, Tea Board, Kolkata | Decoding tea ( <i>Camellia sinensis</i> ) genome for identification of commercially important genes                                                                                                 | T. K. Mondal              | 116.00                  |
| 26.    | ICAR-NASF                                            | Genetic improvement of rice for yield, abiotic and biotic stress tolerance through RNA guided genome editing (CRISPR-Cas9)                                                                          | P. K. Dash                | 78.10                   |
| 27.    | ICAR-NPTC                                            | Bast fibre yield in flax                                                                                                                                                                            | P. K. Dash                | 248.46                  |
| 28.    | ICAR-NHEP                                            | Centre for next generation technology in adaptive agriculture                                                                                                                                       | P. K. Dash                | 88.72                   |
| 29.    | BIRAC-BIPP                                           | A novel micro RNA-induced gene silencing strategy for development of pest & disease tolerant cotton plants                                                                                          | Rohini Sreevathsa         | 96.00                   |
| 30.    | ICAR-NPTC                                            | Transgenics for pod borer resistance in pigeonpea                                                                                                                                                   | Rohini Sreevathsa         | 46.10                   |
| 31.    | DBT-ISCB                                             | Improvement of pigeonpea for plant type, early maturity, pod borer resistance and moisture stress tolerance                                                                                         | Rohini Sreevathsa         | 95.90                   |
| 32.    | DBT                                                  | Maintenance, characterization and use of EMS mutants of upland variety Nagina 22 for functional genomics in rice- Phase II                                                                          | S. V. Amitha C. R. Mithra | 258.98                  |
| 33.    | DBT                                                  | Referral centre for genetic fidelity testing of tissue culture raised plants (NCS-TCP)                                                                                                              | Amolkumar U. Solanke      | 71.37                   |
| 34.    | ICAR-CABin                                           | RiceMetaSys: Understanding rice gene network for blast resistance and drought tolerance through system biology approach                                                                             | Amolkumar U. Solanke      | 13.49                   |
| 35.    | DST-SERB                                             | CRISPR-Cas9 based genome editing of multiple negative regulators for blast resistance in rice                                                                                                       | Amolkumar U. Solanke      | 31.93                   |
| 36.    | DST-SERB                                             | Broadening genetic diversity of Indian mustard ( <i>Brassica juncea</i> ) through resynthesis of amphidiploid genome by crossing parental diploid species                                           | Mahesh Roa                | 33.64                   |
| 37.    | DST-ECR                                              | Molecular dissection of mechanism of resistance to sheath blight in rice                                                                                                                            | D. S. Bisht               | 36.36                   |
| 38.    | DST-SERB                                             | Development of haploid inducer line in <i>Brassica oleracea</i> var. botrytis through CRISPR-Cas mediated engineering of centromeric histone H3 gene                                                | Anshul Watts              | 40.21                   |

# Technology Commercialization

## IPR

The mandate of the Institute Technology Management Unit relates to registration of patents, facilitation of contract research projects and commercialization of IPR enabled technologies of

the centre through Public- Private Partnership.

The following activities were undertaken by the ITMU during the year 2018-19.

### I. Patents Granted:

| S. No. | Title of patent application                                            | Patent Application Number & date of filling | Date of Grant | Granted Patent Number |
|--------|------------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| 1      | Novel trichome specific promoter.                                      | 2251/DEL/2009<br>30.10.2009                 | 09.07.2018    | 298639                |
| 2      | Rice polynucleotide associated with blast resistance and uses thereof. | 241/DEL/2010<br>04.02.2010                  | 05.03.2019    | 308533                |

### II. MoUs /MTAs signed:

1. A Memorandum of Understanding (MoU) has been signed for R & D promotion in partnership mode between ICAR-NIPB, New Delhi and Nand Educational Foundation for Rural Development (NEFORD), Lucknow, U.P. on 01.04.2018 to carrying out joint research programmes in the areas of mutual interest in technology dissemination through in-house and externally funded projects.
2. A Memorandum of Understanding (MoU) has been signed for R & D promotion in partnership mode between ICAR-NIPB, New Delhi and Regional Agricultural Research Station, Assam Agricultural University, North Lakhimpur (RARS-AAU, NL), Assam on 07.06.2018 to carrying out joint research programmes in the areas of Improvement of deep water rice (DWR) of Assam through Marker Assisted Selection (MAS).
3. A Memorandum of Understanding (MoU) has been signed for R & D promotion in partnership mode between ICAR-NIPB, New Delhi and Uttar Banga Krishi Viswavidyalaya (UBKV), Pundibari, Cooch Behar, West Bengal on 25.06.2018 to carrying out joint research programmes in the areas of
4. Development of semi-dwarf high yielding short grain aromatic traditional rice varieties of West Bengal through marker assisted breeding.
5. A Memorandum of Understanding (MoU) has been signed for R & D promotion in partnership mode between ICAR-NIPB, New Delhi and Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, M.P. on 12.11.2018 to carrying out joint research programmes in the areas of Development of semi-dwarf high yielding short grain aromatic traditional rice varieties of West Bengal through marker assisted breeding.
6. Material Transfer Agreement (MTA) with Professor Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad pertaining to sharing of 12 rice varieties developed at NIPB has been signed on 31.07.2018.
7. Material Transfer Agreement (MTA) with Assam Agricultural University, Jorhat, Assam pertaining to sharing Brassica wild species of ICAR-NIPB has been signed on 01.08.2018.
8. Material Transfer Agreement (MTA) with ICAR-NIPB, New Delhi and The Energy and Resources Institute (TERI), New Delhi pertaining to sharing of biological materials PA5 and PB6 constructs for rice transformation developed at TERI, New Delhi has been signed on 01.12.2018.

# Linkages and COLLABORATIONS

The centre has an active and close collaboration with other ICAR institutes and State Agricultural Universities. NRCPB is instrumental in providing gene constructs for transgenic development targeting biotic and abiotic tolerance and quality improvement. It also plays a pivotal role in developing genomic resources, both functional and structural, for a variety of crop species. The scientists at the centre work in close collaboration with different disciplines of the Indian Agricultural Research Institute, like Biochemistry, Plant Physiology, Genetics, Entomology, Pathology, Nematology, Horticulture, Microbiology and Water Technology Centre. With the ever-increasing need of computational and statistical intervention in the area of modern molecular biology and biotechnology, the centre has also developed active collaboration with the Indian Agricultural Statistics Research Institute. Over the years the centre has developed working linkages with the international laboratories.

## Inter-Institutional Linkages

- Delhi University, New Delhi
- Jawaharlal Nehru University, New Delhi
- Chaudhary Sarwan Kumar Krishi Vishwavidyalaya, Palampur
- Punjab Agricultural University, Ludhiana
- Tamil Nadu Agricultural University, Coimbatore

- Central Rice Research Institute, Cuttack
- Directorate of Medicinal and Aromatic Plants, Anand
- University of Agricultural Sciences, Dharwad
- Institute of Himalayan Bioresources and Technology, Palampur
- National Bureau of Plant Genetic Resources, New Delhi
- Bose Institute, Kolkata
- Indian Institute of Technology, Kanpur
- International Centre for Genetic Engineering and Biotechnology, New Delhi
- Twenty three ICAR institutes under the Network project on Transgenics
- Twenty seven ICAR institutes, SAUs and others IITs in NAIP Mega Project entitled “Bioprospecting of genes and allele mining for abiotic stress tolerance”.
- QTL to variety 15 Institutions plus IARI

## International Linkages

- Division of Plant Pathology, UC Davis, California, USA
- International Rice Research Institute, Manila, The Philippines
- Sixteen Institutes under International Wheat Genome Sequencing Project

# Awards and Honours

- Prof. N. K. Singh received ‘Bharat Jyoti Award’ from India International Friendship Society.
- ‘Shikshak Shri Award’ from Indian Society of International Law in 2018.
- Dr. Pranab K. Mandal joined as Member, Editorial Board of the Journal of Wheat and Barley Research, Karnal, India.
- Dr. Debasis Pattanayak became Fellow of National Academy of Agricultural Sciences in January, 2019.
- Dr Subosh K. Sinha received INEW Scientific Exchange Scheme Fellowship to undertake research at Department of Plant Sciences, University of Cambridge, UK with Prof. Howard Griffiths.
- Dr. Navin C. Gupta received “Young Scientist Award” 2018 from Society of Plant Protection Sciences, ICAR-NCIPM, India; and “Adarsh Vidya Saraswati Rashtriya Puraskar” National Award of Excellence-2018, from Glacier Journal Research Foundation, Ahmadabad, India.
- Dr. Navin C. Gupta was elected as “Member of Editorial Board” Glacier Journal of Scientific Research, in the year 2018.
- Dr. Nimmy M. S. received ‘Best Oral Presentation’ award during National Conference on-ICIESTRSP-2019, Sardar Vallabhai Patel University of Agriculture and Technology, Meerut held on 20-21 April, 2019
- Dr. Sandhya got ‘Best Oral Presentation’ award in National Seminar on Innovative Approaches for Rural and Agriculture Advancement (IARAA-2019 at JNKVV, COA, MP.
- Dr Sandhya received Young Scientist Award in International Conference on Advances in Agricultural, Biological and Applied Sciences for Sustainable Future at Meerut.

## Visits Abroad

Prof. N. K. Singh visited San Diego, USA from January 11 to January 18, 2019 to attend Plant & Animal Genome XXVII Conference.

Dr. Pranab K. Mandal visited London, UK from 10-11 September 2018 for the Review meeting of the INEW project and presented the progress of the INEW project of ICAR-NIPB.

Dr. R. C. Bhattacharya visited Yezhin University, Myanmar from 6<sup>th</sup> Sep to 10<sup>th</sup> Oct, 2018 for teaching a full course.

- Dr. Kishor Gaikwad visited Yezhin University, Myanmar from 6<sup>th</sup> Sep to 10<sup>th</sup> Oct, 2018 for teaching a full course on “Genomics and Bioinformatics” to PG Students, under the IM ACARE project.
- Dr. Monika Dalal visited Bologna, Italy for attending ‘Training course on high-throughput wheat phenotyping’ organized by Wheat Initiative at Cosmopolitan Hotel, Via del Commercio Associato 9, Bologna, Italy from 17-18 September 2018.
- Tapan K. Mondal visited Myanmar to teach the M. Sc. and Ph. D. students of Molecular Biology

and Biotechnology a course entitled as, “Plant gene expression and regulation (MBB-624)” under Indo-Myanmar project at ACARE, Yezin Agricultural University, Myanmar from 15<sup>th</sup> June to 9<sup>th</sup> July, 2018.

- Dr. Prasanta Dash visited Yezhin University, Myanmar from 22<sup>nd</sup> Feb 2019 to 25<sup>th</sup> March, 2019 for teaching, and attended Plant and Animal Genome conference-2019 at San Diego, USA
- Dr SK Sinha visited University of Cambridge, Cambridge, UK to undertake training on N uptake at anthesis and post-anthesis in different wheat varieties grown under six different N-levels in the field under Prof. Howard Griffiths during 30.04.2018 to 10.06.2018.
- Amitha Mithra Sevanthi attended the International Symposium on Plant Mutation Breeding and Biotechnology held during 27– 31 August 2018 at IAEA, Vienna, Austria and made an oral presentation in the topic entitled “National repository of rice EMS mutants of an upland rice cultivar Nagina 22: progress update on characterization and utilization”.

# Recruitments/Promotions/

## Recruitments

1. Mr. Deepak Singh Rathore joined as Technical Assistant (T-3), ICAR-NIPB on 17<sup>th</sup> November, 2018.
2. Ms. Rita joined as Technical Assistant (T-3), ICAR-NIPB on 20<sup>th</sup> November, 2018.
3. Ms. Megha joined as Technical Assistant (T-3), ICAR-NIPB on 21<sup>st</sup> December, 2018.
4. Mr. Anshul Kumar Verma joined as Technical Assistant (T-3), ICAR-NIPB on 06<sup>th</sup> March, 2019.
5. Ms. Priyanka Nair joined as Lower Division Clerk, ICAR-NIPB on 25<sup>th</sup> October, 2018.
6. Mr. Mittravesh Choudhary joined as Lower Division Clerk, ICAR-NIPB on 05<sup>th</sup> December, 2018.

## Promotions

1. Dr. Rohit Chamola & Dr. Pankaj Kumar were promoted from ACTO to CTO.
2. Dr. Vandana Rai was promoted from Senior Scientist to Principal Scientist.

## Retirements

1. Dr. Ranjeet Singh Jaat, Scientist superannuated from 30<sup>th</sup> November, 2018.
2. Sh. Rakesh Kumar Narula, CTO superannuated from 30<sup>th</sup> November, 2018.

## Tansfers

1. Dr. Basavaprabhu L. Patil, Senior Scientist has been transfered from ICAR – NIPB, New Delhi to ICAR – IIHR, Bangalore.

## Other Activities

### Vigilance Awareness week:

ICAR-NIPB observed 'Vigilance Awareness Week' from 29<sup>th</sup> October, 2018 to 3<sup>rd</sup> November 2018. Director, NIPB administered pledge to the staff on 29<sup>th</sup> October, 2018.



### Hindi Activities 2018-19

#### Hindi Month:

Competitions in Hindi essay writing, technical work writing, declamation, poetry, recitation and noting and drafting were conducted for the staff of the NIPB as part of Hindi month from 14<sup>th</sup> September, 2018 to 15<sup>th</sup> October, 2018.



Prize distribution in Hindi month

### Hindi Workshop:

Following Hindi workshops were conducted in 2018-19.

| S. No. | Date       | Quarter           | Guest speaker          |
|--------|------------|-------------------|------------------------|
| 1.     | 26-06-2018 | April- June       | Mr. Arvind Kumar Singh |
| 2.     | 26-09-2018 | July - September  | Mr. Prem Pal Sharma    |
| 3.     | 29-12-2018 | October- December | Prof. N. K. Singh      |

### Swachh Bharat Abhiyan:

Swachta Pakhwada organized by the Centre from December 16-30, 2018 under Swachh Bharat Abhiyan. All staff members of NRCPB participated as per the schedule provided by ICAR. Display of Banner at Prominent places, taking Swachhta Pledge and cleanliness drive including cleaning of offices, corridors, premises, public places, community market places etc.



# Participation in Conference/

1. Dr. Sanjay Singh participated in the 56th All India Wheat and Barley Research Workers Meeting held at BHU UP from August 25-28, 2017
2. Dr. Jasdeep C. Padaria attended 57<sup>th</sup> All India Wheat and Barley Research Workers' Meet held on August 24-26, 2017, BIRSA, BAU, Ranchi, Jharkhand.
3. Dr. Jasdeep C. Padaria attended National Symposium on "Connecting Innovations in Plant Biology for Food and nutritional security", Division of Biochemistry, IARI, New Delhi on 19th Nov. 2018
4. Dr. Jasdeep C. Padaria attended National Genetic Congress on Genetics for Sustainable Food, Health and Nutrition Security, IARI, December 14-16, 2018,
5. Dr. Jasdeep C. Padaria attended 4th **International Group Meeting on 'Wheat productivity enhancement through climate smart practices'**, CSK HPKV, Palampur 14-16th Feb 2019
6. Dr. Jasdeep C. Padaria attended National Conference on "Resilience And Resource Management Including ICT for Sustainable Agriculture & Biotechnology" at Dr. MPS Group of Institutions Dr. Bhim Rao Ambedkar University, Agra, 23- 24 February 2019
7. Dr. Jasdeep C. Padaria attended 54th Annual Group Meeting of ICAR-All India Coordinated Research Project on Pearl Millet, Genetics Division IARI, New Delhi 14th March 2019
8. Dr. Pranab K. Mandal attended the Review meeting of the INEW project held at London from 10-11 September 2018 and presented the progress of the INEW project of our centre and discussed about future plan.
9. Dr. Pranab K. Mandal attended International workshop on 'Wheat Genomic data analysis for improvement of nitrogen use' on 28- February to 2nd March 2019 at ICAR\_NBPGR, New Delhi and also delivered an invited talk.
10. Dr. Pranab K. Mandal attended the Review meeting of the INEW project held at PAU, Ludhiana from 5-6 March 2019 and presented the progress of the INEW project of our centre and discussed about future plan.
11. Dr. Pranab K. Mandal attended a workshop on 'Long-Look' & Review Meeting of Indo-UK Virtual Joint Centers on Agricultural N at ICRISAT, Hyderabad during Aug 30-31, 2018 and delivered a talk overall progress of INEW project
12. Dr. Pranab K. Mandal attended 4th International Group Meeting (IGM) 2019 on February 14-16, 2019, CSK HPKV Palampur, HP, and delivered an invited talk.
13. Dr. Pranab K. Mandal delivered Lecture entitled "Wheat Nutrition and Celiac Disease" at Centre for Advanced Faculty Training on Biochemistry of food Crops: from Omics studies to Nutrient analysis on 27th September 2018 at Division of Biochemistry, IARI, New Delhi.
14. Dr. Kishor Gaikwad delivered a lecture at a CAFT sponsored program on Whole genome sequencing of plant pathogens: Methods and Applications on 12th Jan, 2018 at Division of Plant Pathology, IARI, New Delhi.
15. Dr. Kanika attended National Symposium on "Connecting Innovations in Plant Biology for Food and Nutritional Security" organized at ICAR-IARI, New Delhi, on 19th November, 2018.
16. Dr. Kanika attended International Salinity Conference on "Resilient Agriculture in Saline Environments under Changing Climate: Challenges & Opportunities" held from February 7-9, at ICAR- CSSRI, Karnal, Haryana
17. Dr. Monika Dalal attended *National Symposium On Connecting Innovations in Plant Biology for Food and Nutritional Security*, held on November 19, 2018 at IARI, New Delhi National seminar Nov 19th 2018. Feb 22-25

18. Dr. Monika Dalal attended *XIV Agriculture Science Congress* pp110, held on 20-23 Feb, 2019 at NASC complex, New Delhi
19. Dr Subosh K Sinha attended 3rd National Workshop of Officer In-charge, Data Management (ICAR Research Data Repository for Knowledge Management; KRISHI) on 4-5, December, 2018, NASC, New Delhi-12 as Officer In-Charge/Nodal Officer, NRCPB.
20. Dr. NC Gupta attended the International conference on emerging issues in agricultural, environmental, & applied sciences for sustainable development (EIAEASSD-2018). 27-29 November 2018, SHUATS, Allahabad.
21. DR.NC Gupta attended the 1st International conference on climate change and adaptive crops protection for sustainable Agri-horticulture landscape, NRCSS, Ajmer, Rajasthan.
22. Dr. NC Gupta attended the XIVth Agricultural Science Congress, February 20-23, 2019, NAASC, New Delhi.
23. Dr. A. Watts attended 4th international plant physiology congress-2018 from 2nd -5th December 2018 at Lucknow
24. Dr. A. Watts attended India International Science festival-2018 from 5-8th October 2018 held at Lucknow
25. Dr. A.Watts attended International Conference on food security from 7-8th December 2018 held at ThaparUniversity, Patiala
26. Dr. Sandhya delivered a lecture on NGS platform: Sequencing chemistry and library preparation for Illumina Hiseq. in 21 days CAFT training program at NIPB. Aug 28th- 17th Sept 2018
27. Dr Sandhya delivered a lecture on Ionomics: A newer approach for discovery of novel genes regulating plant mineral and quality traits in national conference at Dr. B.R Ambedkar University, 23-24 February 2019, Agra
28. Dr Sandhya attended 2 days National seminar on Innovative Approaches for Rural and Agriculture Advancement (IARAA-2019) at JNKVV College of Agriculture Tikamgarh M.P. 10-11 Jan 2019
29. Dr. Sandhya attended 3 Days National Genetics Congress at IARI, New Delhi, 14– 16th Dec 2018
30. Dr. Sandhya attended 3 days international conference on ‘Advance in Agricultural, Biological and Applied Science for Sustainable Future (ABAS-2018) at Subharti University, Meerut, UP. 20-22 Oct 2018

# Publications

## Research Papers

1. Ali S, Mir ZA, Bhat JA, Tyagi A, Chandrashekar N, Yadav P, Rawat S, Sultana M & Grover A. 2018. Isolation and characterization of systemic acquired resistance marker gene *PR1* and its promoter from *Brassica juncea*. *3 Biotech* **8**: 10.
2. Awana M, Yadav K, Rani K, Gaikwad K, Praveen S, Kumar S & Singh A. 2019. Insights into salt stress-induced biochemical, molecular and epigenetic regulation of spatial responses in pigeonpea (*Cajanus cajan* L.). *J Plant Growth Regulation* (<https://doi.org/10.1007/s00344-019-09955-4>).
3. Banerjee S, Gill S, Gawade B, Jain PK, Subramaniam K & Sirohi A. 2018. Host delivered RNAi of two cuticle collagen genes, *Mi-col-1* and *Lemmi-5*, hampers structure and fecundity in *Meloidogyne incognita*. *Front Plant Sci* **8**: 2266.
4. Bharti S & Sinha SK. 2018. Biochemical characterization of building block of condensed tannin in faba bean (*Vicia faba* L.). *Vegetos* **31**: 25-27.
5. Bhradwaj N, Gogoi N, Barthakur S & Basumatary N. 2018. Morpho-physiological responses in different mungbean genotypes under drought stress. *Res J Recent Sci* **7**:10-18.
6. Chandrashekar N, Ali S & Grover A. 2018. Exploring expression patterns of PR-1, PR-2, PR-3, and PR-12 like genes in *Arabidopsis thaliana* upon *Alternaria brassicae* inoculation. *3 Biotech* **8**: 230.
7. Chowrasia S, Panda AK, Rawal HC, Kaur H & Mondal TK. 2018. Identification of jumonjiC domain containing gene family among the *Oryza* species and their expression analysis in FL478, a salt tolerant rice genotype. *Plant Physiol Biochem* **130**: 43-53.
8. Dalal M, Sahu S, Tiwari S, Rao AR & Gaikwad K. 2018. Transcriptome analysis reveals interplay between hormones, ROS metabolism and cell wall biosynthesis for drought-induced root growth in wheat. *Plant Physiol Biochem* **130**: 482-492.
9. Dash PK, Gupta P, Jailani AK & Rai R. 2018. Hydropenia induces expression of drought responsive genes (DRGs) *erd1*, *hat*, *plD*, and *zfa* in *Linum usitatissimum* L. *Ind J Exp Biol* **56**: 743-749
10. Dash PK, Rai R, Rai V & Pasupalak S. 2018. Drought induced signaling in rice: delineating canonical and non-canonical pathways. *Front Chem* **6**: 264.
11. Divte P, Yadav P, Jain PK, Paul S & Singh B. 2019. Ethylene regulation of root growth and phytosiderophore biosynthesis determines iron deficiency tolerance in wheat (*Triticum spp*). *Environ Exp Bot* **162**: 1-13.
12. Dubey H, Kiran K, Jaswal R, Jain P, Kayastha AM, Bhardwaj SC, Mondal TK & Sharma TR. 2019. Discovery and profiling of small RNAs from *Puccinia triticina* by deep sequencing and identification of their potential targets in wheat. *Funct Integr Genomics* **7**: 1-7.
13. Ganie SA, Molla KA, Henry RJ, Bhat KV & Mondal TK. 2019. Advances in understanding salt tolerance in rice. *Theor Appl Genet* **132**: 851-70.
14. Gaurav AK, Namita DR, Raju DV, Singh M, Singh B, Krishnan SG, Mithra SV, Panwar S, Singh M & Dhiman MR. 2018. Genetic characterization of Rosa species using morphological markers. *Indian J Agr Sci* **88**: 1396-402.
15. Gayatri, Rani M, Mahato AK, Sinha SK, Dalal M, Singh NK & Mandal PK. 2018. Homeologue specific gene expression analysis of two vital carbon metabolizing enzymes, citrate synthase and NADP-isocitrate dehydrogenase, from wheat (*Triticum aestivum* L.) under nitrogen stress. *Appl Biochem Biotech* (<https://doi.org/10.1007/s12010-018-2912-2>).
16. Gogoi N, Farooq M, Barthakur S, Baroowa B, Paul S, Bharadwaj N & Ramanjulu S. 2018. Thermal stress impacts on reproductive development and grain yield in grain legumes. *J Plant Biol* **61**: 265-291.
17. Gupta NC, Gupta RK, Rao M & Kumar A. 2019. Combining the sclerotinia stem rot disease

- resistance trait in interspecific hybrids of *Brassica napus* and *Brassica carinata*. *Ann Pl Protec Sci* **27**: 70-76.
18. Gupta OP, Dahuja A, Sachdev A, Jain PK, Kumari S & Praveen S. 2019. Cytosine methylation of isoflavone synthase gene in the genic region positively regulates its expression and isoflavone biosynthesis in soybean seeds. *DNA Cell Biol* (doi: 10.1089/dna.2018.4584).
  19. Gupta OP, Dahuja A, Sachdev A, Kumari S, Jain PK, Vinutha T & Praveen S. 2019. Conserved miRNAs modulate the expression of potential transcription factors of isoflavonoid biosynthetic pathway in soybean seeds. *Mol Biol Rep* (<https://doi.org/10.1007/s11033-019-04814-7>).
  20. Kaur A, Vishwakarma H, Maibam A & Padaria JC. 2018. Cloning, characterization and *in-silico* studies on abiotic stress responsive *Hsp17.9* gene from *Prosopis cineraria*. *Indian J Plant Physiol* (<https://doi.org/10.1007/s40502-018-0414-4>).
  21. Kohli D, Chidambaranathan P, Tej Kumar JP, Singh AK, Kumar A, Subramaniam K, Srinivasan R, Bharadvaja N & Jain PK. 2018. Host-mediated RNAi of a Notch-like receptor gene in *Meloidogyne incognita* induces nematode resistance. *Parasitology* (<https://doi.org/10.1017/S0031182018000641>).
  22. Kumar S, Dhar S, Barthakur S, Chandrakala M, Kochewad S A, Meena LR & Kumar D. 2018. Effect of integrated potassium management on soil biological properties and yields of corn under corn-wheat cropping system. *Int J Curr Microbiol App Sci* **7**: 1855-1866.
  23. Kumar S, Dhar S, Barthakur S, Kumar S, Mondal B, Kumar D, Kochewad SA & Ram L. 2018. Integrated K management exhibit a key role in potassium uptake transporter (*ZmKUP*) expression to improve growth and yield of corn. *Int J Curr Microbiol App Sci* **7**: 1867-1887.
  24. Kumari S, Kumar J, Kumar RR, Kumar A, Nimmy MS, Ranjan T & Kumar V. 2019. Genetic diversity analysis in grasspea (*Lathyrus sativus* L.) using SSR molecular markers. *Curr J Appl Sci Technol* **33**: 1-6.
  25. Latha GM, Raman KV, Lima GM, Pattanayak D, Singh AK, Chinnusamy V, Bansal KC, Rao KRSS & Mohapatra T. 2019. Genetic engineering of *indica* rice with *AtDREB1A* gene for enhanced abiotic stress tolerance. *Plant Cell Tissue Org Cult* **136**: 173-188.
  26. Mehraj U, Panwar S, Singh KP, Pandey R, Solanke AU & Mallick N. 2018. Development of protocol for *in vitro* rooting and hardening of doubled haploid line of *Tagetes erecta* L. derived through ovule culture. *Indian J Hort* **75**: 651-656.
  27. Mishra P, Jain A, Takabe T, Tanaka Y, Negi M, Singh N, Jain N, Mishra V, Maniraj R & Sreevathsa R. 2019. Heterologous expression of serine hydroxymethyltransferase3 from rice confers tolerance to salinity stress in *E. coli* and *Arabidopsis*. *Front Plant Sci* **10**: 217.
  28. Mishra P, Singh N, Jain A, Jain N & Mishra V. 2018. Identification of cis-regulatory elements associated with salinity and drought stress tolerance in rice from co-expressed gene interaction networks. *Bioinformation* **14**: 123.
  29. Mittal S, Banduni P, Mallikarjuna MG, Rao AR, Rao AR, Jain PA, Dash PK & Thirunavukkarasu N. 2018. Structural, functional and evolutionary characterization of major drought transcription factors families in maize. *Front Chem* **6**: 177.
  30. Mondal TK, Rawal HC, Chowrasia S, Varshney D, Panda AK, Mazumdar A, Kaur H, Gaikwad K, Sharma TR & Singh NK. 2018. Draft genome sequence of first monocot-halophytic species *Oryza coarctata* reveals stress-specific genes. *Sci Rep* **8**: 13698.
  31. Nagar CK, Gayatri, Bharati A, Sinha SK, Venkatesh K & Mandal PK. 2018. Nitrogen stress leads to induce change in expression of genes for nitrate transporter in wheat genotypes. *Intl J Curr Microbiol Appl Sci* **7**: 06.
  32. Nagar CK, Gayatri, Bharati A, Sinha SK, Venkatesh K & Mandal PK. 2018. Nitrogen stress induced changes in root system architecture (RSA) in diverse wheat (*T. aestivum* L.) genotypes at seedling stage. *Wheat Barley Res* **10**: 93-101.
  33. Ngangkham U, Nath M, Dokku P, Mithra SA, Ramamurthy S, Singh NK, Sharma RP & Mohapatra T. 2018. An EMS-induced new sequence variant, TEMS5032, in the coding region of *SRS3* gene leads to shorter grain length in rice (*Oryza sativa* L.). *J App Genet* **59**: 377-389.
  34. Padaria JC, Tarafdar A, Sharma V, Vishwakarma

- H, Singh GP & Singh S. 2019. Cultivar specific response of callus induction and plant regeneration from mature embryos in different elite Indian wheat. *Res J Biotech* **14**: 1-8.
35. Panwar BS, Kaur J, Kumar P & Kaur S. 2018. A novel *cry52Ca1* gene from an Indian *Bacillus thuringiensis* isolate is toxic to *Helicoverpa armigera* (cotton boll worm). *J Invert Pathol* **159**:137-140.
36. Panwar BS, Narula RK & Kaur S. 2018. Theoretical 3D modelling of a novel Cry **toxin isolated from native *Bacillus thuringiensis* isolate SK711**. *Indian J Entomol* **80**: 108-113.
37. Panwar BS, Ram C, Narula RK & Kaur S. 2018. Pool deconvolution approach for high-throughput gene mining from *Bacillus thuringiensis*. *Appl Microbiol Biotechnol* **102**: 1467-1482.
38. Rai S, Kaushik M, Sinha SK, Gupta RK, Mahendru A & Mandal PK. 2018. Transcriptome data of cultivated tetraploid and hexaploid wheat variety during grain development. *Data in Brief* **22**:551-556.
39. Ram H, Dhar S, Choudhary H, Sharma RK, Prasad L, Solanke AU & Tomar BS. 2018. Estimates of genetic components of variation for earliness and yield traits in garden pea (*Pisum sativum* L.). *Chem Sci Rev Lett* **7**: 379-382.
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41. Ramakrishna G, Kaur P, Nigam D, Chaduvula PK, Yadav S, Talukdar A, Singh NK & Gaikwad K. 2018. Genome-wide identification and characterization of InDels and SNPs in *Glycine max* and *Glycine soja* for contrasting seed permeability traits. *BMC Plant Biol* **18**: 141.
42. Raman KV, Agarwal D, Rao SR, Sreevathsa R, Singh AK, Abidin MZ, Mohapatra T & Pattanayak D. 2018. Rapid and efficient *Agrobacterium*-mediated transformation of early scutellum derived calli of *indica* rice. *Indian J Exp Biol* **56**: 20-28.
43. Rathinam M, Kesiraju K, Singh S, Thimmegowda V, Rai V, Pattanayak D & Sreevathsa R. 2019. Molecular interaction-based exploration of the broad spectrum efficacy of a *Bacillus thuringiensis* insecticidal chimeric protein, Cry1AcF. *Toxins* **11**: 143.
44. Rawal HC, Mithra SA, Arora K, Kumar V, Goel N, Mishra DC, Chaturvedi KK, Rai A, Devi SV, Sharma TR & Solanke AU. 2018. Genome-wide analysis in wild and cultivated *Oryza* species reveals abundance of NBS genes in progenitors of cultivated rice. *Plant Mol Biol Rep* **36**: 373-386.
45. Reddy SH, Kambalimath SK, Singhal RK, Chikkakariyappa MK, Muthurajan R, Rajanna MP, Sreevathsa R, Sevanthi AM, Mohapatra T, Sarla N & Chinnusamy V. 2018. Allele-specific analysis of single parent backcross population identifies HOX10 transcription factor as a candidate gene regulating rice root growth. *Physiol Plant*.
46. Sahu S, Rao AR, Pandey J, Gaikwad K, Ghoshal S & Mohapatra T. 2018. Genome-wide identification and characterization of lncRNAs and miRNAs in cluster bean (*Cyamopsis tetragonoloba*). *Gene* **667**: 112-121
47. Saini RP, Raman V, Dhandapani G, Malhotra EV, Sreevathsa R, Kumar PA, Sharma TR & Pattanayak D. 2018. Silencing of *HaAce1* gene by host-delivered artificial microRNA disrupts growth and development of *Helicoverpa armigera*. *PLoS ONE* **13**: e0194150.
48. Sanand S, Kumar S, Bara N & Kaul G. 2018. Comparative evaluation of half-maximum inhibitory concentration and cytotoxicity of silver nanoparticles and multiwalled carbon nanotubes using buffalo bull spermatozoa as a cell model. *Toxicol Indus Health* (DOI: 10.1177/0748233718783389).
49. Saxena S, Kaila T, Chaduvula PK, Singh A, Singh NK & Gaikwad K. 2019. Novel chloroplast microsatellite markers in pigeonpea (*Cajanus cajan* L. Millsp.) and their transferability to wild *Cajanus* species. *AJCS* **13**: 185-191.
50. Sevanthi AM, Kandwal P, Kale PB, Prakash C, RamkumarMK, YadavN, MahatoAK, Sureshkumar V, Behera M, Deshmukh RK & Jeyaparakash P. 2018. Whole genome characterization of a few EMS-induced mutants of upland rice variety Nagina 22 reveals a staggeringly high frequency of SNPs which show high phenotypic plasticity towards the wild-type. *Front Plant Sci* **9**.
51. Sharma P, Rao M & Gupta NC. 2018. Molecular aspects of sclerotinia stem rot disease management in oilseed crops. *Int J Microbiol Res* **4**: 1166-1170.

52. Sharma P, Samkumar A, Rao M, Mishra DC, Singh VV, Prasad L & Gupta NC. 2018. Genetic diversity studies based on morphological variability, pathogenicity and molecular phylogeny of the *S.sclerotiorum* population from Indian mustard (*Brassica juncea*). *Front Microbiol* **9**:1169.
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54. Sharma S, Kaur R, Solanke AK, Dubey H, Tiwari S & Kumar K. 2019. Transcriptome sequencing of Himalayan raspberry (*Rubus ellipticus*) and development of simple sequence repeat markers. *3 Biotech* **9**: 161.
55. Shingote PR, Mithra SA, Sharma P, Devanna NB, Arora K, Holkar SK, Khan S, Singh J, Kumar S, Sharma TR & Solanke AU. 2019. LTR retrotransposons and highly informative ISSRs in combination are potential markers for genetic fidelity testing of tissue culture-raised plants in sugarcane. *Mol Breed* **39**: 25.
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57. Singh S, Khan S, Padaria JC & Solanke AU. 2018. Evaluation of finger millet as heat tolerant crop using physiological and biochemical assays. *J AgriSearch* **5**:110-115.
58. Singh S, Kumar NR, Maniraj R, Lakshmikanth R, Rao KYS, Muralimohan N, Arulprakash T, Karthik K, Shashibhushan NB, Vinutha T, Pattanayak D, Dash PK, Kumar PA & Sreevathsa R. 2018. Expression of Cry2Aa, a *Bacillus thuringiensis* insecticidal protein in transgenic pigeonpea confers resistance to gram pod borer, *Helicoverpa armigera*. *Sci Rep* **8**: 8820.
59. Sinha S, Sevanthi V, Chaudhary S, Tyagi P, Venkadesan S, Rani M & Mandal P. 2018. Transcriptome analysis of two rice varieties contrasting for nitrogen use efficiency under chronic N starvation reveals differences in chloroplast and starch metabolism-related genes. *Genes* **9**: 206.
60. Sinha SK & Kumar A. 2018. Condensed tannin: a major anti-nutritional constituent of faba bean (*Vicia faba* L.). *Hort Int J* **2**:32-33.
61. Sinha SK, Rani M, Kumar A, Kumar S, Venkatesh K & Mandal PK. 2018. Natural variation in root system architecture in diverse wheat genotypes grown under different nitrate conditions and root growth media. *Theor Exp Plant Physiol* **30**: 223-234.
62. Sinha SK, Tyagi A & Mandal PK. 2018. External nitrogen and carbon source-mediated response on modulation of root system architecture and nitrate uptake in wheat seedlings. *J Plant Growth Regulation* (<https://doi.org/10.1007/s00344-018-9840-9>).
63. Sreeman S, Vijayaraghavareddy P, Sreevathsa R, Rajareddy S, Arakesh S, Bharti P, & Soolanayakanahally R. 2018. Introgression of physiological traits for a comprehensive improvement of drought adaptation in crop plants. *Front Chem* **6**: 92.
64. Sureshkumar V, Dutta B, Kumar V, Prakash G, Mishra DC, Chaturvedi KK, Rai A, Sevanthi AM & Solanke AU. 2019. RiceMetaSysB: a database of blast and bacterial blight responsive genes in rice and its utilization in identifying key blast-resistant WRKY genes. *Database*.
65. Taloh A, Dantuluri DS, Kumar G, Padaria JC, Kumar S & Rao GP. 2018. Elimination of 'Candidatus Phytoplasma aurantifolia' from *Chrysanthemum morifolium* cv. Ajay Orange by application of oxytetracycline by *in vivo* and *in vitro* treatments. *Phytopath Mollicutes* **8**: 1-7.
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in genetic diversity analysis. *Indus Crops Products* **133**: 221-231.

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## Review Articles

1. Gupta NC, Rao M & Sharma P. 2018. Molecular aspects of *Sclerotinia* stem rot disease management

in oilseed crops. *Int J Microbiol Res* **4**: 1166-1170.

2. Kumar PR, Singh CB, Gupta AK, Kumar K. 2018. Quality status of wheat seed in eastern India and the early vegetative growth of wheat in relation to the size of the seed. *J Comm Mob & Sustain Dev* **13**: 98-106.
3. Sajad A, Ganai BA, Kamili A, Bhat AA, Mir ZA, Bhat JA, Tyagi A, Islam ST, Mushtaq M, Yadav P, Rawat S & Grover A. 2018. Pathogenesis-related proteins and peptides as promising tools for engineering plants with multiple stress tolerance. *Microbiol Res* **212**: 29-37.
4. Tribhuvan KU, Sandhya, Kumar K, Sevanthi AM & Gaikwad K. 2018. MutMap: a versatile tool for identification of mutant loci and mapping of genes. *Ind J Plant Physiol* **23**: 612-621.
5. Watts A, Kumar V, Raipuria RK & Bhattacharya RC. 2018. *In Vivo* haploid production in crop plants: methods and challenges. *Plant MolBiol Rep* **36**: 685-694.

## Book Chapters

1. Barthakur S & Kulshreshta N. 2018. An optimized universal protocol for isolation of DNA from varied agricultural waste biomass and biocompost samples for downstream molecular analyses In: *Advances in Compost Production Technology*, pp. 212-220, ICAR-IARI, New Delhi.
2. Jain PK, Bhattacharya RC, Kohli D, Aminedi R & Agrawal PK. 2018. RNAi for resistance against biotic stresses in crop plants. In: *Biotechnologies of Crop Improvement* (Gosal SS & Wani SH, eds.), Springer International Publishing AG ([https://doi.org/10.1007/978-3-319-90650-8\\_4](https://doi.org/10.1007/978-3-319-90650-8_4)).
3. Prakash C, Sevanthi AM & Shanmugavadivel PS. 2019. Use of QTLs in developing abiotic stress tolerance in Rice. . In: *Advances in Rice Research for Abiotic Stress Tolerance* (Hasanuzzaman M, Fujita M, Nahar K & Biswas JK, eds.), pp. 942, Woodhead Publishing, Elsevier.
4. Sandhya, Tyagi A & Kumar S. 2018. Smart mesoporous nanomaterials with improved therapeutic applications. In: *Multifunctional Nanocarriers for Contemporary Healthcare Applications*, pp. 431-447, IGI Global.
5. Sevanthi AM, Prakash C & Shanmugavadivel PS. 2019. Recent progress in rice varietal

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# Important Committees

## Research Advisory Committee (RAC)

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